ASYMCHEM	Technical Summary for LSN4182913 (ASYM-140019) Step1 PAR Study	Document No.: TR-1400190305.01

Technical Summary for LSN4182913 (ASYM-140019) Step 1 PAR Study

Document No.: TR-1400190301 Version No.01

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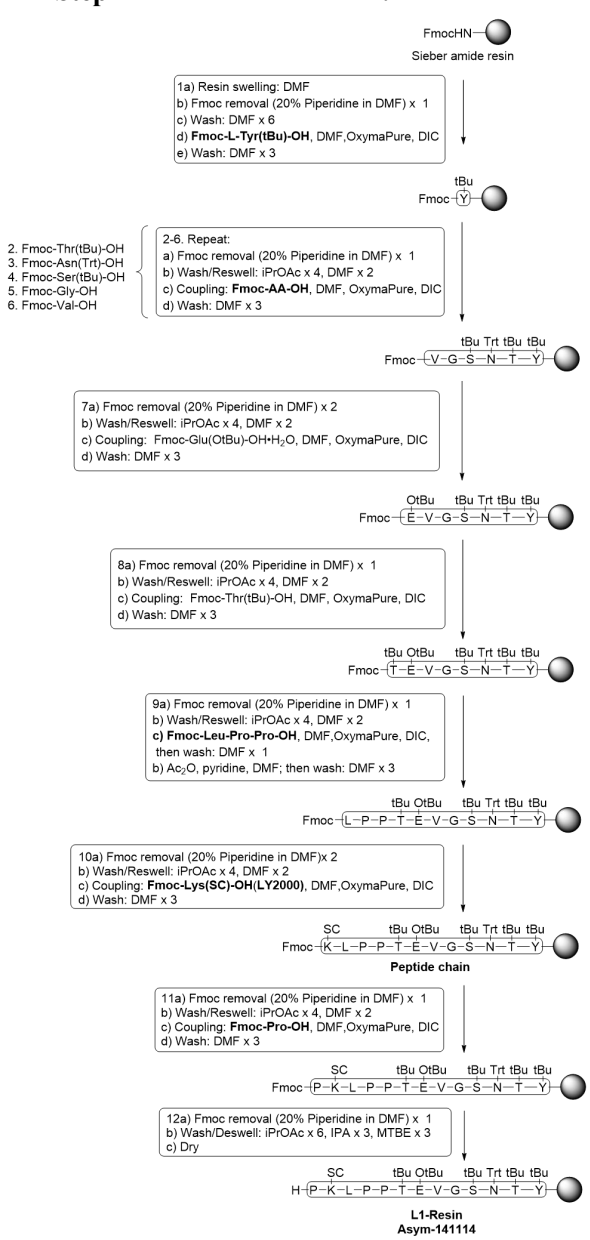
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1. Introduction

This technical summary documents the PAR and hold time studies for Step 1 of LSN4182913 (ASYM-140019). The content covers baseline process overview, PAR study experimental design, experimental data, mass spectrometric impurity analysis, and corresponding conclusions.

1.1 Process Flow Diagram

Step 1 peptide-on-resin ASYM-141114 was synthesized from Sieber Amide resin and amino acids (Fmoc-L-Tyr(tBu)-OH, Fmoc-L-Thr(tBu)-OH, Fmoc-L-Asn(Trt)-OH, Fmoc-L-Ser(tBu)-OH, Fmoc-Gly-OH, Fmoc-L-Val-OH, Fmoc-L-Glu-(OtBu)-OH·H_2O, Fmoc-L-Thr(tBu)-OH Fmoc-L-Leu-L-Pro-L-Pro-OH, Side chain LY2000/ASYM-140400, Fmoc-L-Pro-OH,) in multiple cycles mediated by condensing reagents. The synthetic route of **Step 1** is shown as Scheme **1.**



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2. Study Objective

Based on Lilly's experimental design table, an Initial Experiment Design was employed to evaluate some of key factors: resin substitution level, amino acid coupling reagent equivalents, solvent volume, coupling temperature, Fmoc-deprotection solvent volume, deprotection temperature and deprotection time. A series of experimental runs were conducted—including 16 SPPS reactions and corresponding 16 soft cleavage reactions, and data were collected as well. The related experimental data are summarized as follows.

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2.1 SPPS Experimental Setup Diagram

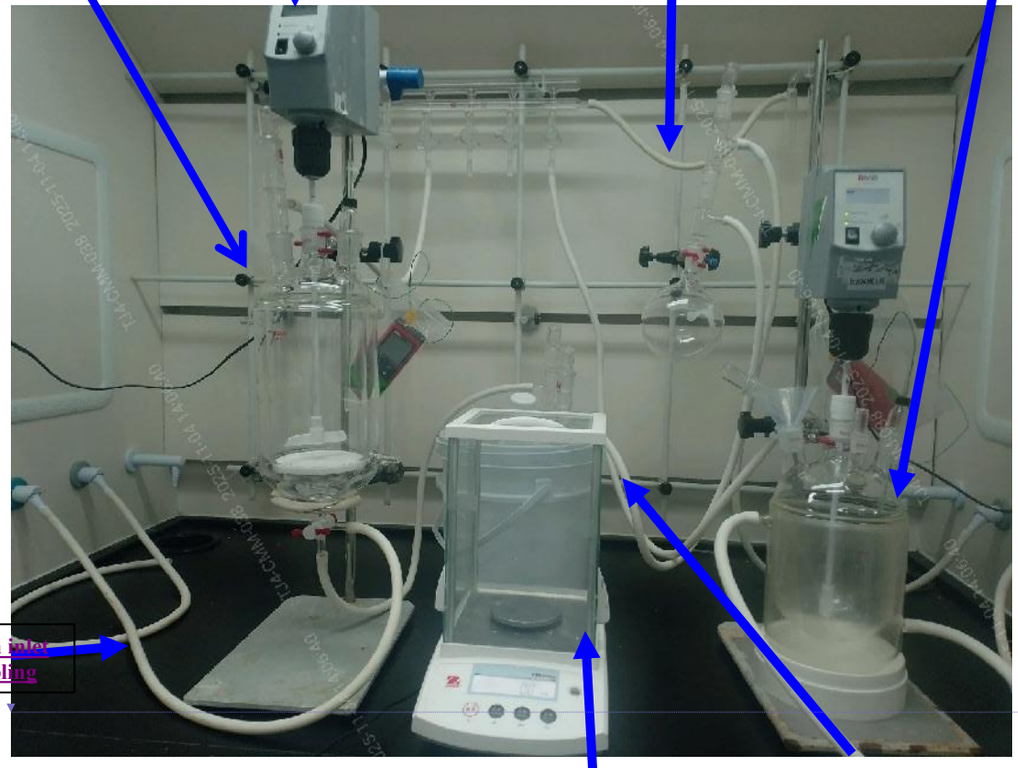
SPPS synthesizer(2L) with thermometer and jacket

Overhead agitator with stirring paddle and stirring speed (0-700rpm).

Nitrogen protection for SPPS and activation vessel.

Activation vessel with thermometer and jacket(1L)

High and low temperature circulator





Nitrogen inlet for bubbling

Balance

vacuum pumping line with 3L flask for waste filtrate collecting

(medium: Ethylene glycol: water=1:1; -30 ~ 80℃)

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Piperidine solution makeup vessel type: Firstly prepare bulk uses of 20% piperidine/DMF solution in 3-neck flask, then transfer to teflon barrel for storage.
Piperidine solution vessel size: 5L/10L/20L
How is it measured: Perform weighing with an electronic balance.
How is it transferred to synthesizer: Charge through peristaltic pump tubing.
How is it drained: Drained by vacuum through the bottom connecting pipeline under nitrogen protection, the filtrate was collected in 5L flask through pipeline.
How is temperature controlled: Controlled jacket temperature through dedicated circulator (high and low temperature circulation device).

2. DeFmoc Washes

How is it measured: Perform weighing with an electronic balance.
How is it transferred to synthesizer: Charge through peristaltic pump tubing.
How is it drained: Drained by vacuum through the bottom connecting pipeline. The filtrate was collected in 5L flask through pipeline.
How is temperature controlled: Controlled jacket temperature through dedicated circulator.

3. Activation

Activation vessel type: 4-neck jacketed glass bottle
Activation vessel size: 1L
How is AA measured: weight from balance
How is AA transferred to activation vessel: Charge through peristaltic pump tubing
How is oxyma measured: weight from balance.
How is oxyma transferred to activation vessel: Poured through funnel.
How is DIC measured: weight from balance
How is DIC transferred to activation vessel: graduated dropper
How is it mixed: overhead agitator.
How is temperature controlled: Controlled jacket temperature through dedicated circulator.
How is it washed (is the wash part of the coupling wash?)
Yes, rinse the inner walls of the reactor with DMF using a wash bottle, directing the flow along the walls. Stir for 3~7 minutes, then transfer the resulting wash solution to the reactor for washing subsequently to the reactor was wash waste container for disposal. Repeat this procedure three times.

4. Coupling

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How is activated solution transferred to synthesizer: Charge through peristaltic pump tubing
How is it mixed: Mixed through mechanical stirring device(overhead agitator) in 4-neck jacketed glass bottle
How is temperature controlled: Controlled jacket temperature through dedicated circulator.
How is it drained: Drained by vacuum through the bottom connecting pipeline.
How is sample taken: use graduated dropper to absorb mixture and then transfer to test tube.

5. Coupling Washes

How is it measured: Measured through glass graduated cylinder.
How is it transferred to activation vessel: Poured through funnel.
How is it transferred to synthesizer: Weigh on an electronic scale.

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3. Study Review

3.1 Initial Step 1 Baseline process description

Table 3.1.1 Summary of SPPS Process PAR Experiment Description

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#	Unit operation name	Process Description
1	Prepare 20% Piperidine/DMF Bulk Solution	Prepare 20% piperidine of DMF (v/v%) solution:
2		Add DMF (75.844g/g) to a 10L flask, then start the agitator, and charge (17.26g/g) piperidine. Stir for 30min at 20 °C. Transfer the piperidine/DMF solution into 5L/10L PTFE bottle (N_2 displacement), store at 20 °C.
3	Equipment chain rinsing	1. Add 8.0 V (7.584 g/g) DMF into the 1L activation flask, and then transfer into the 2L synthetic flask. Start the agitator, bubble with nitrogen for NLT 8 min. Stop stirring, turn off the nitrogen, and drain until there is no obvious dripping. 2. Repeat this operation once again.

#	Unit operation name	Process Description
4	Resin swelling& Resin wash	1. Add 1.0 g/g Sieber Amide resin into 2L synthetic flask jacket temp to 25°C. 2. Add 8V (7.584 g/g) DMF into the 2L synthetic flask. Start the agitator, bubble with nitrogen for NLT 8 min. Stop stirring, turn off the nitrogen, and drain until there is no obvious dripping. Repeat again. 3. Add 7V (6.636 g/g) DMF into the 2L synthetic flask. Start the agitator and bubble with nitrogen for NLT 5 min. Turn off the nitrogen. 4. Rinse 2L synthetic flask wall with 1V DMF. Adjust the mixture to 25°C, stop stirring, turn off the nitrogen, and drain until there is no obvious dripping.
5	DeFmoc (pre Tyr37)	1. Add 6V (5.585g/g) piperidine in DMF into the 2L synthetic flask. Start the agitator and bubble with nitrogen to 25℃. 2. After stirring at 25°C for 120 min, stop stirring and turn off the nitrogen. Drain until there is no obvious dripping
6	Post de-Fmoc washing	1. Set jacket temp to 20℃. 2. Add 8V (7.584 g/g) DMF into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 20 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 3. Repeat Instruction operation 6-2 five more times.
7	Tyr37 activation	1. Add 6V (5.688g/g) DMF into the 1L activation flask to 20℃. Start the agitator. 2. Add 2.0eq Fmoc-L-Tyr (tBu)-OH and 2.0eq OxymaPure with solid feeding funnel into 1L activation flask wall 3. Rinse the solid feeding funnel and 1L activation flask wall with 0.5 V (0.474g/g) DMF. Stir until solids dissolve clearly by visual check. 4. Adjust the internal TEMP in 1L activation flask to 18℃. 5. Add 2.2eq DIC to the mixture, maintaining the internal temperature not exceeding 18 °C. At production scale, the reference addition time is 3-5 minutes. 6. Rinse the feeding funnel and 1L activation flask wall with 0.5V (0.474g/g) DMF. Adjust internal temp to 20 ℃. Stir for 30min
8	Tyr37 coupling	1. Transfer the mixture in 1L activation flask into 2L synthetic flask. Rinse the wall of 1L activation flask using 1 V (0.948g/g) DMF, and transfer solution into 2L synthetic flask. 2. Start the stirrer. Bubble mixture with nitrogen for NLT 5min, adjust the mixture to 25°C. 3. Stir for 240 min, then stop stirring and turn off the nitrogen. 4. Sample for Ninhydrin test analysis. 5. Drain until there is no reaction mixture drip.

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#	Unit operation name	Process Description
9	Post coupling washing	1. After draining resin post-coupling, rinse the 1L activation flask with 7V (6.636 g/g) DMF and use washing liquor to wash resin and 2L synthetic flask walls. Bubble with nitrogen, after 5min, start the agitator. After stirring for 7min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 2. Repeat Instruction operation 1 twice.
10	DeFmoc (pre Thr36)	1. Add 6V (5.585g/g) piperidine in DMF into the 2L synthetic flask. Start the agitator and bubble with nitrogen to 25℃. 2. After stirring at 25°C for 120 min, stop stirring and turn off the nitrogen. Drain until there is no obvious dripping
11	Post de-Fmoc washing	1. Set jacket temp to 20℃. 2. Add 10V (8.730 g/g) IPAC into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 10 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 3. Repeat Instruction operation 2 two more times. 4. Add 10V (9.480 g/g) DMF into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 20 min, stop stirring, turn off the nitrogen, drain until there is no obvious drip. 5. Repeat Instruction operation 4-5 one more time.
12	Thr36 activation	1. Add 6V (5.688g/g) DMF into the 1L activation flask to 20℃. Start the agitator. 2.Add 2.0eq Fmoc-L-Thr (tBu)-OH and 2.0eq OxymaPure with solid feeding funnel into 1L activation flask wall 3. Rinse the solid feeding funnel and 1L activation flask wall with 0.5 V (0.474g/g) DMF. Stir until solids dissolve clearly by visual check. 4. Adjust the internal TEMP in 1L activation flask to 18℃ 5. Add 2.2eq DIC to the mixture, maintaining the internal temperature not exceeding 18 °C. At production scale, the reference addition time is 3-5 minutes. 6. Rinse the feeding funnel and 1L activation flask wall with 0.5V (0.474g/g) DMF. Adjust internal temp to 20 ℃. Stir for 30min
13	Thr36 coupling	1. Transfer the mixture in 1L activation flask into 2L synthetic flask. Rinse the wall of 1L activation flask using 1 V (0.948g/g) DMF, and transfer solution into 2L synthetic flask. 2. Start the stirrer. Bubble mixture with nitrogen for NLT 5min, adjust the mixture to 25°C. 3. Stir for 180 min, then the sample for Ninhydrin test analysis(IPC). Take a sample every 60 minutes. The reaction is considered complete after two consecutive passing results. Record the time, then stop stirring and turn off the nitrogen. 4. Drain until there is no reaction mixture drip.

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#	Unit operation name	Process Description
14	Post coupling washing	1. After draining resin post-coupling, rinse the 1L activation flask with 7V (6.636 g/g) DMF and use washing liquor to wash resin and 2L synthetic flask walls. Bubble with nitrogen, after 5min, start the agitator. After stirring for 7min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 2. Repeat Instruction operation 1 twice.
15	DeFmoc (pre Asn35)	1. Add 6V (5.585g/g) piperidine in DMF into the 2L synthetic flask. Start the agitator and bubble with nitrogen to 25℃. 2. After stirring at 25°C for 120 min, stop stirring and turn off the nitrogen. Drain until there is no obvious dripping
16	Post de-Fmoc washing	1. Set jacket temp to 20℃. 2. Add 10V (8.730 g/g) IPAC into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 10 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 3. Repeat Instruction operation 2 two more times. 4. Add 10V (9.480 g/g) DMF into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 20 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 5. Repeat Instruction operation 4 one more time.
17	Asn35 activation	1. Add 6V (5.688g/g) DMF into the 1L activation flask to 20℃. Start the agitator. 2. Add 2.0eq Fmoc-L-Asn(Trt)-OH and 2.0eq OxymaPure with solid feeding funnel into 1L activation flask wall 3. Rinse the solid feeding funnel and 1L activation flask wall with 0.5 V (0.474g/g) DMF. Stir until solids dissolve clearly by visual check. 4. Adjust the internal TEMP in 1L activation flask to 18℃ 5. Add 2.2eq DIC to the mixture, maintaining the internal temperature not exceeding 18 °C. At production scale, the reference addition time is 3-5 minutes. 6. Rinse the feeding funnel and 1L activation flask wall with 0.5V (0.474g/g) DMF. Adjust internal temp to 20 ℃. Stir for 30min
18	Asn35 coupling	1. Transfer the mixture in 1L activation flask into 2L synthetic flask. Rinse the wall of 1L activation flask using 1 V (0.948g/g) DMF, and transfer solution into 2L synthetic flask. 2. Start the stirrer. Bubble mixture with nitrogen for NLT 5min, adjust the mixture to 25°C. 3. Stir for 180 min, then stop stirring and turn off the nitrogen. 4. Sample for Ninhydrin test analysis 5. Take a sample every 60 minutes. The reaction is considered complete after two consecutive passing results. Record the time.

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#	Unit operation name	Process Description
		6. Drain until there is no reaction mixture drip.
19	Post coupling washing	1. After draining resin post-coupling, rinse the 1L activation flask with 7V (6.636 g/g) DMF and use washing liquor to wash resin and 2L synthetic flask walls. Bubble with nitrogen, after 5min, start the agitator. After stirring for 7min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 2. Repeat Instruction operation 1 twice.
20	DeFmoc (pre Ser34)	1. Add 6V (5.585g/g) piperidine in DMF into the 2L synthetic flask. Start the agitator and bubble with nitrogen to 25℃. 2. After stirring at 25°C for 120 min, stop stirring and turn off the nitrogen. Drain until there is no obvious dripping
21	Post de-Fmoc washing	1. Set jacket temp to 20℃. 2. Add 10V (8.730 g/g) IPAC into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 10 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 3. Repeat Instruction operation 2 two more times. 4. Add 10V (9.480 g/g) DMF into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 20 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 5. Repeat Instruction operation 4 one more time.
22	Ser34 activation	1. Add 6V (5.688g/g) DMF into the 1L activation flask to 20℃. Start the agitator. 2. Add 2.0eq Fmoc-L-Ser(tBu)-OH and 2.0eq OxymaPure with solid feeding funnel into 1L activation flask wall 3. Rinse the solid feeding funnel and 1L activation flask wall with 0.5 V (0.474g/g) DMF. Stir until solids dissolve clearly by visual check. 4. Adjust the internal TEMP in 1L activation flask to 18℃ 5. Add 2.2eq DIC to the mixture, maintaining the internal temperature not exceeding 18 °C. At production scale, the reference addition time is 3-5 minutes. 6. Rinse the feeding funnel and 1L activation flask wall with 0.5V (0.474g/g) DMF. Adjust internal temp to 20 ℃. Stir for 30min
23	Ser34 coupling	1. Transfer the mixture in 1L activation flask into 2L synthetic flask. Rinse the wall of 1L activation flask using 1 V (0.948 g/g) DMF, and transfer solution into 2L synthetic flask. 2. Start the stirrer. Bubble mixture with nitrogen for NLT 5min, adjust the mixture to 25°C. 3. Stir for 180 min, then stop stirring and turn off the nitrogen. 4. Sample for Ninhydrin test analysis

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#	Unit operation name	Process Description
		5. Take a sample every 60 minutes. The reaction is considered complete after two consecutive passing results. Record the time. 6. Drain until there is no reaction mixture drip.
24	Post coupling washing	1. After draining resin post-coupling, rinse the 1L activation flask with 7V (6.636 g/g) DMF and use washing liquor to wash resin and 2L synthetic flask walls. Bubble with nitrogen, after 5min, start the agitator. After stirring for 7min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 2. Repeat Instruction operation 1 twice.
25	DeFmoc (pre Gly33)	1. Add 6V (5.585g/g) piperidine in DMF into the 2L synthetic flask. Start the agitator and bubble with nitrogen to 25℃. 2. After stirring at 25°C for 120 min, stop stirring and turn off the nitrogen. Drain until there is no obvious dripping
26	Post de-Fmoc washing	1. Set jacket temp to 20℃. 2. Add 10V (8.730 g/g) IPAC into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 10 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 3. Repeat Instruction operation 2 two more times. 4. Add 10V (9.480 g/g) DMF into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 20 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 5. Repeat Instruction operation 4 one more times.
27	Gly33 activation	1. Add 6V (5.688g/g) DMF into the 1L activation flask to 20℃. Start the agitator. 2. Add 2.0eq Fmoc-L-Gly-OH and 2.0eq OxymaPure with solid feeding funnel into 1L activation flask wall 3. Rinse the solid feeding funnel and 1L activation flask wall with 0.5 V (0.474g/g) DMF. Stir until solids dissolve clearly by visual check. 4. Adjust the internal TEMP in 1L activation flask to 18℃ 5. Add 2.2eq DIC to the mixture, maintaining the internal temperature not exceeding 18 °C. At production scale, the reference addition time is 3-5 minutes. 6. Rinse the feeding funnel and 1L activation flask wall with 0.5V (0.474g/g) DMF. Adjust internal temp to 20 ℃. Stir for 30min
28	Gly33 coupling	1. Transfer the mixture in 1L activation flask into 2L synthetic flask. Rinse the wall of 1L activation flask using 1 V (0.948g/g) DMF, and transfer solution into 2L synthetic flask. 2. Start the stirrer. Bubble mixture with nitrogen for NLT 5min, adjust the mixture to 25°C. 3. Stir for 180 min, then stop stirring and turn off the nitrogen.

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#	Unit operation name	Process Description
		4. Sample for Ninhydrin test analysis 5. Take a sample every 60 minutes. The reaction is considered complete after two consecutive passing results. Record the time. 6. Drain until there is no reaction mixture drip.
29	Post coupling washing	1. After draining resin post-coupling, rinse the 1L activation flask with 7V (6.636 g/g) DMF and use washing liquor to wash resin and 2L synthetic flask walls. Bubble with nitrogen, after 5min, start the agitator. After stirring for 7min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 2. Repeat Instruction operation 1 twice.
30	DeFmoc (pre Val32)	1. Add 6V (5.585g/g) piperidine in DMF into the 2L synthetic flask. Start the agitator and bubble with nitrogen to 25℃. 2. After stirring at 25°C for 120 min, stop stirring and turn off the nitrogen. Drain until there is no obvious dripping
31	Post de-Fmoc washing	1. Set jacket temp to 20℃. 2. Add 10V (8.730 g/g) IPAC into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 10 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 3. Repeat Instruction operation 2 two more times. 4. Add 10V (9.480 g/g) DMF into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 20 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 5. Repeat Instruction operation 4 one more time.
32	Val32 activation	1. Add 6V (5.688g/g) DMF into the 1L activation flask to 20℃. Start the agitator. 2. Add 2.0eq Fmoc-L-Val-OH and 2.0eq OxymaPure with solid feeding funnel into 1L activation flask wall 3. Rinse the solid feeding funnel and 1L activation flask wall with 0.5 V (0.474g/g) DMF. Stir until solids dissolve clearly by visual check. 4. Adjust the internal TEMP in 1L activation flask to 18℃ Add 2.2eq DIC to the mixture, maintaining the internal temperature not exceeding 18 °C. At production scale, the reference addition time is 3-5 minutes. Rinse the feeding funnel and 1L activation flask wall with 0.5V (0.474g/g) DMF. Adjust internal temp to 20 ℃. Stir for 30min
33	Val32 coupling	1. Transfer the mixture in 1L activation flask into 2L synthetic flask. Rinse the wall of 1L activation flask using 1 V (0.948g/g) DMF, and transfer solution into 2L synthetic flask. 2. Start the stirrer. Bubble mixture with nitrogen for NLT 5min, adjust the mixture to 25°C.

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#	Unit operation name	Process Description
		3. Stir for 180 min, then stop stirring and turn off the nitrogen. 4. Sample for Ninhydrin test analysis 5. Take a sample every 60 minutes. The reaction is considered complete after two consecutive passing results. Record the time. 6. Drain until there is no reaction mixture drip.
33	Post coupling washing	1. After draining resin post-coupling, rinse the 1L activation flask with 7V (6.636 g/g) DMF and use washing liquor to wash resin and 2L synthetic flask walls. Bubble with nitrogen, after 5min, start the agitator. After stirring for 7min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 2. Repeat Instruction operation 1 twice.
34	DeFmoc (pre Glu31)	1. Add 6V (5.585g/g) piperidine in DMF into the 2L synthetic flask. Start the agitator and bubble with nitrogen to 25℃. 2. After stirring at 25°C for 30 min, stop stirring and turn off the nitrogen. Drain until there is no obvious dripping 3. Add 6V (5.585g/g) piperidine in DMF into the 2L synthetic flask. Start the agitator and bubble with nitrogen to 25℃. 4. After stirring at 25°C for 30 min, stop stirring and turn off the nitrogen. Drain until there is no obvious dripping
35	Post de-Fmoc washing	1. Set jacket temp to 20℃. 2. Add 10V (8.730g/g) IPAC into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 10 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 3. Repeat Instruction operation 2 two more times. 4. Add 10V (9.480g/g) DMF into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 20 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 5. Repeat Instruction operation 4 one more time.
36	Glu31 activation	1. Add 6V (5.688g/g) DMF into the 1L activation flask to 20℃. Start the agitator. 2. Add 2.0eq Fmoc-Glu(OtBu)-OH•H2O and 2.0eq OxymaPure with solid feeding funnel into 1L activation flask wall 3. Rinse the solid feeding funnel and 1L activation flask wall with 0.5 V (0.474g/g) DMF. Stir until solids dissolve clearly by visual check. 4. Adjust the internal TEMP in 1L activation flask to 18℃ 5. Add 2.2eq DIC to the mixture, maintaining the internal temperature not exceeding 18 °C. At production scale, the reference addition time is 3-5 minutes. 6. Rinse the feeding funnel and 1L activation flask wall with 0.5V (0.474g/g) DMF. Adjust internal temp to 20 ℃. Stir for 30min

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#	Unit operation name	Process Description
37	Glu31 coupling	1. Transfer the mixture in 1L activation flask into 2L synthetic flask. Rinse the wall of 1L activation flask using 1 V (0.948g/g) DMF, and transfer solution into 2L synthetic flask. 2. Start the stirrer. Bubble mixture with nitrogen for NLT 5min, adjust the mixture to 25°C. 3. Stir for 180 min, then stop stirring and turn off the nitrogen. 4. Sample for Ninhydrin test analysis 5. Take a sample every 60 minutes. The reaction is considered complete after two consecutive passing results. Record the time. 6. Drain until there is no reaction mixture drip.
38	Post coupling washing	1. After draining resin post-coupling, rinse the 1L activation flask with 7V (6.636 g/g) DMF and use washing liquor to wash resin and 2L synthetic flask walls. Bubble with nitrogen, after 5min, start the agitator. After stirring for 7min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 2. Repeat Instruction operation 1 twice.
39	DeFmoc (pre Thr30)	1. Add 6V (5.585g/g) piperidine in DMF into the 2L synthetic flask. Start the agitator and bubble with nitrogen to 25℃. 2. After stirring at 25°C for 120 min, stop stirring and turn off the nitrogen. Drain until there is no obvious dripping
40	Post de-Fmoc washing	1. Set jacket temp to 20℃. 2. Add 10V (8.730 g/g) IPAC into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 10 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 3. Repeat Instruction operation 2 two more times. 4. Add 10V (9.480 g/g) DMF into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 20 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 5. Repeat Instruction operation 4 one more time.
41	Thr30 activation	1. Add 6V (5.688g/g) DMF into the 1L activation flask to 20℃. Start the agitator. 2. Add 2.0eq Fmoc-Thr(tBu)-OH and 2.0eq OxymaPure with solid feeding funnel into 1L activation flask wall 3. Rinse the solid feeding funnel and 1L activation flask wall with 0.5 V (0.474g/g) DMF. Stir until solids dissolve clearly by visual check. 4. Adjust the internal TEMP in 1L activation flask to 18℃ 5. Add 2.2eq DIC to the mixture, maintain the internal temperature not exceeding 18 °C. At production scale, the reference addition time is 3-5 minutes.

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#	Unit operation name	Process Description
		5. Rinse the feeding funnel and 1L activation flask wall with 0.5V (0.474g/g) DMF. Adjust internal temp to 20 ℃. Stir for 30min
42	Thr30 coupling	1. Transfer the mixture in 1L activation flask into 2L synthetic flask. Rinse the wall of 1L activation flask using 1 V (0.948g/g) DMF, and transfer solution into 2L synthetic flask. 2. Start the stirrer. Bubble mixture with nitrogen for NLT 5min, adjust the mixture to 25°C. 3. Stir for 180 min, then stop stirring and turn off the nitrogen. 4. Sample for Ninhydrin test analysis 5. Take a sample every 60 minutes. The reaction is considered complete after two consecutive passing results. Record the time. 6. Drain until there is no reaction mixture drip.
43	Post coupling washing	1. After draining resin post-coupling, rinse the 1L activation flask with 7V (6.636 g/g) DMF and use washing liquor to wash resin and 2L synthetic flask walls. Bubble with nitrogen, after 5min, start the agitator. After stirring for 7min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 2. Repeat Instruction operation 1 twice.
44	DeFmoc (pre-Leu27-Pro28-Pro29)	1. Add 6V (5.585g/g) piperidine in DMF into the 2L synthetic flask. Start the agitator and bubble with nitrogen to 25℃. 2. After stirring at 25°C for 120 min, stop stirring and turn off the nitrogen. Drain until there is no obvious dripping.
45	Post de-Fmoc washing	1. Set jacket temp to 20℃. 2. Add 10V (8.730 g/g) IPAC into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 10 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 3. Repeat Instruction operation 2 two more times. 4. Add 10V (9.480 g/g) DMF into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 20 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 5. Repeat Instruction operation 4 one more time.
46	Leu27-Pro28-Pro29 activation	1. Add 6V (5.688g/g) DMF into the 1L activation flask to 20℃. Start the agitator. 2. Add 2.0eq Fmoc-Leu-Pro-Pro-OH and 3.0eq OxymaPure with solid feeding funnel into 1L activation flask wall 3. Rinse the solid feeding funnel and 1L activation flask wall with 0.5 V (0.474g/g) DMF. Stir until solids dissolve clearly by visual check. 4.Adjust the internal TEMP in 1L activation flask to 18℃

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#	Unit operation name	Process Description
		5. Add 3.3eq DIC to the mixture, maintaining the internal temperature not exceeding 18 °C. At production scale, the reference addition time is 3-5 minutes. 6. Rinse the feeding funnel and 1L activation flask wall with 0.5 V (0.474g/g) DMF. Adjust internal temp to 20 ℃. Stir for 30min
47	Leu27-Pro28-Pro29 coupling	1. Transfer the mixture in 1L activation flask into 2L synthetic flask. Rinse the wall of 1L activation flask using 1 V (0.948g/g) DMF, and transfer solution into 2L synthetic flask. 2. Start the stirrer. Bubble mixture with nitrogen for NLT 5min, adjust the mixture to 35°C. 3. Stir for 360 min, then stop stirring and turn off the nitrogen. 4. Sample for Ninhydrin test analysis 5. Take a sample every 60 minutes. The reaction is considered complete after two consecutive passing results. Record the time. 6. Drain until there is no reaction mixture drip. 7. This is stated below: After draining resin post-coupling, rinse the 1L activation flask with DMF and use washing liquor to wash resin and 2L synthetic flask walls. Bubble with nitrogen, after 5min, start the agitator. After stirring for 7min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip.
48	Post coupling washing	1. After draining resin post-coupling, rinse the 1L activation flask with 7V (6.636 g/g) DMF and use washing liquor to wash resin and 2L synthetic flask walls. Bubble with nitrogen, after 5min, start the agitator. After stirring for 7min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip.
49	Ac_2O Capping Solution	1. Add 5V (4.740g/g) DMF into 1L flask. Start the stirrer. 2. Add 1.25eq acetic anhydride into 1L flask 3. Add 1.5eq pyridine into 1L activation flask 4. Bubble with nitrogen, after 5min, start the agitator. Warm the mixture to Cap to 25℃ 5. After 7 min stir time, transfer the mixture in 1L activation flask into 2L synthetic flask. Rinse the wall of 1L activation flask using 1V (0.948g/g) DMF, and transfer solution into 2L synthetic flask.
50	Ac_2O Capping& Post capping washing	1. Bubble mixture with nitrogen for 5min, then start the stirrer. Warm the mixture to 25℃, stir for 60min. Stop stirring, and turn off the nitrogen. Drain until there is no drip. 2. After draining resin post-coupling, rinse the 1L activation flask with 7V (6.636g/g) DMF and use washing liquor to wash resin and 2L synthetic flask walls. Bubble with nitrogen, after 5min, start the agitator. After stirring for 7min, stop stirring, turn off the nitrogen. Drain until

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#	Unit operation name	Process Description
		there is no obvious drip. 3. Repeat operation 2 twice
51	2 x deFmoc (pre-LY2000 - Lys26)	1. Add 6V (5.585g/g) piperidine in DMF into the 2L synthetic flask. Start the agitator and bubble with nitrogen to 25℃. 2. After stirring at 25°C for 30 min, stop stirring and turn off the nitrogen. Drain until there is no obvious dripping 3. Repeat operation 2 one more time.
52	Post deFmoc washing	1. Set jacket temp to 20℃. 2. Add 10V (8.730 g/g) IPAC into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 10 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 3. Repeat Instruction operation 2 two more times 4. Add 10V (9.480 g/g) DMF into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 20 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 5. Repeat Instruction operation 4 one more time
53	LY2000 - Lys26 activation	1. Add 6V (5.688g/g) DMF into the 1L activation flask to 20℃. Start the agitator. 2. Add 1.5eq LY2000 and 3.0eq OxymaPure with solid feeding funnel into 1L activation flask wall 3. Rinse the solid feeding funnel and 1L activation flask wall with 1V (0.948g/g) DMF. Stir until solids dissolve clearly by visual check. 4. Adjust the internal TEMP in 1L activation flask to 18℃ 5. Add 3.3eq DIC to the mixture, maintaining the internal temperature not exceeding 18 °C. At production scale, the reference addition time is 3-5 minutes. 6. Rinse the feeding funnel and 1L activation flask wall with 0.5V (0.474g/g) DMF. Adjust internal temp to 20 ℃. Stir for 30min
54	LY2000 - Lys26 coupling	1. Transfer the mixture in 1L activation flask into 2L synthetic flask. Rinse the wall of 1L activation flask using 1 V (0.948g/g) DMF, and transfer solution into 2L synthetic flask. 2. Start the stirrer. Bubble mixture with nitrogen for NLT 5min, adjust the mixture to 30°C. 3. Stir for 480 min, then stop stirring and turn off the nitrogen. 4. Sample for Ninhydrin test analysis 5. Take a sample every 60 minutes. The reaction is considered complete after two consecutive passing results. Record the time. 6. Drain until there is no reaction mixture drip.

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#	Unit operation name	Process Description
55	Post coupling washing	1. After draining resin post-coupling, rinse the 1L activation flask with 7V (6.636 g/g) DMF and use washing liquor to wash resin and 2L synthetic flask walls. Bubble with nitrogen, after 5min, start the agitator. After stirring for 7min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 2. Repeat Instruction operation 1 twice.
56	DeFmoc (pre Pro25)	1. Add 6V (5.585g/g) piperidine in DMF into the 2L synthetic flask. Start the agitator and bubble with nitrogen to 25℃. 2. After stirring at 25°C for 120 min, stop stirring and turn off the nitrogen. Drain until there is no obvious dripping
57	Post de-Fmoc washing	1. Set jacket temp to 20℃. 2. Add 10V (8.730 g/g) IPAC into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 10 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 3. Repeat Instruction operation 2 two more times. 4. Add 10V (9.480 g/g) DMF into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 20 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 5. Repeat Instruction operation 4 one more time.
58	Pro activation	1. Add 6V (5.688g/g) DMF into the 1L activation flask to 20℃. Start the agitator. 2. Add 2.0eq Fmoc-Pro-OH and 2.0eq OxymaPure with solid feeding funnel into 1L activation flask wall 3. Rinse the solid feeding funnel and 1L activation flask wall with 0.5 V (0.474g/g) DMF. Stir until solids dissolve clearly by visual check. 4. Adjust the internal TEMP in 1L activation flask to 18℃ 5. Add 2.2eq DIC to the mixture, maintaining the internal temperature not exceeding 18 °C. At production scale, the reference addition time is 3-5 minutes. 6. Rinse the feeding funnel and 1L activation flask wall with 0.5V (0.474g/g) DMF. Adjust internal temp to 20 ℃. Stir for 30min
59	Pro coupling	1. Transfer the mixture in 1L activation flask into 2L synthetic flask. Rinse the wall of 1L activation flask using 1 V (0.948g/g) DMF, and transfer solution into 2L synthetic flask. 2. Start the stirrer. Bubble mixture with nitrogen for NLT 5min, adjust the mixture to 25°C. 3. Stir for 180 min, then stop stirring and turn off the nitrogen. 4. Sample for Ninhydrin test analysis 5. Take a sample every 60 minutes. The reaction is considered complete after two consecutive passing results. Record the time.

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#	Unit operation name	Process Description
		6. Drain until there is no reaction mixture drip.
60	Post coupling washing	1. After draining resin post-coupling, rinse the 1L activation flask with 7V (6.636 g/g) DMF and use washing liquor to wash resin and 2L synthetic flask walls. Bubble with nitrogen, after 5min, start the agitator. After stirring for 7min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 2. Repeat Instruction operation 1 twice.
61	Final deFmoc	1. Add 6V (5.585g/g) piperidine in DMF into the 2L synthetic flask. Start the agitator and bubble with nitrogen to 25℃. 2. After stirring at 25°C for 120 min, stop stirring and turn off the nitrogen. Drain until there is no obvious dripping
62	Post de-Fmoc washing	1. Set jacket temp to 20℃. 2. Add 10V (8.730 g/g) IPAC into 2L synthetic flask. Bubble with nitrogen, after 5 min, start the agitator. After stirring for NLT 10 min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 3. Repeat Instruction operation 2 five more times.
63	De-swell - IPA	1. Add 8V (6.280g/g) IPA into 2L synthetic flask, washing the flask walls. Bubble with nitrogen, after 5min, start the agitator. After stirring for NLT 10min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 2. Repeat Instruction operation twice.
64	De-swell - MTBE	1. Add 8V (5.928kg/kg) MTBE into 2L synthetic flask, washing the flask walls. Bubble with nitrogen, after 5min, start the agitator. After stirring for NLT 10min, stop stirring, turn off the nitrogen. Drain until there is no obvious drip. 2. Repeat Instruction operation twice.
65	Dry	1. Sweep with nitrogen above, nitrogen flow is 10 m^3/h±4m^3/h (target: 10 m^3/h), and vacuum underneath at Tjacket≤30℃. 2. 12h later, sample the solid for LOD analysis every 6±4h until LOD≤10.0% or the difference between two consecutive samples is ≤1.0%. 3. After drying 6h, begin to stir the solid, 12h later, stop stirring. After sampling at 12 hours, stir for 2~5min every 4±1 h at 10±5rpm (reference).
66	Collect	Put In-IPC intermediate (Step 1 product peptidyl resin ASYM-141114) into double LDPE bags, inner layer of low-density polyethylene bag, purge the inside of the package with nitrogen. Label and weigh, then put into package drum(s) for temporary storage. Then placed in cold storage where the temperature is minus 2~8ºC

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Table 3.1.2 Summary of In-Process Controls (IPCs) in Step 1

Sample #	Operation NO.	In-Process Control	Tests	Acceptance Criteria
1	7	Ninhydrin test in amino acid coupling (Fmoc-L-Tyr (tBu)-OH)	Ninhydrin test	Negative
2	11	Ninhydrin test in amino acid coupling (Fmoc-L-Thr (tBu)-OH)	Ninhydrin test	Negative
3	15	Ninhydrin test in amino acid coupling (Fmoc-L-Asn(Trt)-OH and 2.0eq)	Ninhydrin test	Negative
4	19	Ninhydrin test in amino acid coupling (Fmoc-L-Ser(tBu)-OH)	Ninhydrin test	Negative
5	23	Ninhydrin test in amino acid coupling (Fmoc-L-Gly-OH)	Ninhydrin test	Negative
6	27	Ninhydrin test in amino acid coupling (Fmoc-L-Val-OH)	Ninhydrin test	Negative
7	31	Ninhydrin test in amino acid coupling (Fmoc-Glu(OtBu)-OH•H2O)	Ninhydrin test	Negative
8	35	Ninhydrin test in amino acid coupling (Fmoc-L-Thr (tBu)-OH)	Ninhydrin test	Negative
9	39	Ninhydrin test in amino acid coupling (Fmoc-Leu-Pro-Pro-OH)	Ninhydrin test	Negative
10	45	Ninhydrin test in amino acid coupling (LY2000)	Ninhydrin test	Negative

11	49	Ninhydrin test in amino acid coupling (Fmoc-Pro-OH)	Ninhydrin test	Negative
12	55	LOD under drying	LOD	≤ 10.0%
13	55	Appearance test of step 1 product peptidyl resin ASYM-141114 after collecting	Appearance	Light yellow to yellow solid

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3.2 PAR study summary

3.2.1 Initial Experiment Design

The Initial Experiment Design are referring to Lilly's L1 SPPS PAR Design and presented in **Table 3.2.1** and **Table 3.2.2**.

Table 3.2.1 Initial PAR Experiment Design of Step 1

PAR Experimental Design																				
Unit Operation	**Unit**	**Set point (@)**	**Lower limit (-)**	**Upper limit (+)**	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Piperidine %	%	20.00	15.00	25.00	@	@	+	@	-	@	@	@	@	@	@	@	+	@	@	@
Piperidine/DMF vol	V	6	5.244	6.756	@	@	+	@	-	@	@	@	@	@	@	@	+	@	@	@
DeFmoc Temp	℃	25	13	37	@	@	@	@	-	+	@	@	@	@	@	@	@	@	+	@
DeFmoc stir time (min) - except Val and trimer	Min	120	60	180	@	@	@	@	-	+	@	@	@	@	@	@	@	@	+	@
DeFmoc stir time (min) – only Val and trimer	Min	30	15	45	@	@	@	@	-	+	@	@	@	@	@	@	@	@	+	@
Post-DeFmoc wash vol (DMF)	V	8	6.992	8	@	@	@	-	@	@	@	@	@	@	@	@	-	@	@	@
Post-DeFmoc wash vol (IPAC)	V	10	8.740	10	@	@	@	-	@	@	@	@	@	@	@	@	-	@	@	@
Post-DeFmoc DMF resewell vol	V	10	8.740	10	@	@	@	@	@	@	@	@	-	@	@	@	@	@	@	@
AA eq- except LY2000	Eq	2.00	1.874	2.126	@	@	+	-	@	-	@	@	@	@	@	@	-	@	@	+
Oxyma eq- except trimer & LY2000	Eq	2.00	1.748	2.252	@	@	+	-	@	-	@	@	@	@	@	@	-	@	@	+
DIC eq- except trimer & LY2000	Eq	2.20	1.9228	2.4772	@	@	+	-	@	+	@	@	@	@	@	@	-	@	@	+
AA eq- LY2000	Eq	1.50	1.4055	1.5945	@	@	+	-	@	+	@	@	@	@	@	@	-	@	@	+
Oxyma eq- trimer & LY2000	Eq	3.00	2.622	3.378	@	@	+	-	@	+	@	@	@	@	@	@	-	@	@	+
DIC eq- trimer & LY2000	Eq	3.30	2.8842	3.7158	@	@	+	-	@	+	@	@	@	@	@	@	-	@	@	+

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PAR Experimental Design																				
Unit Operation	**Unit**	**Set point (@)**	**Lower limit (-)**	**Upper limit (+)**	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Coupling DMF Vol- except LY2000	V	8	6.992	9.008	@	@	-	+	@	@	@	@	@	@	@	@	@	@	@	@
Coupling DMF Vol-LY2000	V	9	7.866	10.134	@	@	-	+	@	@	@	@	@	@	@	@	@	@	@	@
Target activation time (min)	Min	30	0	70	@	@	@	-	@	+	@	@	@	@	@	@	@	+	@	@
Activation Temp	℃	20	20	32	@	@	@	@	@	+	@	@	@	@	@	@	@	+	@	@
Total couple time (min) - except trimer &LY2000	min	240	240	600	@	@	+	@	@	+	@	@	@	+	@	@	@	+	@	@
Total couple time (min) - trimer	Min	480	480	1020	@	@	+	@	@	+	@	@	@	+	@	@	@	+	@	@
Total couple time (min) -LY2000	Min	480	480	1560	@	@	+	@	@	+	@	@	@	+	@	@	@	+	@	@
Coupling Temp (non-trimer& LY2000)	℃	25	17	33	@	@	@	@	@	+	@	@	-	@	@	@	@	+	@	@
Coupling Temp (LY2000)	℃	30	22	38	@	@	@	@	@	+	@	@	-	@	@	@	@	+	@	@
Coupling Temp (trimer)	℃	35	17	41	-	-	-	-	-	-	-	-	-	@	@	@	@	+	-	@
e	V	7	6.118	7	@	@	@	@	@	@	@	@	@	-	@	@	@	@	@	@
Post Leu-Pro-Pro Capping Ac2O eq	Eq	1.25	1.0925	1.4075	@	@	+	-	-	@	@	@	@	@	@	@	-	@	@	+
Post Leu-Pro-Pro Capping pyr eq	Eq	1.50	1.311	1.689	@	@	+	-	-	@	@	@	@	@	@	@	-	@	@	+
Post Leu-Pro-Pro DMF vol	V	6	5.244	6.756	@	@	-	+	-	@	@	@	@	@	@	@	@	-	@	@
Post Leu-Pro-Pro Capping time	min	60	0	120	@	@	+	-	-	@	@	@	@	@	@	@	@	@	@	+
Post Leu-Pro-Pro Capping temp	℃	25	13	37	@	@	+	-	-	@	@	@	@	@	@	@	@	@	@	+
Post Leu-Pro-Pro Capping DMF vol	V	7	7	7	@	@	@	@	@	@	@	@	@	@	@	@	@	@	@	@
Final washing vol	V	8	7	8	@	@	@	-	@	@	@	@	@	@	@	@	@	@	@	@
Final washing temp	℃	20	8	20	@	@	@	@	@	@	@	@	@	@	@	@	@	@	-	@
Drying temp - split A	℃	30	30	30	@	@	@	@	@	@	@	@	@	@	@	@	@	@	@	@
Drying temp - split B	℃	30	30	42	@	+	@	@	@	@	@	@	@	@	@	@	@	@	@	@

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Table 3.2.2 Initial PAR study Experiment Design subject of Step 1

PAR study Objective	Baseline 1	Baseline large scale for cleavage PARs	high conversion build	low coupling conversion build	low deFmoc conversion build	Racemization/ Epimerization/ DKP Build	Baseline 2	high resin loading	cold coupling	coupling long hold	Baseline Trimer 35°C	Baseline large scale Trimer 35°C	Deconvol PAR 4 (low equiv only)	Epim	DKP	Deconvol PAR 3 (high equiv)
PAR STUDY NUMBER:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
e	**Planned 6σ direction for PAR builds**										**use 35 °C trimer coupling**					
Piperidine %	20.00%	20.00%	25.00%	20.00%	15.00%	20.00%	20.00%	20.00 %	20.00%	20.00%	20.00%	20.00%	25.00%	20.00%	20.00%	20.00%
Piperidine/DMF vol	6V	6V	6.756V	6V	5.244V	6V	6V	6V	6V	6V	6V	6V	6.756V	6V	6V	6V
DeFmoc Temp - except last 2	25℃	25℃	25℃	25℃	13℃	37℃	25℃	25℃	25℃	25℃	25℃	25℃	25℃	25℃	37℃	25℃
DeFmoc Temp - last 2 only	25℃	25℃	25℃	25℃	13℃	37℃	25℃	25℃	25℃	25℃	25℃	25℃	25℃	25℃	33℃	25℃
DeFmoc stir time (min) - except last 2	120min	120min	120min	120min	60min	180min	120min	120min	120min	120min	120min	120min	120min	120min	180min	120min
DeFmoc stir time (min) - last 2 only	120min	120min	120min	120min	60min	180min	120min	120min	120min	120min	120min	120min	120min	120min	160min	120min
Post-DeFmoc wash vol (DMF)	8V	8V	8V	6.992V	8V	8V	8V	8V	8V	8V	8V	8V	6.992V	8V	8V	8V
Post-DeFmoc wash vol (IPAC)	10V	10V	10V	8.740V	10V	10V	10V	10V	10V	10V	10V	10V	8.740V	10V	10V	10V
Post-DeFmoc DMF resewell vol	10V	10V	10V	10V	10V	10V	10V	10V	8.740V	10V	10V	10V	10V	10V	10V	10V
AA eq- except LY2000	2.00	2.00	2.126	1.874	2.00	1.874	2.00	2.00	2.00	2.00	2.00	2.00	1.874	2.00	2.00	2.126
Oxyma eq- except trimer & LY2000	2.00	2.00	2.252	1.748	2.00	1.748	2.00	2.00	2.00	2.00	2.00	2.00	1.748	2.00	2.00	2.252
DIC eq- except trimer & LY2000	2.20	2.20	2.4772	1.9228	2.20	2.4772	2.20	2.20	2.20	2.20	2.20	2.20	1.9228	2.20	2.20	2.4772
AA eq- LY2000	1.50	1.50	1.5945	1.4055	1.50	1.5945	1.50	1.50	1.50	1.50	1.50	1.50	1.4055	1.50	1.50	1.5945

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PAR study Objective	Baseline 1	Baseline large scale for cleavage PARs	high conversion build	low coupling conversion build	low deFmoc conversion build	Racemization/ Epimerization/ DKP Build	Baseline 2	high resin loading	cold coupling	coupling long hold	Baseline Trimer 35°C	Baseline large scale Trimer 35°C	Deconvol PAR 4 (low equiv only)	Epim	DKP	Deconvol PAR 3 (high equiv)
PAR STUDY NUMBER:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Oxyma eq- trimer &LY2000	3.00	3.00	3.378	2.622	3.00	3.378	3.00	3.00	3.00	3.00	3.00	3.00	2.622	3.00	3.00	3.378
DIC eq- trimer & LY2000	3.30	3.30	3.7158	2.8842	3.30	3.7158	3.30	3.30	3.30	3.30	3.30	3.30	2.8842	3.30	3.30	3.7158
Coupling DMF Vol- except LY2000	8V	8V	6.992V	9.008V	8V	8V	8V	8V	8V	8V	8V	8V	8V	8V	8V	8V
Coupling DMF Vol-LY2000	9V	9V	7.866V	10.134V	9V	9V	9V	9V	9V	9V	9V	9V	9V	9V	9V	9V
Target activation time (min)	30min	30min	30min	0min	30min	30min	30min	30min	30min	30min	30min	30min	30min	70min	30min	30min
Activation Temp	20℃	20℃	20℃	20℃	20℃	32℃	20℃	20℃	20℃	20℃	20℃	20℃	20℃	28℃	20℃	20℃
Total couple time (min) - except trimer &LY2000	240min	240min	300min	240min	240min	300min	240min	240min	240min	600min	240min	240min	240min	280min	240min	240min
Total couple time (min) - trimer	480min	480min	1020min	480min	480min	1020min	480min	480min	480min	2040min	480min	480min	480min	520min	480min	480min
Total couple time (min) -LY2000	480min	480min	780min	480min	480min	780min	480min	480min	480min	1560min	480min	480min	480min	520min	480min	480min
Coupling Temp (non-trimer& LY2000)	25℃	25℃	25℃	25℃	25℃	33℃	25℃	25℃	17℃	25℃	25℃	25℃	25℃	33℃	25℃	25℃
Coupling Temp (LY2000)	30℃	30℃	30℃	30℃	30℃	38℃	30℃	30℃	22℃	30℃	30℃	30℃	30℃	38℃	30℃	30℃
Coupling Temp (trimer)	25℃	25℃	25℃	25℃	25℃	33℃	25℃	25℃	17℃	25℃	35℃	35℃	35℃	41℃	29℃	35℃
Post-couple wash vol	7V	7V	7V	7V	7V	7V	7V	7V	7V	6.118V	7V	7V	7V	7V	7V	7V
Post Leu-Pro-Pro Capping Ac2O eq	1.25eq	1.25eq	1.4075eq	1.0925eq	1.0925eq	1.25eq	1.25eq	1.25eq	1.25eq	1.25eq	1.25eq	1.25eq	1.0925eq	1.25eq	1.25eq	1.4075eq
Post Leu-Pro-Pro Capping pyr eq	1.50eq	1.50eq	1.689eq	1.311eq	1.311eq	1.50eq	1.50eq	1.50eq	1.50eq	1.50eq	1.50eq	1.50eq	1.311eq	1.50eq	1.50eq	1.689eq
Post Leu-Pro-Pro DMF vol	6V	6V	5.244V	6.756V	5.244V	6V	6V	6V	6V	6V	6V	6V	6V	5.244V	6V	6V

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PAR study Objective	Baseline 1	Baseline large scale for cleavage PARs	high conversion build	low coupling conversion build	low deFmoc conversion build	Racemization/ Epimerization/ DKP Build	Baseline 2	high resin loading	cold coupling	coupling long hold	Baseline Trimer 35°C	Baseline large scale Trimer 35°C	Deconvol PAR 4 (low equiv only)	Epim	DKP	Deconvol PAR 3 (high equiv)
PAR STUDY NUMBER:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Post Leu-Pro-Pro Capping time	60min	60min	120min	0min	0min	60min	60min	60min	60min	60min	60min	60min	60min	60min	60min	120min
Post Leu-Pro-Pro Capping temp	25℃	25℃	37℃	13℃	13℃	25℃	25℃	25℃	25℃	25℃	25℃	25℃	25℃	25℃	25℃	37℃
Post Leu-Pro-Pro Capping DMF vol	7V	7V	7V	7V	7V	7V	7V	7V	7V	6.118V	7V	7V	7V	7V	7V	7V
Final washing vol	8V	8V	8V	7V	8V	8V	8V	8V	8V	8V	8V	8V	7V	8V	8V	8V
Final washing temp	20℃	20℃	20℃	20℃	20℃	20℃	20℃	20℃	20℃	20℃	20℃	20℃	20℃	20℃	8℃	20℃
Drying temp - split A	30℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃
Drying temp - split B	30℃	42℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃	30℃
Purity (%)	91.16%	90.95%	90.45%	88.82%	90.09%	89.52%	91.50%	91.19 %	90.83%	91.28%	90.88%	91.86%	92.39%	90.90%	89.85%	92.49%
LOD (%)	7.58	7.36	7.14	7.41	6.67	6.93	6.93	7.01	7.14	5.99	7.09	6.73	5.04	6.62	7.28	6.98
Gain-in Yield (%)	93.9	92.9	99.3	92.3	96.1	94.2	90.6	93.0	96.0	98.7	99.6	98.1	96.2	97.9	96.7	97.8

Table 3.2.3 Actual Execution Parameters of Step 1

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76.1%
75.7%
76.1%
79.1%
76.4%
78.6%
78.5%
76.9%
79.1%
77.1%
77.9%
79.3%
77.7%
78.0%
79.6%
81.6%

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PAR Experimental Design

Unit Operation		Unit	Set point (@)	Lower limit (-)	Upper limit (+)	1 Baseline 1	2 Baseline large scale for cleavage PARs	3 high conversion build	4 low coupling conversion build	5 low deFmoc conversion build	6 Racemization/ Epimerization/ DKP Build	7 Baseline 2	8 high resin loading	9 cold coupling	10 coupling long hold	11 Baseline Trimer 35°C	12 Baseline large scale Trimer 35°C	13 Decon PAR 4 (low equiv only)	14 Epim	15 DKP
Piperidine %		%	20.0	15.00	25.00	20.0	20.0	25.0	20.0	15.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	25.0	20.0	20.0
Piperidine/DMF		V	6	5.244	6.756	6	6	6.756	6	5.244	6	6	6	6	6	6	6	6	6	6
DeFmoc Temp	Tyr	℃	25.0	13	37	25.3	25.0	25.1	26.1	13.1	37.0	25.0	25.0	25.0	25.0	25.0	25.1	25.0	25.0	37.0
	Thr	℃	25.0	13	37	25.2	25.0	24.8	25.5	13.0	37.0	25.0	25.0	25.0	25.1	25.3	25.0	25.0	25.0	37.0
	Asn	℃	25.0	13	37	25.0	25.0	25.0	26.3	12.7	37.0	25.0	25.0	25.0	24.7	24.6	25.1	25.0	25.0	37.0
	Ser	℃	25.0	13	37	25.3	25.1	25.1	24.8	12.9	37.2	25.1	25.0	25.0	25.0	25.4	25.0	25.0	25.3	37.0
	Gly	℃	25.0	13	37	25.0	25.0	24.9	25.0	13.0	37.1	25.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0	37.0
	Val	℃	25.0	13	37	24.8	25.0	24.8	24.8	15.4	37.1	25.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0	37.0
	Glu	℃	25.0	13	37	25.0	25.0	25.0	24.8	13.3	37.1	25.0	25.0	25.0	25.0	25.1	25.0	25.0	25.0	37.0
	Thr	℃	25.0	13	37	25.0	24.9	25.0	24.9	14.2	37.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0	37.0
	Leu-Pro-Pro	℃	25.0	13	37	25.0	25.1	25.0	24.9	13.4	37.1	25.0	25.0	25.0	24.9	25.0	25.0	25.0	25.3	37.0
	LY2000	℃	25.0	13	37	25.1	25.0	25.0	24.9	37.0	37.0	25.1	25.2	25.0	25.0	25.0	25.0	25.0	25.2	37.0
	Pro	℃	25.0	13	37	24.9	24.9	25.0	24.8	37.0	37.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0	25.2	33
DeFmoc stir time	Tyr	Min	120	60	180	124	122	122	123	63	187	122	122	124	126	128	126	126	120	189
	Thr	Min	120	60	180	125	129	120	123	80	185	125	123	125	122	126	125	122	120	191
	Asn	Min	120	60	180	120	124	121	125	73	187	121	120	121	126	122	121	122	120	192
	Ser	Min	120	60	180	120	126	123	125	73	185	122	125	123	127	125	125	124	120	188
	Gly	Min	120	60	180	127	124	120	122	60	188	122	126	128	124	128	122	124	120	186
	Val	Min	120	60	180	122	122	121	122	62	185	121	124	128	129	124	130	121	120	189
	Glu	Min	30	15	45	30	32	31	31	23	95	34	31	34	33	31	31	34	30	49
	Thr	Min	120	60	180	123	121	121	120	61	188	125	130	125	127	137	128	122	120	186

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PAR Experimental Design

Unit Operation		Unit	Set point (@)	Lower limit (-)	Upper limit (+)	1 Baseline 1	2 Baseline large scale for cleavage PARs	3 high conversion build	4 low coupling conversion build	5 low deFmoc conversion build	6 Racemization/ Epimerization/ DKP Build	7 Baseline 2	8 high resin loading	9 cold coupling	10 coupling long hold	11 Baseline Trimer 35°C	12 Baseline large scale Trimer 35°C	13 Decon PAR 4 (low equiv only)	14 Epim	15 DKP	Decon P
	Leu-Pro-Pro	Min	120	60	180	126	120	121	123	60	189	124	123	133	126	126	123	121	120	186	
	LY2000	Min	30	15	45	30	32	30	33	16	97	32	31	32	32	32	30	35	30	51	33
	Pro	Min	120	60	180	120	123	121	120	60	191	124	122	126	128	136	124	123	121	167	127
Post-DeFmoc wash (DMF)	Tyr	g	8V (455.04)	6.992 (397.70)	8V (455.04)	454.30	1894.2	455.08	455.00	455.44	455.30	455.54	455.76	455.20	455.26	455.98	1895.1	397.98	455.21	455.52	
Post-DeFmoc wash (IPAC)	Thr	g	10V (523.80)	8.740 (457.10)	10V (523.80)	524.46	2183.1	524.30	458.20	523.81	522.9	523.93	523.85	523.70	522.49	523.63	2182.6	458.25	524.01	523.94	
	Asn	g	10V (523.80)	8.740 (457.10)	10V (523.80)	524.2	2188.4	524.0	457.71	523.48	523.87	525.33	523.93	523.69	524.05	523.83	2182.1	457.92	524.03	524.20	523.64
	Ser	g	10V (523.80)	8.740 (457.10)	10V (523.80)	522.9	2182.3	523.5	457.78	523.93	523.76	523.70	523.67	524.90	523.34	523.99	2181.8	458.08	524.11	524.92	523.84
	Gly	g	10V (523.80)	8.740 (457.10)	10V (523.80)	523.80	2181.8	523.81	457.79	524.18	523.72	523.68	522.28	523.41	524.09	527.55	2182.7	458.32	524.49	523.84	524.38
	Val	g	10V (523.80)	8.740 (457.10)	10V (523.80)	523.3	2182.8	523.97	457.22	523.92	523.87	523.83	523.58	523.78	524.01	525.90	2182.8	457.87	525.54	524.39	524.00
	Glu	g	10V (523.80)	8.740 (457.10)	10V (523.80)	523.2	2181.5	524.03	457.74	523.87	524.00	522.43	524.16	523.74	523.86	524.74	2180.9	457.93	524.08	523.92	524.60
	Thr	g	10V (523.80)	8.740 (457.10)	10V (523.80)	524.2	2183.6	525.07	457.13	523.69	521.95	523.67	523.85	523.56	524.09	525.10	2181.9	457.80	524.13	524.32	523.71
	Leu-Pro-Pro	g	10V (523.80)	8.740 (457.10)	10V (523.80)	524.3	2188.8	524.98	458.20	522.63	523.50	524.47	523.61	524.17	523.48	523.77	2183.2	458.23	524.00	524.16	524.13
	LY2000	g	10V (523.80)	8.740 (457.10)	10V (523.80)	523.86	2189.3	523.81	456.70	523.32	523.84	522.81	524.15	523.91	524.14	524.94	2184.3	458.49	523.84	523.79	523.89
	Pro	g	10V (523.80)	8.740 (457.10)	10V (523.80)	523.91	2178.9	524.37	458.23	523.73	523.67	524.32	523.77	524.44	524.19	523.75	2183.7	458.05	524.79	524.84	524.13
Post-DeFmoc DMF reswell	Thr	g	10V (568.80)	8.740 (497.13)	10V (568.80)	568.60	2369.7	569.40	569.13	568.75	569.5	569.32	569.29	497.81	568.96	568.71	2370.3	568.89	568.85	570.11	569.76
	Asn	g	10V (568.80)	8.740 (497.13)	10V (568.80)	568.6	2372.4	568.4	568.79	568.54	570.41	568.39	571.12	497.38	567.48	569.56	2369.1	569.59	570.91	569.11	568.96
	Ser	g	10V (568.80)	8.740 (497.13)	10V (568.80)	569.0	2371.9	567.4	569.35	568.39	568.80	568.59	568.26	503.45	568.41	569.12	2366.5	568.08	569.62	568.94	569.31
	Gly	g	10V (568.80)	8.740 (497.13)	10V (568.80)	568.3	2368.1	569.35	568.60	568.46	569.97	566.71	569.19	497.56	569.34	570.70	2369.9	568.93	567.86	570.34	
	Val	g	10V (568.80)	8.740 (497.13)	10V (568.80)	570.2	2379.2	569.04	568.63	568.92	569.11	569.22	568.93	496.53	569.20	569.76	2366.9	568.72	569.15	568.86	

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PAR Experimental Design

Unit Operation		Unit	Set point (@)	Lower limit (-)	Upper limit (+)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
						Baseline 1	Baseline large scale for cleavage PARs	high conversion build	low coupling conversion build	low deFmoc conversion build	Racemization/ Epimerization/ DKP Build	Baseline 2	high resin loading	cold coupling	coupling long hold	Baseline Trimer 35°C	Baseline large scale Trimer 35°C	Decon PAR 4 (low equiv only)	Epim	DKP	
	Glu	g	10V (568.80)	8.740 (497.13)	10V (568.80)	573.5	2374.5	568.54	568.74	569.05	569.20	568.13	569.27	497.01	568.88	568.10	2367.9	569.33	568.38	569.70	
	Thr	g	10V (568.80) (568.80)	8.740 (497.13)	10V (568.80)	569.0	2373.7	569.11	568.8	568.87	570.07	569.11	569.14	499.65	568.71	568.13	2372.4	568.75	568.60	570.13	
	Leu-Pro-Pro	g	10V (568.80)	8.740 (497.13)	10V (568.80)	569.0	2373.1	570.88	568.91	568.97	568.95	569.28	569.47	496.73	568.78	570.84	2372.3	569.09	569.03	569.73	
	LY2000	g	10V (568.80)	8.740 (497.13)	10V (568.80)	568.40	2372.0	568.92	568.82	568.92	568.97	568.84	569.30	497.23	569.41	568.97	2377.8	569.34	569.20	568.47	
	Pro	g	10V (568.80)	8.740 (497.13)	10V (568.80)	569.54	2367.7	569.24	569.32	568.70	568.88	569.45	568.92	497.87	568.91	568.53	2377.1	569.05	568.70	569.69	
AA eq	Tyr	g	2.00 eq (41.35)	1.874 (38.74)	2.126 (45.82)	41.37	173.21	43.97	38.81	41.42	38.78	40.84	48.55	41.37	41.36	41.37	172.35	38.78	41.36	41.39	
	Thr	g	2.00 eq (35.78)	1.874 (38.74)	2.126 (45.82)	35.79	149.10	38.07	33.53	35.79	33.56	35.42	42.01	35.72	35.77	35.79	149.07	33.52	35.78	35.79	
	Asn	g	2.00 eq (53.70)	1.874 (38.74)	2.126 (45.82)	53.90	224.10	57.11	50.33	53.71	50.33	53.03	63.02	53.71	53.75	53.70	223.81	50.34	53.70	53.70	
	Ser	g	2.00 eq (34..51)	1.874 (38.74)	2.126 (45.82)	34.60	143.86	36.70	32.35	34.59	32.38	34.07	40.49	34.50	34.52	34.53	143.80	32.35	34.52	34.60	
	Gly	g	2.00 eq (26.76)	1.874 (38.74)	2.126 (45.82)	26.80	111.52	28.48	25.07	26.77	25.12	26.42	31.43	26.78	26.79	26.76	111.52	25.07	26.76	26.74	
	Val	g	2.00 eq (30.54)	1.874 (38.74)	2.126 (45.82)	30.55	127.50	32.50	28.68	30.56	28.63	30.16	35.86	30.50	30.58	30.58	127.41	28.62	30.55	30.54	
	Glu	g	2.00 eq (39.91)	1.874 (38.74)	2.126 (45.82)	39.96	166.32	42.46	37.40	39.94	37.41	39.40	46.85	39.93	40.03	39.92	166.32	37.40	39.94	39.93	
	Thr	g	2.00 eq (35.78)	1.874 (38.74)	2.126 (45.82)	35.79	149.07	38.10	33.53	35.77	33.52	35.31	42.00	35.80	35.79	35.82	149.09	33.52	35.77	35.79	
	Leu-Pro-Pro	g	2.00 eq (49.29)	1.874 (38.74)	2.126 (45.82)	49.30	205.39	52.40	46.19	49.30	46.20	48.63	57.84	49.32	49.29	49.30	205.42	46.19	49.30	49.30	
	LY2000	g	1.50 eq (75.56)	1.874 (38.74)	2.126 (45.82)	75.55	314.90	80.44	70.80	75.59	70.81	74.62	88.91	75.58	75.58	75.49	314.92	70.81	75.57	75.49	
	Pro	g	2.00 eq (30.36)	1.874 (38.74)	2.126 (45.82)	30.37	127.11	32.30	28.46	30.41	28.46	29.99	35.66	30.37	30.37	30.36	126.60	28.46	30.37	30.37	
Oxyma eq	Tyr	g	2.00 eq (12.79)	1.748 (11.18)	2.252 (14.40)	12.80	53.36	14.43	11.22	12.84	11.17	12.66	15.00	12.81	12.81	12.81	53.31	11.19	12.79	12.83	
	Thr	g	2.00 eq (12.79)	1.748 (11.18)	2.252 (14.40)	12.81	53.30	14.42	11.19	12.81	11.18	12.64	15.03	12.78	12.79	12.81	53.32	11.18	12.79	12.80	
	Asn	g	2.00 eq (12.79)	1.748 (11.18)	2.252 (14.40)	12.80	52.30	14.41	11.19	12.80	11.20	12.64	15.03	12.78	12.79	12.80	53.32	11.19	12.79	12.80	

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PAR Experimental Design

Unit Operation		Unit	Set point (@)	Lower limit (-)	Upper limit (+)	1 Baseline 1	2 Baseline large scale for cleavage PARs	3 high conversion build	4 low coupling conversion build	5 low deFmoc conversion build	6 Racemization/ Epimerization/ DKP Build	7 Baseline 2	8 high resin loading	9 cold coupling	10 coupling long hold	11 Baseline Trimer 35°C	12 Baseline large scale Trimer 35°C	13 Decon PAR 4 (low equiv only)	14 Epim	15 DKP
	Ser	g	2.00 eq (12.79)	1.748 (11.18)	2.252 (14.40)	12.80	53.32	14.40	11.19	12.80	11.20	12.65	15.02	12.76	12.79	12.81	53.29	11.19	12.79	12.80
	Gly	g	2.00 eq (12.79) (12.79)	1.748 (11.18)	2.252 (14.40)	12.80	53.83	14.46	11.18	12.79	11.19	12.64	15.07	12.79	12.85	12.87	53.29	11.18	12.79	12.80
	Val	g	2.00 eq (12.79) (12.79)	1.748 (11.18)	2.252 (14.40)	12.80	53.51	14.40	11.21	12.81	11.18	12.60	15.03	12.76	12.80	12.79	53.50	11.21	12.80	12.80
	Glu	g	2.00 eq (12.79)	1.748 (11.18)	2.252 (14.40)	12.81	53.30	14.44	11.19	12.80	11.24	12.63	15.03	12.82	12.81	12.81	53.47	11.19	12.79	12.79
	Thr	g	2.00 eq (12.79)	1.748 (11.18)	2.252 (14.40)	12.77	53.31	14.51	11.20	12.81	11.19	12.63	15.05	12.83	12.81	12.80	53.30	11.20	12.79	12.79
	Leu-Pro-Pro	g	3.00 eq (18.94)	1.748 (11.18)	2.252 (14.40)	19.20	79.97	21.61	16.79	19.20	16.76	18.94	22.57	19.25	19.19	19.18	79.86	16.77	19.19	19.18
	LY2000	g	3.00 eq (18.94)	1.748 (11.18)	2.252 (14.40)	19.19	80.04	21.64	16.80	19.20	16.77	18.98	22.76	19.18	19.20	19.17	79.99	16.78	19.18	19.17
	Pro	g	2.00 eq (12.79)	1.748 (11.18)	2.252 (14.40)	12.79	53.31	14.46	11.20	12.84	11.21	12.64	15.02	12.81	12.80	12.80	53.40	11.19	12.79	12.84
DIC eq	Tyr	g	2.20 eq (12.49)	1.9228 (10.92)	2.4772 (15.47)	12.54	52.19	14.14	10.94	12.51	14.19	12.28	14.69	12.55	12.51	12.57	52.09	10.95	12.43	12.49
	Thr	g	2.20 eq (12.49)	1.9228 (10.92)	2.4772 (15.47)	12.49	52.00	14.12	10.92	12.53	14.09	12.68	14.73	12.51	12.52	12.53	52.11	10.93	12.50	12.53
	Asn	g	2.20 eq (12.49)	1.9228 (10.92)	2.4772 (15.47)	12.40	52.30	14.13	10.95	12.49	14.04	12.33	14.68	12.49	12.43	12.47	52.21	11.03	12.51	12.50
	Ser	g	2.20 eq (12.49)	1.9228 (10.92)	2.4772 (15.47)	12.50	52.11	14.10	10.92	12.63	14.03	12.44	14.65	12.54	12.56	12.49	52.17	10.93	12.50	12.57
	Gly	g	2.20 eq (12.49)	1.9228 (10.92)	2.4772 (15.47)	12.50	52.22	14.12	10.95	12.39	14.06	12.30	14.62	12.53	12.58	12.41	52.08	10.97	12.49	12.50
	Val	g	2.20eq (12.49)	1.9228 (10.92)	2.4772 (15.47)	12.54	52.21	14.10	10.93	12.47	14.14	12.40	14.58	12.55	12.54	12.50	53.50	10.99	12.50	12.57
	Glu	g	2.20 eq (12.49)	1.9228 (10.92)	2.4772 (15.47)	12.53	51.94	14.14	10.90	12.49	14.10	12.33	14.76	12.51	12.53	12.50	52.05	10.87	12.50	12.49
	Thr	g	2.20 eq (12.49)	1.9228 (10.92)	2.4772 (15.47)	12.56	52.34	14.14	10.94	12.53	14.01	12.31	14.69	12.48	12.51	12.54	52.09	10.91	12.68	12.52
	Leu-Pro-Pro	g	3.30 eq (18.74)	1.9228 (10.92)	2.4772 (15.47)	18.90	78.11	21.13	16.51	18.80	21.11	18.51	22.00	18.74	18.85	18.79	77.96	16.35	18.81	18.75
	LY2000	g	3.30 eq (18.74)	1.9228 (10.92)	2.4772 (15.47)	18.74	78.16	21.12	16.38	18.83	21.30	18.47	21.99	18.65	18.88	18.81	78.15	16.39	18.74	18.80
	Pro	g	2.20 eq (12.49)	1.9228 (10.92)	2.4772 (15.47)	12.45	52.22	14.10	10.97	12.51	14.08	12.37	14.56	12.53	12.52	12.55	52.11	10.95	12.52	12.51

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PAR Experimental Design

Unit Operation		Unit	Set point (@)	Lower limit (-)	Upper limit (+)	1 Baseline 1	2 Baseline large scale for cleavage PARs	3 high conversion build	4 low coupling conversion build	5 low deFmoc conversion build	6 Racemization/ Epimerization/ DKP Build	7 Baseline 2	8 high resin loading	9 cold coupling	10 coupling long hold	11 Baseline Trimer 35°C	12 Baseline large scale Trimer 35°C	13 Decon PAR 4 (low equiv only)	14 Epim	15 DKP	
Coupling DMF	Tyr	g	8V (455.04)	6.992 (397.70)	9.008 (512.37)	456.16	1907.50	399.10	514.44	455.51	456.09	456.28	455.79	455.53	454.32	458.36	1895.09	456.43	455.57	456.37	4
	Thr	g	8V (455.04)	6.992 (397.70)	9.008 (512.37)	454.55	1895.10	398.59	513.11	455.54	454.89	456.90	457.97	455.07	456.35	455.47	1906.10	455.24	455.98	456.35	4
	Asn	g	8V (455.04)	6.992 (397.70)	9.008 (512.37)	455.90	1902.40	398.27	513.22	455.88	455.46	457.05	455.36	456.11	456.83	455.11	1898.90	456.12	455.78	456.75	4
	Ser	g	8V (455.04)	6.992 (397.70)	9.008 (512.37)	455.90	1897.35	400.04	515.35	456.05	455.32	455.57	456.61	455.78	456.12	455.27	1900.47	455.39	457.74	454.39	4
	Gly	g	8V (455.04)	6.992 (397.70)	9.008 (512.37)	457.80	1899.21	398.09	513.35	456.12	458.90	456.77	455.55	457.51	456.83	458.07	1899.60	455.29	455.32	455.51	4
	Val	g	8V (455.04)	6.992 (397.70)	9.008 (512.37)	456.98	1907.88	397.47	512.60	455.99	454.72	456.26	454.36	455.90	456.22	455.43	1900.00	455.17	456.36	456.77	4
	Glu	g	8V (455.04)	6.992 (397.70)	9.008 (512.37)	456.30	1900.19	399.35	513.90	455.87	456.40	456.23	456.98	454.93	454.59	457.88	1904.41	455.83	454.82	456.83	4
	Thr	g	8V (455.04)	6.992 (397.70)	9.008 (512.37)	457.81	1904.10	400.44	512.43	455.85	458.21	455.13	458.98	457.59	456.43	457.10	1897.10	455.60	454.85	457.00	4
	Leu-Pro-Pro	g	8V (455.04)	6.992 (397.70)	9.008 (512.37)	455.70	1899.59	402.11	512.82	455.12	456.39	455.00	457.08	455.29	455.25	456.41	1898.36	456.31	458.82	456.31	4
	LY2000	g	9V (511.92)	6.992 (397.70)	9.008 (512.37)	513.52	2144.68	448.98	579.23	512.98	511.78	511.70	516.00	513.88	511.84	513.73	2135.79	512.65	511.73	512.63	5
	Pro	g	8V (455.04)	6.992 (397.70)	9.008 (512.37)	456.81	1894.80	399.31	514.04	456.75	455.91	456.63	454.86	455.36	455.61	457.10	1898.36	456.29	455.47	456.41	4
Target activation time	Tyr	Min	30	22	38	30	30	90	0	30	90	30	30	30	30	30	30	30	70	30	
	Thr	Min	30	22	38	33	30	90	0	30	96	32	30	30	30	30	30	30	70	30	
	Asn	Min	30	22	38	30	30	90	0	31	90	30	30	31	30	30	30	30	70	30	
	Ser	Min	30	22	38	35	31	90	0	30	90	31	30	31	31	30	30	30	70	30	
	Gly	Min	30	22	38	30	30	90	0	30	90	30	30	30	30	30	30	30	70	30	
	Val	Min	30	22	38	30	30	90	0	30	93	30	30	30	31	30	30	30	70	30	
	Glu	Min	30	22	38	30	30	93	0	30	90	30	30	30	30	30	30	30	70	30	
	Thr	Min	30	22	38	30	30	90	0	30	90	30	30	30	33	30	30	30	70	30	
	Leu-Pro-Pro	Min	30	22	38	30	30	90	0	30	90	31	30	30	30	30	30	30	70	30	

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PAR Experimental Design

Unit Operation		Unit	Set point (@)	Lower limit (-)	Upper limit (+)	1 Baseline 1	2 Baseline large scale for cleavage PARs	3 high conversion build	4 low coupling conversion build	5 low deFmoc conversion build	6 Racemization/ Epimerization/ DKP Build	7 Baseline 2	8 high resin loading	9 cold coupling	10 coupling long hold	11 Baseline Trimer 35°C	12 Baseline large scale Trimer 35°C	13 Decon PAR 4 (low equiv only)	14 Epim	15 DKP	Decon P
	LY2000	Min	30	22	38	30	30	90	0	30	90	30	30	30	30	30	30	30	70	30	
	Pro	Min	30	22	38	30	30	90	0	30	90	30	30	30	31	30	30	30	73	30	30
Activation Temp	Tyr	℃	20	20	32	20.0	20.3	20.0	20.0	19.7	32.0	20.0	20.0	19.9	19.8	20.0	20.0	20.0	28.0	20.0	20.0
	Thr	℃	20	20	32	20.0	20.0	20.0	20.0	19.8	31.9	20.1	20.0	20.0	20.1	20.0	20.1	20.0	28.0	20.0	20.0
	Asn	℃	20	20	32	20.0	20.1	20.0	20.0	19.7	31.7	20.0	20.1	20.0	20.1	20.0	20.0	20.0	27.9	20.0	20.1
	Ser	℃	20	20	32	20.0	19.7	20.0	20.0	20.0	32.0	20.0	20.0	20.0	20.1	20.1	20.0	20.0	28.0	20.0	19.9
	Gly	℃	20	20	32	20.0	19.9	20.0	20.0	19.8	32.5	20.0	20.0	20.0	20.2	20.0	20.0	20.0	28.0	20.0	20.0
	Val	℃	20	20	32	20.0	20.1	20.0	20.0	19.8	32.1	20.0	20.0	20.0	20.0	20.0	20.0	20.0	28.0	20.0	20.0
	Glu	℃	20	20	32	20.0	20.0	20.0	20.0	19.8	32.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	28.0	20.0	20.0
	Thr	℃	20	20	32	20.0	20.0	20.0	20.0	19.8	32.0	20.0	20.0	20.0	19.9	20.0	20.2	20.0	27.9	20.0	20.0
	Leu-Pro-Pro	℃	20	20	32	20.0	20.0	19.8	20.0	20.0	32.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	28.0	20.0	20.1
	LY2000	℃	20	20	32	20.0	19.9	20.0	20.0	20.0	32.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	28.0	20.0	20.0
	Pro	℃	20	20	32	20.0	20.0	20.1	20.0	19.8	32.4	20.0	20.0	20.0	20.0	20.0	20.0	20.0	28.0	20.0	20.0
Total couple time	Tyr	min	240	240	600	256	247	300	254	253	300	240	248	245	603	245	240	240	280	251	250
	Thr	min	240	240	600	262	247	309	255	264	303	245	247	240	600	240	245	249	280	247	255
	Asn	min	240	240	600	250	245	300	262	265	300	249	250	243	600	251	240	240	280	249	250
	Ser	min	240	240	600	261	240	300	255	260	300	251	241	255	602	240	243	240	280	244	240
	Gly	min	240	240	600	249	243	302	249	249	300	244	241	248	602	245	245	242	280	240	240
	Val	min	240	240	600	253	240	300	252	256	300	240	241	240	615	240	248	240	280	252	249
	Glu	min	240	240	600	256	240	300	250	248	300	240	247	240	600	240	242	243	280	249	240
	Thr	min	240	240	600	245	250	300	247	250	302	250	240	245	600	240	240	243	280	240	240

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PAR Experimental Design

Unit Operation		Unit		Set point (@)	Lower limit (-)	Upper limit (+)	1 Baseline 1	2 Baseline large scale for cleavage PARs	3 high conversion build	4 low coupling conversion build	5 low deFmoc conversion build	6 Racemization/ Epimerization/ DKP Build	7 Baseline 2	8 high resin loading	9 cold coupling	10 coupling long hold	11 Baseline Trimer 35°C	12 Baseline large scale Trimer 35°C	13 Decon PAR 4 (low equiv only)	14 Epim	15 DKP	Decon P
	Leu-Pro-Pro	min		480	480	1020	480	490	1020	515	480	1024	482	480	480	2040	502	480	480	520	480	
	LY2000	min		480	480	1560	480	480	786	480	480	787	481	480	488	1568	480	480	480	520	480	480
	Pro	min		240	240	600	240	253	300	255	240	300	243	240	248	601	240	243	240	280	240	250
Coupling Temp	Tyr	℃	Max	25	17	33	25.4	25.0	25.1	25.1	25.0	33.0	25.1	25.2	17.1	25.1	24.6	25.1	25.0	33.1	25.0	25.1
			Min		17	33	25.0	24.8	24.8	24.9	24.5	32.5	24.9	24.9	17.0	24.9	24.5	24.8	24.9	33.0	24.9	25.1
	Thr		Max	25	17	33	25.4	25.2	25.0	25.0	25.0	33.2	25.0	25.0	17.0	24.7	25.0	25.1	25.2	33.1	25.0	25.1
			Min		17	33	25.0	24.2	22.1	24.8	24.8	32.2	24.8	24.9	16.8	24.5	24.9	25.0	25.0	32.8	24.7	24.9
	Asn		Max	25	17	33	25.3	25.0	25.3	25.1	24.9	32.9	25.0	25.0	17.1	25.1	25.0	25.1	25.0	33.1	25.0	25.1
			Min		17	33	24.5	24.9	24.8	24.3	23.6	32.6	24.9	24.9	16.8	24.8	25.0	25.0	25.0	32.9	25.0	25.0
	Ser		Max	25	17	33	25.3	25.4	25.3	25.0	24.9	33.0	25.1	25.0	17.0	25.1	25.1	24.9	25.0	33.0	25.0	25.0
			Min		17	33	25.0	23.3	24.9	24.9	24.8	32.8	25.0	24.7	16.8	24.9	25.1	22.3	25.0	33.0	24.9	25.0
	Gly		Max	25	17	33	25.0	25.2	25.5	25.0	25.1	33.0	25.0	25.0	17.1	25.1	25.0	25.1	25.0	33.1	25.0	25.0
			Min		17	33	24.3	25.0	24.9	24.8	24.1	32.7	24.9	24.8	17.0	24.8	23.7	25.0	25.0	32.9	25.0	25.0
	Val		Max	25	17	33	25.5	25.1	25.1	25.0	25.0	33.1	25.2	25.0	17.0	25.1	25.0	25.1	25.9	33.1	25.0	25.0
			Min		17	33	25.0	25.0	25.0	24.8	23.8	32.9	24.9	24.8	16.8	24.7	24.9	23.2	25.0	33.0	25.0	25.0
	Glu		Max	25	17	33	25.3	25.0	25.0	25.1	25.1	33.0	25.0	25.0	17.0	25.1	25.0	25.1	25.3	33.0	25.0	25.0
			Min		17	33	25.0	24.9	24.7	24.9	24.9	32.8	25.0	24.9	16.8	24.8	25.0	24.5	25.0	30.0	25.0	25.0
	Thr		Max	25	17	33	25.0	24.9	25.2	25.0	25.0	33.0	25.0	25.0	17.1	25.1	25.0	25.2	25.0	33.0	25.0	25.0
			Min		17	33	25.0	24.8	24.8	24.7	24.7	32.8	24.8	25.0	16.9	24.8	24.9	25.0	24.9	32.9	25.0	25.0
	Leu-Pro-Pro		Max	35	17	41	25.1	25.0	25.8	25.1	25.2	33.0	25.1	25.1	17.1	25.3	35.1	35.0	35.0	41.1	29.0	35.1
			Min		17	41	25.0	24.0	24.9	24.8	23.6	32.8	24.9	24.8	17.0	24.8	34.8	34.7	34.9	40.9	29.0	35.0

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PAR Experimental Design

Unit Operation		Unit	Set point (@)	Lower limit (-)	Upper limit (+)	1 Baseline 1	2 Baseline large scale for cleavage PARs	3 high conversion build	4 low coupling conversion build	5 low deFmoc conversion build	6 Racemization/ Epimerization/ DKP Build	7 Baseline 2	8 high resin loading	9 cold coupling	10 coupling long hold	11 Baseline Trimer 35°C	12 Baseline large scale Trimer 35°C	13 Decon PAR 4 (low equiv only)	14 Epim	15 DKP	Decon P
	LY2000	Max	30	22	38	30.1	30.7	30.5	30.1	30.0	38.0	30.1	30.2	22.2	30.3	30.2	30.1	30.1	38.1	30.0	
		Min		22	38	29.8	29.6	28.5	29.9	29.8	37.9	29.8	30.0	22.0	30.0	27.1	29.9	29.9	37.8	29.9	30.0
	Pro	Max	25	17	33	25.1	25.0	25.1	25.0	25.0	33.0	25.1	25.1	17.1	25.2	25.0	25.0	25.0	33.2	25.0	25.0
		Min		17	33	24.7	24.9	24.8	24.8	24.9	32.7	25.0	25.0	17.0	24.9	24.8	24.5	25.0	33.0	25.0	25.0
Post-couple wash	Tyr	g	7V (398.16)	6.118 (347.99)	7V (398.16)	399.15	1668.8	399.40	396.99	398.36	399.40	398.38	398.13	398.33	348.12	398.99	1658.2	399.01	398.16	399.48	398.20
	Thr	g	7V (398.16)	6.118 (347.99)	7V (398.16)	400.80	1665.1	398.70	398.31	398.20	398.01	399.18	398.81	398.16	348.12	398.92	1658.2	398.11	398.23	398.49	398.48
	Asn	g	7V (398.16)	6.118 (347.99)	7V (398.16)	398.30	1661.4	398.80	398.32	398.14	398.16	398.22	399.26	398.44	348.69	397.93	1658.3	398.88	397.96	399.82	398.02
	Ser	g	7V (398.16)	6.118 (347.99)	7V (398.16)	398.20	1662.8	397.66	399.35	399.77	398.77	396.35	398.45	398.44	348.46	398.25	1661.0	399.71	399.17	398.47	399.29
	Gly	g	7V (398.16)	6.118 (347.99)	7V (398.16)	398.20	1660.3	397.74	398.04	398.09	398.26	398.48	398.59	398.45	348.09	395.20	1658.8	398.32	397.84	398.66	398.27
	Val	g	7V (398.16)	6.118 (347.99)	7V (398.16)	398.10	1659.3	398.16	398.05	397.85	399.21	399.20	398.30	401.92	348.86	398.52	1659.2	399.83	397.88	399.33	399.27
	Glu	g	7V (398.16)	6.118 (347.99)	7V (398.16)	396.40	1657.5	398.71	398.77	398.13	399.82	397.95	398.56	397.96	348.00	399.20	1660.4	398.74	397.84	399.39	398.75
	Thr	g	7V (398.16)	6.118 (347.99)	7V (398.16)	397.20	1659.7	399.78	397.91	398.18	399.26	398.34	398.43	400.17	346.80	398.09	1658.9	398.20	398.25	398.68	398.49
	Leu-Pro-Pro	g	7V (398.16)	6.118 (347.99)	7V (398.16)	398.50	1669.6	398.93	398.22	398.16	398.07	399.22	399.37	397.94	348.21	400.64	1659.7	398.04	398.43	399.02	399.24
	LY2000	g	7V (398.16)	6.118 (347.99)	7V (398.16)	398.53	1664.2	397.94	398.46	398.35	397.99	398.13	398.22	398.58	347.98	402.56	1658.8	398.67	398.10	398.54	398.38
	Pro	g	7V (398.16)	6.118 (347.99)	7V (398.16)	397.85	1660.3	398.57	398.07	398.99	398.31	400.10	398.78	397.86	347.58	398.03	1658.9	399.18	398.13	399.10	397.60
Post Leu-Pro-Pro Capping Ac2O eq		g	1.25 eq (5.74)	1.0925 (5.02)	1.4075 (6.46)	5.73	23.99	6.52	5.08	4.95	5.74	5.77	6.80	5.76	5.75	5.74	23.97	5.03	5.74	5.75	6.48
Post Leu-Pro-Pro Capping py eq		g	1.50 eq (5.34)	1.311 (2.04)	1.689 (6.012)	5.33	22.26	6.08	4.64	4.67	5.40	5.41	6.30	5.36	5.37	5.40	22.40	4.57	5.35	5.35	6.02
Post Leu-Pro-Pro washing DMF		g	7V (398.16)	7V (398.16)	7V (398.16)	398.39	1659.7	348.04	347.83	398.22	398.06	398.57	398.32	399.06	398.60	398.96	1659.7	398.04	400.64	399.02	399.24
Post Leu-Pro-Pro Capping time		min	480	480	2040	480	490	1020	515	480	1024	482	480	480	2040	502	480	480	520	480	480
Post Leu-Pro-Pro Capping temp		℃	35	17	41	25	25	25.7	25	24.6	32.9	25	25	17	25.3	34.8	35	35	41	29	35.1

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PAR Experimental Design																			
Unit Operation	Unit	Set point (@)	Lower limit (-)	Upper limit (+)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
					Baseline 1	Baseline large scale for cleavage PARs	high conversion build	low coupling conversion build	low deFmoc conversion build	Racemization/ Epimerization/ DKP Build	Baseline 2	high resin loading	cold coupling	coupling long hold	Baseline Trimer 35°C	Baseline large scale Trimer 35°C	Decon PAR 4 (low equiv only)	Epim	DKP	Decon P
Post Leu-Pro-Pro capping wash	g	7V (398.16)	7V (398.16)	8 (455.04)	397.85	1660.3	398.57	398.07	398.99	398.31	400.10	398.78	397.86	347.58	398.03	1659.8	399.18	398.13	399.10	
Final washing	g	7V (398.16)	7V (398.16)	8 (455.04)	374.94	1567.6	376.54	457.79	377.32	376.94	375.98	377.19	376.38	376.41	380.39	1573.1	330.38	355.71	375.40	376.92
Final washing temp	℃	20	8	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
Drying temp - split A	℃	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	
Drying temp - split B	℃	30	30	42	N/A	42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Based on the experimental design in **Table 3.2.3**, each PAR study experiments was performed with 60g scale, followed by 30g soft cleavage with obtained peptidyl resin ASYM-141114 based on soft cleavage process to obtain product LSN4182913 / ASYM-140019. By analyzing the yield and impurities level of the product LSN4182913 / ASYM-140019, identify the impact of parameters.

3.3. Execution of Designed Soft cleavage Experiments

3.3.1 Soft Cleavage Experimental Setup Diagram

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Intermediate detection

The peptidyl resin ASYM-141114 obtained from PAR

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Process description:

1st soft cleavage: 20.0 vol. of 2.5%v/v TCA in

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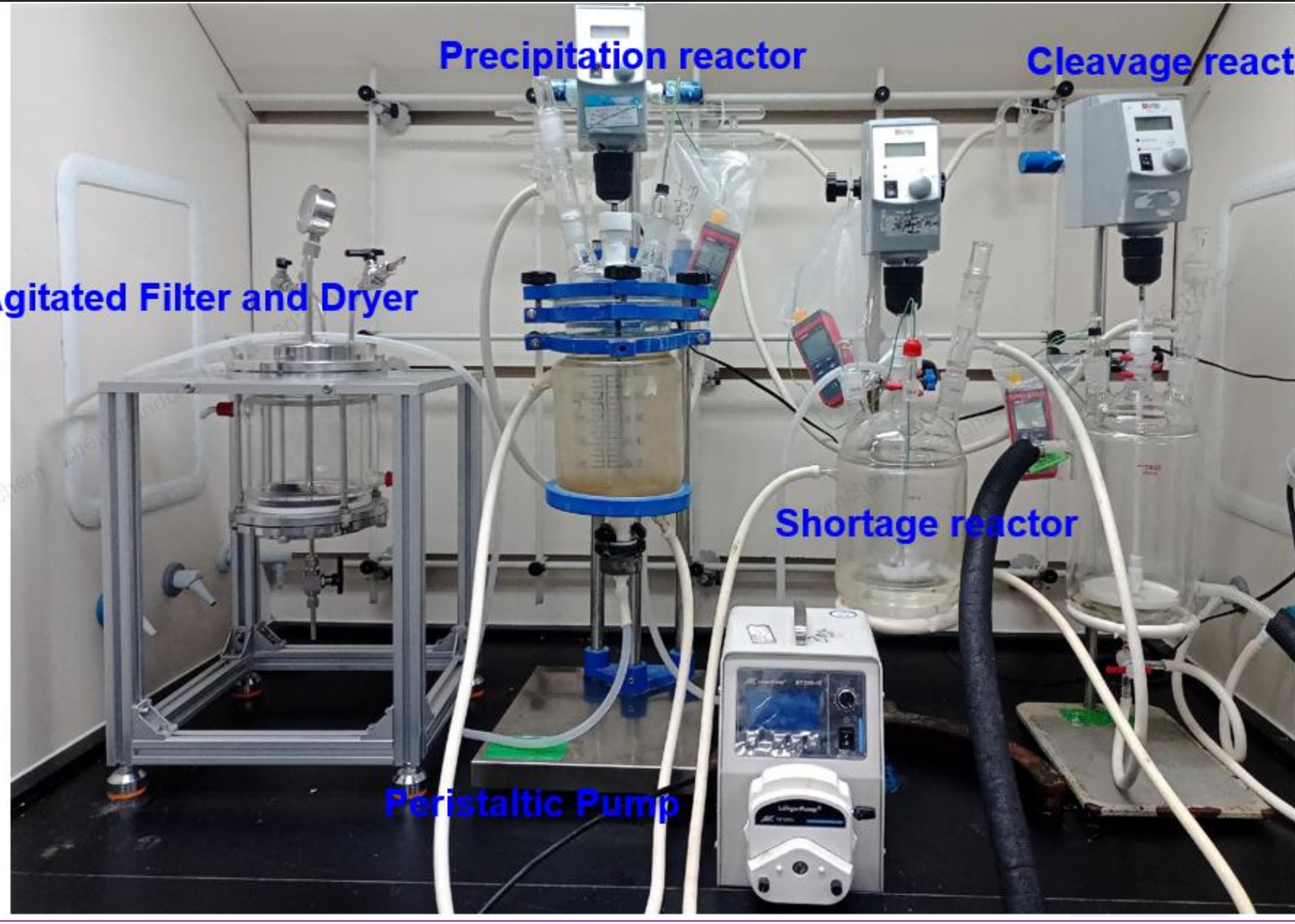
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The peptidyl resin ASYM-141114 obtained from PAR experiments was cleaved under the process conditions of CPo140019-03 Step 2, with TCA used for cleavage to assess the purity of the target intermediate LSN4182913 (ASYM-140019). The specific process is as follows:

Process description：

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Table 3.3.2 Summary of Soft Cleavage Baseline Process PAR Experiment Description

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#	Unit operation name	Process Description
1	Prepare bulk TCA/Toluene solution	Add 23V (19.963 g/g) toluene into a 2L flask Bulk Soln Flask, start agitation. Add 2.037 g/g (relative to Peptides-on-resin) TCA and stir until dissolved.
2	Cleavage solution	1. Transfer the17.649g/g of TCA-toluene solution from Bulk Solution Flask to Soln Prep Flask. Adjust solution temperature to 15℃. 2. Add 0.310 g/g tBuOH to Soln Prep Flask to 15℃.
3	Resin Cleavage	1. Transfer all the cleavage soln from Soln Prep Flask to Flask A and maintain 15℃. 2. Add 30.0g (1.00 g/g) Peptide-Resin in 4 approx. equal portions with ~3-5 min between portions maintaining 15℃ (Adiabatic temp rise is 4.8 °C) 3. Rinse the wall of Flask A with 0.5V (0.43g/g)Toluene using a spray bottle 4. Adjust mixture to 22℃ and record time as temperature reaches 22℃ 5. Maintain the 22℃ for a reaction time of 14 hours. 6. Adjust the temperature of the mixture in the reaction flask to 5 ℃, and set the jacket temperature of the receiving flask to 5 ℃. Stop agitation, filter the mixture, and transfer the filtrate to the holding flask. Maintain the jacket temperature of the holding flask at NMT 5 ℃
4	Prepare bulk TCA/Toluene solution	1. Add 8.4V (7.282 g/g) toluene into a 2L flask Bulk Soln Flask, start agitation. Add 0.720 g/g TCA and stir until dissolved. 2. Transfer the necessary amount of 5.43 g/g TCA-toluene solution from Bulk Solution Flask to Soln Prep Flask. Adjust solution temperature to 22℃. 3. Add 0.093 g/g tBuOH to Soln Prep Flask, maintaining 22℃.
5	Resin Cleavage	1. Add all the cleavage soln from Soln Prep Flask to Flask A. 2. Adjust mixture to 22℃ (Adiabatic temp rise is 11.9 °C) and record time as temperature reaches 22℃ 3. Maintain the 22℃ for a reaction time of 2 hours. 4. Adjust the temperature of the mixture in the reaction flask to 5 ℃, and set the jacket temperature of the receiving flask to 5 ℃. Stop agitation, filter the mixture, and transfer the filtrate to the holding flask. Maintain the jacket temperature of the holding flask at NMT 5 ℃.
6	Precipitation (Co-Addition)	1. Add 15V 10.26 g/g) n-heptane to filter flask, stir and cool to 5 ℃. 2. Simultaneously add lutidine and the cleavage solution dropwise to the precipitation flask over approximately 2 hours, while maintaining the temperature at 5±5℃ 3. Rinse the holding flask with 1V (0.866 g/g) toluene. 4. Stir for 30min at 5℃.
7	Filter and reslurry	1. Reduce the stirring speed to just enough to maintain suspension. Transfer the slurry to a filter-dryer. Allow the mixture to precipitate the mixture at 5 ℃ for 30min 2. Perform pressure filtration on the system, and record the filtration time and pressure. 3. Add 1V (0.866 g/g) n-heptane to the filter-dryer, agitate for 30 min, and then filter.

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8	First Drying	Drying 30℃ with nitrogen sweep and vacuum. Sample at certain intervals to determine LOD% and until residual solvent IPC passes. Also, test these samples for purity. Record Drying 1 TIME.
9	First water slurry wash/filter	1. Add 10V (10.0 g/g) purified water to the filter-dryer. 2. Stir at an internal temperature of 25 ℃ for 180 minutes. 3. Filter the mixture and record the filtration pressure.
10	Bicarbonate solution prep	Add 35.8V (35.8 g/g) purified water to Flask. Set internal temp 25°C. Add 0.731 g/g sodium bicarbonate to Flask. Stir until completely dissolved. Measure density and report it.
11	First bicarbonate slurry wash/filter	1. Add filter-dryer to the 10V (10.152g/g) Pre-prepare Bicarbonate solution. 2. Stir at an internal temperature of 25℃ for 180 minutes. 3. Filter the mixture and record the filtration pressure.
12	Second bicarbonate slurry wash/filter	1. Add filter-dryer to the 10V (10.152g/g) Pre-prepare Bicarbonate solution. 2. Stir at an internal temperature of 25℃ for 180 minutes. 3. Filter the mixture and record the filtration pressure. 4. Add 5V (5.0 g/g) purified water to filter-dryer and filter the solution at 25°C
13	Second water slurry wash/filter	1. Add 10V (10.0 g/g) purified water to the filter-dryer. 2. Stir at an internal temperature of 25 ℃ for 180 minutes. 3. Filter the mixture and record the filtration pressure.
14	Second Drying	Drying 30℃ with nitrogen sweep and vacuum. Sample at certain intervals to determine LOD% and until residual solvent IPC passes. Also test these samples for purity (FIO). Record Drying 2 TIME.
15	Store	Weigh and store the dried solid at freezer conditions under N2 blanket
16	Dry Spent resin	After finishing cleavage, to dry resin, charge MTBE, stir for 10 minutes and drain until there is no dripping. Repeat it two more times. Dry resin at jacket NMT 40°C

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Table 3.3.3 Summary of In-Process Controls (IPCs) in Step 2

No.	Test Item	Specification	Analytical Method
8	Residual organic Solvents by HS-GC	residual Toluene ≤2.0%	AM-140019-005

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(The first drying)	(% w/w)	n-heptane ≤2.0%	AM-140019-005
14 (The Second Drying)	Residual organic Solvents by HS-GC (% w/w)	residual Toluene ≤0.5%	AM-140019-005
		n-heptane ≤0.5%	AM-140019-005
		t-butanol ≤0.5%	AM-140019-005
	Water content by KF (% w/w)	water ≤2.0%	AM-140019-003

4. Execution of Designed Experiments

Table 4.0.1 Date Summary for peptidyl resin and cleaved by TCA isolated product

PAR study Number	Step 1				Step 2 cleaved by TCA				Attachment
	Scale (Resin amount*Sub)	Net weight of peptide-resin	LOD	Gain-in Yield Uncorrected with LOD	Input peptidyl resin	Output L1/ ASYM-140019	Assay	Two steps overall yield[1]	chromatogram
Run1	60*0.75 (Resin: Nankai)	176.33 g	7.58%	93.9%	30g	16.04 g	89.72%	76.69%	step1_PAR1_COA_of_CDo140019-0
Run2	250*0.75 (Resin: Nankai)	726.6 g	7.36%	92.9%	30g	15.42 g	88.96%	73.52%	step1_PAR2_COA_of_CDo140019-0
Run3	60*0.75 (Resin: Nankai)	181.9 g	7.14%	99.3%	30g	15.23 g	88.30%	74.85%	step1_PAR3_COA_of_CDo140019-0
Run4	60*0.75 (Resin: Nankai)	174.08 g	7.41%	92.3%	30g	16.69 g	87.74%	77.74%	step1_PAR4_COA_of_CDo140019-0

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PAR study Number	Step 1				Step 2 cleaved by TCA				Attachment
	Scale (Resin amount*Sub)	Net weight of peptide-resin	LOD	Gain-in Yield Uncorrected with LOD	Input peptidyl resin	Output L1/ ASYM-140019	Assay	Two steps overall yield[1]	chromatogram
Run5	60*0.75 (Resin: Nankai)	177.19 g	6.67%	96.1%	30g	15.75 g	89.24%	77.53%	step1_PAR5_COA_of_CDo140019-0
Run6	60*0.75 (Resin: Nankai)	175.44 g	6.93%	94.2%	30g	16.80 g	89.51%	78.56%	step1_PAR6_COA_of_CDo140019-0
Run7	60*0.74 (Resin: Sunresin)	169.78 g	6.93%	90.6%	30g	16.87 g	89.33%	78.56%	step1_PAR7_COA_of_CDo140019-0
Run8	60*0.88 (Resin: Sunresin)	193.22 g	7.01%	93.0%	30g	17.12 g	90.22%	76.90%	step1_PAR8_COA_of_CDo140019-0
Run9	60*0.75 (Resin: Nankai)	177.9 g	7.14%	96.0%	30g	16.28 g	90.14%	79.12%	step1_PAR9_COA_of_CDo140019-0
Run10	60*0.75 (Resin: Nankai)	178.92 g	5.99%	98.7%	30g	15.61 g	90.43%	77.74%	step1_PAR10_COA_of_CDo140019-
Run11	60*0.75 (Resin: Nankai)	182.15 g	7.09%	99.6%	30g	15.41 g	90.93%	78.20%	step1_PAR11_COA_of_CDo140019-
Run12	250*0.75 (Resin: Nankai)	748.68 g	6.73%	98.1%	30g	16.10 g	90.44%	80.25%	step1_PAR12_COA_of_CDo140019-

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PAR study Number	Step 1				Step 2 cleaved by TCA				Attachment
	Scale (Resin amount*Sub)	Net weight of peptide-resin	LOD	Gain-in Yield Uncorrected with LOD	Input peptidyl resin	Output L1/ ASYM-140019	Assay	Two steps overall yield[1]	chromatogram
Run13	60*0.75 (Resin: Nankai)	174.27 g	5.04%	96.2%	30g	16.33 g	91.96%	79.83%	step1_PAR13_COA_of_CDo140019-
Run14	60*0.75 (Resin: Nankai)	179.15 g	6.62%	97.9%	30g	15.91 g	87.29%	76.33%	step1_PAR14_COA_of_CDo140019-
Run15	60*0.75 (Resin: Nankai)	179.08 g	7.28%	96.7%	30g	16.29 g	90.13%	80.14%	step1_PAR15_COA_of_CDo140019-
Run16	60*0.75 (Resin: Nankai)	179.73 g	6.98%	97.8%	30g	16.67 g	91.74%	84.09%	step1_PAR16_COA_of_CDo140019-

4.1 Baseline experiments

The baseline experiment was an experiment run in which all parameters are set to the target process values (Run11, Run12). The purpose is to simulate the production results in the laboratory, and the experimental results are used for the comparative analysis of the subsequent PAR study experiments. At the same time, three experiments were conducted to examine the robustness of the process.

4.2 Experimental Details of the Baseline and Trimer Coupling at 35 °C:

Objective: To investigate the impact of varying the trimer coupling temperature on the product under the same process conditions as the baseline, PAR 7 validates resins (0.74 mmol/g) from different suppliers with similar substitution degrees.

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Table 4.2.1 Data Summary for peptidyl resin and carry forward to Step 2 yield

PAR study Number	Step 1				Step 2 cleaved by TCA			
	Scale (Resin amount*Sub)	Net weight of peptide-resin	LOD	Gain-in Yield Uncorrected with LOD	Input peptidyl resin	Output L1/ ASYM-140019	Assay	Two steps overall yield[2]
Run1	60*0.75 (Resin: Nankai)	176.33 g	7.58%	93.9%	30g	16.04 g	89.72%	76.69%
Run2	250*0.75 (Resin: Nankai)	726.6 g	7.36%	92.9%	30g	15.42 g	88.96%	73.52%
Run7	60*0.74 (Resin: Sunresin)	169.78 g	6.93%	90.6%	30g	16.87 g	89.33%	78.56%
Run11	60*0.75 (Resin: Nankai)	182.15 g	7.09%	99.6%	30g	15.41 g	90.93%	78.20%
Run12	250*0.75 (Resin: Nankai)	748.68 g	6.73%	98.1%	30g	16.10 g	90.44%	80.25%

Note [1]: 1. Step 1 Gain-in Yield Uncorrected with LOD = ((M1 (net weight of peptide-on-resin)(1-LOD)-Resin amount)/ (M0 (theoretical yield) -Resin amount) *100%*

*Step 1 Theoretical calculation formula (M0) = Resin amount + Resin amount*Sub × 2436.23/1000 kg/mol*

*2. Two steps overall yield (Step 1 & 2) = (M2 * Assay+ Product total weight in sample) / M2*100%*

*The theoretical yield of Step 1&2 M3 is calculated: M3= Resin amount*Sub * FW of L1 / 1000 = Input peptidyl-on-resin/ Net weight of peptide-on-resin (M1)* Resin amount*Sub*2675.51/1000 kg/mol*

Table 4.2.2 Summary 1 of Baseline Matrix PAR experiments for Investigated Parameters

Unit Operation	Unit	Set point (@)	Lower limit (-)	Upper limit (+)	1	2	7	11	12
Piperidine %	%	20.00	15.00	25.00	@	@	@	@	@

Unit Operation	Unit	Set point (@)	Lower limit (-)	Upper limit (+)	1	2	7	11	12
Piperidine/DMF vol	V	6	5.244	6.756	@	@	@	@	@
DeFmoc Temp	℃	25	13	37	@	@	@	@	@
DeFmoc stir time (min) - except Val and Leu-Pro-Pro	Min	120	60	180	@	@	@	@	@
DeFmoc stir time (min) – only Val and Leu-Pro-Pro	Min	30	15	45	@	@	@	@	@
Post-DeFmoc wash vol (DMF)	V	8	6.992	8	@	@	@	@	@
Post-DeFmoc wash vol (IPAC)	V	10	8.740	10	@	@	@	@	@
Post-DeFmoc DMF reswell vol	V	10	8.740	10	@	@	@	@	@
AA eq- except LY2000	Eq	2.00	1.874	2.126	@	@	@	@	@
Oxyma eq- except trimer & LY2000	Eq	2.00	1.748	2.252	@	@	@	@	@
DIC eq- except trimer & LY2000	Eq	2.20	1.9228	2.4772	@	@	@	@	@
AA eq- LY2000	Eq	1.50	1.4055	1.5945	@	@	@	@	@
Oxyma eq- trimer & LY2000	Eq	3.00	2.622	3.378	@	@	@	@	@
DIC eq- trimer & LY2000	Eq	3.30	2.8842	3.7158	@	@	@	@	@
Coupling DMF Vol- except LY2000	V	8	6.992	9.008	@	@	@	@	@
Coupling DMF Vol- LY2000	V	9	7.866	10.134	@	@	@	@	@
Target activation time (min)	Min	30	0	70	@	@	@	@	@
Activation Temp	℃	20	20	32	@	@	@	@	@
Total couple time (min) - except trimer & LY2000	min	240	240	600	@	@	@	@	@
Total couple time (min) - trimer	Min	480	480	1020	@	@	@	@	@
Total couple time (min) - LY2000	Min	480	480	1560	@	@	@	@	@
Coupling Temp (non-trimer& LY2000)	℃	25	17	33	@	@	@	@	@
Coupling Temp (LY2000)	℃	30	22	38	@	@	@	@	@
Coupling Temp (trimer)	℃	35	17	41	25℃	25℃	25℃	@	@
Post-couple wash vol	V	7	6.118	7	@	@	@	@	@
Post Leu-Pro-Pro Capping Ac2O eq	Eq	1.25	1.0925	1.4075	@	@	@	@	@

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Unit Operation	Unit	Set point (@)	Lower limit (-)	Upper limit (+)	1	2	7	11	12
Post Leu-Pro-Pro Capping pyr eq	Eq	1.50	1.311	1.689	@	@	@	@	@
Post Leu-Pro-Pro DMF vol	V	6	5.244	6.756	@	@	@	@	@
Post Leu-Pro-Pro Capping time	min	60	0	120	@	@	@	@	@
Post Leu-Pro-Pro Capping temp	℃	25	13	37	@	@	@	@	@
Post Leu-Pro-Pro Capping DMF vol	V	7	7	7	@	@	@	@	@
Final washing vol	V	8	7	8	@	@	@	@	@
Final washing temp	℃	20	8	20	@	@	@	@	@
Drying temp - split A	℃	30	30	30	@	@	@	@	@
Purity (%)	-	-	-	-	91.16%	90.95%	91.50%	90.88%	91.86%
LOD (%)	-	-	-	-	7.58	7.36	6.93	7.09	6.73
Gain-in Yield (%)	-	-	-	-	93.9	92.6	90.6	99.6	98.1

Results & Discussion: The standard experimental setup outlined in Section 3 was used for the baseline experimental design, with the exception that the coupling temperature for the trimer (tripeptide Fmoc-L-Leu-L-Pro-L-Pro-OH) in Experiments run 1, 2, and 7 was set at 25°C. During the trimer coupling reaction, it was observed that the 25°C temperature condition led to a failure in the first or second Ninhydrin test. furthermore, the purity analysis of final product after soft cleavage revealed an increase in the Ac end-capping 8mer impurity. Therefore, supplementary PAR Experiments 11 and 12 were conducted by raising the trimer coupling temperature to 35°C. The detailed experimental matrix is presented in **Table 4.2.2.**

Table 4.2.3 Brief summary 1 of impurity profile from Baseline Matrix PAR experiments

Exp.No.	Fraction No.	No.	1	2	3	4	5	6	7	8	9	10	11	12	13
		RT [min]	14.734	15.045	16.164	16.475	18.651	22.692	28.287	30.308	30.557	30.774	30.898	31.085	31.334
		RRT	0.474	0.484	0.520	0.530	0.600	0.730	0.910	0.975	0.983	0.990	0.994	1.000	1.008

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		Calculate mass	1487.88	977.5486	1698.0318	1795.0935	1824.163	1432.854	2617.6830	2516.636	2601.701	2488.639	e	2770.799	2530.655	2463.6050	2673.7374	2687.7608
		Structure	9mer	Des-LY2000 Leu-Pro-Pro	Des-ProLY2000	Des-LY2000	DIC-endcapping-11mer	Ac end-capping-8mer	Tyr(Des-tBu)	Des-Thr30	Tyr(Otbu) to Phe	-Glu(OtBu)	+Gly	+Pro28 or 29	-Ser(tBu)	-LeuPro	LSN4182913/140019	M+14
step1_PAR1	CDo140019-03-3104-008-15	%Area	-	0.10%	0.06%	0.10%	0.15%	0.2957%	1.69%	0.28%		-		0.09%		0.22%	91.16%	0.20%
step1_PAR2	CDo140019-03-3104-009-15	e	-	0.13%	0.10%	0.24%	0.12%	0.2202%	1.44%	0.28%		-		0.11%		0.35%	90.95%	0.18%
step1_PAR7	CDo140019-03-3104-002-15	%Area	-	0.13%	0.11%	0.16%	0.13%	0.1932%	1.61%	0.32%		0.18%		0.15%		0.34%	91.50%	0.14%
step1_PAR11	CDo140019-03-3104-006-15	%Area	-	0.10%	0.11%	0.24%	0.15%	-	1.41%	0.27%		0.10%		0.09%		0.27%	90.88%	0.16%
step1_PAR12	CDo140019-03-3104-007-15	%Area	-	0.13%	0.06%	0.09%	0.13%	0.1720%	1.70%	0.22%		0.09%		0.09%		0.23%	91.86%	0.20%

Conclusion: The data for the peptide-on-resin and cleaved isolated product impurities are summarized in **Appendix VI**. The yields of these batches were all within expectations. However, based on the impurity profile, increasing the temperature to 35°C during the tripeptide coupling reaction significantly accelerated the reaction rate (basically the first sample Ninhydrin test passed) and reduced the content of the Ac end-capping impurity.

PAR 7 involves testing resin batches from different suppliers to qualify an alternative supplier, with similar substitution degree to 0.75mmol/g.

4.3 Experimental Details of high conversion build:

Objective: To investigate the impact of enhanced reaction conditions aimed at improving conversion on the product quality.

Table 4.3.1 Data Summary for peptidyl resin and carry forward to Step 2 yield

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删除[Xinran Geng18625]: s
删除[Angao Yuan]: 8mer
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删除[Angao Yuan]:

PAR study Number	Step 1				Step 2 cleaved by TCA			
	Scale (Resin amount*Sub)	Net weight of peptide-resin	LOD	Gain-in Yield Uncorrected with LOD	Input peptidyl resin	Output L1/ ASYM-140019	Assay	Two steps overall yield
Run3	60*0.75 (Resin: Nankai)	181.9 g	7.14%	99.3%	30g	15.23 g	88.30%	74.85%
Run16	60*0.75 (Resin: Nankai)	179.73 g	6.98%	97.8%	30g	16.67 g	91.74%	84.09%

Table 4.3.2 Summary 3 of high conversation build Matrix PAR experiments for Investigated Parameters

Unit Operation	Unit	Set point (@)	Lower limit (-)	Upper limit (+)	3	16
Piperidine %	%	20.00	15.00	25.00	+	@
Piperidine/DMF vol	V	6	5.244	6.756	+	@
DeFmoc Temp	℃	25	13	37	@	@
DeFmoc stir time (min) - except Val and trimer	Min	120	60	180	@	@
DeFmoc stir time (min) – only Val and trimer	Min	30	15	45	@	@
Post-DeFmoc wash vol (DMF)	V	8	6.992	8	@	@
Post-DeFmoc wash vol (IPAC)	V	10	8.740	10	@	@
Post-DeFmoc DMF resewell vol	V	10	8.740	10	@	@
AA eq- except LY2000	Eq	2.00	1.874	2.126	+	+
Oxyma eq- except trimer & LY2000	Eq	2.00	1.748	2.252	+	+
DIC eq- except trimer & LY2000	Eq	2.20	1.9228	2.4772	+	+
AA eq- LY2000	Eq	1.50	1.4055	1.5945	+	+
Oxyma eq- trimer & LY2000	Eq	3.00	2.622	3.378	+	+
DIC eq- trimer & LY2000	Eq	3.30	2.8842	3.7158	+	+
Coupling DMF Vol- except LY2000	V	8	6.992	9.008	-	@

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Unit Operation	Unit	Set point (@)	Lower limit (-)	Upper limit (+)	3	16
Coupling DMF Vol-LY2000	V	9	7.866	10.134	-	@
Target activation time (min)	Min	30	0	70	+	@
Activation Temp	℃	20	20	32	@	@
Total couple time (min) - except trimer &LY2000	min	240	240	600	+	@
Total couple time (min) - trimer	Min	480	480	1020	+	@
Total couple time (min) -LY2000	Min	480	480	1560	+	@
Coupling Temp (non-trimer&LY2000)	℃	25	17	33	@	@
Coupling Temp (LY2000)	℃	30	22	38	@	@
Coupling Temp (trimer)	℃	35	17	41	-	@
Post-couple wash vol	V	7	6.118	7	@	@
Post Leu-Pro-Pro Capping Ac2O eq	Eq	1.25	1.0925	1.4075	+	+
Post Leu-Pro-Pro Capping pyr eq	Eq	1.50	1.311	1.689	+	+
Post Leu-Pro-Pro DMF vol	V	6	5.244	6.756	-	@
Post Leu-Pro-Pro Capping time	min	60	0	120	+	+
Post Leu-Pro-Pro Capping temp	℃	25	13	37	+	+
Post Leu-Pro-Pro Capping DMF vol	V	7	7	7	-	@
Final washing vol	V	8	7	8	@	@
Final washing temp	℃	20	8	20	@	@
Drying temp - split A	℃	30	30	30	@	@
Drying temp - split B	℃	30	30	42	@	@
Purity (%)	-	-	-	-	90.45%	92.49%
LOD (%)	-	-	-	-	7.14	6.98
Gain-in Yield (%)	-	-	-	-	99.3	97.8

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Results & Discussion:

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The baseline experiments utilized the standard experimental setup outlined in Section 3. With the exception, that Experiments 3 and 16 were designed to improve reaction conversion by increasing the reagent equivalents and concentration, coupling temperature of trimer in experiment 3 was set as 25^{E}C. The detailed experimental matrix is presented in **Table 4.3.2**.

Table 4.3.3 Data Summary 3 of impurity profile from high conversion build Matrix PAR experiments

Exp.No.	Fraction No.	No.	1	2	3	4	5	6	7	8		9		10		11	12	13
		RT [min]	14.734	15.045	16.164	16.475	18.651	22.692	28.287	30.308		30.557		30.774		30.898	31.085	31.334
		RRT	0.474	0.484	0.520	0.530	0.600	0.730	0.910	0.975		0.983		0.990		0.994	1.000	1.008
		Calculate mass	1487.88	977.5486	1698.0318	1795.0935	1824.163	1432.854	2617.6830	2516.636	2601.701	2488.639	2730.771	2770.799	2530.655	2463.6050	2673.7374	2687.7608
		Structure	9mer	Des-LY2000 Leu-Pro-Pro	Des-ProLY2000	Des-LY2000	DIC-endcapping-11mer	Ac end-capping-8mer	Tyr(Des-tBu)	Des-Thr30	Tyr(Otbu) to Phe	-Glu(OtBu)	+Gly	+Pro 28 or 29	-Ser(tBu)	-LeuPro	LSN4182913/140019	M+14
step1_PAR3	CDo140019-03-3104-010-15	%Area	-	0.08%	0.20%	0.55%	0.10%	0.1767%	1.57%	0.18%		-		0.10%		0.26%	90.45%	0.17%
step1_PAR16	CDo140019-03-3104-019-15	%Area	-	0.12%	0.07%	0.05%	0.19%	0.1789%	1.60%	0.20%		-		0.08%		0.21%	92.49%	0.01%

Conclusion: The purity and yield data for the peptide-on-resin and cleaved isolated products are summarized in **Table 4.3.3.** The yields of these batches were comparable to those of the baseline experiments. However, in PAR 3, increasing the reagent equivalents for amino acids other than LY2000 did not result in any significant changes in the impurity profile. Conversely, reducing the coupling solvent (high concentration) for LY2000 caused the reaction mixture to become highly

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viscous, leading to mixing difficulties and insufficient reaction between the resin and the amino acids. This ultimately resulted in an increase in the Des-LY2000 impurity level.

In PAR 16, when the coupling solvent volume was restored to the baseline level, the level of the Des-LY2000 impurity also returned to normal. This indicates that the solvent volume for LY2000 coupling is the primary influencing factor. Therefore, it is recommended to set the lower limit for the DMF volume at 9 V.

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4.4 Experimental Details of low couping conversion build/ coupling long hold/ Deconvol (low equiv only):

Objective: To investigate the impact on the product under the following experimental conditions: weakened coupling conditions, extended reaction time, intensified deprotection conditions, and reduced coupling conditions.

Table 4.4.1 Data Summary for peptidyl resin and carry forward to Step 2 yield

PAR study Number	Step 1				Step 2 cleaved by TCA			
	Scale (Resin amount*Sub)	Net weight of peptide-resin	LOD	Gain-in Yield Uncorrected with LOD	Input peptidyl resin	Output L1/ ASYM-140019	Assay	Two steps overall yield
Run4	60*0.75 (Resin: Nankai)	174.08 g	7.41%	92.3%	30g	16.69 g	87.74%	77.74%
Run10	60*0.75 (Resin: Nankai)	178.92 g	5.99%	98.7%	30g	15.61 g	90.43%	77.74%
Run13	60*0.75 (Resin: Nankai)	174.27 g	5.04%	96.2%	30g	16.33 g	91.96%	79.83%

Table 4.4.2 Summary 4: Investigated Parameters Matrix of low coupling conversion build/ coupling long hold/ Deconvol (low equiv only)

Unit Operation	Unit	Set point (@)	Lower limit (-)	Upper limit (+)	4	10	13
Piperidine %	%	20.00	15.00	25.00	@	@	+
Piperidine/DMF vol	V	6	5.244	6.756	@	@	+
DeFmoc Temp	℃	25	13	37	@	@	@
DeFmoc stir time (min) - except Val and trimer	Min	120	60	180	@	@	@
DeFmoc stir time (min) – only Val and trimer	Min	30	15	45	@	@	@
Post-DeFmoc wash vol (DMF)	V	8	6.992	8	-	@	-
Post-DeFmoc wash vol (IPAC)	V	10	8.740	10	-	@	-
Post-DeFmoc DMF resewell vol	V	10	8.740	10	@	@	10V
AA eq- except LY2000	Eq	2.00	1.874	2.126	-	@	-
Oxyma eq- except trimer & LY2000	Eq	2.00	1.748	2.252	-	@	-
DIC eq- except trimer & LY2000	Eq	2.20	1.9228	2.4772	-	@	-
AA eq- LY2000	Eq	1.50	1.4055	1.5945	-	@	-
Oxyma eq- trimer & LY2000	Eq	3.00	2.622	3.378	-	@	-
DIC eq- trimer & LY2000	Eq	3.30	2.8842	3.7158	-	@	-
Coupling DMF Vol- except LY2000	V	8	6.992	9.008	+	@	@
Coupling DMF Vol- LY2000	V	9	7.866	10.134	+	@	@
Target activation time (min)	Min	30	0	70	-	@	@
Activation Temp	℃	20	20	32	@	@	@
Total couple time (min) - except trimer & LY2000	min	240	240	600	@	+	@
Total couple time (min) - trimer	Min	480	480	1020	@	+	@
Total couple time (min) - LY2000	Min	480	480	1560	@	+	@

删除[Handong Cheng15615]: 1400190301

删除[Angao Yuan]: **Matrix for Investigated Parameters**

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

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删除[Handong Cheng15615]: Lys

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Unit Operation	Unit	Set point (@)	Lower limit (-)	Upper limit (+)	4	10	13
Coupling Temp (non-trimer&LY2000)	℃	25	17	33	@	@	@
Coupling Temp (LY2000)	℃	30	22	38	@	@	@
Coupling Temp (trimer)	℃	35	17	41	-	-	@
Post-couple wash vol	V	7	6.118	7	@	-	@
Post Leu-Pro-Pro Capping Ac2O eq	Eq	1.25	1.0925	1.4075	-	@	-
Post Leu-Pro-Pro Capping pyr eq	Eq	1.50	1.311	1.689	-	@	-
Post Leu-Pro-Pro DMF vol	V	6	5.244	6.756	+	@	@
Post Leu-Pro-Pro Capping time	min	60	0	120	-	@	@
Post Leu-Pro-Pro Capping temp	℃	25	13	37	-	@	@
Post-capping wash vol	V	7	7	7	@	@	@
Final washing vol (IPA&MTBE)	V	8	7	8	-	@	@
Final washing temp	℃	20	8	20	@	@	@
Drying temp - split A	℃	30	30	30	@	@	@
Purity (%)	-	-	-	-	88.82%	91.28%	92.39%
LOD (%)	-	-	-	-	7.41	5.99	5.04
Gain-in Yield (%)	-	-	-	-	92.3	98.7	96.2

Results & Discussion:

PAR experiment 4 is to study the impact of low conversion conditions on the reaction and product quality including decreasing washing volume, E the AA/capping reagent eq, increasing reaction solvent volume and decreasing activation/end capping time. PAR 10 was designed to study the stability of long coupling and washing volume after coupling. PAR 13 Deconvol was just to study low equiv AA and capping reagent and increasing Piperidine% to 25% as well to ensure de-Fmoc completely, which can avoid the influence by incomplete defmoc.

删除[Handong Cheng15615]: 1400190301

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: @

删除[Angao Yuan]: The standard experimental setup outlined in Section 3 was used to investigate the impact of coupling reaction time on impurities by shortening or extending the duration.

删除[Xinran Geng18625]: e

删除[Xinran Geng18625]: ,

删除[Xinran Geng18625]: e

删除[Xinran Geng18625]: meanwhile

删除[Xinran Geng18625]:

删除[Xinran Geng18625]: of

删除[Nil Tandogan Gray]: ly

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Table 4.4.3 Data Summary 4 of impurity profile from low coupling conversion build/ coupling long hold/ Deconvol (low equiv only)

Matrix

Exp.No.-	Fraction No.	No.	1	2	3	4	5	6	7	8		9		10		11	12	13
		RT [min]	14.734	15.045	16.164	16.475	18.651	22.692	28.287	30.308		30.557		30.774		30.898	31.085	31.334
		RRT	0.474	0.484	0.520	0.530	0.600	0.730	0.910	0.975		0.983		0.990		0.994	1.000	1.008
		Calculate mass	1487.88	977.5486	1698.0318	1795.0935	1824.163	1432.854	2617.6830	2516.636	2601.701	2488.639	2730.771	2770.799	2530.655	2463.6050	2673.7374	2687.7608
		Structure	9mer	Des-LY2000-Leu-Pro-Pro	Des-Pro-LY2000	Des-LY2000	DIC-endcapping-11mer	Ac end-capping-8mer	Tyr(Des-tBu)	Des-Thr30	Tyr(Otbu) to Phe	-Glu(OtBu)	+Gly	+Pro28 or 29	-Ser(tBu)	-LeuPro	LSN4182913/140019	M+14
step1_PAR4	CDo140019-03-3104-011-15	%Area	0.07%	0.18%	0.06%	0.07%	0.31%	1.1403%	1.58%	0.57%		0.09%		0.22%		0.28%	88.82%	0.12%
step1_PAR10	CDo140019-03-3104-005-15	%Area	-	0.09%	0.12%	0.33%	0.16%	0.0929%	1.54%	0.17%		0.12%		0.08%		0.31%	91.28%	0.12%
step1_PAR13	CDo140019-03-3104-016-15	%Area	-	0.10%	0.17%	-	0.25%	0.1405%	1.24%	0.21%		-		0.14%		0.17%	92.39%	0.02%

Conclusion:

The data for the peptide-on-resin and cleaved isolated products are summarized in Table 4.4.2. In PAR 4, both the reagent equivalents (excluding the coupling solvent) and the reaction temperature were set lower than the baseline process parameters. It was observed that the level of the Ac end-capping-8mer impurity increased, indicating the trimer coupling was not complete under these conditions. But for other AA coupling, current low conversion coupling condition works well. PAR 13 was conducted as a supplementary experiment to PAR 4. Under normal capping conditions, the capping equivalents were lowered and other experimental conditions were also reduced. This adjustment showed no significant effect on the impurity profile or product purity.

删除[Handong Cheng15615]: 1400190301

删除[Angao Yuan]: **cleaved isolated Solid-Part 2**

设置格式[Handong Cheng15615]: 字体颜色: 自动设置

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设置格式[Handong Cheng15615]: 字体颜色: 自动设置

删除[Handong Cheng15615]: Lys

删除[Angao Yuan]: Fmoc-endcapping-8mer

删除[Handong Cheng15615]: (SC)

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: (SC)

删除[Handong Cheng15615]: (SC)

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带格式表格[Handong Cheng15615]

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删除[Handong Cheng15615]: 48

设置格式[Handong Cheng15615]: 字体颜色: 自动设置

Conversely, PAR 10 was designed as the opposite of PAR 4 by extending the coupling time. However, this extension did not yield any significant positive effect on the conversion rate.

4.5 Experimental Details:

Objective: To separately investigate the impact of mild deprotection conditions and mild coupling conditions on the product.

Table 4.5.1 Data Summary for peptidyl resin and carry forward to Step 2 yield

PAR study Number	Step 1				Step 2 cleaved by TCA			
	Scale (Resin amount*Sub)	Net weight of peptide-resin	LOD	Gain-in Yield Uncorrected with LOD	Input peptidyl resin	Output L1/ ASYM-140019	Assay	Two steps overall yield
Run5	60*0.75 (Resin: Nankai)	177.19 g	6.67%	96.1%	30g	15.75 g	89.24%	77.53%
Run9	60*0.75 (Resin: Nankai)	177.9 g	7.14%	96.0%	30g	16.28 g	90.14%	79.12%

Table 4.5.2 Summary 5: Matrix for Investigated Parameters of low deFmoc conversion build / cold coupling

Unit Operation	Unit	Set point (@)	Lower limit (-)	Upper limit (+)	5	9
Piperidine %	%	20.00	15.00	25.00	-	@

删除[Handong Cheng15615]: 1400190301

删除[Handong Cheng15615]: ; instead, it led to the formation of a new impurity

删除[Handong Cheng15615]:

删除[Angao Yuan]: Therefore, it is recommended that the coupling time should not be less than 30 minutes, and the lower limit for the reagent equivalents should be established based on the data from PAR 4.

删除[Angao Yuan]: **of**

删除[Angao Yuan]:

删除[Angao Yuan]: **Matrix for Investigated Parameters**

Unit Operation	Unit	Set point (@)	Lower limit (-)	Upper limit (+)	5	9
Piperidine/DMF vol	V	6	5.244	6.756	-	@
DeFmoc Temp	℃	25	13	37	-	@
DeFmoc stir time (min) - except Val and trimer	Min	120	60	180	-	@
DeFmoc stir time (min) – only Val and trimer	Min	30	15	45	-	@
Post-DeFmoc wash vol (DMF)	V	8	6.992	8	@	@
Post-DeFmoc wash vol (IPAC)	V	10	8.740	10	@	@
Post-DeFmoc DMF reswell vol	V	10	8.740	10	@	-
AA eq- except LY2000	Eq	2.00	1.874	2.126	@	@
Oxyma eq- except trimer & LY2000	Eq	2.00	1.748	2.252	@	@
DIC eq- except trimer & LY2000	Eq	2.20	1.9228	2.4772	@	@
AA eq- LY2000	Eq	1.50	1.4055	1.5945	@	@
Oxyma eq- trimer & LY2000	Eq	3.00	2.622	3.378	@	@
DIC eq- trimer & LY2000	Eq	3.30	2.8842	3.7158	@	@
Coupling DMF Vol- except LY2000	V	8	6.992	9.008	@	@
Coupling DMF Vol- LY2000	V	9	7.866	10.134	@	@
Target activation time (min)	Min	30	0	70	@	@
Activation Temp	℃	20	20	32	@	@
Total couple time (min) - except trimer & LY2000	min	240	240	600	@	@
Total couple time (min) - trimer	Min	480	480	1020	@	@
Total couple time (min) - LY2000	Min	480	480	1560	@	@
Coupling Temp (non-trimer& LY2000)	℃	25	17	33	@	-
Coupling Temp (LY2000)	℃	30	22	38	@	-
Coupling Temp (trimer)	℃	35	17	41	-	-
Post-couple wash vol	V	7	6.118	7	@	@
Post Leu-Pro-Pro Capping Ac2O eq	Eq	1.25	1.0925	1.4075	-	@

删除[Handong Cheng15615]: 1400190301

删除[Nil Tandogan Gray]: e

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

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删除[Handong Cheng15615]: Lys

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删除[Handong Cheng15615]: Lys

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Unit Operation	Unit	Set point (@)	Lower limit (-)	Upper limit (+)	5	9
Post Leu-Pro-Pro Capping pyr eq	Eq	1.50	1.311	1.689	-	@
Post Leu-Pro-Pro DMF vol	V	6	5.244	6.756	-	@
Post Leu-Pro-Pro Capping time	min	60	0	120	-	@
Post Leu-Pro-Pro Capping temp	℃	25	13	37	-	@
Post Leu-Pro-Pro Capping DMF vol	V	7	7	7	@	@
Final washing vol	V	8	7	8	@	@
Final washing temp	℃	20	8	20	@	@
Drying temp - split A	℃	30	30	30	@	@
Drying temp - split B	℃	30	30	42	@	@
Purity (%)	-	-	-	-	90.09%	90.83%
LOD (%)	-	-	-	-	6.67	7.14
Gain-in Yield (%)	-	-	-	-	96.1	96.0

删除[Handong Cheng15615]: 1400190301

删除[Handong Cheng15615]: Post-capping wash vol

Results & Discussion:

The standard experimental setup outlined in Section 3 was used to investigate the impact of low De-Fmoc conversion build / cold coupling conditions. The detailed experimental matrix is presented in **Table 4.5.2.**

删除[Angao Yuan]: coupling reaction time on impurities by reducing the

Table 4.5.3 Data Summary 5: impurity profile from low deFmoc conversion build / cold coupling Matrix soft cleavage solid LC test

删除[Angao Yuan]:

删除[Angao Yuan]: **of**

删除[Angao Yuan]: **cleaved isolated Solid-Part 2**

Exp.No.	Fraction No.	No.	1	2	3	4	5	6	7	8	9	10	11	12	13
		RT [min]	14.734	15.045	16.164	16.475	18.651	22.692	28.287	30.308	30.557	30.774	30.898	31.085	31.334
		RRT	0.474	0.484	0.520	0.530	0.600	0.730	0.910	0.975	0.983	0.990	0.994	1.000	1.008

		Calculate mass	1487.88	977.5486	1698.0318	1795.0935	1824.163	1432.854	2617.6830	2516.636	2601.701	2488.639	2730.771	2770.799	2530.655	2463.6050	2673.7374	2687.7608
		Structure	9mer	Des-LY2000 Leu-Pro-Pro	Des-ProLY2000	Des-LY2000	DIC-endcapping-11mer	Ac end-capping-8mer	Tyr(Des-tBu)	Des-Thr30	Tyr(Otbu) to Phe	-Glu(OtBu)	+Gly	+Pro 28 or 29	-Ser(tBu)	-LeuPro	LSN4182913/140019	M+14
step1_PAR5	CDo140019-03-3104-012-15	%Area	1.41%	0.06%	0.22%	0.11%	0.18%	0.2306%	1.56%	0.21%		-		0.08%		-	90.09%	0.42%
step1_PAR9	CDo140019-03-3104-004-15	%Area	-	0.12%	0.21%	0.28%	0.16%	0.8544%	1.65%	0.49%		0.16%		0.14%		0.22%	90.83%	0.46%

Conclusion:

In PAR 5, it was observed that the 9-mer impurity level was very high compared to other batches. We suspect low de-Fmoc conditions caused the incomplete de-fmoc after Thr30 coupling because peptide chain occurred β-fold upon completion of the coupling. Then in the following AA de-fmoc after trimer and sidechain, fmoc group was removed and Pro coupled, which resulted in a significant increase in the 9mer impurity.

In PAR 9, reducing the coupling temperature directly influence the tripeptide coupling reaction efficiency, leading to an increase in the Ac end-capping-8mer impurity. These observations are consistent with the conclusions from our previous baseline experiments in PAR 1 and 11.

4.6 Experimental Details:

Objective: To evaluate the impact on the product when using a high-substitution-degree resin under baseline process conditions

Table 4.6.1 Data Summary for peptidyl resin and carry forward to Step 2 yield

删除[Handong Cheng15615]: 1400190301
删除[Angao Yuan]: Fmoc-endcapping-8mer
删除[Handong Cheng15615]: Lys
删除[Handong Cheng15615]: (SC)
删除[Handong Cheng15615]: Lys
删除[Handong Cheng15615]: Lys
删除[Handong Cheng15615]: (SC)
删除[Handong Cheng15615]: (SC)
删除[Xinran Geng18625]: than
删除[Nil Tandogan Gray]: ly
删除[Xinran Geng18625]: , b
删除[Angao Yuan]:
删除[Angao Yuan]: to
删除[Xinran Geng18625]: was
删除[Xinran Geng18625]:
删除[Angao Yuan]: octapeptide
删除[Xinran Geng18625]: but
删除[Angao Yuan]: leading to difficulties in Fmoc remov
删除[Xinran Geng18625]: this
删除[Angao Yuan]: impaired
删除[Angao Yuan]: Fmoc-endcapping-8mer
删除[Handong Cheng15615]: , therefore, it is recommer

PAR study Number	Step 1				Step 2 cleaved by TCA			
	Scale (Resin amount*Sub)	Net weight of peptide-resin	LOD	Gain-in Yield Uncorrected with LOD	Input peptidyl resin	Output L1/ ASYM-140019	Assay	Two steps overall yield
Run8	60*0.88 (Resin: Sunresin)	193.22 g	7.01%	93.0%	30g	17.12 g	90.22%	76.90%

Table 4.6.2 Summary 6 investigated parameters of high loading resin (0.88 mmol/g) use

Unit Operation	Unit	Set point (@)	Lower limit (-)	Upper limit (+)	8
Piperidine %	%	20.00	15.00	25.00	@
Piperidine/DMF vol	V	6	5.244	6.756	@
DeFmoc Temp	℃	25	13	37	@
DeFmoc stir time (min) - except Val and trimer	Min	120	60	180	@
DeFmoc stir time (min) – only Val and trimer	Min	30	15	45	@
Post-DeFmoc wash vol (DMF)	V	8	6.992	8	@
Post-DeFmoc wash vol (IPAC)	V	10	8.740	10	@
Post-DeFmoc DMF resewell vol	V	10	8.740	10	@
AA eq- except LY2000	Eq	2.00	1.874	2.126	@
Oxyma eq- except trimer & LY2000	Eq	2.00	1.748	2.252	@
DIC eq- except trimer & LY2000	Eq	2.20	1.9228	2.4772	@
AA eq- LY2000	Eq	1.50	1.4055	1.5945	@
Oxyma eq- trimer & LY2000	Eq	3.00	2.622	3.378	@
DIC eq- trimer & LY2000	Eq	3.30	2.8842	3.7158	@
Coupling DMF Vol- except LY2000	V	8	6.992	9.008	@

删除[Handong Cheng15615]: 1400190301

删除[Angao Yuan]: **Matrix**

删除[Angao Yuan]: **for Investigated Parameters**

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

Unit Operation	Unit	Set point (@)	Lower limit (-)	Upper limit (+)	8
Coupling DMF Vol-LY2000	V	9	7.866	10.134	@
Target activation time (min)	Min	30	0	70	@
Activation Temp	℃	20	20	32	@
Total couple time (min) - except trimer &LY2000	min	240	240	600	@
Total couple time (min) - trimer	Min	480	480	1020	@
Total couple time (min) -LY2000	Min	480	480	1560	@
Coupling Temp (non-trimer&LY2000)	℃	25	17	33	@
Coupling Temp (LY2000)	℃	30	22	38	@
Coupling Temp (trimer)	℃	35	17	41	-
Post-couple wash vol	V	7	6.118	7	@
Post Leu-Pro-Pro Capping Ac2O eq	Eq	1.25	1.0925	1.4075	@
Post Leu-Pro-Pro Capping pyr eq	Eq	1.50	1.311	1.689	@
Post Leu-Pro-Pro DMF vol	V	6	5.244	6.756	@
Post Leu-Pro-Pro Capping time	min	60	0	120	@
Post Leu-Pro-Pro Capping temp	℃	25	13	37	@
Post Leu-Pro-Pro Capping DMF vol	V	7	7	7	@
Final washing vol	V	8	7	8	@
Final washing temp	℃	20	8	20	@
Drying temp - split A	℃	30	30	30	@
Drying temp - split B	℃	30	30	42	@
Purity (%)	-	-	-	-	91.19%
LOD (%)	-	-	-	-	7.01
Gain-in Yield (%)	-	-	-	-	93.0

删除[Handong Cheng15615]: 1400190301

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Post-capping wash vol

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Results & Discussion: The standard experimental setup outlined in Section 3 was used, with the exception that a resin with a higher substitution level (0.88 mmol/g) was employed for the coupling. The detailed experimental matrix is presented in Table 4.6.1.

Table 4.6.3 Data Summary 6: impurity profile of cleaved isolated Solid of high resin loading

Exp.No.	Fraction No.	No.	1	2	3	4	5	6	7	8		9		10		11	12	13
		RT [min]	14.734	15.045	16.164	16.475	18.651	22.692	28.287	30.308		30.557		30.774		30.898	31.085	31.334
		RRT	0.474	0.484	0.520	0.530	0.600	0.730	0.910	0.975		0.983		0.990		0.994	1.000	1.008
		Calculate mass	1487.88	977.5486	1698.0318	1795.0935	1824.163	1432.854	2617.6830	2516.636	2601.701	2488.639	2730.771	2770.799	2530.655	2463.6050	2673.7374	2687.7608
		Structure	9mer	Des-LY2000 Leu-Pro-Pro	Des-ProLY2000	Des-LY2000	DIC-endcapping-11mer	Ac end-capping-8mer	Tyr(Des-tBu)	Des-Thr30	Tyr(Otbu) to Phe	-Glu(OtBu)	+Gly	+Pro 28 or 29	-Ser(tBu)	-LeuPro	LSN4182913/140019	M+14
step1_PAR8	CDo140019-03-3104-003-15	%Area	-	0.12%	0.17%	0.32%	0.22%	0.42%	1.72%	0.37%		0.17%		0.14%		0.23%	91.19%	0.49%

Conclusion: The use of a high-substitution resin (0.88 mmol/g) had an impact on the impurity profile and product purity that remained within acceptable limits, and the product still met the quality specification of LSN4182913 / ASYM-140019. Therefore, it is recommended to set the upper limit for the resin substitution level at 0.88 mmol/g

4.7 Experimental Details of DKP Build:

Table 4.7.1 Data Summary for peptide-on-resin and carry forward to Step 2 yield

删除[Handong Cheng15615]: 1400190301

删除[Angao Yuan]: **Data Summary**

删除[Angao Yuan]: **-Part 2**

删除[Angao Yuan]: Fmoc-endcapping-8mer

删除[Handong Cheng15615]: ys(SC)

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: (SC)

删除[Handong Cheng15615]: (SC)

删除[Handong Cheng15615]: 90

删除[Xinran Geng18625]: **yl**

PAR study Number	Step 1				Step 2 cleaved by TCA			
	Scale (Resin amount*Sub)	Net weight of peptide-resin	LOD	Gain-in Yield Uncorrected with LOD	Input peptidyl resin	Output L1/ ASYM-140019	Assay	Two steps overall yield
Run6	60*0.75 (Resin: Nankai)	175.44 g	6.93%	94.2%	30g	16.80 g	89.51%	78.56%
Run14	60*0.75 (Resin: Nankai)	179.15 g	6.62%	97.9%	30g	15.91 g	87.29%	76.33%
Run15	60*0.75 (Resin: Nankai)	179.08 g	7.28%	96.7%	30g	16.29 g	90.13%	80.14%

删除[Handong Cheng15615]: 1400190301

Table 4.7.2 Summary 6: Investigated Parameters Matrix of Racemization/ Epimerization/ DKP Build

删除[Angao Yuan]: **Summary of Matrix for Investigated Parameters**

Unit Operation	Unit	Set point (@)	Lower limit (-)	Upper limit (+)	6	14	15
Piperidine %	%	20.00	15.00	25.00	@	@	@
Piperidine/DMF vol	V	6	5.244	6.756	@	@	@
DeFmoc Temp	℃	25	13	37	+	@	+
DeFmoc stir time (min) - except Val and trimer	Min	120	60	180	+	@	+
DeFmoc stir time (min) – only Val and trimer	Min	30	15	45	+	@	+
Post-DeFmoc wash vol (DMF)	V	8	6.992	8	@	@	@
Post-DeFmoc wash vol (IPAC)	V	10	8.740	10	@	@	@
Post-DeFmoc DMF resewell vol	V	10	8.740	10	@	@	@
AA eq- except LY2000	Eq	2.00	1.874	2.126	-	@	@
Oxyma eq- except trimer & LY2000	Eq	2.00	1.748	2.252	-	@	@
DIC eq- except trimer & LY2000	Eq	2.20	1.9228	2.4772	+	@	@

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Unit Operation	Unit	Set point (@)	Lower limit (-)	Upper limit (+)	6	14	15
AA eq-LY2000	Eq	1.50	1.4055	1.5945		@	@
Oxyma eq- trimer &LY2000	Eq	3.00	2.622	3.378		@	@
DIC eq- trimer &LY2000	Eq	3.30	2.8842	3.7158	+	@	@
Coupling DMF Vol- except LY2000	V	8	6.992	9.008	8V	@	@
Coupling DMF Vol-LY2000	V	9	7.866	10.134	9V	@	@
Target activation time (min)	Min	30	0	70	+	+	@
Activation Temp	℃	20	20	32	+	+	@
Total couple time (min) - except trimer &LY2000	min	240	240	600	+	+	@
Total couple time (min) - trimer	Min	480	480	1020	+	+	@
Total couple time (min) -LY2000	Min	480	480	1560	+	+	@
Coupling Temp (non-trimer&LY2000)	℃	25	17	33	+	+	@
Coupling Temp (LY2000)	℃	30	22	38	+	+	@
Coupling Temp (trimer)	℃	35	17	41	-	+	-
Post-couple wash vol	V	7	6.118	7	@	@	@
Post Leu-Pro-Pro Capping Ac2O eq	Eq	1.25	1.0925	1.4075	@	@	@
Post Leu-Pro-Pro Capping pyr eq	Eq	1.50	1.311	1.689	@	@	@
Post Leu-Pro-Pro Capping DMF vol	V	6	5.244	6.756	@	-	@
Post Leu-Pro-Pro Capping time	min	60	0	120	@	@	@
Post Leu-Pro-Pro Capping temp	℃	25	13	37	@	@	@
Post-capping wash vol	V	7	7	7	@	@	@
Final washing vol	V	8	7	8	@	@	@
Final washing temp	℃	20	8	20	@	@	-
Drying temp - split A	℃	30	30	30	@	@	@
Drying temp - split B	℃	30	30	42	@	@	@
Purity (%)	-	-	-	-	89.52%	90.90%	89.85%

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Unit Operation	Unit	Set point (@)	Lower limit (-)	Upper limit (+)	6	14	15
LOD (%)	-	-	-	-	6.93	6.62	7.28
Gain-in Yield (%)	-	-	-	-	94.2	97.9	96.7

Results & Discussion:

The standard experimental setup outlined in Section 3 was used to investigate the impurity profile associated with DKP formation by intensifying the amino acid reaction conditions. PAR 15 was conducted as a supplementary experiment to PAR 6. The detailed experimental matrix is presented in Table 4.7.1. PAR 14 was designed to investigate the generation and profile of isomers by applying elevated temperatures.

Table 4.7.3 Data Summary 7: impurity profile of Racemization/ Epimerization/ DKP Build

Exp.No.	Fraction No.	No.	1	2	3	4	5	6	7	8		9		10		11	12	13
		RT [min]	14.734	15.045	16.164	16.475	18.651	22.692	28.287	30.308		30.557		30.774		30.898	31.085	31.334
		RRT	0.474	0.484	0.520	0.530	0.600	0.730	0.910	0.975		0.983		0.990		0.994	1.000	1.008
		Calculate mass	1487.88	977.5486	1698.0318	1795.0935	1824.163	1432.854	2617.6830	2516.636	2601.701	2488.639	2730.771	2770.799	2530.655	2463.6050	2673.7374	2687.7608
		Structure	9mer	Des-LY2000-Leu-Pro-Pro	Des-Pro-LY2000	Des-LY2000	DIC-endcapping-11mer	Ac end-capping-8mer	Tyr(Des-tBu)	Des-Thr30	Tyr(Otbu) to Phe	-Glu(OtBu)	+Gly	+Pro 28 or 29	-Ser(tBu)	-LeuPro	LSN4182913/140019	M+14
step1_PAR6	CDo140019-03-3104-001-15	%Area	-	0.20%	0.17%	0.16%	0.23%	0.0620%	1.55%	0.21%		0.21%		0.12%		1.27%	89.52%	0.45%
step1_PAR14	CDo140019-03-3104-017-15	%Area	-	0.13%	0.17%	0.07%	0.15%	0.1665%	0.87%	0.18%		0.07%		0.10%		0.55%	90.90%	-

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删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: Lys

删除[Handong Cheng15615]: (SC)

删除[Handong Cheng15615]: (SC)

step1_PAR15	CDo140019-03-3104-018-15	%Area	-	0.37%	0.14%	0.07%	0.16%	0.2376%	1.59%	0.30%	0.05%	0.13%	0.70%	89.85%	-

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Conclusion:

In PAR 6, the elevated temperature and extended de-fmoc reaction time exacerbated the DKP effect, leading to a significant increase in the -LeuPro impurity. In the supplementary experiment, PAR 15, the -LeuPro impurity remained markedly elevated even under mitigated reaction conditions. Therefore, using the PAR 15 data as a reference for the upper limit, it is recommended to set the upper limit for the de-Fmoc reaction temperature at 30°C.

In PAR 14, no significant changes in isomeric impurities were observed despite the intensified reaction conditions. However, the elevated temperature led to an increase in DKP impurities. Therefore, it is recommended to set the upper limit for the amino acid activation temperature at 23°C.

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4.8. Stability Holds Study

Experimental Details:

To support manufacturing, informal stability data for the peptidyl resin (ASYM-141114) of LSN4182913/ASYM-140019 were collected in the laboratory. After 6 months of storage at 2–8 °C in brown medicinal glass bottles, soft cleavage was performed according to the standard process described in Section 5. The detailed peptidyl resin source, purity, and impurity profile at time zero (release results) and 6 months are presented in **Table 4.8.1** below.

Table 4.8.1. Summary of Demonstrated Stability Holds Duration

No.	RT	RRT	Cal Mass	Impurity structure	PAR2 (Baseline large scale for cleavage PARs)	
					Peptidyl resin Batch: CDo140019-03-1577-002-01 (LOD: 7.36%)	
					Soft cleavage 2026/01/07	Soft cleavage 2026/07/22
1	4.010	0.129	no MS Signal	Unknown	0.0990%	

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2	8.206	0.264	354.166	Unknown		
3	9.046	0.291	892.521	4mer		
4	11.066	0.356	no MS Signal	Unknown		
5	12.341	0.397	1075.6592	DIC encapping 5mer		
6	13.708	0.441	1018.6378	DIC encapping 4mer		
7	14.175	0.456	1390.8316	8mer		
8	14.206	0.457	791.4366	Ac encapping 3mer		
9	14.734	0.474	1487.8800	9mer		
10	14.890	0.479	1159.6577	Pyr encapping 6mer		
11	15.045	0.484	977.5486	Formyl encapping 5mer	0.1259%	0.12%
12	16.164	0.520	1698.0120	Des-ProLY2000	0.0979%	0.10%
13	16.475	0.530	1795.0656	Des-LY2000	0.2410%	0.25%
15	17.687	0.569	1516.9518	DIC encapping 8mer		

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16	18.651	0.600	1824.1630	DIC-endcapping-11mer	0.1186%	0.12%
17	22.443	0.722	1130.6216	Fmoc(OH)-endcapping-4mer	0.1480%	0.13%
18	22.692	0.730	1432.8544	Ac-endcapping-8mer	0.2202%	0.22%
19	22.847	0.735	1726.0408	Formyl encapping 11mer	-	0.10%
20	23.096	0.743	1740.0936	Ac endcapping-11mer	-	-
21	24.371	0.784	1799.2174	Pro25-Gly33	-	-
22	24.464	0.787	1505.8671	Unknown	-	-
23	24.681	0.794	1783.0996	Unknown	-	-
24	25.334	0.815	1462.8656	8mer+72	0.1012%	0.11%
25	25.863	0.832	1770.0593	11mer+72	-	-
26	26.764	0.861	2898.9012	+ProLY2000	0.1739%	0.23%
27	26.920	0.866	2617.6830	SC(Des-tBu)	0.2183%	0.19%
28	27.044	0.870	2431.6326	Des-Trt	0.1031%	0.11%

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29	28.008	0.901	2310.4696	Unknown	-	-
E	28.287	0.910	2617.6830	Tyr(Des-tBu)	1.4371%	1.47%
31	29.220	0.940	2617.6830	Glu(Des-tBu)	0.5339%	0.64%
32	29.437	0.947	2617.6830	Thr36(Des-tBu)	0.4482%	0.53%
33	29.655	0.954	2617.6844	Thr30(Des-tBu)	0.1864%	0.23%
34	29.842	0.960	2617.6840	Ser(Des-tBu)	0.4767%	0.60%
35	30.090	0.968	2516.6346	Des-Thr36(OtBu)	-	0.09%
36	30.308	0.975	2516.6362	Des-Thr30	0.2234%	0.28%
			2601.7014	Tyr(Otbu) to Phe	0.0521%	0.04%
			2769.6794	+TFA	-	-
37	30.432	0.979	2744.7622	+βAla	-	-
38	30.463	0.980	2488.6392	-Glu(OtBu)	-	0.12%
	30.557	0.983	2730.7708	+Gly	-	

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39	30.774	0.990	2770.7994	+Pro28 or 29	0.0795%	0.13%--
			2530.6546	-Ser(tBu)	0.0266%	-
40	30.898	0.994	2463.6050	-LeuPro	0.3479%	0.24%
41	31.085	1.000	2673.7374	LSN4182913/140019	91.8657%	91.78%
42	31.334	1.008	2687.7608	M+14 (Val to Ile/Leu)	0.4937%	0.46%
43	31.582	1.016	2576.6936	-Pro28 or Pro29	0.1053%	0.08%
			2883.8846	+Leu27Pr28	0.0387%	0.01%
			2674.7484	Deamidation	0.0849%	0.15%
			2673.7476	Leu isomer	0.1177%	0.14%
44	31.707	1.020	2858.8558	+Glu(OtBu)	0.1405%	0.06%
			2517.6570	-Val32Gly33	0.0737%	0.05%
45 1	1.89323.906	026	2771.7976	unknown	0.1752%	0.03%
			2616.7240	-Gly	0.2314%	0.07%

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46	32.018	1.030	2702.7894	DIC encapping 12mer	-	-
47	32.328	1.040	2851.8276	Fm endcapping	0.2268%	0.28%
48	33.261	1.070	2754.7480	Fm encapping 12mer	0.1017%	0.05%
49	36.680	1.180	2687.7034	+CH2	-	-
50	37.084	1.193	2701.7430	Formyl encapping 13mer	-	0.08%
51	37.209	1.197	2604.6888	Formyl encapping 12mer	-	-
			2715.7336	Ac encapping	-	-
52	38.608	1.242	2911.7812	Fmoc(OH)	-	-
53	38.887	1.251			-	-
54	40.815	1.313	2588.5998	Formaldehyde encapping 12mer	-	-
Chromatogram						

4.9 Drying stability study

For verification purposes, the PAR2 peptidyl resin that had passed the drying acceptance criteria was subjected to a 7-day drying stress test at 42°C under a nitrogen atmosphere. Detailed information is provided in Table 7.3.1.

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Table 4.9.1 Summary of Drying Stability

Remove 10g resin and dry it at 42 °C for the duration.					PAR 2 Split 14.1.1 drying at 42°C	
Time					0d	6d18h
No.	RT [min]	RRT	Calculate mass	Structure	%Area	%Area
1	4.010	0.129	no MS Signal	Unknown	0.0990%	-
2	15.045	0.484	977.5486	Formyl encapping 5mer	0.1259%	0.1220%
3	16.164	0.520	1698.0120	Des-ProLY2000	0.0979%	0.1264%
4	16.475	0.530	1795.0656	Des-LY2000	0.2410%	0.2187%
5	18.651	0.600	1824.1630	DIC-endcapping-11mer	0.1186%	0.1746%
6	22.443	0.722	1130.6216	Fmoc(OH)-endcapping-4mer	0.1480%	0.1211%
7	22.692	0.730	1432.8544	Ac-endcapping-8mer	0.2202%	0.2493%
8	25.334	0.815	1462.8656	8mer+72	0.1012%	0.0975%
9	26.764	0.861	2898.9012	+ProLY2000	0.1739%	0.1514%

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10	26.920	0.866	2617.6830	SC(Des-tBu)	0.2183%	0.2389%
11	27.044	0.870	2431.6326	Des-Trt	0.1031%	0.1089%
12	28.287	0.910	2617.6830	Tyr(Des-tBu)	1.4371%	1.5565%
13	29.220	0.940	2617.6830	Glu(Des-tBu)	0.5339%	0.5631%
14	29.437	0.947	2617.6830	Thr36(Des-tBu)	0.4482%	0.4618%
15	29.655	0.954	2617.6844	Thr30(Des-tBu)	0.1864%	0.2170%
16	29.842	0.960	2617.6840	Ser(Des-tBu)	0.4767%	0.5170%
17	30.308	0.975	2516.6362	Des-Thr30	0.2234%	0.2311%
			2601.7014	Tyr(Otbu) to Phe	0.0521%	0.0547%
			2769.6794	+TFA	-	-
19	30.774	0.990	2770.7994	+Pro28 or 29	0.0795%	0.0936%
			2530.6546	-Ser(tBu)	0.0266%	0.0270%

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20	30.898	0.994	2463.6050	-LeuPro	0.3479%	0.1887%
21	31.085	1.000	2673.7374	LSN4182913/140019	91.8657%	91.1687%
22	31.334	1.008	2687.7608	M+14 (Val to Ile/Leu)	0.4937%	0.3597%
23	31.582	1.016	2576.6936	-Pro28 or Pro29	0.1053%	0.1304%
			2883.8846	+Leu27Pr28	0.0387%	0.0542%
			2674.7484	Deamidation	0.0849%	0.1401%
			2673.7476	Asn isomer	0.1177%	0.1357%
24	31.707	1.020	2858.8558	+Glu(OtBu)	0.1405%	-
			2517.6570	-Val32Gly33	0.0737%	-
25	31.893	1.026	2771.7976	unknown	0.1752%	0.0527%
			2616.7240	-Gly	0.2314%	0.0669%
26	32.328	1.040	2851.8276	Fm endcapping	0.2268%	0.2605%

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Asymchem Confidential Information

27	33.261	1.070	2754.7480	Fm encapping 12mer	0.1017%	0.0523%
29	38.608	1.242	2911.7812	Fmoc(OH)	-	0.1390%
Chromatogram					PAR2	CDo140019-03-1 577-017-02

Conclusion: No abnormal impurities were observed after the resin was stored at 2–8°C for more than 6 months and subjected to cleavage;

Conclusion: After drying the peptide resin at 42°C for 9720 min, an increase in isomer impurities was observed, while the distribution of other impurities showed no significant changes. Therefore, it is recommended that the drying temperature should not exceed 35°C.

Note [2]: The experiment was carried out in accordance with the standard cleavage procedure and equipment

5. Information for Material Input

All materials and reagents used for the PAR experiments are production quality.
The analysis data for this lot of are listed in **Table 5.0.1&5.0.2.**

Table 5.0.1 Summary of all Material Lots for Step 1 PAR Study

Materials	Runs	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
SieberAmide Resin	lotNo.	L125060454	L125060454	L125060454	L125060454	L125060454	L125060454	L225120950	L225120836	L125060454	L125060454	L125060454	L125060454	L125060454	L125060454	L125060454	L125060454

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Remove 10g resin and dry it at 42 °C for the duration.
Drying TEMP
42℃

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37#Fmoc-Tyr(tBu)-OH	lotNo.	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL12407 0130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002
36#Fmoc-L-Thr(tBu)-OH	lotNo.	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	[redacted][3]	L225111436	L225111436	CPr138389-05-04-01	CPr138389-05-04-01[3]
35#Fmoc-L-Asn(Trt)-OH	lotNo.	L125070301	L125070301	L125070301	L125070301	L125070301	L125070301	L125070301	L125070301	L125070301	L125070301	CPS138519-03-02P-02	L125070301	L225120049	L225120049	CPr138519-03-02p-02	CPr138519-03-02p-02
34#Fmoc-L-Ser(tBu)-OH	lotNo.	L125040586	L125040586	L125040586	L125040586	L125040586	L125040586	L125040586	L125040586	L125040586	L125040586	L225120045	L125040586	L125040586	L125040586	L125040586	L125040586
33#Fmoc-Gly-OH	lotNo.	L225120048	L225120048	L225120048	L225120048	L225120048	L225120048	L225120048	L225120048	L225120048	L225120048	L225111434	L225120048	L225111702	L225111702	CPr138200-04-01-01	CPr138200-04-01-01
32#Fmoc-L-Val-OH	lotNo.	CPS138202-02-01-01	CPS138202-02-01-01	CPS138202-02-01-01	CPS138202-02-01-01	CPS138202-02-01-01	CPS138202-02-01-01	CPS138202-02-01-01	CPS138202-02-01-01	CPS138202-02-01-01	CPS138202-02-01-01	L225120043	CPS138202-02-01-01	L126010746	L126010746	CPr138202-04-01-01	CPr138202-04-01-01
31#Fmoc-Glu(OtBu)-OH•H2O	lotNo	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053
30#Fmoc-L-Thr(tBu)-OH	lotNo.	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L225111436	L225111436	CPr138389-05-04-01	L124090070
27#28#29#Fmoc-Leu-Pro-Pro-OH	lotNo.	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534
LY2000	lotNo.	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02
Fmoc-Pro-OH	lotNo.	L125070093	L225120917	L225120917	L125070093	L225111090	L225120917	L225120917	L225120917	L225120917	L225120917	L225121595	L225121595	L225121595	L225121595	CPr140730-01-01-03	CPr140730-01-01-03
Ac2O	lotNo.	L125080638	L125080638	L125080638	L125080638	L125080638	L125080638	L125080638	L125080638	L125080638	L125080638	L125080638	L125080638	L125080417	L125080417	L125080417	L125080417

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pyridine	lotNo.	L125080127	L125080127	L125080127	L125080127	L125080127	L125080127	L125080127	L125080127	L125080127	L125080127	L125110406	L125080127	L125080127	L125080127	L125080127	L125080127
Piperidine	lotNo.	L125110087-m-TF30	L125110087-m-TF30	L125110087-m-TF30	L125110087-m-TF30	L125110087-m-TF30	L125110087-m-TF30	L125110087-m-TF30	L125110087-m-TF30	L125110087-m-TF30	L125110087-m-TF30	L125100624	L125110087-m-TF30	L226020623	L226020623	L226020623	L226020623
Oxyma	lotNo.	L225111981	L225111981	L225111981	L225111981	L225111981	L225111981	L225111981	L225111981	L225111981	L225111984	L225111984	L225120161	L225111942	L225120161	L225120161	L225120161
DIC	lotNo.	L225111983	L225111983	L225111983	L225111983	L225111983	L225111983	L225111983	L225111983	L225111983	L225111983	L125120162	L225120001	FJL124070059-012	FJL124070059-012	L225120001	L225120001
DMF	lotNo.	L125120281-m-TF01	L125120281-m-TF01	L125120281-m-TF01	L125120281-m-TF01	L125120281-m-TF01	L125120281-m-TF01	L125120281-m-TF01	L125120281-m-TF01	L125120281-m-TF01	L125120281-m-TF01	L125120281-m-TF01	L125120281-m-TF01	L125120281-m-TF01	L125120281-m-TF01	L226012197	L226012197
IPAC	lotNo.	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L125120520	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L226011711	L226011711
IPA	lotNo.	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L226010079	L226010079
MTBE	lotNo.	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L225111835	L225111835

Note [3]: In experiments PAR12 and PAR16, two different lots of threonine were used for the coupling step.

Table 5.0.2 Material Lot Information and COAs for Step 1 PAR Study

S. No.	Raw material	Asymchem/ Supplier Lot	Supplier	Purity	COA	Specifications
1	Sieber amide resin	L125060454	Tianjin Nankai Hecheng Science & Technology Co., Ltd	N/A	L125060454.pdf	Table 5.1.1

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		L225120836	Sunresin New Materials Co., Ltd		10-240205.pdf	
		L225120950			L225120950.pdf	
2	Fmoc-Tyr(tBu)-OH	L124070130	Chengdu ZY Biochemical Technoliogy Co.,Ltd	99.88%	L124070130	
3	Oxyma	L225111981	Suzhou Highfine Biotech Co., Ltd	99.99%	L225111981.pdf	
		L225111984	Huanggang Luban Pharmaceutical Co., Ltd.	100%	L225111984.pdf	
		L225120161	Liren Medical Technology	99.9%	L225120161.pdf	
4	DIC	L124070059	Suzhou Highfine Biotech Co., Ltd	99.89%	L124070059.pdf	
		L225111983				
		L225120001	Zhejiang Bulk Chemical Co.Ltd	99.78%	DIC20251103..pdf	

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		L225120162	Liren Medical Technology	99.7%	N,N'-二异丙基碳二亚胺（693-13-0	
5	Fmoc-L-Thr(tBu)-OH	L225120052	Chengdu ZY Biochemical Technoliogy Co.,Ltd	99.92%	Fmoc-L-Thr(tBu)-OH ZYF07025041	
		L225111436	Liaoning Asymchem Laboratories Co.,Ltd	99.89%	YL003563.pdf	
		CPr138389-05-04P-01		100%	CPr138389-05-04-01-a COA-01-生	
6	Fmoc-L-Asn(Trt)-OH	L125070301	Chengdu ZY Biochemical Technoliogy Co.,Ltd	99.61%	L125070301.pdf	
		L225120049		99.60%	L225120049.pdf	
		CPs138519-03-02p-02		99.7%	L225111704.pdf	
7	Fmoc-L-Ser(tBu)-OH	L225120045	Chengdu ZY Biochemical Technoliogy Co.,Ltd	99.88%	L225120045.pdf	
		L125040586	Liaoning Asymchem Laboratories Co.,Ltd	99.9%	L125040586.pdf	

8	Fmoc-L-Gly-OH	L225120048	Chengdu ZY Biochemical Technoliogy Co.,Ltd	99.9%	L225120048.pdf	
		L225111702	Liaoning Asymchem Laboratories Co.,Ltd	99.7%	L225111702.pdf	
		L225111434				
		CPr138200-04-01-01				
9	Fmoc-L-Val-OH	CPs138202-02-01-01	Liaoning Asymchem Laboratories Co.,Ltd	99.9%	L225111703.pdf	
		CPr138202-04-01-01		99.9%	CPr138202-04-01-01	
		L126010746	Chengdu ZY Biochemical Technoliogy Co.,Ltd	99.92%	L126010746.pdf	
		L225120043		99.79%	L225120043.pdf	
10	Fmoc-L-Glu(OtBu)-OH.H2O	L125070053	Liaoning Asymchem Laboratories Co.,Ltd	99.94%	L125070053.pdf	

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11.	Fmoc-Leu-Pro-Pro-OH	L125090534	Jilin Asymchem Laboratories Co.,Ltd	99.7%	L125090534.pdf	
12.	Ac_2O	L125080638	TIANJIN YIFANG SCIENCE & TECHNOLOGY, CO., LTD.	99.9%	L125080638.pdf	
		L125080417	Tianjin Rmwell Chemical Industry Limited Company	99.51%	L125080417.pdf	
13.	pyridine	L125080127	Tianjin Rmwell Chemical Industry Limited Company	99.91%	L125080127.pdf	
14.	ASYM-140400/LY2000	CPr140400-01-03-02	Asymchem Laboratories (Tianjin) Co.,Ltd (TJ3)	99%	COA of LY2000 FMOC-LYS (CPr14	
15.	Fmoc-Pro-OH	CPr140730-01-01-03	Liaoning Asymchem Laboratories Co.,Ltd	100%	CPr140730-01-01 -03 COA-01-生效_	
		L225120917	ChengDu Chron Chemicals Co,.Ltd	99.9%	L225120917.pdf	
		L225121595	Chengdu ZY Biochemical Technoliogy Co.,Ltd	99.94	Package	
		L225120917	Zenji Pharmaceuticals(SU ZHOU) Ltd	100%	L225120917	

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16	Piperidine	L125110087	Tianjin Tairong Chemical Trading Co., Ltd	99.65%	L125110087.pdf	
		L226020623	Tianjin Bohua chemical reagents Co., Ltd	99.5%	L226020623.pdf	
17	DMF	L125120281	Tianjin Tairong Chemical Trading Co., Ltd	99.97%	L125120281.pdf	
		L226012197	Tianjin Yubo Fine Chemical Co., Ltd	99.99%	L226012197.png	
18	IPAC	L125090783	Tianjin Yubo Fine Chemical Co., Ltd	99.96%	L125090783.pdf	
		L226011711		99.86%	L226011711.png	
19	IPA	L125080512	Tianjin Yubo Fine Chemical Co., Ltd	99.96%	L125080512.pdf	
		L226010079		99.98%	L226010079.png	
		L125120520	Tianjin Tairong Chemical Trading Co., Ltd	99.92%	L125120520.pdf	

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20	MTBE	L125110298	Tianjin Tairong Chemical Trading Co., Ltd	99.67%	L125110298.pdf	
		L225111835	Tianjin Yubo Fine Chemical Co., Ltd	99.47%%	L225111835.png	

5.1. Summmary of raw material Analytical Method

All the samples were run using qualified methods. The data could be easily traceable.

The method parameters are shown in **Table 5.1.1.**

Table 5.1.1 Summary for Analytical Method

Raw Material Name	Current Status		
	Analytical test	Acceptance criteria	Test method
Fmoc-L-Tyr(tBu)-OH	Appearance	White to off-white solid	AM-001
	Identification by 1H-NMR	Conforms to structure	AM-429
	Identification by HPLC	Retention time conforms to reference	AM-YL003427-06
	Water content (% w/w)	≤1.0%	AM-YL003427-03
	D-isomer impurity (% area/area)	≤0.2%	AM-YL003427-02
	De-Fmoc impurity	≤0.1%	AM-YL003427-010
	CBZ impurity (%)	N/A	N/A
	Purity (% area/area)	≥99.0%	AM-YL003427-06
	Unspecified Impurity (% area/area)	≤0.2%	AM-YL003427-06
	Fmoc-β-Ala-OH	≤0.2%	AM-YL003427-06
	Assay	≥97.0%	AM-YL003427-06
Fmoc-L-Ser(tBu)-OH	Appearance	White to off-white solid	AM-001

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Intermediate/Reagent/solvent
Lot no.
Supplier
COA

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Raw Material Name	Current Status		
	Analytical test	Acceptance criteria	Test method
	Identification by ^{1}H-NMR	Conforms to structure	AM-429
	Identification by HPLC	Retention time conforms to reference	AM-YL003428-06
	Water content (% w/w)	≤1.0%	AM-YL003428-04
	D-isomer impurity (% area/area)	≤0.2%	AM-YL003428-02
	Purity (% area/area)	≥99.0%	AM-YL003428-06
	Unspecified Impurity (% area/area)	≤0.2%	AM-YL003428-06
	Fmoc-β-Ala-OH	≤0.2%	AM-YL003428-06
	De-Fmoc impurity (% w/w)	≤0.1%	AM-YL003428-010
	Assay	≤97%	AM-YL003428-06
Fmoc-Gly-OH	Appearance	White to off-white solid	AM-001
	Identification by H-NMR	Conforms with structure	AM-429
	Identification by HPLC	Retention time conforms to reference	AM-YL004846-05
	Purity	≥99.0%	AM-YL004846-05
	Unspecific impurity	≤0.2%	AM-YL004846-05
	Fmoc-β-Ala-Gly -OH	≤0.2%	AM-YL004846-05
	Fmoc-Gly-Gly-OH	≤0.1%	AM-YL004846-05
	Assay	≥97.0%	AM-YL004846-05
	Water content	≤1.0%	AM-YL004846-03
	De-Fmoc impurity	≤0.1%	AM-YL004846-012
Fmoc-L-Thr(tBu)-OH	Appearance	White to off-white solid	AM-001
	Identification by H-NMR	Conforms with structure	AM-429
	Identification by HPLC	Retention time conforms to reference	AM-YL003563-06
	Assay	≥97.0%	AM-YL003563-06
	Water content	≤1.0%	AM-YL003563-03
	Purity	≥99.0%	AM-YL003563-06
	Unspecified impurity	≤0.2%	AM-YL003563-06

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Raw Material Name	Current Status		
	Analytical test	Acceptance criteria	Test method
	D-isomer impurity	≤0.2%	AM-YL003563-05
	CBZ impurity	N/A	N/A
	Fmoc-β-Ala-OH	≤0.2%	AM-YL003563-06
	De-Fmoc impurity	≤0.1%	AM-YL003563-009
Fmoc-L-Asn(Trt)-OH	Appearance	White to off-white solid	AM-001
	Identification by H-NMR	Conforms with structure	AM-429
	Identification by HPLC	Retention time conforms to reference	AM-YL004843-003
	Assay	≥97.0%	AM-YL004843-003
	Water content	≤1.0%	AM-431
	Purity	≥99.0%	AM-YL004843-003
	D-isomer impurity	≤0.2%	AM-YL004843-004
	De-Fmoc impurity	≤0.1%	AM-YL004843-003
	Fmoc-β-Ala-OH	≤0.2%	AM-YL004843-003
	Unspecific impurity	≤0.2%	AM-YL004843-003
Fmoc-L-Val-OH	Appearance	White to off-white solid	AM-001
	Identification by ^{1}H-NMR	Conforms to structure	AM-YL003110-09
	Identification by HPLC	Retention time conforms to reference	AM-YL003110-08
	Water content (% w/w)	≤1.0%	AM-YL003110-05
	D-isomer impurity (% area/area)	≤0.2%	AM-YL003110-04
	Purity (% area/area)	≥99.0%	AM-YL003110-08
	Unspecified Impurity (% area/area)	≤0.2%	AM-YL003110-08
	Fmoc-β-Ala-OH	≤0.2%	AM-YL003110-08
	Fmoc-Leu-OH	≤0.1%	AM-YL003110-08
	Fmoc-Ile-OH	≤0.1%	AM-YL003110-08
	De-Fmoc impurity (% w/w)	≤0.1%	AM-YL003110-07
	Assay	≥97.0%	AM-YL003110-08
Fmoc-L-Glu-(OtBu)-OH·H_2O	Appearance	White to off-white solid	AM-001

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Raw Material Name	Current Status		
	Analytical test	Acceptance criteria	Test method
	Identification by ^{1}H-NMR	Conforms to structure	AM-429
	Identification by HPLC	Retention time conforms to reference	AM-YL005741-05
	Water content (% w/w)	≤6.0%	AM-YL005741-02
	D-isomer impurity (% area/area)	≤0.2%	AM-YL005741-01
	Purity (% area/area)	≥99.0%	AM-YL005741-05
	Unspecified Impurity (% area/area)	≤0.2%	AM-YL005741-05
	Fmoc-β-Ala-OH	≤0.2%	AM-YL005741-05
	Fmoc-β-Ala-Glu(OtBu)-OH	≤0.2%	AM-YL005741-05
	De-Fmoc impurity (% w/w)	≤0.1%	AM-YL005741-010
	Assay	≥97.0%	AM-YL005741-05
	CBZ impurity (%)	N/A	N/A
Fmoc-L-Pro-OH	Appearance	White to off-white solid	AM-001
	Identification by ^{1}H-NMR	Conforms to structure	AM-429
	Identification by HPLC	Retention time conforms to reference	AM-YL008688-004
	Water content (% w/w)	≤1.0%	AM-430
	D-isomer impurity (% area/area)	≤0.2%	AM-YL003423-02
	Purity (% area/area)	≥99.0%	AM-YL008688-004
	Unspecified Impurity (% area/area)	≤0.2%	AM-YL008688-004
	Fmoc-β-Ala-OH	≤0.2%	AM-YL008688-004
	De-Fmoc impurity (% w/w)	≤0.1%	AM-YL003423-013
	Assay	≥97.0%	AM-YL008688-004
(((9H-fluoren-9-yl)methoxy)carbonyl)-L-prolyl-L-proline/Fmoc-Pro-Pro-OH	Appearance	White to off-white solid	AM-001

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Raw Material Name	Current Status		
	Analytical test	Acceptance criteria	Test method
	Identification	Conforms to structure	AM-014
	Assay	≥97.0%	AM-014
	Purity	≥99.0%	AM-122935-004
	Fmoc-L-Pro-Pro-Pro-OH	≤0.1%	
	Fmoc-L-Pro-OH	≤0.1%	
	Unspecific impurity	≤0.2%	
	Fmoc-L-Pro-D-Pro-OH	≤0.2%	AM-122935-005
	Fmoc-D-Pro-D-Pro-OH	≤0.2%	
	Fmoc-D-Pro-L-Pro-OH	≤0.2%	
	Free Pro-OH	≤0.1%	AM-122935-006
	DCM	≤600ppm	AM-122935-003
	MeOH	≤3000ppm	
	EtOH	≤5000ppm	
	IPAc	≤5000ppm	
	ROI	≤5000ppm	AM-041
	Water content	≤1.0%	AM-003
Fmoc-L-Leu-OH	Appearance	White to off-white solid	AM-001
	Identification/	Conforms to structure	AM-014
	Assay	≥97.0%	AM-014
	Purity	≥99.0%	AM-YL003425-010
	Fmoc-L-lle-OH	≤0.1%	
	Unspecific impurity	≤0.2%	
	D-isomer impurity	≤0.2%	AM-YL003425-011
	De-Fmoc impurity	≤0.1%	AM-YL003425-013

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Raw Material Name	Current Status		
	Analytical test	Acceptance criteria	Test method
	Water content	≤1.0%	AM-003
Sieber Amide resin	Appearance	Pale white or light yellow beads	AM-001
	Identification by IR	The IR spectrum of the sample is consistent with that of the reference standard spectrum	AM-YL004889-003
	Substitution degree	0.7~0.9 mmol/g	AM-YL004889-002
	AM-YL004889-004	4.0~7.0	Based on the COA provided by the qualified supplier
	AM-YL004889-005	DV (90) :105μm-180μm	

6. Summary of Impurity Profiles

The data for the peptidyl resin and the cleaved isolated product carried forward to Step 2 are summarized in **Table 3.4.1.** Runs 11 and 12 exhibited comparable purity and yield. However, some batches of the isolated solid carried forward to Step 2 failed to meet the quality specifications (LSN4182913 / ASYM-140019 specifications).

Table 6.0.1 Date Summary for peptide resin and cleaved by TCA isolated product

No.	RT [min]	RRT	Cal mass	Structure	step1_PAR1	step1_PAR2	step1_PAR3	step1_PAR4	step1_PAR5	step1_PAR6	step1_PAR7	step1_PAR8
					Baseline 1	Baseline large scale for cleavage PARs	high conversion build	low coupling conversion build	low deFmoc conversion build	Racemization/ Epimerization/ DKP Build	Baseline 2	high resin loading
1	4.010	0.129	no MS Signal	Unknown		0.0990%	0.1071%	-	-	-	-	-
2	8.206	0.264	354.166	Unknown		-	-	-	-	-	-	-
3	9.046	0.291	892.521	4mer	-	-	-	-	-	-	-	-
4	11.066	0.356	no MS Signal	Unknown	-	-	-	-	-	0.1260%	-	-
5	12.341	0.397	1075.6592	DIC encapping 5mer	-	-	-	0.2212%	-	-	-	-
6	13.708	0.441	1018.6378	DIC encapping	-	-	-	0.1092%	-	-	-	-

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No.	RT [min]	RRT	Cal mass	Structure	step1_PAR1	step1_PAR2	step1_PAR3	step1_PAR4	step1_PAR5	step1_PAR6	step1_PAR7	step1_PAR8
					Baseline 1	Baseline large scale for cleavage PARs	high conversion build	low coupling conversion build	low deFmoc conversion build	Racemization/ Epimerization/ DKP Build	Baseline 2	high resin loading
				4mer								
7	14.175	0.456	1390.8316	8mer	-	-	-	-	0.1116%	-	-	-
8	14.206	0.457	791.4366	Ac encapping 3mer	-	-	-	-	-	-	-	-
9	14.734	0.474	1487.8800	9mer	-	-	-	-	1.4113%	-	-	-
10	14.890	0.479	1159.6577	Pyr encapping 6mer	-	-	-	-	-	0.1978%	-	-
11	15.045	0.484	977.5486	Formyl encapping 5mer	0.1011%	0.1259%	-	0.1773%	-	0.3252%	0.1245%	0.1174%
12	15.988	0.518	1359.8406	Unknown	-	-	-	0.1112%	-	-	-	-
13	16.164	0.520	1698.0120	Des-ProLY2000	-	0.0979%	0.2034%	-	-	0.1687%	0.1083%	0.1653%
14	16.475	0.530	1795.0656	Des-LY2000	0.1021%	0.2410%	0.5547%	-	0.2224%	0.1566%	0.1576%	0.3166%
15	17.532	0.564	2102.2716	Unknown	-	-	-	-	-	-	-	-
16	17.687	0.569	1516.9518	DIC encapping 8mer	-	-	-	0.3188%	0.1099%	-	-	-
17	18.651	0.600	1824.1630	DIC-endcapping-11mer	0.1500%	0.1186%	0.0975%	0.3331%	0.1817%	0.2292%	0.1301%	0.2372%
18	22.443	0.722	1130.6216	Fmoc(OH)-endcapping-4mer	0.1259%	0.1480%	0.1562%	0.1121%	0.1297%	0.1300%	0.1272%	0.1306%
19	22.692	0.730	1432.8544	Ac-endcapping-8mer	0.2957%	0.2202%	0.1767%	1.1486%	0.2306%	-	0.1932%	0.4205%
20	22.847	0.735	1726.0408	Formyl encapping 11mer	-	-	-	-	-	0.1821%	-	-

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No.	RT [min]	RRT	Cal mass	Structure	step1_PAR1	step1_PAR2	step1_PAR3	step1_PAR4	step1_PAR5	step1_PAR6	step1_PAR7	step1_PAR8
					Baseline 1	Baseline large scale for cleavage PARs	high conversion build	low coupling conversion build	low deFmoc conversion build	Racemization/ Epimerization/ DKP Build	Baseline 2	high resin loading
21	23.096	0.743	1740.0936	Ac endcapping-11mer	-	-	-	-	-	-	-	-
22	23.614	0.765	1471.7830	Unknown	-	-	-	0.0966%	-	-	-	-
23	24.371	0.784	1799.2174	Pro25-Gly33	-	-	-	-	-	-	-	-
24	24.464	0.787	1505.8671	Unknown	-	-	-	0.1035%	-	-	-	-
25	24.681	0.794	1783.0996	Unknown	-	-	-	-	-	0.1142%	-	-
26	25.334	0.815	1462.8656	8mer+72	-	0.1012%	0.3297%	0.1339%	-	0.1217%	-	0.1430%
27	25.863	0.832	1770.0593	11mer+72	-	-	0.2596%	-	-	-	-	0.1008%
28	26.764	0.861	2898.9012	+ProLys	0.1721%	0.1739%	0.1584%	0.2015%	0.1542%	0.1313%	0.1778%	0.1626%
29	26.920	0.866	2617.6830	SC(Des-tBu)	0.2415%	0.2183%	0.2263%	0.2031%	0.2358%	0.2307%	0.2477%	0.2292%
30	27.044	0.870	2431.6326	Des-Trt	0.1091%	0.1031%	0.1126%	0.1138%	0.1201%	0.1133%	0.1880%	0.1402%
31	28.008	0.901	2310.4696	Unknown	-	-	-	-	0.1139%	-	-	-
32	28.287	0.910	2617.6830	Tyr(Des-tBu)	1.6928%	1.4371%	1.5658%	1.6117%	1.5559%	1.5354%	1.5822%	1.7040%
33	29.220	0.940	2617.6830	Glu(Des-tBu)	0.5887%	0.5339%	0.5780%	0.6312%	0.5293%	0.5730%	0.6126%	0.5613%
34	29.437	0.947	2617.6830	Thr36(Des-tBu)	0.5126%	0.4482%	0.4906%	0.4676%	0.4563%	0.5035%	0.5372%	0.4753%
35	29.655	0.954	2617.6844	Thr30(Des-tBu)	0.2128%	0.1864%	0.1991%	0.2254%	0.2039%	0.2295%	0.3025%	0.2666%
36	29.842	0.960	2617.6840	Ser(Des-tBu)	0.5235%	0.4767%	0.5153%	0.4921%	0.4725%	0.5176%	0.5790%	0.5339%
37	30.090	0.968	2516.6346	Des-Thr36(OtBu)	-	-	-	-	-	-	0.1051%	-
38	30.308	0.975	2516.6362	Des-Thr30	0.2198%	0.2234%	0.1305%	0.4639%	0.1653%	0.1530%	0.2526%	0.3027%

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No.	RT [min]	RRT	Cal mass	Structure	step1_PAR1	step1_PAR2	step1_PAR3	step1_PAR4	step1_PAR5	step1_PAR6	step1_PAR7	step1_PAR8
					Baseline 1	Baseline large scale for cleavage PARs	high conversion build	low coupling conversion build	low deFmoc conversion build	Racemization/ Epimerization/ DKP Build	Baseline 2	high resin loading
			2601.7014	Tyr(Otbu) to Phe	0.0592%	0.0521%	0.0491%	0.0438%	0.0492%	0.0571%	0.0759%	0.0656%
			2769.6794	+TFA	-	-	-	-	-	-	-	-
39	30.432	0.979	2744.7622	+βAla	-	-	-	-	-	-	-	-
40	30.463	0.980	2488.6392	-Glu(OtBu)	-	-	-	-	-	0.1136%	0.1327%	0.1104%
	30.557	0.983	2730.7708	+Gly						0.1006%	-	-
41	30.774	0.990	2770.7994	+Pro28 or 29	-	0.0795%	0.0743%	0.0675% 0.1377%	-	0.0647%	0.0820%	0.0743%
			2530.6546	-Ser(tBu)		0.0266%	0.0232%			0.0514%	0.0721%	0.0724%
42	30.898	0.994	2463.6050	-LeuPro	0.2249%	0.3479%	0.2558%	0.3146%	-	1.2570%	0.3448%	0.2313%
43	31.085	1.000	2673.7374	LSN4182913/ 140019	91.9695%	91.8657%	91.3574%	89.2805%	90.9550%	89.5280%	91.4934%	91.1520%
44	31.334	1.008	2673.7374	LSN4182913/ 140019	0.3151%	0.3153%	0.3088%	0.2226%	0.2839%	0.3177%	0.3530%	0.3134%
			2687.7608	M+14 (Val to Ile/Leu)	0.1961%	0.1784%	0.1724%	0.1164%	0.1338%	0.1301%	0.1429%	0.1717%
45	31.582	1.016	2576.6936	-Pro28 or Pro29	0.1130%	0.1053%	0.1304%	0.1053%	0.1764%	0.0904%	0.1631%	0.1786%
			2883.8846	+Leu27Pr28	0.0401%	0.0387%	0.0425%	0.0617%	0.0560%	0.0247%	0.0448%	0.0455%
			2674.7484	Deamidation	0.0579%	0.0849%	0.1385%	0.1062%	0.0092%	0.1151%	0.0716%	0.0350%
			2673.7476	Leu isomer	0.1462%	0.1177%	0.1798%	0.1912%	0.2288%	0.1809%	0.1656%	0.0911%
46	31.707	1.020	2858.8558	+Glu(OtBu)	0.1297%	0.1405%	0.0271%	0.0829%	0.0473%	0.0367%	0.1062%	0.0971%
			2517.6570	-Val32Gly33	0.0576%	0.0737%	0.0188%	0.0376%	0.0164%	0.3104%	0.0643%	0.1071%
47	31.893	1.026	2771.7976	unknown	0.2436%	0.1752%	0.1443%	0.1327%	0.1523%	0.1142%	0.0579%	0.0709%

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No.	RT [min]	RRT	Cal mass	Structure	step1_PAR1	step1_PAR2	step1_PAR3	step1_PAR4	step1_PAR5	step1_PAR6	step1_PAR7	step1_PAR8
					Baseline 1	Baseline large scale for cleavage PARs	high conversion build	low coupling conversion build	low deFmoc conversion build	Racemization/ Epimerization/ DKP Build	Baseline 2	high resin loading
			2616.7240	-Gly	0.2354%	0.2314%	0.0795%	0.2157%	0.1623%	0.1088%	0.0935%	0.1015%
48	32.018	1.030	2702.7894	DIC encapping 12mer	-	-	-	0.1448%	-	-	-	-
49	32.328	1.040	2851.8276	Fm endcapping	0.2123%	0.2268%	0.1913%	0.1973%	0.0993%	0.5530%	0.1974%	0.2195%
50	33.261	1.070	2754.7480	Fm encapping 12mer	-	0.1017%	-	-	-	-	-	-
51	36.680	1.180	2687.7034	+CH2	-	-	-	-	-	-	-	-
52	37.084	1.193	2701.7430	Formyl encapping 13mer	-	-	-	-	-	0.1804%	-	-
53	37.209	1.197	2604.6888	Formyl encapping 12mer	-	-	-	0.0829% -	-	0.2236%	-	-
			2715.7336	Ac encapping						-		
54	37.201	1.206	2687.6886	unknown				0.0186%		-		
			3552.3480	unknown	-	-	-	0.0550%	-	-	-	-
55	38.608	1.242	2911.7812	Fmoc(OH)	-	-	-	0.0421%	-	-	-	-
56	38.887	1.251			-	-	-	-	-	0.1234%	0.1081%	-
57	38.658	1.252	1832.4925	unknown	-	-	0.0979%	-	-	-	-	-
58	38.886	1.260	1379.8209	unknown	-	-	-	-	-	-	-	-
59			1420.8240	unknown	-	-	-	-	-	-	-	-
60	40.815	1.313	2588.5998	Formaldehyde encapping 12mer	-	-	-	-	-	-	-	-

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Table 6.0.2 Summary of impurities in the SPPS of fragment L1

No.	RT [min]	RRT	Cal mass	Structure	step1_PAR9	step1_PAR10	step1_PAR11	step1_PAR12	step1_PAR13	step1_PAR14	step1_PAR15	step1_PAR16
					cold coupling	coupling long hold	Baseline Trimer 35°C	Baseline large scale Trimer 35°C	Deconvol PAR 4 (low equiv only)	Epim	DKP	Deconvol PAR 3
1	4.010	0.129	no MS Signal	Unknown	-	0.0965%	0.0979%	-	-	-	-	-
2	8.206	0.264	354.166	Unknown	-	-	0.2394%	-	-	-	-	-
3	9.046	0.291	892.521	4mer	-	-	-	-	-	0.2090%	-	-
4	11.066	0.356	no MS Signal	Unknown	-	-	0.6102%[1]	-	-	-	-	-
5	12.341	0.397	1075.6592	DIC encapping 5mer	-	-	-	-	-	-	-	-
6	13.708	0.441	1018.6378	DIC encapping 4mer	-	-	-	-	-	-	-	-
7	14.175	0.456	1390.8316	8mer	-	-	-	-	-	-	-	-
8	14.206	0.457	791.4366	Ac encapping 3mer	-	-	-	-	-	0.1159%	-	-
9	14.734	0.474	1487.8800	9mer	-	-	-	-	-	-	-	-
10	14.890	0.479	1159.6577	Pyr encapping 6mer	-	-	-	-	-	-	0.2095%	-
11	15.045	0.484	977.5486	Formyl encapping 5mer	0.1171%	-	0.0955%	0.1254%	-	0.1261%	0.3782%	0.1166%
12	15.988	0.518	1359.8406	Unknown	-	-	-	-	-	-	-	-
13	16.164	0.520	1698.0120	Des-ProLY2000	0.2048%	0.1203%	0.1076%	-	0.1060%	0.1193%	-	-
14	16.475	0.530	1795.0656	Des-LY2000	0.2753%	0.3265%	0.2436%	-	0.1959%	0.1799%	0.1656%	-
15	17.532	0.564	2102.2716	Unknown	-	-	0.3047%	-	-	-	-	-
16	17.687	0.569	1516.9518	DIC encapping 8mer	-	-	-	-	-	-	-	-

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No.	RT [min]	RRT	Cal mass	Structure	step1_PAR9	step1_PAR10	step1_PAR11	step1_PAR12	step1_PAR13	step1_PAR14	step1_PAR15	step1_PAR16
					cold coupling	coupling long hold	Baseline Trimer 35°C	Baseline large scale Trimer 35°C	Deconvol PAR 4 (low equiv only)	Epim	DKP	Deconvol PAR 3
17	18.651	0.600	1824.1630	DIC-endcapping-11mer	0.1586%	0.1564%	0.1474%	0.1295%	0.2307%	0.1588%	0.2284%	0.1877%
18	22.443	0.722	1130.6216	Fmoc(OH)-endcapping-4mer	0.1431%	0.1323%	0.1428%	0.1465%	0.1298%	0.1214%	0.1254%	0.1142%
19	22.692	0.730	1432.8544	Ac-endcapping-8mer	0.8544%	-	-	0.1720%	0.1425%	0.1785%	0.2471%	0.1863%
20	22.847	0.735	1726.0408	Formyl encapping 11mer	-	0.0986%	-	-	-	-	-	-
21	23.096	0.743	1740.0936	Ac endcapping-11mer	-	-	-	0.1332%	0.1196%	0.2107%	0.1022%	0.1504%
22	23.614	0.765	1471.7830	Unknown	-	-	-	-	-	-	-	-
23	24.371	0.784	1799.2174	Pro25-Gly33	-	-	-	-	-	0.0980%	-	-
24	24.464	0.787	1505.8671	Unknown	-	-	-	-	-	-	-	-
25	24.681	0.794	1783.0996	Unknown	-	-	-	-	-	-	-	-
26	25.334	0.815	1462.8656	8mer+72	0.1440%	0.1580%	-	-	-	0.1074%	-	-
27	25.863	0.832	1770.0593	11mer+72	-	-	-	-	-	-	-	-
28	26.764	0.861	2898.9012	+ProLys	0.1563%	0.1769%	0.1809%	0.1739%	0.1691%	0.2285%	0.1886%	0.2030%
29	26.920	0.866	2617.6830	SC(Des-tBu)	0.2226%	0.2221%	0.2097%	0.2395%	0.2123%	0.1687%	0.1941%	0.1954%
30	27.044	0.870	2431.6326	Des-Trt	0.1173%	-	-	0.1166%	0.1029%	0.1048%	0.1113%	0.1073%
31	28.008	0.901	2310.4696	Unknown	-	-	-	-	-	-	-	-
32	28.287	0.910	2617.6830	Tyr(Des-tBu)	1.6083%	1.5378%	1.4105%	1.6987%	1.2527%	0.8889%	1.6065%	1.6019%

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No.	RT [min]	RRT	Cal mass	Structure	step1_PAR9	step1_PAR10	step1_PAR11	step1_PAR12	step1_PAR13	step1_PAR14	step1_PAR15	step1_PAR16
					cold coupling	coupling long hold	Baseline Trimer 35°C	Baseline large scale Trimer 35°C	Deconvol PAR 4 (low equiv only)	Epim	DKP	Deconvol PAR 3
33	29.220	0.940	2617.6830	Glu(Des-tBu)	0.5637%	0.5768%	0.5250%	0.5967%	0.4827%	0.8108%	0.5581%	0.5602%
34	29.437	0.947	2617.6830	Thr36(Des-tBu)	0.4710%	0.4414%	0.4660%	0.4876%	0.3655%	0.3476%	0.4364%	0.4427%
35	29.655	0.954	2617.6844	Thr30(Des-tBu)	0.2277%	0.1928%	0.1800%	0.2120%	0.2385%	0.2202%	0.2553%	0.2560%
36	29.842	0.960	2617.6840	Ser(Des-tBu)	0.5285%	0.4982%	0.4748%	0.5244%	0.4244%	0.6063%	0.4858%	0.5028%
37	30.090	0.968	2516.6346	Des-Thr36(OtBu)	-	-	-	-	-	-	-	-
38	30.308	0.975	2516.6362	Des-Thr30	0.4341%	0.1065%	0.2086%	0.1573%	0.1344%	0.0725%	0.2741%	0.1492%
			2601.7014	Tyr(Otbu) to Phe	0.0601%	0.0590%	0.0573%	0.0623%	0.0426%	0.0356%	0.0759%	0.0427%
			2769.6794	+TFA	-	-	-	-	-	0.0334%	-	-
39	30.432	0.979	2744.7622	+βAla	-	-	-	-	0.1218%	-	0.0954%	-
40	30.463	0.980	2488.6392	-Glu(OtBu)	0.0971%	0.0916%	0.0649%	-	-	-	-	-
	30.557	0.983	2730.7708	+Gly	-	0.0245%	0.0303%		-	0.1006%	-	-
41	30.774	0.990	2770.7994	+Pro28 or 29	0.0879%	-	-	-	0.0905%	-	-	-
			2530.6546	-Ser(tBu)	0.0615%		0.0232%		0.0489%	0.0514%	0.0721%	0.0724%
42	30.898	0.994	2463.6050	-LeuPro	0.2224%	0.3106%	0.2669%	0.2277%	0.1929%	0.5901%	0.7073%	0.2267%
43	31.085	1.000	2673.7374	LSN4182913/140019	90.8718%	92.1295%	91.8124%	92.5643%	92.9092%	91.3533%	90.4357%	93.2262%
44	31.334	1.008	2673.7374	LSN4182913/140019	0.2704%	0.3603%	0.3127%	0.3007%	0.1346%	0.1345%	0.1522%	0.1195%
			2687.7608	M+14 (Val to Ile/Leu)	0.1918%	0.1246%	0.1603%	0.2005%	0.0208%	0.0277%	0.0163%	0.0110%
45	31.582	1.016	2576.6936	-Pro28 or Pro29	0.1838%	0.0934%	0.1196%	0.0679%	0.1285%	0.1713%	0.1683%	0.1092%

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No.	RT [min]	RRT	Cal mass	Structure	step1_PAR9	step1_PAR10	step1_PAR11	step1_PAR12	step1_PAR13	step1_PAR14	step1_PAR15	step1_PAR16
					cold coupling	coupling long hold	Baseline Trimer 35°C	Baseline large scale Trimer 35°C	Deconvol PAR 4 (low equiv only)	Epim	DKP	Deconvol PAR 3
			2883.8846	+Leu27Pr28	0.0497%	0.0258%	0.0526%	0.0107%	0.0558%	0.0381%	0.0701%	0.0386%
			2674.7484	Deamidation	0.1111%	0.0805%	0.1196%	0.1715%	0.0935%	0.1002%	0.1154%	0.1025%
			2673.7476	Leu isomer	0.1507%	0.1224%	0.1354%	0.1072%	0.1806%	0.2326%	0.2167%	0.2111%
46	31.707	1.020	2858.8558	+Glu(OtBu)	0.0421%	0.1256%	0.0315%	0.0743%	0.0500%	0.0457%	0.0677%	0.0390%
			2517.6570	-Val32Gly33	0.0197%	0.1290%	0.0214%	0.0843%	0.0319%	0.0414%	0.0960%	0.0169%
47	31.893	1.026	2771.7976	unknown	0.0469%	0.0772%	0.0717%	0.0823%	0.2885%	0.1292%	0.2987%	0.1538%
			2616.7240	-Gly	0.1127%	0.0527%	0.0442%	0.1042%	0.2723%	0.0875%	0.4400%	0.0580%
48	32.018	1.030	2702.7894	DIC encapping 12mer	-	-	-	-	-	-	-	-
49	32.328	1.040	2851.8276	Fm endcapping	0.2157%	0.2453%	0.2409%	0.1905%	0.2051%	-	0.3767%	0.2204%
50	33.261	1.070	2754.7480	Fm encapping 12mer	-	0.1195%	0.1145%	-	-	-	-	-
51	36.680	1.180	2687.7034	+CH2	-	-	-	-	-	0.0973%	-	-
52	37.084	1.193	2701.7430	Formyl encapping 13mer	-	-	-	-	-	0.1778%	-	-
53	37.209	1.197	2604.6888	Formyl encapping 12mer	-	-	-	-	0.0436%	0.0731%	0.1411%	-
			2715.7336	Ac encapping			-		0.0470%	0.0679%		
54	37.201	1.206	2687.6886	unknown			-	0.0186%	0.0097%	0.1055%		
			3552.3480	unknown	-	-	-	-	-	0.1163%	-	-
55	38.608	1.242	2911.7812	Fmoc(OH)	-	-	-	-	-	0.0186%	-	-

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No.	RT [min]	RRT	Cal mass	Structure	step1_PAR9	step1_PAR10	step1_PAR11	step1_PAR12	step1_PAR13	step1_PAR14	step1_PAR15	step1_PAR16
					cold coupling	coupling long hold	Baseline Trimer 35°C	Baseline large scale Trimer 35°C	Deconvol PAR 4 (low equiv only)	Epim	DKP	Deconvol PAR 3
56	38.887	1.251			0.1153%	-	-	-	-	-	-	-
57	38.658	1.252	1832.4925	unknown	-	-	-	-	-	-	-	-
58	38.886	1.260	1379.8209	unknown	-	-	-	-	-	0.1027%	-	-
59			1420.8240	unknown	-	-	-	-	0.0477%	-	-	-
60	40.815	1.313	2588.5998	Formaldehyde encapping 12mer			-	-	-	-	-	-

Note [4]: The impurity at RT11.006min (RRT 0.35) is indeed present in the PAR11 product, but no mass spectrometry signal was detected. It is speculated that this impurity may originate from equipment or reagent contamination during the cleavage process, as this impurity was not detected in the small-scale experiment. Recommend the PAR12 data be used as typical data. For RRT1.00 and RRT1.02, RRT 1.00 is located on the tail of the main peak, and the integration is performed by directly splitting the peak, which may lead to an overestimation of the detection result. For RRT 1.02, there may be co-eluting components. The column efficiency will go down when testing this sample. MS ration of each impurities have calculated and will do for future production samples.

Table 6.0.3. Data Summary of In-process Control

Test Item	Specification	Analytical Method
Appearance	White or practically white solid	AM-140019-002
Identification by UHPLC	Retention time conforms to reference	AM-140019-001
Identification by MS	Consistent with the exact mass	AM-013
Purity	NLT 90.0%	AM-140019-001

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Total Impurities	NMT 10.0%	AM-140019-001
ASYM-146983,(Des-tBu of Tyr)	NMT 2.0%	AM-140019-001
Any Unspecified Impurity	NMT 1.0%	AM-140019-001
Individual impurities NLT 0.10%	Report by RRT	AM-140019-001
Water content by KF	NMT 3.0%	AM-140019-003
Assay by UHPLC	NLT 80.0 and NMT 110.0 (%w/w)	AM-140019-001
Trichloroacetic acid content by HPLC	NMT 2.0%	AM-140019-004
Residual 2,6-Lutidine by HS-GC	Report	AM-140019-018
Residual Piperidine by HS-GC	Report	AM-140019-006
Residual organic Solvents by HS-GC	Isopropanol: Report t-Butanol: NMT 5000ppm Methyl tert-Butyl Ether: Report Isopropyl acetate: NMT 5000ppm n-Heptane: Report Toluene: Report N, N-Dimethylformamide: NMT 880ppm Acetonitrile: Report	AM-140019-005

6.1 PAR Experiment Conclusions for SPPS of Fragment L1

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In the PAR experiments 1~10, the coupling temperature of the trimer Fmoc-L-Leu-L-Pro-L-Pro-OH is 25°C, which is lower than the process temperature of 35°C. Therefore, the content of Ac-endcapping-8mer is slightly higher. We suggest setting the coupling temperature of the trimer Fmoc-L-Leu-L-Pro-L-Pro-OH to 30–35°C.

1. In PAR experiment 3, increasing the reagent equivalents for amino acids other than LY2000 did not result in any significant changes in the impurity profile. Conversely, reducing the coupling solvent (high concentration) for LY2000 caused the reaction mixture to become highly viscous, leading to mixing difficulties and insufficient reaction between the resin and the amino acids. This ultimately resulted in an increase in the Des-LY2000 impurity level. So the DMF volume of LY2000 coupling should be NLT 8V.

2. In PAR experiment 4, reducing equivalents, concentration and other low-conversion conditions has little impact on amino acid coupling except for Fmoc-L-Leu-L-Pro-L-Pro-OH. However, low-conversion conditions result in a lower conversion and Fmoc-L-Leu-L-Pro-L-Pro-OH coupling is not so good as baseline experiment. As a result, the content of Ac-endcapping-8mer increase to 1.14%.

3. In PAR experiment 5, due to the low deFmoc conversion condition and lower Ac_2O equivalent, the **9mer** impurity is significantly increased to 1.41%.

4. In PAR experiment 6, due to the high DeFmoc temperature and DeFmoc stir time, DKP impurity des-LeuPro impurity and Fm-end-capping impurity were higher than baseline batch.

5. In PAR experiment 8, the use of high-substitution resin resulted in no abnormalities in the weight-gain yield of the peptide-resin or the impurity profile of the product, indicating no significant impact on the reaction.

6. In PAR experiment 9, lowering the coupling temperature leads to incomplete trimer coupling, resulting in an increase in the Ac-end-capping 8mer impurity, while showing no significant impact on other amino acids.

7. In PAR experiment 10, simply extending the coupling reaction time has no impact on the product's purity, impurity profile, or yield.

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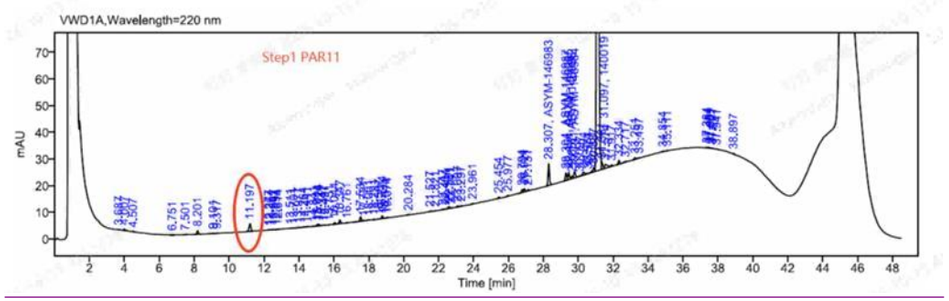
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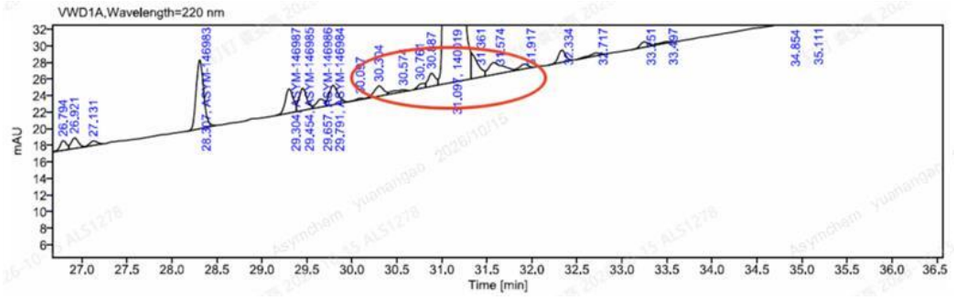
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8. In baseline PAR experiment 11 and experiment 12, the higher impurity levels observed at RRT 0.35, RRT 1.00, and RRT 1.02 compared with the other PAR studies. After data review, it is confirmed that the impurity with RRT 0.35 is indeed present in the PAR11 product, but no mass spectrometry signal was detected. It is speculated that this impurity may originate from equipment or reagent contamination during the cleavage process, as this impurity was not detected in the small-scale experiment. It is suggesting the PAR12 data be used as typical data.



9. For RRT1.00 and RRT1.02, RRT 1.00 is located on the tail of the main peak, and the integration is performed by directly splitting the peak, which may lead to an overestimation of the detection result. For RRT 1.02, there may be co-eluting components as follows chromatography and table show. It should be the column efficiency go down when testing this sample. MS ration of each impurities have calculated and will do for future samples.



Exp.No.	Fraction No.	No.	43				44	
		RT [min]	31.582				31.707	
		RRT	1.016				1.020	
		m/z	1289.3468	1442.9423	1338.3742	1337.8738	1430.4279	1259.8285
		Charge	2	2	2	2	2	2
		Calculate mass	2576.6936	2883.8846	2674.7484	2673.7476	2858.8558	2517.6570
		Delta mass	-97.046	210.145	1.009	0.008	185.116	-156.083
		Structure	-Pro28 or Pro29	+Leu27Pr28	Deamidation	Leu isomer	+Glu(OtBu)	-Val32Gly33
step1_PAR11	CDo140019-03-3104-006-15	%Area	0.4801%					
		MS %Area	0.1196%	0.0526%	0.1196%	0.1354%	0.0315%	0.0214%
step1_PAR12	CDo140019-03-3104-007-15	%Area	0.3572%				0.1586%	
		MS %Area	0.0679%	0.0107%	0.1715%	0.1072%	0.0743%	0.0843%

10. In PAR experiment 14, within a certain range, increasing the amino acid equivalents, extending the reaction time, raising the reaction temperature, and reducing the capping reagent equivalents have no significant impact on the product's purity.

11. In PAR experiment 15, due to the high DeFmoc temperature and DeFmoc stir time, DKP impurity and Fm-end-capping impurity were higher than baseline batch.

12. In PAR experiment 16, within a certain range, independently increasing the equivalents of the capping reagent and trimer, extending the reaction time, and raising the reaction temperature have no significant impact on the product's purity. When the coupling solvent volume was restored to the baseline level, the level of the Des-LY2000 impurity also returned to normal. This indicates that the solvent volume for LY2000 coupling is the primary influencing factor. Therefore, it is recommended to set the lower limit for the DMF volume at 9 V

Conclusions: The baseline experiments demonstrated the peptide resin ASYM-141114 and carry forward to Step2 isolated ASYM-140019 were as expected and met all specifications. The process could be used in the following PAR experiments and the results could be used as reference for comparison.

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删除[Handong Cheng15615]: In PAR experiment 13, simply extending the amino acid equivalents has no impact on the product's purity, impurity profile, or yield.

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7. Study on Impurity Spiking Experiment

Experimental Plan: Based on the test results of typical batches of ASYM - 140019, impurities RRT 0.910; Tyr37 (Des-tBu)；RRT 0.960; Ser34 (Des-tBu)；RRT 0.947; Thr36 (Des-tBu)；RRT 0.954; Thr30 (Des-tBu)；RRT 0.940; Glu31 (Des-tBu)；RRT 0.860; SC26 (Des-tBu)；RRT 0.722; Ac-endcapping-8mer; can participate in downstream reactions and be introduced into the API. Therefore, a spiking experiment is needed to confirm the purification method's ability to remove impurity derivatives. Synthesize the derivative markers of impurities RRT 0.910; Tyr37 (Des-tBu); RRT 0.960; Ser34 (Des-tBu); RRT 0.947; Thr36 (Des-tBu); RRT 0.954; Thr30 (Des-tBu); RRT 0.940; Glu31 (Des-tBu); RRT 0.860; SC26 (Des-tBu); RRT 0.722; Ac-endcapping-8mer separately. After obtaining the products, conduct spiking experiments. Separate impurity derivative markers of RRT 0.910; Tyr37 (Des - tBu), RRT 0.960; Ser34 (Des - tBu), RRT 0.947; Thr36 (Des - tBu), RRT 0.954; Thr30 (Des - tBu), RRT 0.940; Glu31 (Des - tBu), RRT 0.860; SC26 (Des - tBu) are to be mixed with the crude API 142526 at a mass ratio of 1:1 respectively, and then purified to confirm the purification method's capacity to remove impurity derivatives. The derivative marker of RRT 0.722; Ac - endcapping - 8mer is to be added alone to the crude API 142526, with a required proportion reaching 2.2% area, and then purified to confirm the purification method's ability to eliminate impurity derivatives. The design of the spiking experiment is shown in **Table 8.0.1**.

Table 7.0.1 Design of Impurity Spiking Experiment

Experiment No.	Name or Code	Main impurities before spiking (Marker)		Main impurities after spiking (Marker+ASYM-142526)		Suggested Spiking Amount	Real Spiking Amount
Experiment 1	RRT 0.910; Tyr37 (Des-tBu) derivative marker	RRT 0.886: 3.35%Area RRT 1.077: 8.00% Area	CDo140019-03-3040-087-05.pdf	RRT 0.886: 2.04%Area RRT 1.077: 4.87% Area	CDo140019-03-3040-087-04.pdf	50%(w/w)	50.01%(w/w)
Experiment 2	RRT 0.960; Ser34 (Des-tBu) derivative marker	RRT 0.886: 2.19% Area RRT 1.075: 1.65% Area	CDo140019-03-3040-096-02.pdf	RRT 0.886: 1.36%Area RRT 1.077: 1.46% Area	CDo140019-03-3040-096-05.pdf	50%(w/w)	50.04%(w/w)
Experiment 3	RRT 0.947; Thr36 (Des-tBu) derivative marker	RRT 0.886: 4.00 % Area RRT 1.077: 0.55% Area	CDo140019-03-3040-082-02.pdf	RRT 0.886: 2.18 % Area RRT 1.077: 2.10% Area	CDo140019-03-3040-082-04.pdf	50%(w/w)	50.00%(w/w)

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Experiment No.	Name or Code	Main impurities before spiking (Marker)		Main impurities after spiking (Marker+ASYM-142526)		Suggested Spiking Amount	Real Spiking Amount
Experiment 4	RRT 0.954; Thr30 (Des-tBu) derivative marker	RRT 0.885: 0.99% Area RRT 1.074 4.80% Area	CDo140019-03-3040-095-02.pdf	RRT 0.885: 0.81% Area RRT 1.074: 3.66% Area	CDo140019-03-3040-095-05.pdf	50%(w/w)	50.02%(w/w)
Experiment 5	RRT 0.940; Glu31 (Des-tBu) derivative marker	RRT 0.886: 1.36% Area RRT 1.048: 3.58% Area RRT 1.074: 5.43% Area RRT 1.107: 2.75% Area RRT 1.621: 8.78% Area	CDo140019-03-3040-080-03.pdf	RRT 0.886: 0.99% Area RRT 1.048: 2.44% RRT 1.074: 3.53% Area RRT 1.107: 1.58 % Area RRT 1.404: 5.64% Area	CDo140019-03-3040-080-05.pdf	50%(w/w)	50.00%(w/w)
Experiment 6	RRT 0.860; SC26 (Des-tBu) derivative marker	RRT 0.886: 3.72% Area RRT 1.048: 1.58% Area RRT 1.077: 1.60% Area RRT 1.202: 9.13% Area RRT 1.422: 5.10% Area	CDo140019-03-3040-081-04.pdf	RRT 0.886: 2.24% Area RRT 1.048: 2.15% Area RRT 1.077: 1.49% Area RRT 1.202: 5.22% Area RRT 1.422: 3.09% Area	CDo140019-03-3040-081-05.pdf	50%(w/w)	50.03%(w/w)
Experiment 7	RRT 0.722; Ac-endcapping-8mer derivative marker	RRT 0.327: 75.69% Area	CDo140019-03-3040-051-02.pdf	RRT 0.327: 2.50% Area RRT 1.033: 1.87% Area RRT 1.077: 1.18% Area	CDo140019-03-3040-051-07.pdf	2.2% Area	2.49% Area
Experiment 8	RTT 1.26; 2M+100	RRT 1.860: 81.38% Area	CDo140019-03-3040-104-13.pdf	RRT 0.962&0.973: 1.31% Area RRT 1.037:2.45% Area RRT 1.094:1.05%Area RRT 1.630&1.649: 3.09% Area RRT 1.653: 2.25% Area RRT1.860: 1.60%	CDo140019-03-3040-104-16.pdf	1.5% Area	1.60 %Area

Sample processing strategy: Dissolve the crude active pharmaceutical ingredient (API) in a 50% acetonitrile - water solution at a concentration of 40 g/L. Adjust the pH of the system to 8.00 - 9.00 using an aqueous sodium hydroxide solution of 1 mol/L (4% by mass), with the target pH controlled at 8.25. Subsequently, stir the mixture at 10 °C for 2 hours. After stirring, adjust the pH to 2.80 - 3.20 using a 2.5 mol/L citric acid solution, with the target pH being 3.00. Then, add an equal volume of a 10% acetonitrile - water solution to the system for a 1:1 dilution, so that the final proportion of acetonitrile in the system is 30%. The final concentration of the diluted crude API is 20 g/L.

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RP1 Collection Strategy: The purification conditions and collection strategy of RP1 are shown in Table 2. Purity and impurity profile of RP1 fractions is shown in Table 3.

3 injections (finished);

New analysis test method is performed to the fractions. (18min for single fraction, 50min for mock pool.)

RP1 Collection trigger: start collect when 280nm threshold reach up to 350mAu.

1. Fraction 1 collection 0.56min (0.1CV）;

2. Fraction 2 collection 1.38min (0.25CV);

3. Fraction 3 collection 1.38min (0.25CV);

4. Fraction 4 collection 3.19min (0.575CV);

5. Fraction 5 collection 3.19min (0.575CV);

6. Fraction 6 collection 3.19min (0.575CV);

Following every fraction collection 1.11min until UV＜500mAU (reference), stop(The last is subject to the actual collection time). An estimated 2 to 20 fractions will be collected per injection. Stop at 500mAu·

RP2 Collection Strategy: 3 injections (finished);

New analysis test method is performed to the fractions. (18min for single fraction, 50min for mock pool.)

RP2 Collection trigger: start collect when 280nm threshold reach up to 450mAu.

1. Fraction 1 collection 0.56min (0.1CV）;

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2. Fraction 2 collection 0.56min (0.1CV)；

3. Fraction 3 collection 0.84min (0.15CV)；

4. Fraction 4 collection 2.13min (0.38CV)；

5. Fraction 5 collection 2.13min (0.38CV)；

6. Fraction 6 collection 2.13min (0.38CV)；

Following every fraction collection 1.11min until UV＜500mAU (reference), stop(The last is subject to the actual collection time). An estimated 2 to 20 fractions will be collected per injection.

7.1 Expt.1: Spiking experiment study of Tyr37 (Des-tBu) at RRT 0.910

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- **Step1: L2-L1- Tyr37 (Des-tBu):**

Charge 2-MeTHF (14 V) and DMSO (1 V) into the reaction flask, adjust the temperature to 20 °C and start agitation. Add L1- Tyr37 (Des-tBu) (1 eq) and Fmoc-L2 (1.02 eq), and stir until complete dissolution. Charge Oxyma (1.50 eq) and EDCI (1.50 eq), then rinse the addition vessel with DMSO (1 V). Add 2,4,6-trimethylpyridine (3.00 eq) dropwise. Upon completion of addition, hold the reaction at 20 °C for 3 h. Sample for UPLC monitoring; proceed to the next step when residual L1- Tyr37 (Des-tBu) is <0.5%. Add piperidine (0.634 eq)/water (2 V) solution to the flask and react at 15 °C for 2 h. Monitor the reaction by sampling; once full conversion of Fmoc-L2-L1- Tyr37 (Des-tBu) is confirmed, allow phase separation and discard the aqueous layer. Perform two sequential acid washes: for each wash, add the prepared ~10% dilute HCl to the organic phase, stir for 10 min, settle, separate and remove the aqueous layer while retaining the organic phase. Conduct an alkaline wash by adding the prepared ~8% aqueous $NaHCO_3$, stir for 10 min, settle, separate and discard the aqueous layer. Carry out a brine wash with the prepared 18% aqueous NaCl, stir for 10 min, settle, separate and retain the organic phase.

Transfer the organic phase to a constant-pressure dropping funnel for crystallization. Charge n-heptane (30 V) into a second reaction flask, then add the organic phase dropwise at 20±5℃. After addition is finished, stir the slurry for 1 h and filter. Slurry-wash the filter cake with n-heptane/2-MeTHF mixture (3 V / 1 V) for 0.5 h, filter again, dry the solid and collect the product L2-L1- Tyr37 (Des-tBu).

- **Step2: Boc-L3-L2-L1- Tyr37 (Des-tBu)**

Charge 2-MeTHF (14 V) and DMSO (1 V) into the reaction flask, adjust the temperature to 20 °C and start agitation. Add L2-L1- Tyr37 (Des-tBu) (1.00 eq) and Boc-L3 (1.10 eq), and stir until complete dissolution. Charge Oxyma (1.50 eq) and EDCI (1.50 eq), then rinse the addition vessel with DMSO (1 V). Add 2,4,6-trimethylpyridine (3.00 eq) dropwise. Upon completion of addition, hold the reaction at 20 °C for 3 h. Sample for UPLC monitoring; proceed to the next step when residual L2-L1- Tyr37 (Des-tBu) is <0.5%.

Add purified water (4.58 V) to the batch, stir for 10 min, then allow phase separation and discard the aqueous layer. Perform acid wash: add the prepared ~10% dilute HCl to the organic phase, stir for 10 min, settle, separate and remove the aqueous layer while retaining the organic phase. Conduct brine wash with the prepared 18% aqueous NaCl, stir for 10 min, settle, separate and retain the organic phase.

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Transfer the organic phase to a constant-pressure dropping funnel for crystallization. Charge n-heptane (30 V) into a second reaction flask, then add the organic phase dropwise at 20±5℃. After addition is finished, stir the slurry for 1 h and filter. Slurry-wash the filter cake with n-heptane/2-MeTHF mixture (3 V / 1 V) for 0.5 h, filter again, dry the solid and collect the product Boc-L3-L2-L1- Tyr37 (Des-tBu).

- **Step3: impurity markers (L3-L2-L1)**

Charge toluene (6 V) and anisole (0.25 V) into the reaction flask and maintain the temperature at 20 °C. Add DTT (0.25 eq), purified water (0.25 V) and TIPS (0.193 eq). While holding at 20 °C, charge Boc-L3-L2-L1- Tyr37 (Des-tBu) (1.00 eq). Cool the batch to 10 °C, then add TFA (9.24 V) dropwise. Upon completion of addition, warm the mixture to 20 °C and hold the reaction for 3 h.

Sample the batch to confirm full conversion of Boc-L3-L2-L1- Tyr37 (Des-tBu). Cool the reaction mixture to −15 °C, then add pre-chilled MTBE (48 V) dropwise. After addition is complete, stir the slurry for 1 h.

Filter the solids, slurry-wash the filter cake with MTBE (3 V) for 0.5 h, and filter again. Dry the isolated solid and collect the product L3-L2-L1. Test the purity and submit for MS analysis.

Table 7.1.1 Experiment 1: Synthesis of Tyr37 (Des-tBu) impurity marker

Reaction and solid product purity for each step				
No.	Impurity name	Purity of product solid	Product solid Chromatogram	MS Chromatogram
Step 1	Tyr37 (Des-tBu) impurity Raw material purity: 92.40% Feed amount: 4 g	Step1 Product: 6.93 g purity: 89.62%	CDo140019-03-3 040-085-01	CDo140019-03-3 040-085-01-.PNG

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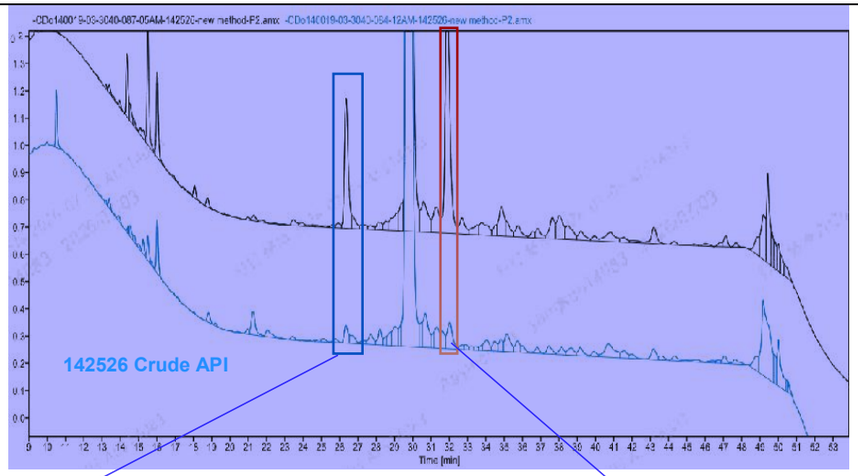
Step 2	Raw material purity：89.62% Feed amount: 6 g	Step2 Product: 7.10 g purity：74.45%	CDo140019-03-3 040-086-01	CDo140019-03-3 040-086-01.PNG
Step 3	Raw material purity：74.45% Feed amount: 6 g	Step3 Product: 4.40 g purity：59.2%	CDo140019-03-3 040-087-01	CDo140019-03-3 040-087-01.PNG

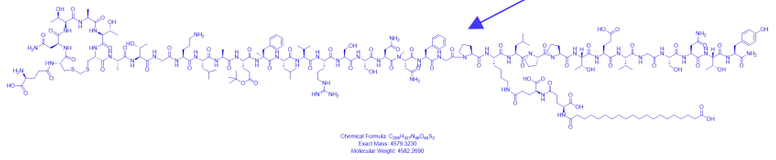
The Tyr37 (Des-tBu) impurity underwent a 3-step reaction (step1, step2, step3) to afford des-tBu-API, with the main peak corresponding to crude API (ASYM-142526). Des-tBu-API was spiked into crude API (ASYM 142526) at a level of 50%. The purity data for the impurity, the crude API (ASYM-142526), and the crude API spiked with des-tBu-API impurity (ASYM-142526) are presented in Table 8.1.2.

Then the Tyr37 (Des-tBu)-API impurity-spiked API sample (ASYM-142526) was adjusted to pH 8~9 with NaOH solution, stirred at 10 ± 5 °C for 2 h to transfer the depsipeptide, then re-adjusted the pH value to 2.8-3.2 with citric acid solution

7.1.1 Expt.1: UPLC Overlay Diagram of Experimental Group and Control Group

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Two major spiked impurities were included: +tBu, L3+L2+L2+L1, with corresponding RRT values of 0.886, 1.077 respectively.

7.1.2. Expt.1: LCMS data comparison of step 3 crude API

RT	14.381	16.014	21.261	26.343	29.036	29.313	29.726	30.469	30.683	31.324	31.616	31.866	32.689	33.748	34.826	35.143	37.448	38.143	38.666	40.691	40.784	43.173	48.798	49.179	49.375	49.814	50.013
RRT	0.484	0.539	0.715	0.886	0.977	0.986	1.000	1.026	1.033	1.054	1.064	1.077	1.100	1.135	1.172	1.182	1.260	1.283	1.301	1.369	1.372	1.452	1.642	1.654	1.661	1.676	1.682

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m/z	1285.634	-	1187.981	1528.449	1475.745	1528.435	1509.761	1475.746	1471.4152	1444.727	1504.758	1221.628	1095.095	1503.754	1456.086	1045.561	1527.4479	1547.46	1082.578	1191.994	1222.183	-	1592.7914	-	-	-	-
charge	2	-	3	3	3	3	3	3	3	3	3	5	2	3	3	2	3	3	3	3	2	-	3	-	-	-	-
Monoisotopic Mass mass	2568.252	-	3559.919	4579.343	4422.210	4580-276	4523.254	4422.210	4409.239	4329.153	4509.248	6103.969	2187.173	4506.237	4363.229	2088.104	4579.344	4636.356	3243.731	3571.956	2441.350	-	4775.374	-	-	-	-
Delta mass	-1955.000	-	-963.340	56.090	-101.040	57.080	0	-101.040	-114.020	-194.100	-14.010	1580.715	-2336.080	-17.020	-160.030	-2435.150	56.050	113.100	-1279.520	-951.300	-2081.900	-	252.080	-	-	-	-
Structure	Glu1-Phe23]+Pip	unknown	Orn11-Tyr37	+ **tBu**	Des Thr9	Endo Gly24	142526	Des Thr30	Des Asn21	DesPro^{28}Pro^{29}	-CH_2	**L3+L2+L2 +L1**	Phe23-Tyr37	Asn22 cyclization	unknown	105+[Pro25-Tyr37]	unknown	+Leu27	[Glu14-Tyr37], and Glu14-H_2O	Formaldehyde+[Orn11-Tyr37]	Formaldehyde+[Asn21-Tyr37]	unknown	+ Pbf	unknown	unknown	unknown	unknown
CPo142526-02-03-01	0.03	0.74	0.69	0.50	0.72	0.37	72.29	0.46	1.37	0.70	0.53	1.08	0.12	0.11	0.36	0.77	0.31	0.27	0.36	0.99	-	0.40	0.55	4.44	-	0.51	1.35
Exp.1(CDo140019-03-3040-087-05)	0.86	1.14	0.14	3.35	0.77	0.69	61.61	1.50		1.10		8.00	0.64	0.71	1.06	-	0.76	0.97	0.69	-	0.64	0.62	0.56	1.35	1.80	-	-

7.1.3 Expt.1: The purification conditions and collection strategy of RP1

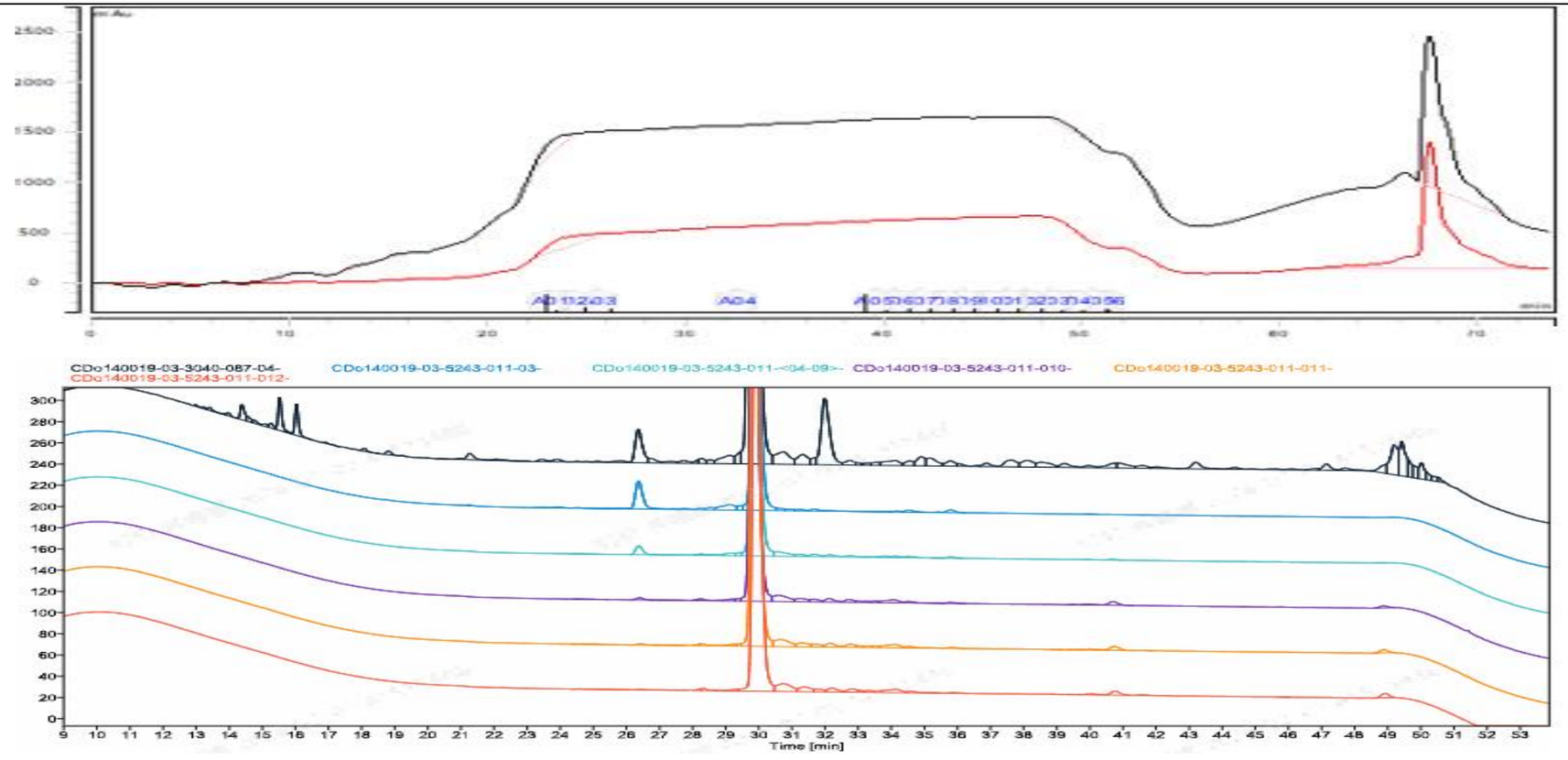
Column	10*250mm bed column=19.63 mL		
Sample	CDo140019-03-3040-087-04 (Purity: 68.61%)		
Resin	Daisogel 120-10-ODS-RPS		
Mobile phase A	90% 50 mM Citric Acid, **pH 3.00**: 10% ACN	Mobile phase B	Acetonitrile
Detection wavelength	230nm	Monitor wavelength	280nm
Flowrate	3.53mL/min(270cm/h)		
Loading amount	**45.0mg/mL, 883 mg used.**		

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Post-Load Wash	30~35.0(RP1)B% for 11.1min(2CV) at 3.53mL/min						
Elution	T(time)	0	62.5	62.5	73.6		
	B (%)	27.8	44.4	66.7	66.7		
UV	Start collection when UV reaches 350mAu						

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7.1.4 Expt.1: Purity and impurity profile of RP1 fractions

RRT	0.886	0.942	0.949	0.958	0.977	0.986	1.000	1.026	1.033	1.054	1.064	1.077	1.100	1.135	1.145	1.160	1.172	1.182	1.372	1.642	Spectrum Attachments
m/z	1528.449	-	-	-	1475.745	1528.435	1509.761	1475.746	1471.415	1444.727	1504.758	1221.628	1095.095	1503.754	-	-	1456.086	1045.561	1222.183	1592.7914	-
charge	3	-	-	-	3	3	3	3	3	3	3	5	2	3	-	-	3	2	2	3	
Monoisotopic Mass mass	4579.3431	-	-	-	4422.2095	4580-2759	4523.2544	4422.210	4409.239	4329.153	4509.248	6103.969	2187.1731	4506.2373	-	-	4363.229	2088.104	2441.3499	4775.3742	
Delta mass	56.09	-	-	-	-101.040	57.080	0	-101.040	-114.020	-194.100	-14.010	1580.715	-2336.080	-17.020	-	-	-160.030	-2435.150	-2081.900	252.0804	
Structure	+ **tBu**	unknown	unknown	unknown	Des Thr9	Endo Gly24	142526	Des Thr30	Des Asn21	$DesPro^{28}Pro^{29}$	$-CH_2$	**L3+L2+L2+L1**	Phe23-Tyr37	Asn22 cyclization	unknown	unknown	unknown	105+[Pro25-Tyr37]	Formaldehyde+[Asn21-Tyr37]	+ Pbf	
Exp.1 CDo140019-03-3040-087-05[3]	3.35	0.14	0.14	0.26	0.77	0.69	61.61	1.50		1.10	-	8.00	0.64	0.71	-	0.26	1.06	0.46	0.17	0.64	0.56	CDo140019-03-3040-087-05.pdf
Before RP1(after spiking) (CDo140019-03-3040-087-04[4]	2.04	-	0.26	0.25	1.12	-	68.61	1.50 -		0.78	0.46	4.74	0.45	-	0.62	0.30	0.34	0.85	0.18	0.36	0.53	CDo140019-03-3040-087-04.pdf
CDo140019-03-5243-011-03[5]	6.60	-	0.36	0.55	2.41	0.70	84.19	0.54 -		-	0.58	0.14	-	-	0.25	0.62	-	0.72	-	-	-	CDo140019-03-5243-011-03.pdf
CDo140019-03-5243-011-04-09[5]	1.72	-	0.25	0.13	0.62	0.41	91.12	1.59 -		0.53	0.58	0.50	0.31	0.27	0.31	0.37	-	0.27	-	-	-	CDo140019-03-5243-011-04-09.pdf
CDo140019-03-5243-011-10[5]	0.45	0.12	0.26	-	0.41	0.29	89.08	2.35 -		0.90	0.62	0.80	0.59	-	1.08	0.36	-	0.22	0.24	-	0.41	CDo140019-03-5243-011-10.pdf
CDo140019-03-5243-011-11[5]	0.22	0.13	0.30	-	0.36	0.21	88.33	2.53 -		1.07	0.61	0.86	0.67	-	1.15	0.37	-	0.23	0.19	-	0.65	CDo140019-03-5243-011-11.pdf
CDo140019-03-5243-011-12[5]	-	0.15	0.32	-	0.32	-	87.50	2.80		1.34	0.59	0.94	0.73	-	1.21	0.37	-	0.20	0.21	-	0.81	CDo140019-03-5243-011-12.pdf
Mock pool (Combine fraction: CDo140019-03-5243-011-03 ~ 12,11-18)[6]	1.95	0.15	0.15	0.26	0.75	0.45	90.22	1.57		0.44	0.60	0.51	0.32	-	0.41	0.24	-	0.34	0.14	-	0.11	CDo140019-03-5243-011-03-11,18-28.pdf

Note [3]: The identifier "Expt.1CDo140019-03-3040-087-05" refers to the synthesis experiment number of the Marker used for the spiking experiment.
Note [4]: Before RP1 (after spiking) — CDo140019-03-3040-087-04 denotes the sample number after spiking.
Note [5]: Fraction Identification for RP1 Step
Note [6]: Fraction Numbers of RP1 Pooled Fractions

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Conclusion：RP1 exhibits poor removal capacity for the impurity with RRT 0.886, while it performs well in eliminating the impurity at RRT 1.077.

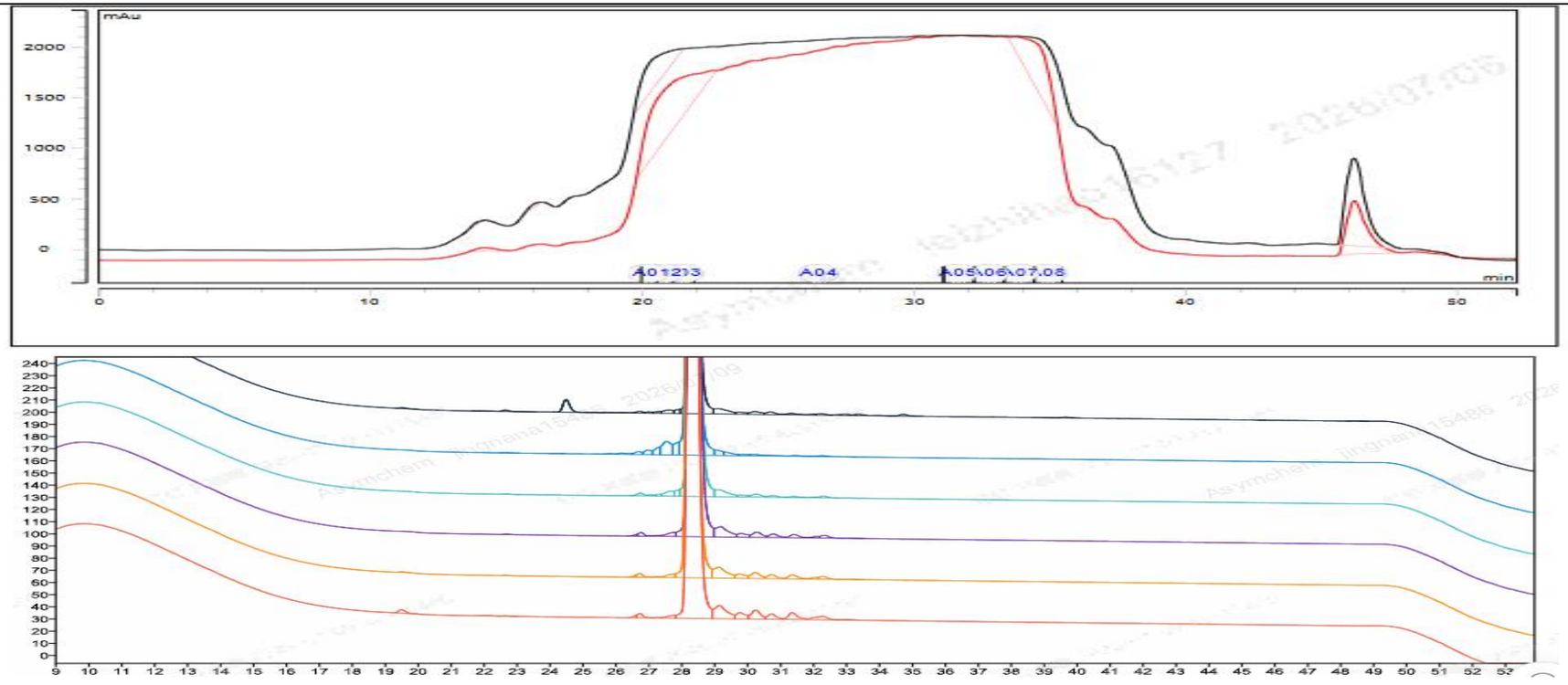
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7.1.5 Expt.1: The purification conditions and collection strategy of RP2

Column	10*250mm bed column=19.63 mL						
Sample	CDo140019-03-5243-017-01						
Resin	Daisogel 120-10-ODS-RPS						
Mobile phase A	90% 50 mM Citric Acid, **pH 6.50**: 10% ACN			Mobile phase B	Acetonitrile		
Detection wavelength	230nm			Monitor wavelength	280nm		
Flowrate	3.53mL/min(270cm/h)						
Loadingamount	**45.0 mg/mL**						
Post-Load Wash	20.0(RP2)B% for 11.1min(2CV) at 3.53mL/min						
Elution	T(time)	0	41.1	41.1	52.2		
	B(%)	20	35.6	66.7	66.7		
Collection	Start collection when the UV280 reaches 450mAu						

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7.1.6 Expt.1: Purity and impurity profile of RP2 fractions

Batch	Fraction Number	Impurity profile (RRT, %area) Only the components with the main - peak purity greater than 80% were counted													
RRt		0.886	0.942	0.949	0.958	0.974	0.987	1	1.024	1.046	1.057	1.063	1.077	1.099	1.145	Spectrum Attachments

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Before RP2[7] (CDo140019-03-5243-011-03 ~ 12,11-18)		1.95	0.15	0.15	0.26	0.75	0.45	90.22	1.57	0.44	-	0.6	0.51	0.32	0.41	CDo140019-03-5243-011-03-11,18-28.pdf
Loading: 45 mg/mL (CDo140019-03-5243-018)	CDo140019-03-5243-018-03	ND	0.19	0.24	0.40	1.55	0.83	94.52	0.47	0.37	0.15	0.22	0.11	-	0.16	CDo140019-03-5243-018-03.pdf
	CDo140019-03-5243-018-04	ND	-	0.22	-	0.53	0.33	95.51	1.08	0.21	-	0.34	0.19	0.14	0.17	CDo140019-03-5243-018-04.pdf
	CDo140019-03-5243-018-05	ND	-	0.26	-	0.37	-	95.73	1.48	-	0.38	0.54	0.35	0.34	0.24	CDo140019-03-5243-018-05.pdf
	CDo140019-03-5243-018-06	ND	-	0.26	-	0.33	-	95.29	1.60	0.45	0.62	-	0.39	0.43	0.26	CDo140019-03-5243-018-06.pdf
	CDo140019-03-5243-018-07	ND	-	0.29	-	0.26	-	93.57	1.94	0.65	-	0.94	0.51	0.71	0.53	CDo140019-03-5243-018-07.pdf
Mock pool (Combine fraction: CDo140019-03-5243-018- 03 ~ 07)		ND	ND	0.28	ND	0.49	ND	96.28	1.11	0.32	ND	0.47	0.26	0.25	0.19	CDo140019-03-5243-018-03-07.pdf

Note [7]: Before RP2 — CDo140019-03-5243-011-03 ~ 12,11-18 denotes the sample number after spiking.
Note [8]: Fraction Identification for RP2 Step
Note [9]: Fraction Numbers of RP2 Pooled Fractions

Conclusion：After two-step purification of RP1 and RP2, the removal rate of the impurity at RRT 0.886 is 100%, while the removal rate of the impurity at RRT 1.077 is 94.66%.

7.2 Expt.2: Spiking experiment study of RRT0.960; Ser34 (Des-tBu)

- **Step1: L2-L1- Ser34 (Des-tBu):**

Charge 2-MeTHF (14 V) and DMSO (1 V) into the reaction flask, adjust the temperature to 20 °C and start agitation. Add L1- Ser34 (Des-tBu) (1 eq) and Fmoc-L2 (1.02 eq), and stir until complete dissolution. Charge Oxyma (1.50 eq) and EDCI (1.50 eq), then rinse the addition vessel with DMSO (1 V). Add 2,4,6-trimethylpyridine (3.00 eq) dropwise. Upon completion of addition, hold the reaction at 20 °C for 3 h. Sample for UPLC monitoring; proceed to the next step when residual L1- Ser34 (Des-tBu) is <0.5%. Add piperidine (0.634 eq)/water (2 V) solution to the flask and react at 15 °C for 2 h. Monitor the reaction by sampling; once full conversion of Fmoc-L2-L1- Ser34 (Des-tBu) is confirmed, allow phase separation and discard the aqueous layer. Perform two sequential acid washes: for each wash, add the prepared ~10% dilute HCl to the organic phase,

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stir for 10 min, settle, separate and remove the aqueous layer while retaining the organic phase. Conduct an alkaline wash by adding the prepared ~8% aqueous $NaHCO_3$, stir for 10 min, settle, separate and discard the aqueous layer. Carry out a brine wash with the prepared 18% aqueous NaCl, stir for 10 min, settle, separate and retain the organic phase.

Transfer the organic phase to a constant-pressure dropping funnel for crystallization. Charge n-heptane (30 V) into a second reaction flask, then add the organic phase dropwise at 20±5℃. After addition is finished, stir the slurry for 1 h and filter. Slurry-wash the filter cake with n-heptane/2-MeTHF mixture (3 V / 1 V) for 0.5 h, filter again, dry the solid and collect the product L2-L1- Ser34 (Des-tBu).

- **Step2: Boc-L3-L2-L1- Ser34 (Des-tBu)**

Charge 2-MeTHF (14 V) and DMSO (1 V) into the reaction flask, adjust the temperature to 20 °C and start agitation. Add L2-L1- Ser34 (Des-tBu) (1.00 eq) and Boc-L3 (1.10 eq), and stir until complete dissolution. Charge Oxyma (1.50 eq) and EDCI (1.50 eq), then rinse the addition vessel with DMSO (1 V). Add 2,4,6-trimethylpyridine (3.00 eq) dropwise. Upon completion of addition, hold the reaction at 20 °C for 3 h. Sample for UPLC monitoring; proceed to the next step when residual L2-L1- Ser34 (Des-tBu) is <0.5%.

Add purified water (4.58 V) to the batch, stir for 10 min, then allow phase separation and discard the aqueous layer. Perform acid wash: add the prepared ~10% dilute HCl to the organic phase, stir for 10 min, settle, separate and remove the aqueous layer while retaining the organic phase. Conduct brine wash with the prepared 18% aqueous NaCl, stir for 10 min, settle, separate and retain the organic phase.

Transfer the organic phase to a constant-pressure dropping funnel for crystallization. Charge n-heptane (30 V) into a second reaction flask, then add the organic phase dropwise at 20±5℃. After addition is finished, stir the slurry for 1 h and filter. Slurry-wash the filter cake with n-heptane/2-MeTHF mixture (3 V / 1 V) for 0.5 h, filter again, dry the solid and collect the product Boc-L3-L2-L1- Ser34 (Des-tBu).

- **Step3: impurity markers (L3-L2-L1)**

Charge toluene (6 V) and anisole (0.25 V) into the reaction flask and maintain the temperature at 20 °C. Add DTT (0.25 eq), purified water (0.25 V) and TIPS (0.193 eq). While holding at 20 °C, charge Boc-L3-L2-L1- Ser34 (Des-tBu) (1.00 eq). Cool the batch to 10 °C, then add TFA (9.24 V) dropwise. Upon completion of addition, warm the mixture to 20 °C and hold the reaction for 3 h.

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Sample the batch to confirm full conversion of Boc-L3-L2-L1- Ser34 (Des-tBu). Cool the reaction mixture to −15 °C, then add pre-chilled MTBE (48 V) dropwise. After addition is complete, stir the slurry for 1 h.

Filter the solids, slurry-wash the filter cake with MTBE (3 V) for 0.5 h, and filter again. Dry the isolated solid and collect the product L3-L2-L1. Test the purity and submit for MS analysis.

Table 7.2.1 Experiment 2: Synthesis of Ser34 (Des-tBu) impurity marker

Reaction and solid product purity for each step				
No.	Impurity name	Purity of product solid	Product solid Chromatogram	MS Chromatogram
Step 1	Ser34 (Des-tBu) impurity Raw material purity: 93.46% Feed amount: 4 g	Step1 Product: 7.36 g purity: 90.79%	CDo140019-03-3 040-065-01	CDo140019-03-3 040-065-01.PNG
Step 2	Raw material purity: 90.79% Feed amount: 6 g	Step2 Product: 7.61 g purity: 85.95%	CDo140019-03-3 040-073-01	CDo140019-03-3 040-073-01.PNG
Step 3	Raw material purity: 85.95% Feed amount: 6 g	Step3 Product: 4.27 g purity: 64.38%	CDo140019-03-3 040-078-01	CDo140019-03-3 040-078-01.PNG

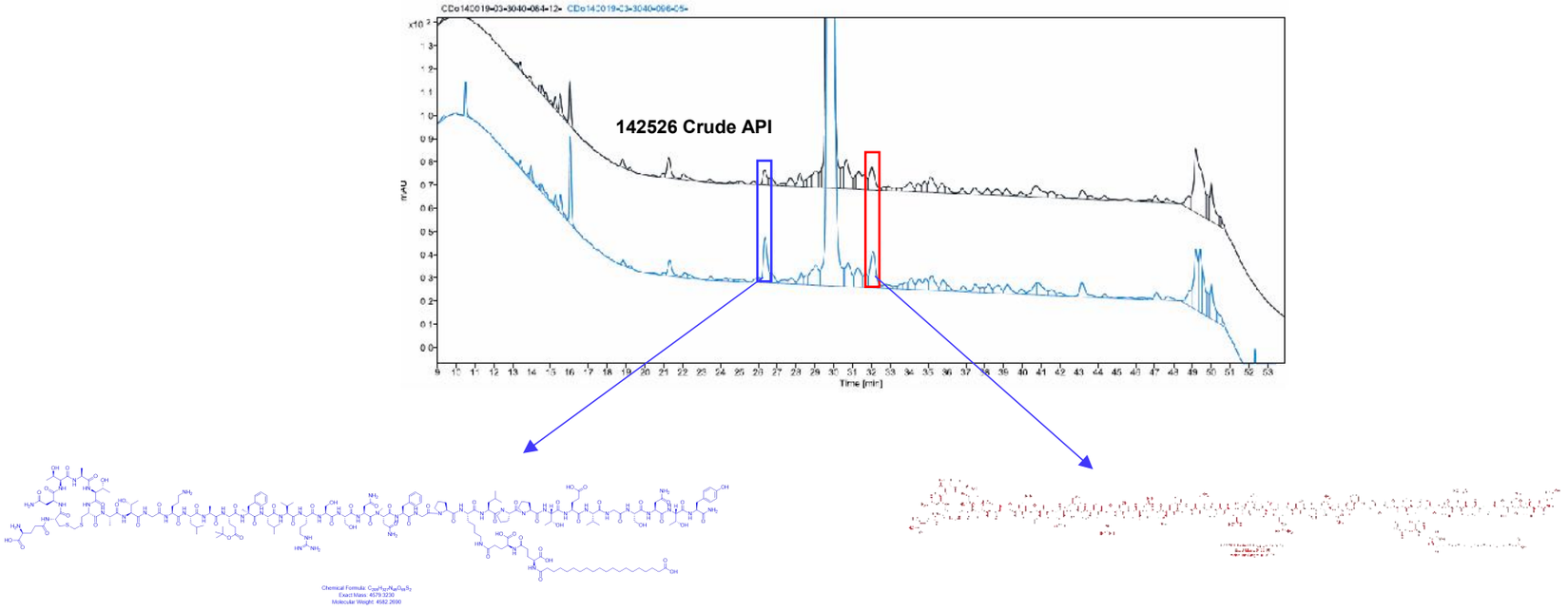
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The Ser34 (Des-tBu) impurity underwent a 3-step reaction (step1, step2, step3) to afford des-tBu-API, with the main peak corresponding to crude API (ASYM-142526). Des-tBu-API was spiked into crude API (ASYM 142526) at a level of 50%(wt). The purity data for the impurity, the crude API (ASYM-142526), and the crude API spiked with des-tBu-API impurity (ASYM-142526) are presented in Table 8.2.2.

Then the Ser34 (Des-tBu)-API impurity-spiked API sample (ASYM-142526) was adjusted to pH 8~9 with NaOH solution, stirred at 10 ± 5 °C for 2 h to transfer the depsipeptide, then re-adjusted the pH value to 2.8-3.2 with citric acid solution.

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7.2.1 Expt.2: UPLC Overlay Diagram of Experimental Group and Control Group



Two major spiked impurities were included: +tBu, L3+L2+L2+L1, with corresponding RRT values of 0.886, 1.077 respectively.

7.2.2 Expt.2: LC/MS data comparison of step 3 crude API

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RT	10.491	16.014	21.261	26.343	28.731	29.036	29.726	30.439	30.683	31.141	31.324	31.616	32.026	35.143	40.691	40.784	43.173	48.798	49.179	49.375	49.583	49.814	50.013	52.258
RRT	0.353	0.539	0.715	0.886	0.967	0.977	1.000	1.026	1.033	1.048	1.054	1.064	1.077	1.182	1.369	1.372	1.452	1.642	1.654	1.661	1.668	1.676	1.682	1.758
m/z	1285.634	-	1187.981	1528.449	-	1475.745	1509.761	1475.746	1471.415		1444.727	1504.758	1221.628	1045.561	1191.994	1222.183	-	1592.7914	-	-	-	-	-	-
charge	2	-	3	3	-	3	3	3	3		3	3	5	2	3	2	-	3	-	-	-	-	-	-
Monoisotopic Mass mass	2568.2522	-	3559.9185	4579.3431	-	4422.2095	4523.2544	4422.210	4409.239	-	4329.153	4509.248	6103.969	2088.104	3571.9561	2441.3499	-	4775.3742	-	-	-	-	-	-
Delta mass	-1955.00	-	-963.34	56.09	-	-101.04	0	-101.040	-114.020	-	-194.100	-14.010	1580.715	-2435.15	-951.30	-2081.90	-	252.0804	-	-	-	-	-	-
Structure	Glu1-Phe23]+Pip	unknown	Orn11-Tyr37	+ **tBu**	unknown	Des Thr9	142526	Des Thr30	Des Asn21	unknown	$DesPro^{28}Pro^{29}$	$-CH_2$	**L3+L2+L2+L1**	105+[Pro25-Tyr37]	Formaldehyde+[Orn11-Tyr37]	Formaldehyde+[Asn21-Tyr37]	unknown	+Pbf	unknown	unknown	unknown	unknown	unknown	unknown
142526 (Crude API CPo142526-02-03-01)	0.72	0.74	0.69	0.50	0.38	0.72	72.29	/	1.37	0.24	0.70	0.53	1.08	0.77	0.99	/	0.40	0.55	4.44	/	/	0.51	1.35	/
Exp.2 (CDo 140019-03-3040-096-02)	/	1.74	0.16	2.19	0.76	0.64	72.07	/	0.76	0.97	/	0.34	1.65	/	/	0.81	0.65	0.65	1.30	1.53	0.84	0.65	0.66	1.30

7.2.3 Expt.2: The purification conditions and collection strategy of RP1

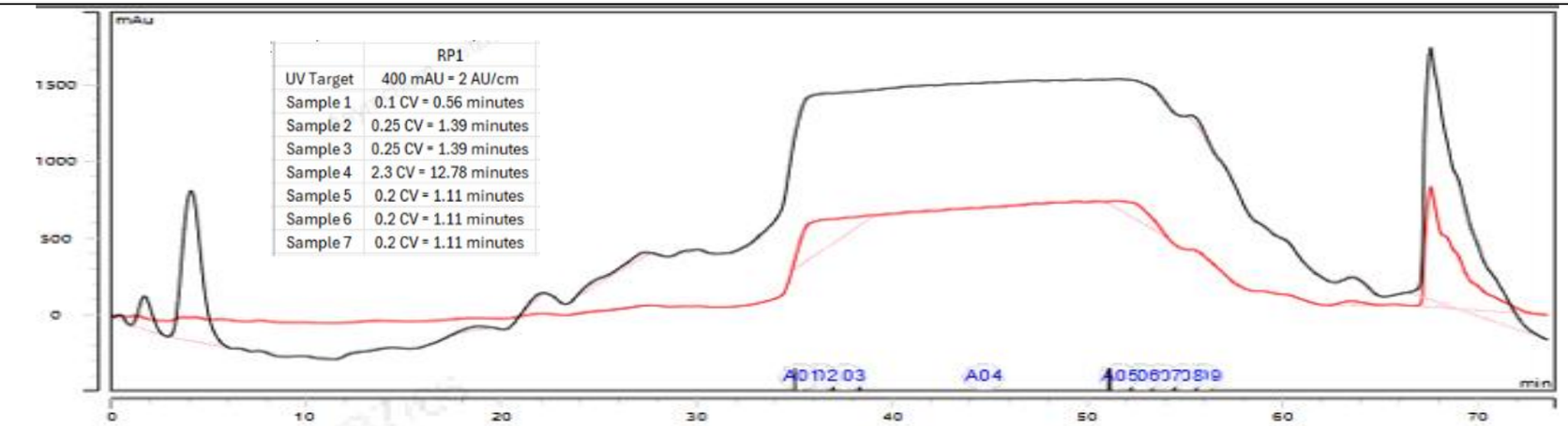
Column	10*250mm bed column=19.63 mL
Sample	CDo140019-03-3040-096-05 (Purity: 72.93%)
Resin	Daisogel 120-10-ODS-RPS

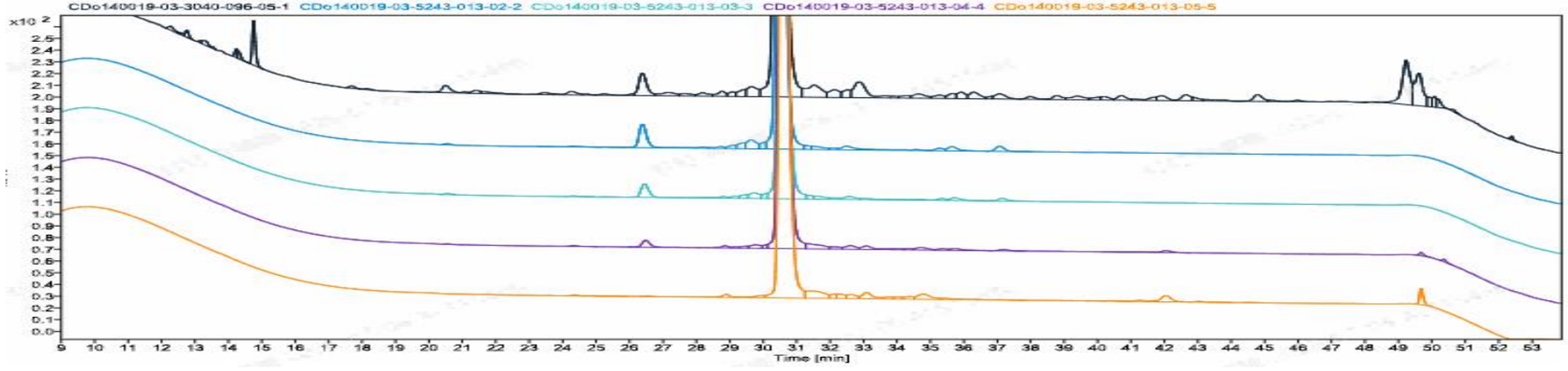
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Mobile phase A	90% 50 mM Citric Acid, **pH 3.00**: 10% ACN			Mobile phase B	Acetonitrile		
Detection wavelength	230nm			Monitor wavelength	280nm		
Flowrate	3.53mL/min(270cm/h)						
Loading amount	**45.0mg/mL, 883 mg used.**						
Post-Load Wash	30~35.0(RP1)B% for 11.1min(2CV) at 3.53mL/min						
Elution	T(time)	0	62.5	62.5	73.6		
	B (%)	27.8	44.4	66.7	66.7		
UV	Start collection when UV reaches 350mAu						

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8.2.4 Expt.2: Purity and impurity profile of RP1 fractions

Batch	Fraction Number	Impurity profile (RRT, %area) Only the components with the main - peak purity greater than 80% were counted															
RRT		0.886	0.967	0.977	0.984	1.000	1.026	1.033	1.048	1.054	1.064	1.077	1.134	1.171	1.212	1.372	Spectrum Attachments
m/z		1528.449	-	1475.745	-	1509.761	1475.746	1471.415		1444.727	1504.758	1221.628	1045.561	1045.561	-	1222.183	
charge		3	-	3	-	3	3	3		3	3	5	2	2	-	2	
Monoisotopic Mass mass		4579.343	-	4422.210	-	4523.254	4422.210	4409.239	-	4329.153	4509.248	6103.969	2088.104	2088.104	-	2441.350	
Delta mass		56.090	-	-101.040	-	0	-101.040	-114.020	-	-194.100	-14.010	1580.715	-2435.150	-2435.150	-	-2081.900	
Structure		+ **tBu**	unknown	Des Thr9	unknown	142526	Des Thr30	Des Asn21	unknown	$DesPro^{28}Pro^{29}$	$-CH_2$	**L3+L2+L2+L1**	105+[Pro25-Tyr37]	105+[Pro25-Tyr37]	unknown	Formaldehyde+[Asn21-Tyr37]	
Exp.2 (CDo140019-03-3040-096-02)		2.19	0.76	0.64	-	72.07	0.76		0.97	-	0.34	1.65	-	-	-	0.81	CDo140019-03-3040-096-02.pdf
Before RP1 (CDo140019-03-3040-096-05)		1.36	-	1.14	-	72.62	-	1.36	0.84	-	0.42	1.46	0.19	0.36	0.16	0.62	CDo140019-03-3040-096-05.pdf
Loading: 45 mg/mL (CDo 140019-03-3040-5243-013)	CDo 140019-03-5243-013-02	3.43	0.77	1.65	0.65	86.80	-	0.66	0.23	-	0.77	0.25	0.18	0.87	0.89	-	CDo140019-03-5243-013-02.pdf
	CDo 140019-03-5243-013-03	1.93	0.46	1.01	0.47	91.08	-	0.58	0.24	-	0.63	0.27	0.18	0.58	0.47	-	CDo140019-03-5243-013-03.pdf
	CDo 140019-03-5243-013-04	0.98	0.23	0.59	0.26	91.67	-	1.32	0.39	-	0.65	0.61	0.51	0.37	0.21	0.33	CDo140019-03-5243-013-04.pdf
	CDo 140019-03-5243-013-05	-	0.12	0.34	-	89.45	-	2.00	0.36	0.53	0.56	0.97	1.06	0.10	-	0.98	CDo140019-03-5243-013-05.pdf
Mock pool (Combine fraction: CDo 140019-03-5243-013-02 ~ 05)		1.10	0.36	0.31	0.57	92.43	-	0.79	0.37	-	0.66	0.57	0.45	0.30	0.24	0.32	CDo140019-03-5243-013_02-05, 11-15.pdf

Conclusion：RP1 exhibits poor removal capacity for the impurity with RRT 0.886, while it performs well in eliminating the impurity at RRT 1.077.

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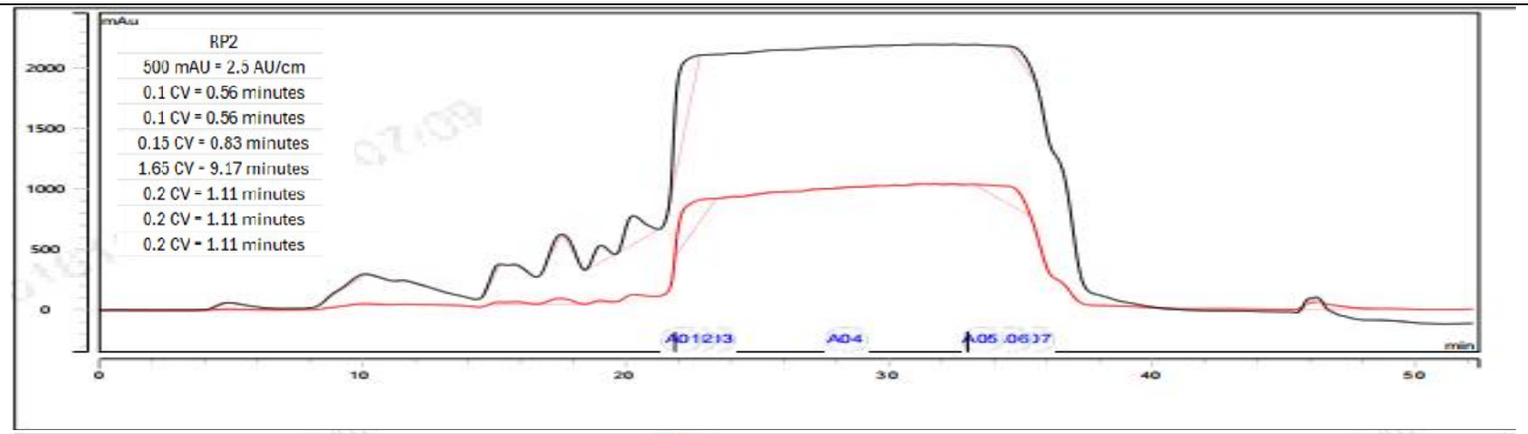
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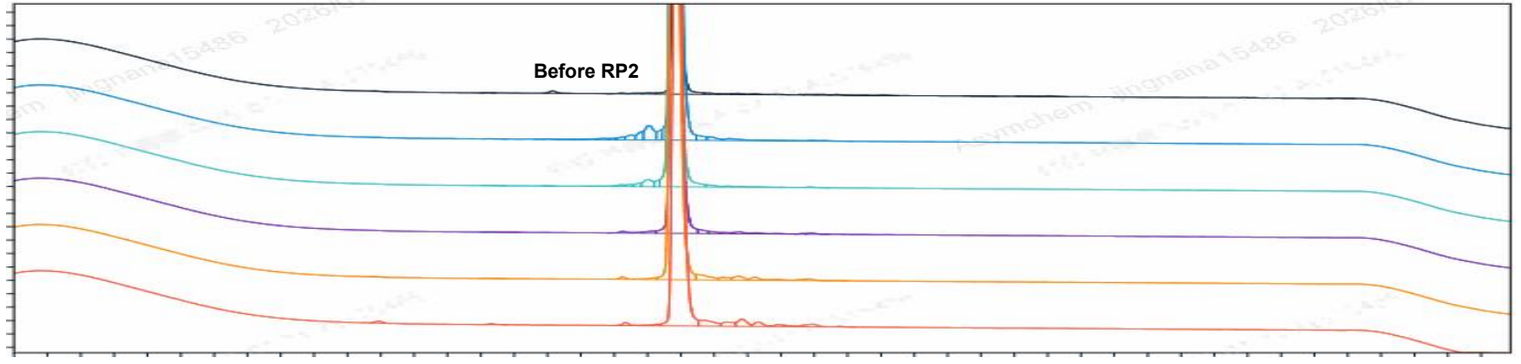
8.2.5 Expt.2: The purification conditions and collection strategy of RP2

Column	10*250mm bed column=19.63 mL						
Sample	CDo140019-03-5243-025-01						
Resin	Daisogel 120-10-ODS-RPS						
Mobile phase A	90% 50 mM Citric Acid, **pH 6.50**: 10% ACN			Mobile phase B	Acetonitrile		
Detection wavelength	230nm			Monitor wavelength	280nm		
Flowrate	3.53mL/min(270cm/h)						
Loadingamount	**45.0 mg/mL**						
Post-Load Wash	20.0(RP2)B% for 11.1min(2CV) at 3.53mL/min						
Elution	T(time)	0	41.1	41.1	52.2		
	B(%)	20	35.6	66.7	66.7		
Collection	Start collection when the UV280 reaches 450mAu						

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7.2.4 Expt.2: Purity and impurity profile of RP2 fractions

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Batch	Fraction Number	Impurity profile (RRT, %area) Only the components with the main - peak purity greater than 80% were counted																
RRt		0.886	0.949	0.954	0.962	0.966	0.975	0.984	1.000	1.026	1.033	1.045	1.061	1.077	1.096	1.126	1.149	Spectrum Attachments
Before RP2		1.10	0.36	-	0.31	0.31	0.57	-	92.43	-	0.79	0.37	0.66	0.57	0.28	0.45	0.45	CDo140019-03-5243-013_02-05，11-15.pdf
Loading: 45 mg/mL (CDo140019-03-5243-026)	CDo140019-03-5243-026-02	ND	0.33	0.71	0.91	0.91	2.49	0.89	91.50	0.70	0.64	0.39	-	-	-	-	0.11	CDo140019-03-5243-026-02.pdf
	CDo140019-03-5243-026-03	ND	0.28	0.27	0.39	0.39	1.22	0.54	96.10	-	0.39	0.17	0.19	0.18	-	-	0.13	CDo140019-03-5243-026-03.pdf
	CDo140019-03-5243-026-04	ND	0.24	-	0.16	0.16	0.40	-	97.09	0.57	0.32	0.21	0.38	0.22	0.11	-	0.18	CDo140019-03-5243-026-04.pdf
	CDo140019-03-5243-026-05	ND	0.31	-	-	-	0.25	-	95.77	-	1.21	0.46	0.70	0.46	0.25	0.15	0.25	CDo140019-03-5243-026-05.pdf
	CDo140019-03-5243-026-06	ND	0.37	-	-	-	0.22	-	93.50	-	1.57	0.85	1.19	0.73	0.27	0.18	0.60	CDo140019-03-5243-026-06.pdf
Mock pool (Combine fraction: CDo140019-03-5243-026-02-06)		ND	0.22	-	0.19	-	0.65	-	96.40	-	0.59	0.39	0.30	0.44	0.25	0.12	0.19	CDo140019-03-5243-026-02-06.pd

Conclusion: After two-step purification of RP1 and RP2, the removal rate of the impurity at RRT 0.886 is 100%, while the removal rate of the impurity at RRT 1.077 is 73.33%.

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7.3. Expt.3: Spiking experiment study of RRT0.947; Thr36 (Des-tBu)

- **Step1: L2-L1- Thr36 (Des-tBu):**

Charge 2-MeTHF (14 V) and DMSO (1 V) into the reaction flask, adjust the temperature to 20 °C and start agitation. Add L1- Thr36 (Des-tBu) (1 eq) and Fmoc-L2 (1.02 eq), and stir until complete dissolution. Charge Oxyma (1.50 eq) and EDCI (1.50 eq), then rinse the addition vessel with DMSO (1 V). Add 2,4,6-trimethylpyridine (3.00 eq) dropwise. Upon completion of addition, hold the reaction at 20 °C for 3 h. Sample for UPLC monitoring; proceed to the next step when residual L1- Thr36 (Des-tBu) is <0.5%. Add piperidine (0.634 eq)/water (2 V) solution to the flask and react at 15 °C for 2 h. Monitor the reaction by sampling; once full conversion of Fmoc-L2-L1- Thr36 (Des-tBu) is confirmed, allow phase separation and discard the aqueous layer. Perform two sequential acid washes: for each wash, add the prepared ~10% dilute HCl to the organic phase, stir for 10 min, settle, separate and remove the aqueous layer while retaining the organic phase. Conduct an alkaline wash by adding the prepared ~8% aqueous $NaHCO_3$, stir for 10 min, settle, separate and discard the aqueous layer. Carry out a brine wash with the prepared 18% aqueous NaCl, stir for 10 min, settle, separate and retain the organic phase.

Transfer the organic phase to a constant-pressure dropping funnel for crystallization. Charge n-heptane (30 V) into a second reaction flask, then add the organic phase dropwise at 20±5℃. After addition is finished, stir the slurry for 1 h and filter. Slurry-wash the filter cake with n-heptane/2-MeTHF mixture (3 V / 1 V) for 0.5 h, filter again, dry the solid and collect the product L2-L1- Thr36 (Des-tBu).

- **Step2: Boc-L3-L2-L1- Thr36 (Des-tBu)**

Charge 2-MeTHF (14 V) and DMSO (1 V) into the reaction flask, adjust the temperature to 20 °C and start agitation. Add L2-L1- Thr36 (Des-tBu) (1.00 eq) and Boc-L3 (1.10 eq), and stir until complete dissolution. Charge Oxyma (1.50 eq) and EDCI (1.50 eq), then rinse the addition vessel with DMSO (1 V). Add 2,4,6-trimethylpyridine (3.00 eq) dropwise. Upon completion of addition, hold the reaction at 20 °C for 3 h. Sample for UPLC monitoring; proceed to the next step when residual L2-L1- Thr36 (Des-tBu) is <0.5%.

Add purified water (4.58 V) to the batch, stir for 10 min, then allow phase separation and discard the aqueous layer. Perform acid wash: add the prepared ~10% dilute HCl to the organic phase, stir for 10 min, settle, separate and remove the aqueous layer while retaining the organic phase. Conduct brine wash with the prepared 18% aqueous NaCl, stir for 10 min, settle, separate and retain the organic phase.

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Transfer the organic phase to a constant-pressure dropping funnel for crystallization. Charge n-heptane (30 V) into a second reaction flask, then add the organic phase dropwise at 20±5℃. After addition is finished, stir the slurry for 1 h and filter. Slurry-wash the filter cake with n-heptane/2-MeTHF mixture (3 V / 1 V) for 0.5 h, filter again, dry the solid and collect the product Boc-L3-L2-L1- Thr36 (Des-tBu).

- **Step3: impurity markers (L3-L2-L1)**

Charge toluene (6 V) and anisole (0.25 V) into the reaction flask and maintain the temperature at 20 °C. Add DTT (0.25 eq), purified water (0.25 V) and TIPS (0.193 eq). While holding at 20 °C, charge Boc-L3-L2-L1- Thr36 (Des-tBu) (1.00 eq). Cool the batch to 10 °C, then add TFA (9.24 V) dropwise. Upon completion of addition, warm the mixture to 20 °C and hold the reaction for 3 h.

Sample the batch to confirm full conversion of Boc-L3-L2-L1- Thr36 (Des-tBu). Cool the reaction mixture to −15 °C, then add pre-chilled MTBE (48 V) dropwise. After addition is complete, stir the slurry for 1 h.

Filter the solids, slurry-wash the filter cake with MTBE (3 V) for 0.5 h, and filter again. Dry the isolated solid and collect the product L3-L2-L1. Test the purity and submit for MS analysis.

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Table 7.3.1 Experiment 3: Synthesis of Thr36 (Des-tBu) impurity marker

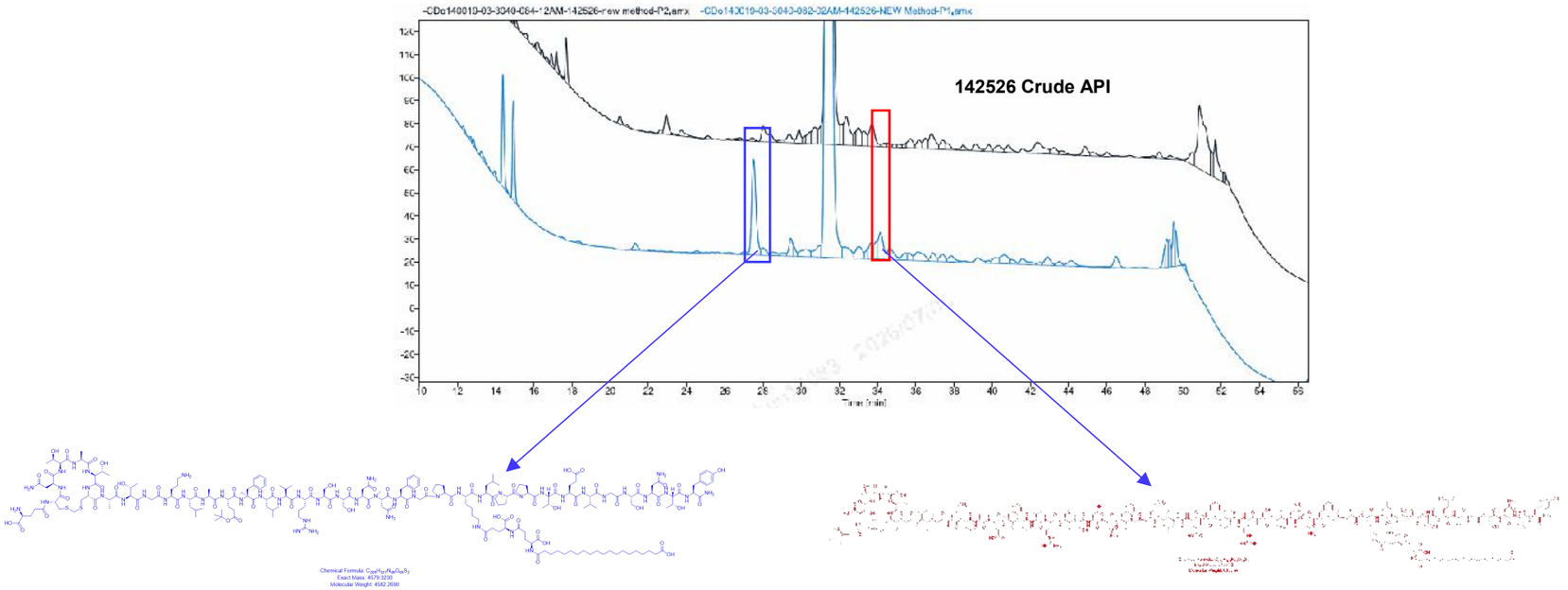
Reaction and solid product purity for each step				
No.	Impurity name	Purity of product solid	Product solid Chromatogram	MS Chromatogram
Step 1	Thr36 (Des-tBu) impurity Raw material purity: 93.57% Feed amount: 4g	Step1 Product: 6.34 g purity: 89.69%	CDo140019-03-3 040-066-01	CDo140019-03-3 040-066-01.PNG
Step 2	Raw material purity: 89.69% Feed amount: 6 g	Step2 Product: 6.25 g purity: 85.42%	CDo140019-03-3 040-076-01	CDo140019-03-3 040-076-01.PNG
Step 3	Raw material purity: 85.42% Feed amount: 6 g	Step3 Product: 4.37 g Purity: 63.18%	CDo140019-03-3 040-082-01	CDo140019-03-3 040-082-01.PNG

The Thr36 (Des-tBu) impurity underwent a 3-step reaction (step1, step2, step3) to afford des-tBu-API, with the main peak corresponding to crude API (ASYM-142526). Des-tBu-API was spiked into crude API (ASYM 142526) at a level of 50%. The purity data for the impurity, the crude API (ASYM-142526), and the crude API spiked with des-tBu-API impurity (ASYM-142526) are presented in Table 8.3.2.

Then the Thr36 (Des-tBu)-API impurity-spiked API sample (ASYM-142526) was adjusted to pH 8~9 with NaOH solution, stirred at 10 ± 5 °C for 2 h to transfer the depsipeptide, then re-adjusted the pH value to 2.8-3.2 with citric acid solution.

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7.3.1 Expt.3: UPLC Overlay Diagram of Experimental Group and Control Group



Two major spiked impurities were included: +tBu, L3+L2+L2+L1, with corresponding RRT values of 0.886, 1.077 respectively.

7.3.2 Expt.3: LCMS data comparison of step 3 crude API:

RT	10.491	13.668	14.150	16.014	21.261	26.343	27.704	28.477	29.036	29.313	29.726	30.469	30.683	31.324	31.616	31.866	32.026	32.312	32.904	34.048	35.143	38.666	40.691	44.054	46.551	46.738	46.908	47.064	48.798

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RRT	0.353	0.460	0.476	0.539	0.715	0.886	0.932	0.958	0.977	0.986	1.000	1.026	1.033	1.054	1.064	1.072	1.077	1.087	1.107	1.145	1.182	1.301	1.369	1.482	1.566	1.572	1.578	1.583	1.642
m-z	1285.634	-	-	-	1187.981	1528.449	-	-	1475.745	1528.435	1509.761	1475.746	1471.415	1444.727	1504.758	-	1221.628	-	1503.754	-	1045.561	1082.578	1191.994	-	-	-	-	-	1592.7914
charge	2	-	-	-	3	3	-	-	3	3	3	3	3	3	3	-	5	-	3	-	2	3	3	-	-	-	-	-	3
Monoisotopic Mass mass	2568.2522	-	-	-	3559.9185	4579.3431	-	-	4422.2095	4580-2759	4523.2544	4422.210	4409.239	4329.153	4509.248	-	6103.969	-	4506.2373	-	2088.104	3243.7314	3571.9561	-	-	-	-	-	4775.3742
Delta mass	-1955.00	-	-	-	-963.34	56.09	-	-	-101.04	57.08	0	-101.040	-114.020	-194.100	-14.010	-	1580.715	-	-17.02	-	-2435.15	-1279.52	-951.30	-	-	-	-	-	252.0804
Structure	Glu1-Phe23]+Pip	unknown	unknown	unknown	Orn11-Tyr37	+tBu	unknown	unknown	Des Thr9	Endo Gly24	142526	Des Thr30	Des Asn21	$DesPro^{28}Pro_{29}$	$-CH_2$	unknown	L3+L2+L2+L1	unknown	Asn22 cyclization	unknown	105+[Pro25/Tyr37]	[Glu14-Tyr37], and Glu14-H2O	Formaldehyde+[Orn11-Tyr37]	unknown	unknown	unknown	unknown	unknown	+Pbf
142526 Crude API CPo142526-02-03-01	0.72	0.01	-	0.74	0.69	0.50	0.37	0.16	0.72	0.37	72.29	0.46	1.37	0.70	0.53	-	1.08	-	0.14	0.43	0.77	0.36	0.99	-	-	0.10	-	0.24	0.55
Exp.3 CDo140019-03-3040-082-02	-	2.20	1.96	-	-	4.00	0.58	0.71	-	0.61	70.31	0.83	-	0.75	0.21	0.68	0.55	1.31	0.74	0.82	-	0.64	0.45	0.56	1.36	0.89	1.23	1.29	-

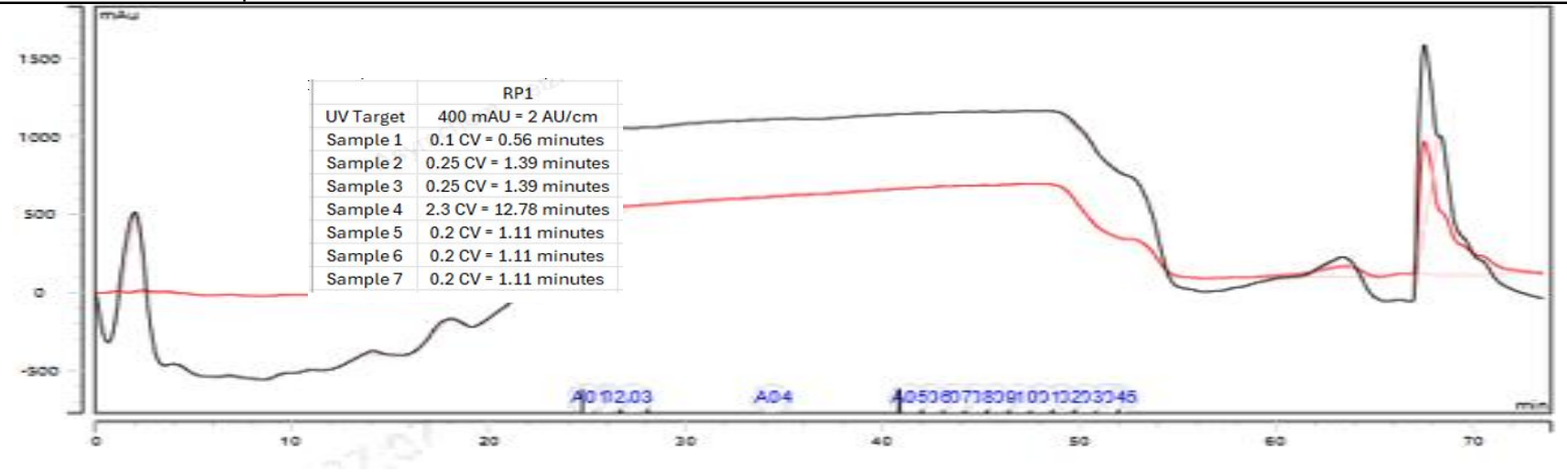
7.3.3 Expt.3: The purification conditions and collection strategy of RP1

Column	10*250mm bed column=19.63 mL		
Sample	CDo140019-03-3040-082-04 (Purity: 79.10%)		
Resin	Daisogel 120-10-ODS-RPS		
Mobile phase A	90% 50 mM Citric Acid, **pH 3.00**: 10% ACN	Mobile phase B	Acetonitrile
Detection wavelength	230nm	Monitor wavelength	280nm

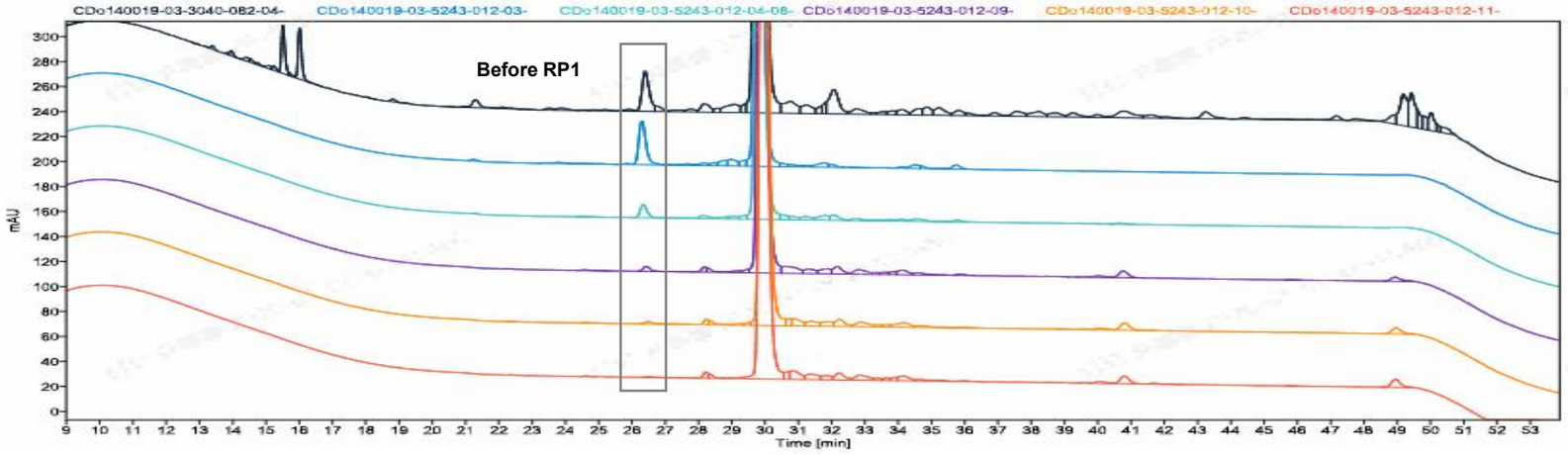
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Flowrate	3.53mL/min(270cm/h)						
Loading amount	**45.0mg/mL, 883 mg used.**						
Post-Load Wash	30~35.0(RP1)B%　for 11.1min(2CV) at 3.53mL/min						
Elution	T(time)	0	62.5	62.5	73.6		
	B (%)	27.8	44.4	66.7	66.7		
UV	Start collection when UV reaches 350mAu						



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7.3.4 Expt.3: Purity and impurity profile of RP1 fractions

Batch	Fraction Number	Impurity profile (RRT, %area) Only the components with the main - peak purity greater than 80% were counted																				
RRT	0.868	0.886	0.977	0.986	1.000	1.026	1.033	1.054	1.064	1.072	1.077	1.087	1.107	1.126	1.135	1.145	1.158	1.170	1.182	1.202	1.301	Spectrum Attachments Spectrum Attachments
m-z		1528.449	1475.745	1528.435	1509.761	1475.746	1471.415	1444.727	1504.758	-	1221.628	-	1503.754	-	-	-	-	-	1045.561	-	1082.578	
charge		3	3	3	3	3	3	3	3	-	5	-	3	-	-	-	-	-	2	-	3	
Monoisotopic Mass mass		4579.343	4422.210	4580-2759	4523.2544	4422.210	4409.239	4329.153	4509.248	-	6103.969	-	4506.2373	-	-	-	-	-	2088.104	-	3243.7314	
Delta mass		56.09	-101.04	57.08	0	-101.040	-114.020	-194.100	-14.010	-	1580.715	-	-17.02	-	-	-	-	-	-2435.15	-	-1279.52	
Structure		+ **tBu**	Des Thr9	Endo Gly24	142526	Des Thr30	Des Asn21	DesPro^{28}Pro^{29}	-CH_2	unknown	**L3+L2+L2+L1**	unknown	Asn22 cyclization	unknown	unknown	unknown	unknown	unknown	105+[Pro25/Tyr37]	unknown	[Glu14-Tyr37], and Glu14-H2O	
Exp.3 CDo140019-03-3040-082-02	-	**4.00**	-	**0.61**	**70.31**	**0.83**	-	**0.75**	**0.21**	**0.68**	**0.55**	**1.31**	**0.74**	-	-	**0.82**	-	-	-	-	**0.64**	CDo140019-03-3040-082-02.pdf
Before RP1 CDo140019-03-3040-082-04	**0.13**	**2.16**	**0.84**	**0.30**	**70.69**	-	**1.26**	**0.78**	**0.28**	-	**2.10**	**0.49**	-	**0.17**	**0.20**	**0.38**	-	**0.91**	**0.61**	**0.35**	-	CDo140019-03-3040-082-02.pdf

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Loading: 45 mg/mL (CDo140019-03-5243-012)	CDo140019-03-5243-012-03	6.70	-	0.72	83.79	0.45	0.43	0.20	0.94	-	0.46	-	-	-	-	0.25	0.67	0.30	-	0.66	-	-	CDo140019-03-5243-012-03.pdf
	CDo140019-03-5243-012-04-08	1.78	-	0.39	90.21	0.64	0.52	0.65	0.96	-	0.87	0.33	0.15	0.19	0.24	0.32	0.64	-	-	0.29	-	0.18	CDo140019-03-5243-012-04-08.pd
	CDo140019-03-5243-012-09	0.61	-	0.29	88.12	-	1.72	0.78	0.85	1.08		0.91	0.19	0.26	0.30	0.74	0.43	-	-	0.22	-	0.99	CDo140019-03-5243-012-09.pdf
	CDo140019-03-5243-012-10	0.34	-	0.23	88.04	0.56	1.06	0.90	0.78	1.11		0.97	0.24	0.26	-	1.07	0.37	-	0.10	0.18	-	1.10	CDo140019-03-5243-012-10.pdf
	CDo140019-03-5243-012-11	0.10	-	-	87.18	0.64	1.38	1.10	0.67	1.14		1.08	0.28	0.26	0.31	0.83	0.21	0.12	0.12	0.12	-	1.17	CDo140019-03-5243-012-11.pdf
Mock pool (Combine fraction: CDo140019-03-5243-012-03 ~ 11)		0.14	2.05	0.83	0.41	89.04	-	1.24	0.50	0.95 -		0.87	0.51	0.17	-	-	0.40	0.23	0.28	0.33	0.32	-	CDo140019-03-5243-012-03-11,17-26.pdf

Conclusion: RP1 exhibits poor removal capacity for the impurity with RRT 0.886, while it performs well in eliminating the impurity at RRT 1.077.

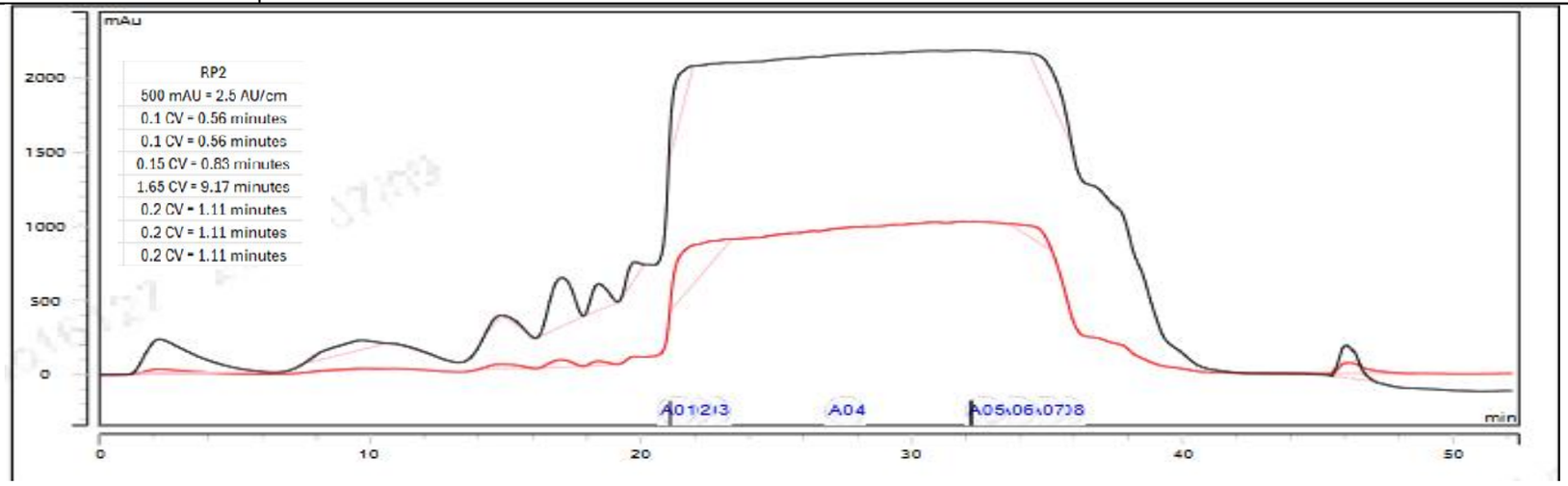
7.3.5 Expt.3: The purification conditions and collection strategy of RP2

Column	10*250mm bed column=19.63 mL		
Sample	CDo140019-03-5243-023-01		
Resin	Daisogel 120-10-ODS-RPS		
Mobile phase A	90% 50 mM Citric Acid, **pH 6.50**: 10% ACN	Mobile phase B	Acetonitrile
Detection wavelength	230nm	Monitor wavelength	280nm

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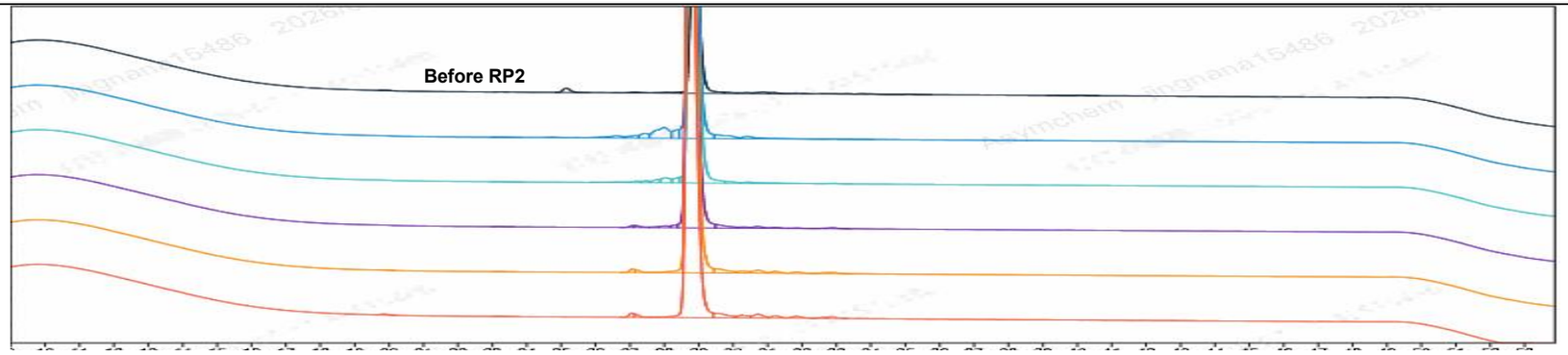
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Flowrate	3.53mL/min(270cm/h)						
Loadingamount	**45.0 mg/mL**						
Post-Load Wash	20.0(RP2)B% for 11.1min(2CV) at 3.53mL/min						
Elution	T(time)	0	41.1	41.1	52.2		
	B(%)	20	35.6	66.7	66.7		
Collection	Start collection when the UV280 reaches 450mAu						



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7.3.6 Expt.3: Purity and impurity profile of RP2 fractions

Batch	Fraction Number		Impurity profile (RRT, %area) Only the components with the main - peak purity greater than 80% were counted																
RRt		0.886	0.908	0.924	0.938	0.943	0.963	0.973	0.985	1.000	1.026	1.033	1.056	1.069	1.077	1.103	1.135	1.157	Spectrum Attachments
Before RP2		2.05	0.14	0.14	0.27	0.26	-	0.83	0.41	89.04	-	1.24	0.5	0.95	0.87	0.51	0.4	0.23	CDo140019-03-5243-012-03-11,17-26.pdf
Loading: 45 mg/mL (CDo140019-03-5243-024)	CDo140019-03-5243-024-02	ND	0.21	0.55	0.56	0.86	-	3.50	1.19	90.38	-	1.36	1.05	-	-	-	-	/	CDo140019-03-5243-024-02.pdf
	CDo140019-03-5243-024-03	ND	-	-	0.33	0.31	0.53	1.08	0.76	95.58	-	0.39	0.25	0.40	-	-	0.13	/	CDo140019-03-5243-024-03.pdf
	CDo140019-03-5243-024-04	ND	-	-	0.16	0.28	0.12	0.32	0.31	96.80	-	0.80	0.23	0.37	0.20	0.14	0.18	/	CDo140019-03-5243-024-04.pdf
	CDo140019-03-5243-024-05	ND	-	-	0.32	0.30	-	-	-	96.17	-	1.06	0.42	0.61	0.35	0.30	0.13	0.20	CDo140019-03-5243-024-05.pdf

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	CDo140019-03-5243-024-06	ND	-	-	0.37	0.32	-	-	-	95.54	-	1.15	0.54	0.71	0.41	0.39	0.14	0.22	CDo140019-03-5243-024-06.pdf
Mock pool (Combine fraction: CDo140019-03-5243-024-02-06)		ND	-	-	0.23	0.25	0.2538	0.52	0.37	95.82	0.59	0.31	0.32	0.42	0.23	0.17	0.17	-	CDo140019-03-5243-024-02-06.pdf

Conclusion: After two-step purification of RP1 and RP2, the removal rate of the impurity at RRT 0.886 is 100%, while the removal rate of the impurity at RRT 1.077 is 89.05%.

7.4. Expt.4: Spiking experiment study of RRT0.954; Thr30 (Des-tBu)

- **Step1: L2-L1- Thr30 (Des-tBu):**

Charge 2-MeTHF (14 V) and DMSO (1 V) into the reaction flask, adjust the temperature to 20 °C and start agitation. Add L1- Thr30 (Des-tBu) (1 eq) and Fmoc-L2 (1.02 eq), and stir until complete dissolution. Charge Oxyma (1.50 eq) and EDCI (1.50 eq), then rinse the addition vessel with DMSO (1 V). Add 2,4,6-trimethylpyridine (3.00 eq) dropwise. Upon completion of addition, hold the reaction at 20 °C for 3 h. Sample for UPLC monitoring; proceed to the next step when residual L1- Thr30 (Des-tBu) is <0.5%. Add piperidine (0.634 eq)/water (2 V) solution to the flask and react at 15 °C for 2 h. Monitor the reaction by sampling; once full conversion of Fmoc-L2-L1- Thr30 (Des-tBu) is confirmed, allow phase separation and discard the aqueous layer. Perform two sequential acid washes: for each wash, add the prepared ~10% dilute HCl to the organic phase, stir for 10 min, settle, separate and remove the aqueous layer while retaining the organic phase. Conduct an alkaline wash by adding the prepared ~8% aqueous $NaHCO_3$, stir for 10 min, settle, separate and discard the aqueous layer. Carry out a brine wash with the prepared 18% aqueous NaCl, stir for 10 min, settle, separate and retain the organic phase.

Transfer the organic phase to a constant-pressure dropping funnel for crystallization. Charge n-heptane (30 V) into a second reaction flask, then add the organic phase dropwise at 20±5℃. After addition is finished, stir the slurry for 1 h and filter. Slurry-wash the filter cake with n-heptane/2-MeTHF mixture (3 V / 1 V) for 0.5 h, filter again, dry the solid and collect the product L2-L1- Thr30 (Des-tBu).

- **Step2: Boc-L3-L2-L1- Thr30 (Des-tBu)**

Charge 2-MeTHF (14 V) and DMSO (1 V) into the reaction flask, adjust the temperature to 20 °C and start agitation. Add L2-L1- Thr30 (Des-tBu) (1.00 eq) and Boc-L3 (1.10

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eq), and stir until complete dissolution. Charge Oxyma (1.50 eq) and EDCI (1.50 eq), then rinse the addition vessel with DMSO (1 V). Add 2,4,6-trimethylpyridine (3.00 eq) dropwise. Upon completion of addition, hold the reaction at 20 °C for 3 h. Sample for UPLC monitoring; proceed to the next step when residual L2-L1- Thr30 (Des-tBu) is <0.5%.

Add purified water (4.58 V) to the batch, stir for 10 min, then allow phase separation and discard the aqueous layer. Perform acid wash: add the prepared ~10% dilute HCl to the organic phase, stir for 10 min, settle, separate and remove the aqueous layer while retaining the organic phase. Conduct brine wash with the prepared 18% aqueous NaCl, stir for 10 min, settle, separate and retain the organic phase.

Transfer the organic phase to a constant-pressure dropping funnel for crystallization. Charge n-heptane (30 V) into a second reaction flask, then add the organic phase dropwise at 20±5℃. After addition is finished, stir the slurry for 1 h and filter. Slurry-wash the filter cake with n-heptane/2-MeTHF mixture (3 V / 1 V) for 0.5 h, filter again, dry the solid and collect the product Boc-L3-L2-L1- Thr30 (Des-tBu).

- **Step3: impurity markers (L3-L2-L1)**

Charge toluene (6 V) and anisole (0.25 V) into the reaction flask and maintain the temperature at 20 °C. Add DTT (0.25 eq), purified water (0.25 V) and TIPS (0.193 eq). While holding at 20 °C, charge Boc-L3-L2-L1- Thr30 (Des-tBu) (1.00 eq). Cool the batch to 10 °C, then add TFA (9.24 V) dropwise. Upon completion of addition, warm the mixture to 20 °C and hold the reaction for 3 h.

Sample the batch to confirm full conversion of Boc-L3-L2-L1- Thr30 (Des-tBu). Cool the reaction mixture to −15 °C, then add pre-chilled MTBE (48 V) dropwise. After addition is complete, stir the slurry for 1 h.

Filter the solids, slurry-wash the filter cake with MTBE (3 V) for 0.5 h, and filter again. Dry the isolated solid and collect the product L3-L2-L1. Test the purity and submit for MS analysis.

Table 7.4.1 Experiment 4: Synthesis of Thr30 (Des-tBu) impurity marker

Reaction and solid product purity for each step				
No.	Impurity name	Purity of product solid	Product solid Chromatogram	MS Chromatogram

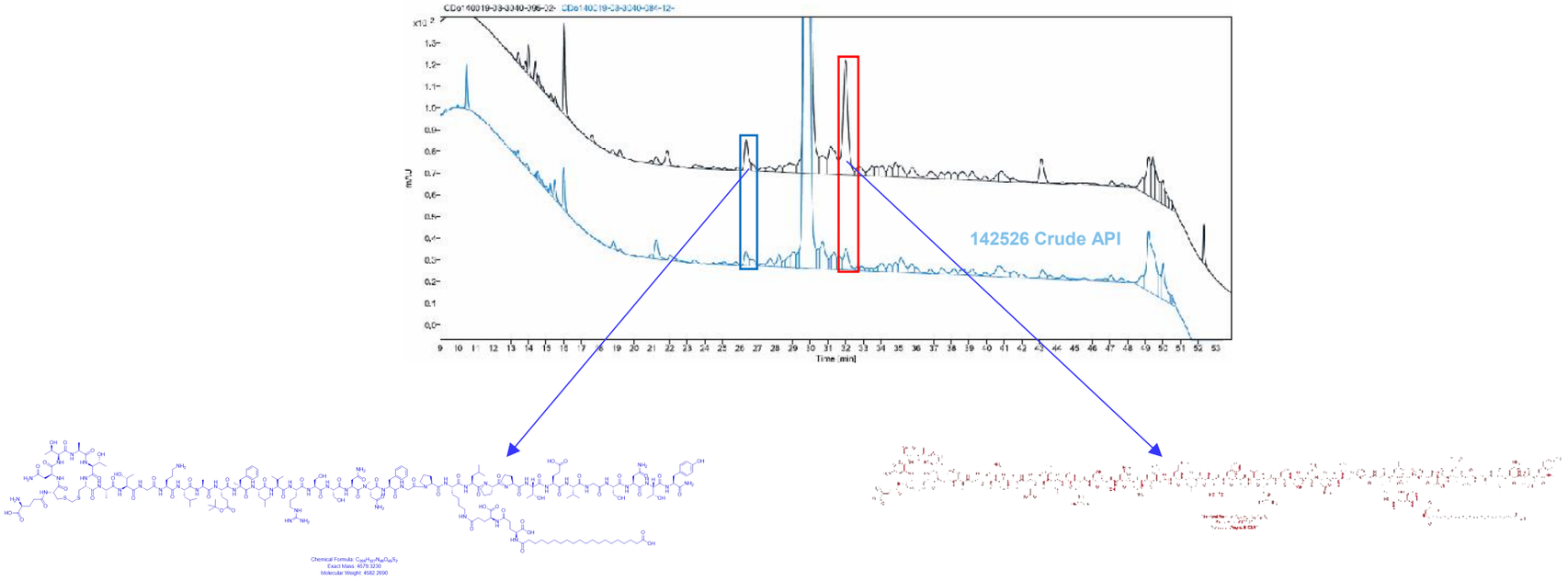
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Step 1	Thr30 (Des-tBu) impurity Raw material purity: 95.50% Feed amount: 4g	Step1 Product: 6.39 g purity: 88.47%	CDo140019-03-3 040-067-01	CDo140019-03-3 040-067-01.PNG
Step 2	Raw material purity: 88.47% Feed amount: 6 g	Step2 Product: 6.87 g purity: 84.04%	CDo140019-03-3 040-077-01	CDo140019-03-3 040-077-01.PNG
Step 3	Raw material purity: 84.04% Feed amount: 6 g	Step3 Product: 4.14 g purity: 58.13%	CDo140019-03-3 040-083-01	CDo140019-03-3 040-083-01.PNG

The Thr30 (Des-tBu) impurity underwent a 3-step reaction (step1, step2, step3) to afford des-tBu-API, with the main peak corresponding to crude API (ASYM-142526). Des-tBu-API was spiked into crude API (ASYM 142526) at a level of 50%. The purity data for the impurity, the crude API (ASYM-142526), and the crude API spiked with des-tBu-API impurity (ASYM-142526) are presented in **Table 7.3.2**.

Then the Thr30 (Des-tBu)-API impurity-spiked API sample (ASYM-142526) was adjusted to pH 8~9 with NaOH solution, stirred at 10 ± 5 °C for 2 h to transfer the depsipeptide, then re-adjusted the pH value to 2.8-3.2 with citric acid solution

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Two major spiked impurities were included: +tBu, L3+L2+L2+L1, with corresponding RRT values of 0.886, 1.077 respectively.

7.4.2 Expt.4: LCMS data comparison of step 3 crude API:

RT	10.491	16.014	21.261	26.343	28.731	29.036	29.726	30.469	30.683	31.141	31.324	31.616	32.026	32.689	35.143	35.729	40.691	40.784	43.043	48.798	49.179	49.375	49.583	49.814	50.013
RRT	0.353	0.539	0.715	0.886	0.967	0.977	1.000	1.026	1.033	1.048	1.054	1.064	1.077	1.100	1.182	1.202	1.369	1.372	1.448	1.642	1.654	1.661	1.668	1.676	1.682
m-z	1285.634	-	1187.981	1528.449	-	1475.745	1509.761	1475.746	1471.415		1444.727	1504.758	1221.628	1095.095	1045.561	-	1191.994	1222.183	-	1592.7914	-	-	-	-	-

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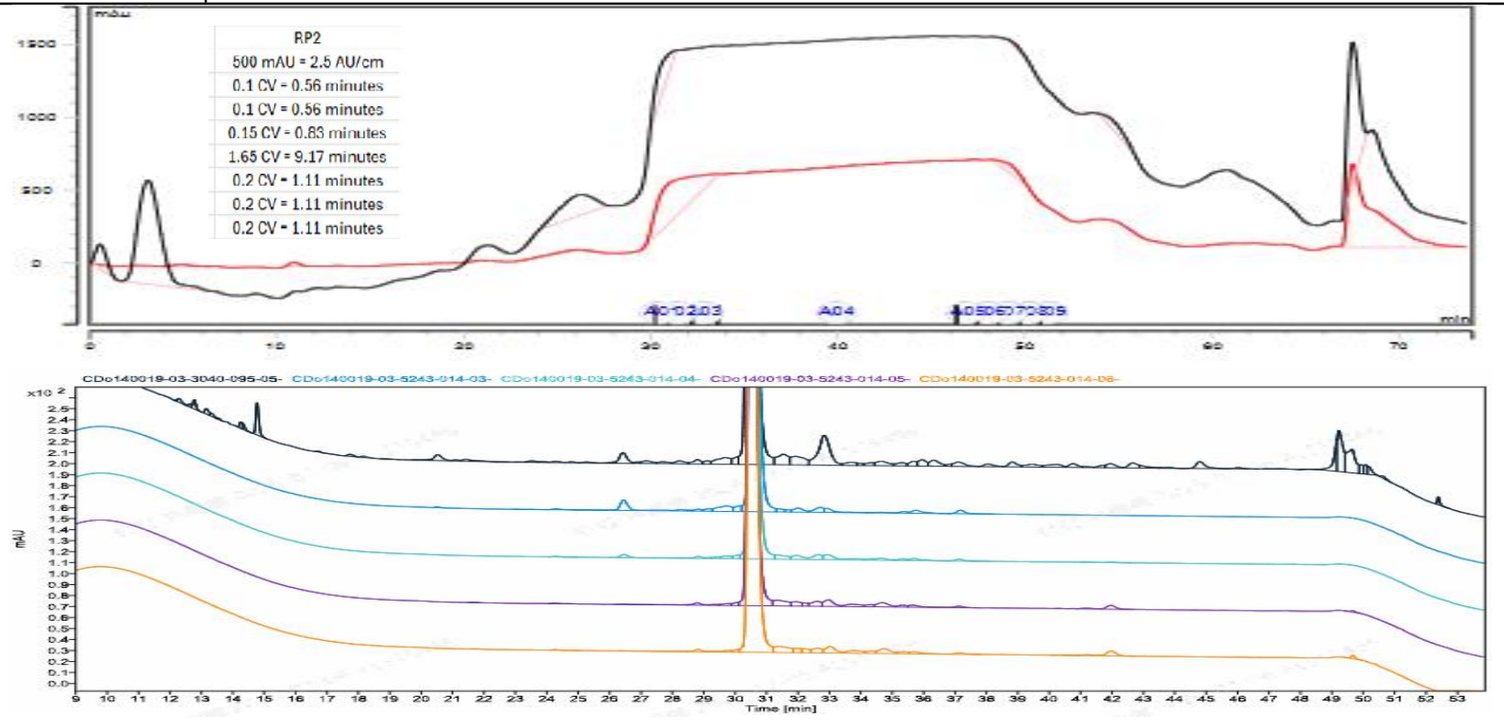
charge	2	-	3	3	-	3	3	3	3		3	3	5	2	2	-	3	2	-	3	-	-	-	-	-
Monoisotopic Mass mass	2568.2522	-	3559.9185	4579.3431	-	4422.2095	4523.2544	4422.210	4409.239	-	4329.153	4509.248	6103.969	2187.1731	2088.104	-	3571.9561	2441.3499	-	4775.3742	-	-	-	-	-
Delta mass	-1955.00	-	-963.34	56.09	-	-101.04	0	-101.040	-114.020	-	-194.100	-14.010	1580.715	-2336.08	-2435.15	-	-951.30	-2081.90	-	252.0804	-	-	-	-	-
Structure	Glu1-Phe23]+Pip	unknown	Orn11-Tyr37	+ **tBu**	unknown	Des Thr9	142526	Des Thr30	Des Asn21	unknown	DesPro^{28}Pro^{29}	-CH_2	**L3+L2+L2+L1**	Phe23-Tyr37	105+[Pro25-Tyr37]	unknown	Formaldehyde+[Orn11-Tyr37]	Formaldehyde+[Asn21-Tyr37]	unknown	+ Pbf	unknown	unknown	unknown	unknown	unknown
142526 Crude API CPo142526-02-03-01	0.72	0.74	0.69	0.50	0.38	0.72	72.29	0.46	1.37	0.24	0.70	0.53	1.08	0.12	0.77	0.38	0.99	-	-	0.55	4.44	-	-	0.51	1.35
Exp.4 CDo140019-03-3040-095-02	0.63	1.37	0.22	0.99	0.74	-	69.79	0.92	-	1.71	-	-	4.90	0.50	-	0.55	0.23	0.57	0.95	0.59	1.29	0.99	0.73	-	0.55

7.4.3 Expt.4: The purification conditions and collection strategy of RP1

Column	10*250mm bed column=19.63 mL		
Sample	CDo140019-03-3040-095-05 (Purity: 71.25%)		
Resin	Daisogel 120-10-ODS-RPS		
Mobile phase A	90% 50 mM Citric Acid, **pH 3.00**: 10% ACN	Mobile phase B	Acetonitrile
Detection wavelength	230nm	Monitor wavelength	280nm
Flowrate	3.53mL/min(270cm/h)		
Loading amount	**45.0mg/mL, 883 mg used.**		
Post-Load Wash	30~35.0(RP1)B% for 11.1min(2CV) at 3.53mL/min		

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Elution	T(time)	0	62.5	62.5	73.6		
	B(%)	27.8	44.4	66.7	66.7		
UV	Start collection when UV reaches 350mAu						



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7.4.4 Expt.4: Purity and impurity profile of RP1 fractions

Batch	Fraction Number	Impurity profile (RRT, %area) Only the components with the main - peak purity greater than 80% were counted																
RRT		0.886	0.977	0.985	1.000	1.026	1.033	1.048	1.054	1.064	1.077	1.100	1.137	1.171	1.192	1.128	1.372	1.629	Spectrum Attachments
m-z		1528.449	1475.745	-	1509.761	1475.746	1471.415		1444.727	1504.758	1221.628	1095.095	-	-	-	-	1222.183	-	-
charge		3	3	-	3	3	3		3	3	5	2	-	-	-	-	2	-	
Monoisotopic Mass mass		4579.343	4422.210	-	4523.2544	4422.210	4409.239	-	4329.153	4509.248	6103.969	2187.173	-	-	-	-	2441.350	-	
Delta mass		56.09	-101.040	-	0	-101.040	-114.020	-	-194.100	-14.010	1580.715	-2336.080	-	-	-	-	-2081.900	-	
Structure		**+ tBu**	Des Thr9	unknown	142526	Des Thr30	Des Asn21	unknown	$DesPro^{28}Pro^{29}$	$-CH_2$	**L3+L2+L2+L1**	Phe23-Tyr37	unknown	unknown	unknown	unknown	Formaldehyde+[Asn21-Tyr37]	unknown	
Exp.4 CDo140019-03-3040-095-02		0.99	-	-	69.79	0.92	-	1.71	-	-	4.90	0.50	-	-	-	-	0.57	-	CDo140019-03-3040-095-02.pdf
Before RP1 CDo140019-03-3040-095-05		0.81	0.95	0.39	71.44	-	1.22	1.21	-	-	3.55	0.34	0.54	0.34	0.58	0.49	0.38	2.14	CDo140019-03-3040-095-05.pdf
CDo140019-03-5243-014	CDo140019-03-5243-014-03	1.73	1.74	0.81	88.22	0.51	0.28	0.93	-	0.99	0.74	0.17	0.21	0.80	-	0.69	-	-	CDo140019-03-5243-014-03.pdf
	CDo140019-03-5243-014-04	0.51	0.52	0.44	91.38	1.02	-	0.89	-	0.95	1.03	0.36	0.44	0.52	-	0.34	-	-	CDo140019-03-5243-014-04.pdf
	CDo140019-03-5243-014-05	-	0.30	0.33	89.75	1.50	-	0.75	0.34	0.91	1.24	0.69	0.96	0.45	-	0.27	0.70	0.15	CDo140019-03-5243-014-05.pdf
	CDo140019-03-5243-014-06	-	0.27	0.26	89.20	1.72	-	0.59	0.55	0.80	1.25	0.75	1.06	0.43	-	0.26	0.87	0.28	CDo140019-03-5243-014-06.pdf
Mock pool (Combine fraction: CDo140019-03-5243-014-03 ~ 06)		0.53	0.44	0.44	90.68	0.93	0.90	1.09	1.07	0.35	0.56	0.44	0.39	0.15	0.44	0.16	0.16	0.12	CDo140019-03-5243-014-03-06, 12-16.pdf

Conclusion: RP1 exhibits poor removal capacity for the impurity with RRT 0.886, while it performs well in eliminating the impurity at RRT 1.077.

7.4.5 Expt.4: The purification conditions and collection strategy of RP2

Column	10*250mm bed column=19.63 mL

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Sample	CDo140019-03-5243-021-01						
Resin	Daisogel 120-10-ODS-RPS						
Mobile phase A	90% 50 mM Citric Acid, **pH 6.50**: 10% ACN			Mobile phase B	Acetonitrile		
Detection wavelength	230nm			Monitor wavelength	280nm		
Flowrate	3.53mL/min(270cm/h)						
Loadingamount	**45.0 mg/mL**						
Post-Load Wash	20.0(RP2)B% for 11.1min(2CV) at 3.53mL/min						
Elution	T(time)	0	41.1	41.1	52.2		
	B(%)	20	35.6	66.7	66.7		
Collection	Start collection when the UV280 reaches 450mAu						

7.4.6. Expt.4: Purity and impurity profile of RP1 fractions

Batch	Fraction Number	Impurity profile (RRT, %area) Only the components with the main - peak purity greater than 80% were counted																	
RRt		0.886	0.913	0.924	0.944	0.953	0.963	0.973	0.984	1.00	1.026	1.033	1.046	1.065	1.077	1.105	1.132	1.141	1.17	Spectrum Attachments
Before RP2 CDo140019-03-5243-014-03 ~ 06		0.53	-	-	-	-	-	0.44	0.44	90.68	0.93	0.9	1.09	1.07	0.56	0.44	0.39	-	0.15	CDo140019-03-5243-014-03-06, 12-16.pdf
CDo140019-03-5243-022	CDo140019-03-5243-022-02	-	0.15	0.35	0.53	0.72	1.07	2.11	1.30	91.17	0.82	0.67	-	0.58	-	0.20	-	0.11	/	CDo140019-03-5243-022-02.pdf

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	CDo140019-03-5243-022-03	-	-	-	0.30	0.26	0.45	1.03	0.82	95.88	0.43	0.40	-	0.49	-	-	-	-	/	CDo140019-03-5243-022-03.pdf
	CDo140019-03-5243-022-04	-	-	-	0.27	-	0.14	0.28	0.36	96.11	0.95	-	0.14	0.47	0.24	0.14	0.14	0.23	0.11	CDo140019-03-5243-022-04.pdf
	CDo140019-03-5243-022-05	-	-	-	0.32	-	-	-	-	95.85	1.30	-	0.26	0.74	0.46	0.23	0.26	0.26	/	CDo140019-03-5243-022-05.pdf
Mock pool (Combine fraction: CDo140019-03-5243-022-02-05)		-	-	-	0.31	-	0.25	0.53	0.45	96.06	0.99	-	0.13	0.46	0.22	0.11	0.11	-	-	CDo140019-03-5243-022-02-05.pdf

Conclusion: After two-step purification of RP1 and RP2, the removal rate of the impurity at RRT 0.886 is 100%, while the removal rate of the impurity at RRT 1.077 is 93.8%.

7.5. Expt.5: Spiking experiment study of RRT0.940; Glu31 (Des-tBu)

- **Step1: L2-L1- Glu31 (Des-tBu):**

Charge 2-MeTHF (14 V) and DMSO (1 V) into the reaction flask, adjust the temperature to 20 °C and start agitation. Add L1- Glu31 (Des-tBu) (1 eq) and Fmoc-L2 (1.02 eq), and stir until complete dissolution. Charge Oxyma (1.50 eq) and EDCI (1.50 eq), then rinse the addition vessel with DMSO (1 V). Add 2,4,6-trimethylpyridine (3.00 eq) dropwise. Upon completion of addition, hold the reaction at 20 °C for 3 h. Sample for UPLC monitoring; proceed to the next step when residual L1- Glu31 (Des-tBu) is <0.5%. Add piperidine (0.634 eq)/water (2 V) solution to the flask and react at 15 °C for 2 h. Monitor the reaction by sampling; once full conversion of Fmoc-L2-L1- Glu31 (Des-tBu) is confirmed, allow phase separation and discard the aqueous layer. Perform two sequential acid washes: for each wash, add the prepared ~10% dilute HCl to the organic phase, stir for 10 min, settle, separate and remove the aqueous layer while retaining the organic phase. Conduct an alkaline wash by adding the prepared ~8% aqueous $NaHCO_3$, stir for 10 min, settle, separate and discard the aqueous layer. Carry out a brine wash with the prepared 18% aqueous NaCl, stir for 10 min, settle, separate and retain the organic phase.

Transfer the organic phase to a constant-pressure dropping funnel for crystallization. Charge n-heptane (30 V) into a second reaction flask, then add the organic phase dropwise

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at 20±5℃. After addition is finished, stir the slurry for 1 h and filter. Slurry-wash the filter cake with n-heptane/2-MeTHF mixture (3 V / 1 V) for 0.5 h, filter again, dry the solid and collect the product L2-L1- Glu31 (Des-tBu).

- **Step2: Boc-L3-L2-L1- Glu31 (Des-tBu)**

Charge 2-MeTHF (14 V) and DMSO (1 V) into the reaction flask, adjust the temperature to 20 °C and start agitation. Add L2-L1- Glu31 (Des-tBu) (1.00 eq) and Boc-L3 (1.10 eq), and stir until complete dissolution. Charge Oxyma (1.50 eq) and EDCI (1.50 eq), then rinse the addition vessel with DMSO (1 V). Add 2,4,6-trimethylpyridine (3.00 eq) dropwise. Upon completion of addition, hold the reaction at 20 °C for 3 h. Sample for UPLC monitoring; proceed to the next step when residual L2-L1- Glu31 (Des-tBu) is <0.5%.

Add purified water (4.58 V) to the batch, stir for 10 min, then allow phase separation and discard the aqueous layer. Perform acid wash: add the prepared ~10% dilute HCl to the organic phase, stir for 10 min, settle, separate and remove the aqueous layer while retaining the organic phase. Conduct brine wash with the prepared 18% aqueous NaCl, stir for 10 min, settle, separate and retain the organic phase.

Transfer the organic phase to a constant-pressure dropping funnel for crystallization. Charge n-heptane (30 V) into a second reaction flask, then add the organic phase dropwise at 20±5℃. After addition is finished, stir the slurry for 1 h and filter. Slurry-wash the filter cake with n-heptane/2-MeTHF mixture (3 V / 1 V) for 0.5 h, filter again, dry the solid and collect the product Boc-L3-L2-L1- Glu31 (Des-tBu).

- **Step3: impurity markers (L3-L2-L1)**

Charge toluene (6 V) and anisole (0.25 V) into the reaction flask and maintain the temperature at 20 °C. Add DTT (0.25 eq), purified water (0.25 V) and TIPS (0.193 eq). While holding at 20 °C, charge Boc-L3-L2-L1- Glu31 (Des-tBu) (1.00 eq). Cool the batch to 10 °C, then add TFA (9.24 V) dropwise. Upon completion of addition, warm the mixture to 20 °C and hold the reaction for 3 h.

Sample the batch to confirm full conversion of Boc-L3-L2-L1- Glu31 (Des-tBu). Cool the reaction mixture to −15 °C, then add pre-chilled MTBE (48 V) dropwise. After addition is complete, stir the slurry for 1 h.

Filter the solids, slurry-wash the filter cake with MTBE (3 V) for 0.5 h, and filter again. Dry the isolated solid and collect the product L3-L2-L1. Test the purity and submit for MS analysis.

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Table 7.5.1 Experiment 5: Synthesis of Glu31 (Des-tBu) impurity marker

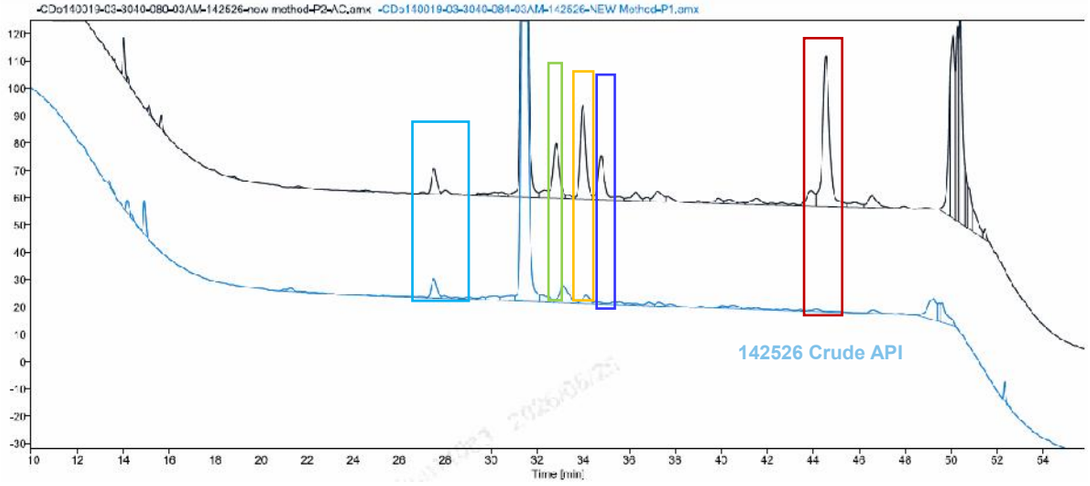
Reaction and solid product purity for each step				
No.	Impurity name	Purity of product solid	Product solid Chromatogram	MS Chromatogram
Step 1	Glu31 (Des-tBu) impurity Raw material purity: 93.50% Feed amount: 4 g	Step1 Product: 4.14 g purity: 80.50%	CDo140019-03-3 040-068-01	CDo140019-03-3 040-068-01.PNG
Step 2	Raw material purity: 80.50% Feed amount: 6 g	Step2 Product: 7.69 g purity: 50.77%	CDo140019-03-3 040-074-01	CDo140019-03-3 040-074-01.PNG
Step 3	Raw material purity: 50.77% Feed amount: 6 g	Step3 Product: 4.44 g Purity: 32.70%	CDo140019-03-3 040-080-01	CDo140019-03-3 040-080-01.PNG

The Glu31 (Des-tBu) impurity underwent a 3-step reaction (step1, step2, step3) to afford des-tBu-API, with the main peak corresponding to crude API (ASYM-142526). Des-tBu-API was spiked into crude API (ASYM 142526) at a level of 50%. The purity data for the impurity, the crude API (ASYM-142526), and the crude API spiked with des-tBu-API impurity (ASYM-142526) are presented in Table 8.3.2.

Then the Glu31 (Des-tBu)-API impurity-spiked API sample (ASYM-142526) was adjusted to pH 8~9 with NaOH solution, stirred at 10 ± 5 °C for 2 h to transfer the depsipeptide, then re-adjusted the pH value to 2.8-3.2 with citric acid solution.

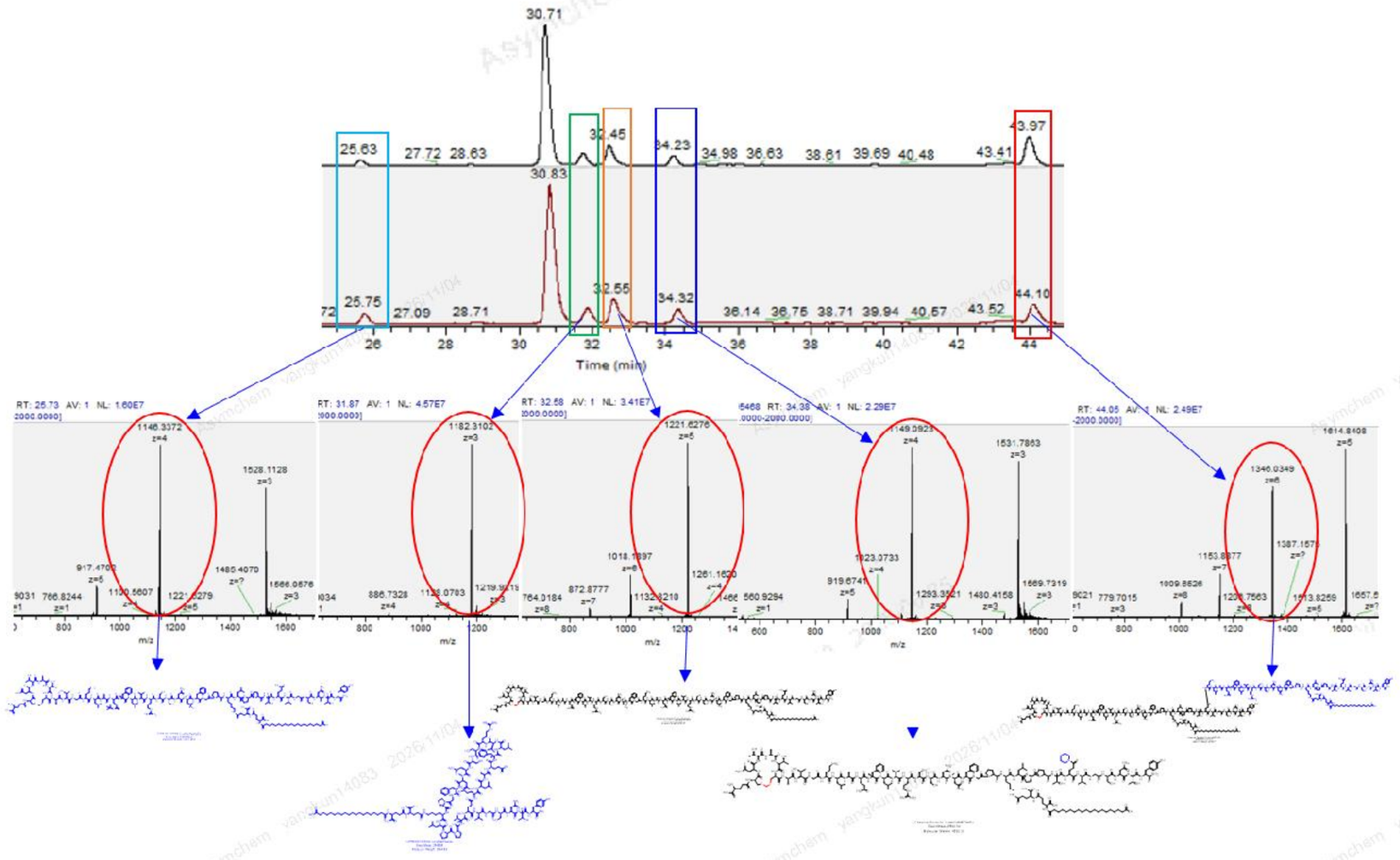
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7.5.1 Expt.5: UPLC Overlay Diagram of Experimental Group and Control Group



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Five major spiked impurities were included: +tBu, L2+L1 cyclization, L3+L2+L2+L1, Piperidine-capped impurity, and L3+L2+L1+L2+L1, with corresponding RRT values of 0.886, 1.048, 1.077, 1.017 and 1.430, respectively.

Table 7.5.2 Expt.5: LCMS data comparison of step 3 crude API

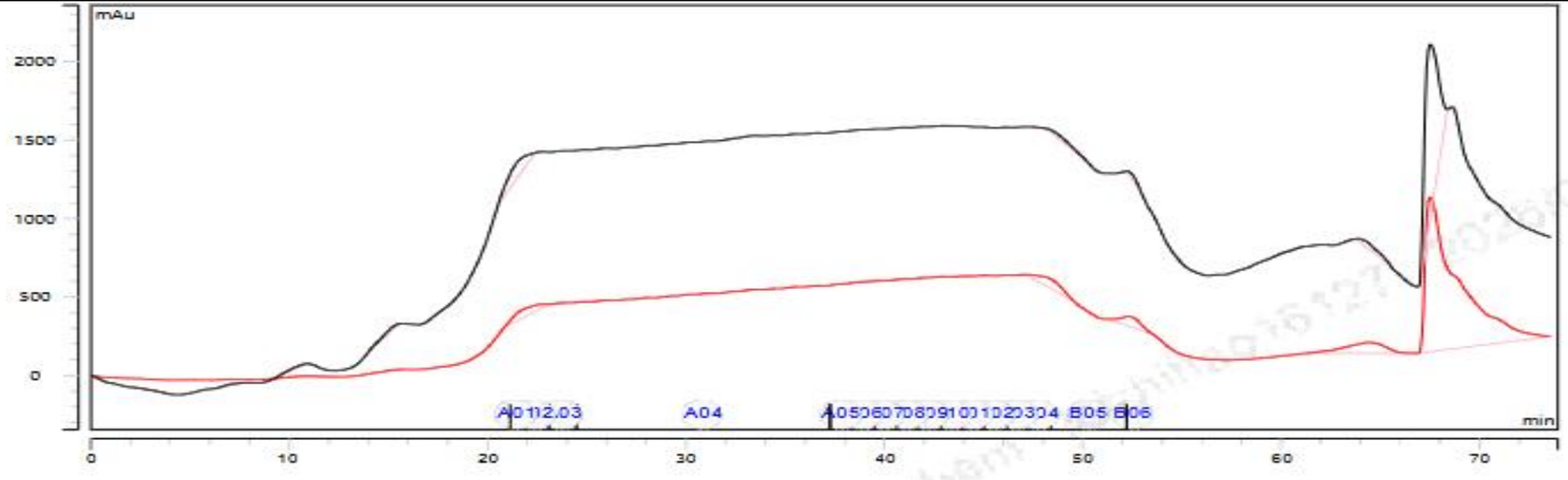
RRT	0.353	0.435	0.539	0.715	0.886	0.977	1.000	1.026	1.033	1.048	1.054	1.064	1.077	1.107	1.160	1.182	1.190	1.331	1.369	1.412	1.430	1.466	1.491	1.604	1.612	1.618	1.621	1.630	1.635	1.642	1.654	1.676	1.682
m/z	1285.634	789.423	-	1187.981	1528.449	1475.746	1509.761	1475.746	1471.415	1444.727	1444.727	1504.758	1221.628	919.4726	1475.747	1045.561	1131.823	-	1191.994	1131.823	1345.206	-	-	-	-	1541.426	-	-	-	1592.791	-		-
charge	2	2	-	3	3	3	3	3	3	3	3	3	5	5	3	2	4	-	3	4	6	-	-	-	-	3	-	-	-	3	-		-
Monoisotopic Mass mass	2568.252	1578.845	-	3559.919	4579.343	4422.210	4523.254	4422.210	4409.239	4329.153	4329.153	4509.248	6103.969	4590.340	4422.214	2088.104	4527.29	-	3571.956	4527.292	8071.235	-	-	-	-	4619.258	-	-	-	4775.374	-		-
Delta mass	-1955.000	-2944.409	-	-963.340	56.09	-101.040	0	-101.040	-114.020	-194.10	-194.100	-14.010	1580.715	67.086	-101.04	-2435.150	4.036	-	-951.300	4.0376	3547.981	-	-	-	-	96.000	-	-	-	252.080	-		-
Structure	Glu1-Phe23]+Pip	unknown	unknown	Orn11-Tyr37	**+tBu**	Des Thr9	142526	Des Thr30	Des Asn21	**L2 + L1 cyclization**	DesPro^{28} Pro^{29}	-CH_2	**L3+L2+ L2+L1**	**Piperidine - capped impurity**	unknown	105+[Pro 25-Tyr37]	unknown	unknown	Formaldehyde+[Orn11-Tyr37]	unknown	**L3+L2+ L1+L2+ L1**	unknown	unknown	unknown	unknown	unknown	unknown	unknown	unknown	+ Pbf	unknown	unknown	unknown
142526 Crude API CPo142526-02-03-01	0.72	0.02	0.74	0.69	0.50	0.72	72.29	0.46	1.37	0.24	0.70	0.53	1.08	0.14	0.33	0.77	-	-	0.99	0.19	-	0.19	0.26	-	-	-	-	-	-	0.55	4.44	0.51	1.35
Exp.5 CDo140019-03-3040-080-03	-	0.88	-	-	1.36	-	33.97	0.56	-	3.58	-	0.22	5.43	2.75	0.71	-	0.90	0.63	0.19	1.32	10.80	0.58	1.17	2.72	8.10	4.41	8.78	1.34	1.42	1.30	-	-	-

7.5.3 Expt.5: The purification conditions and collection strategy of RP1

Column	10*250mm bed column=19.63 mL
Sample	CDo140019-03-3040-080-05 (Purity: 56.73%)
Resin	Daisogel 120-10-ODS-RPS

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Mobile phase A	90% 50 mM Citric Acid, **pH 3.00**: 10% ACN			Mobile phase B	Acetonitrile		
Detection wavelength	230nm			Monitor wavelength	280nm		
Flowrate	3.53mL/min(270cm/h)						
Loading amount	**45.0mg/mL, 883 mg used.**						
Post-Load Wash	30~35.0(RP1)B% for 11.1min(2CV) at 3.53mL/min						
Elution	T(time)	0	62.5	62.5	73.6		
	B(%)	27.8	44.4	66.7	66.7		
Collection	Start collection when UV reaches 350mAu						



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7.5.4 Expt.5: Purity and impurity profile of RP1 fractions

Batch	Fraction Number	Impurity profile (RRT, %area) Only the components with the main - peak purity greater than 80% were counted																							
RRT		0.886	0.941	0.948	0.956	0.977	0.986	1.000	1.026	1.033	1.048	1.054	1.064	1.077	1.107	1.126	1.134	1.145	1.160	1.182	1.190	1.331	1.369	1.430	1.635	Spectrum Attachments
m/z		1528.449	-	-	-	1475.746	-	1509.761	1475.746	1471.415	1444.727	1444.727	1504.758	1221.628	919.4726	1475.747	1087.605	-	-	-	1131.8225	-	1191.994	1345.2059	-	-
charge		3	-	-	-	3	-	3	3	3	3	3	3	5	5	3	3	-	-	-	4	-	3	6	-	
Monoisotopic Mass mass		4579.3431	-	-	-	4422.2095	-	4523.2544	4422.210	4409.239	4329.153	4329.153	4509.248	6103.969	4590.340	4422.2144	3258.7915	-	-	-	4527.29	-	3571.9561	8071.235	-	
Delta mass		56.09	-	-	-	-101.04	-	0	-101.040	-114.020	-194.10	-194.100	-14.010	1580.715	67.086	-101.04	-1264.46 -	-	-	-	4.0356	-	-951.30	3547.981	-	-
Structure		**+ tBu**	unknown	unknown	unknown	Des Thr9	unknown	142526	Des Thr30	Des Asn21	**L2 + L1 cyclization**	DesPro[28]Pro[29]	-CH_2	**L3+L2+L2+ L1**	**Piperidine - capped impurity**	unknown	DIC+[MePhe15-Tyr37]	unknown	unknown	unknown	unknown	unknown	Formaldehyde+[Orn11-Tyr37]	L3+L2+L1+L2+ L1	unknown	
Exp.5 CDo140019-03-3040-080-03		1.36	-	-	-	-	-	33.97	0.56	-	3.58	-	0.22	5.43	2.75	-	-	-	0.71	-	0.90	0.63	0.19	10.80	1.42	CDo140019-03-3040-080-03.pdf
Before RP1 CDo140019-03-3040-080-05		0.99	-	0.21	0.10	0.65	0.37	56.34	-	1.38	-	2.44	0.15	3.53	1.58	0.11	0.12	0.55	0.28	-	0.29	0.15	0.59	5.64	0.26	CDo140019-03-3040-080-05.pdf
CDo140019-03-5243-010	CDo140019-03-5243-010-03	4.86	-	0.39	0.52	2.40	0.96	84.87	-	1.31	-	0.35	0.63	0.12	-	-	-	0.30	0.79	-	0.90	-	-	-	-	CDo140019-03-5243-010-03.pdf
	CDo140019-03-5243-010-04-10	1.06	-	0.32	0.10	0.56	0.44	88.76	-	1.47	-	2.65	0.52	0.50	1.73	0.14	0.20	0.33	0.38	-	0.25	-	0.18	-	-	CDo140019-03-5243-010-04-10.pdf
	CDo140019-03-5243-010-11	0.32	0.18	0.24	-	0.41	0.32	81.20	-	1.93	-	6.06	-	0.80	4.46	0.23	0.22	0.84	0.42	-	0.19	0.27	0.98	-	0.56	CDo140019-03-5243-010-11.pdf
	CDo140019-03-5243-010-12	0.17	0.22	0.24	-	0.30	0.27	81.16	-	1.69	-	5.99	-	0.86	4.66	0.22	0.23	0.83	0.41	-	0.15	0.27	1.00	-	0.77	CDo140019-03-5243-010-12.pdf

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	CDo140019-03-5243-010-13	-	0.29	0.26	-	0.25	0.18	80.34	-	2.25	-	6.03	-	0.92	4.89	0.22	0.23	0.85	0.39	-	0.12	0.24	0.98	-	0.93	CDo140019-03-5243-010-13.pdf
Mock pool (Combine fraction: CDo140019-03-5243-010-03 ~ 13)		1.22	0.17	0.30	0.36	0.62	0.48	88.11	-	0.85	-	2.66	0.64	0.51	1.70	0.37	0.25	0.43	0.34	-	0.17	0.02	0.17	-	0.16	CDo140019-03-5243-010-03-13.pdf

Conclusion: RP1 exhibits poor removal capacity for the impurity with RRT 0.886, 1.054, 1.107; while it performs well in eliminating the impurity at RRT 1.077, 1.430.

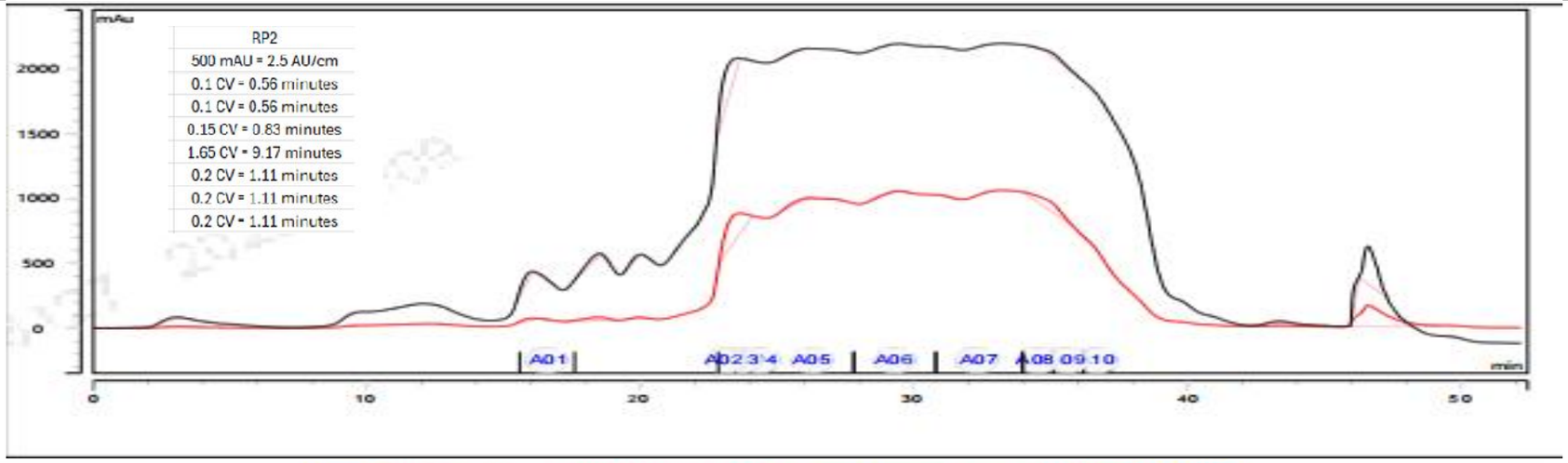
7.5.5 Expt.5: The purification conditions and collection strategy of RP2

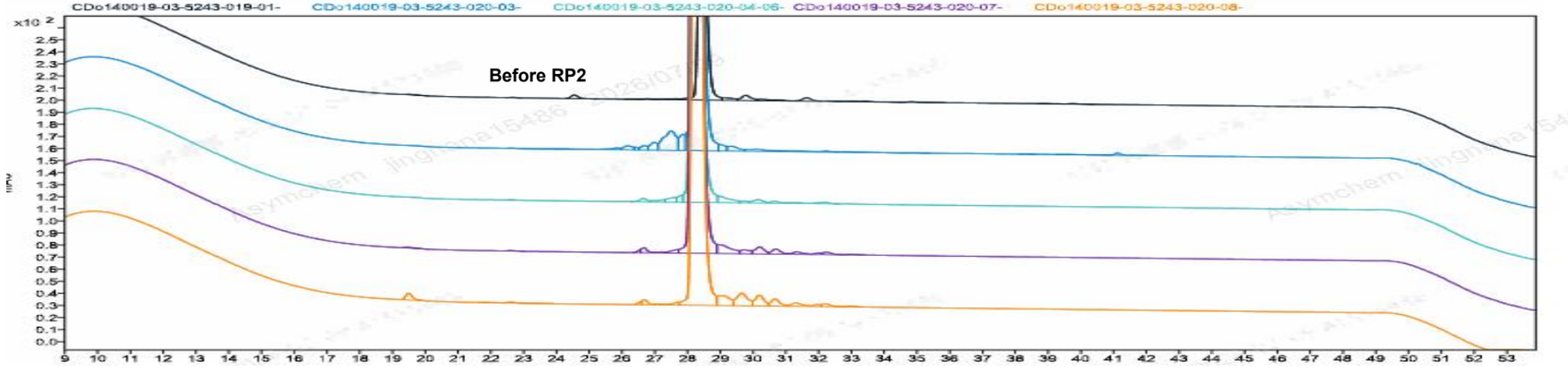
Column	4.6*250mm bed column=4.15 mL						
Sample	CDo140019-03-5243-019-01						
Resin	Daisogel 120-10-ODS-RPS						
Mobile phase A	90% 50 mM Citric Acid, **pH 6.50**: 10% ACN				Mobile phase B	Acetonitrile	
Detection wavelength	230nm				Monitor wavelength	280nm	
Flowrate	0.75mL/min(270cm/h)						
Loadingamount	**45.0 mg/mL**						
Post-Load Wash	20.0(RP2)B% for 11.1min(2CV) at 3.53mL/min						
Elution	T(time)	0	41.1	41.1	52.2		
	B(%)	20	35.6	66.7	66.7		

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Collection	Start collection when the UV280 reaches 450mAu





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7.5.6 Expt.5: Purity and impurity profile of RP2 fractions

Batch	Fraction Number	Impurity profile (RRT, %area) Only the components with the main - peak purity greater than 80% were counted																
RRt		0.690	0.886	0.941	0.948	0.956	0.974	0.986	1	1.026	1.033	1.051	1.063	1.077	1.106	1.126	1.134	Spectrum Attachments
Before RP2 CDo140019-03-5243-010-03-13		-	1.22	0.17	0.3	0.36	0.62	0.48	88.11	-	0.85	2.66	0.64	0.51	1.7	0.37	0.25	CDo140019-03-5243-010-03-13.pdf
CDo140019-03-5243-020	CDo140019-03-5243-020-03	-	-	-	0.40	-	3.42	1.34	90.21	-	0.57	0.72	0.87	-	-	0.15	0.17	CDo140019-03-5243-020-03.pdf
	CDo140019-03-5243-020-04-06	-	-	-	0.29	0.21	0.44	0.38	96.74	-	0.86	0.17	0.37	0.21	0.10	-	0.14	CDo140019-03-5243-020-04-06.pdf
	CDo140019-03-5243-020-07	-	-	-	0.29	-	0.24	-	95.99	-	1.15	0.37	0.64	0.44	0.24	0.10	0.20	CDo140019-03-5243-020-07.pdf
	CDo140019-03-5243-020-08	0.52	-	-	0.33	-	0.21	-	92.77	-	1.40	1.64	1.10	0.72	0.44	0.16	0.28	CDo140019-03-5243-020-08.pdf
Mock pool (Combine fraction: CDo140019-03-5243-020-03-08)		ND	ND	ND	0.38	ND	0.58	0.41	95.41	-	0.68	0.36	0.48	0.29	0.18	ND	0.07	CDo140019-03-5243-020-03-08.pdf

Conclusion: After two-step purification of RP1 and RP2, the removal rate of the impurity at RRT 0.886 is 100%, while the removal rate of the impurity at RRT 1.077 is 91.78%.

7.6. Expt.6: Spiking experiment study of RRT0.860; SC 26(C20diacid Des-tBu)

- **Step1: L2-L1- SC26(C20diacid Des-tBu):**

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Charge 2-MeTHF (14 V) and DMSO (1 V) into the reaction flask, adjust the temperature to 20 °C and start agitation. Add L1- SC26(C20diacid Des-tBu) (1 eq) and Fmoc-L2 (1.02 eq), and stir until complete dissolution. Charge Oxyma (1.50 eq) and EDCI (1.50 eq), then rinse the addition vessel with DMSO (1 V). Add 2,4,6-trimethylpyridine (3.00 eq) dropwise. Upon completion of addition, hold the reaction at 20 °C for 3 h. Sample for UPLC monitoring; proceed to the next step when residual L1- SC26(C20diacid Des-tBu) is <0.5%.

Add piperidine (0.634 eq)/water (2 V) solution to the flask and react at 15 °C for 2 h. Monitor the reaction by sampling; once full conversion of Fmoc-L2-L1- SC26(C20diacid Des-tBu) is confirmed, allow phase separation and discard the aqueous layer. Perform two sequential acid washes: for each wash, add the prepared ~10% dilute HCl to the organic phase, stir for 10 min, settle, separate and remove the aqueous layer while retaining the organic phase. Conduct an alkaline wash by adding the prepared ~8% aqueous $NaHCO_3$, stir for 10 min, settle, separate and discard the aqueous layer. Carry out a brine wash with the prepared 18% aqueous NaCl, stir for 10 min, settle, separate and retain the organic phase.

Transfer the organic phase to a constant-pressure dropping funnel for crystallization. Charge n-heptane (30 V) into a second reaction flask, then add the organic phase dropwise at 20±5℃. After addition is finished, stir the slurry for 1 h and filter. Slurry-wash the filter cake with n-heptane/2-MeTHF mixture (3 V / 1 V) for 0.5 h, filter again, dry the solid and collect the product L2-L1- SC26(C20diacid Des-tBu).

- **Step2: Boc-L3-L2-L1- SC26(C20diacid Des-tBu)**

Charge 2-MeTHF (14 V) and DMSO (1 V) into the reaction flask, adjust the temperature to 20 °C and start agitation. Add L2-L1- SC26(C20diacid Des-tBu) (1.00 eq) and Boc-L3 (1.10 eq), and stir until complete dissolution. Charge Oxyma (1.50 eq) and EDCI (1.50 eq), then rinse the addition vessel with DMSO (1 V). Add 2,4,6-trimethylpyridine (3.00 eq) dropwise. Upon completion of addition, hold the reaction at 20 °C for 3 h. Sample for UPLC monitoring; proceed to the next step when residual L2-L1- SC26(C20diacid Des-tBu) is <0.5%.

Add purified water (4.58 V) to the batch, stir for 10 min, then allow phase separation and discard the aqueous layer. Perform acid wash: add the prepared ~10% dilute HCl to the organic phase, stir for 10 min, settle, separate and remove the aqueous layer while retaining the organic phase. Conduct brine wash with the prepared 18% aqueous NaCl, stir for 10 min, settle, separate and retain the organic phase.

Transfer the organic phase to a constant-pressure dropping funnel for crystallization. Charge n-heptane (30 V) into a second reaction flask, then add the organic phase dropwise

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at 20±5℃. After addition is finished, stir the slurry for 1 h and filter. Slurry-wash the filter cake with n-heptane/2-MeTHF mixture (3 V / 1 V) for 0.5 h, filter again, dry the solid and collect the product Boc-L3-L2-L1- SC26(C20diacid Des-tBu).

- **Step3: impurity markers (L3-L2-L1)**

Charge toluene (6 V) and anisole (0.25 V) into the reaction flask and maintain the temperature at 20 °C. Add DTT (0.25 eq), purified water (0.25 V) and TIPS (0.193 eq). While holding at 20 °C, charge Boc-L3-L2-L1- SC26(C20diacid Des-tBu) (1.00 eq). Cool the batch to 10 °C, then add TFA (9.24 V) dropwise. Upon completion of addition, warm the mixture to 20 °C and hold the reaction for 3 h.

Sample the batch to confirm full conversion of Boc-L3-L2-L1- SC26(C20diacid Des-tBu). Cool the reaction mixture to −15 °C, then add pre-chilled MTBE (48 V) dropwise. After addition is complete, stir the slurry for 1 h.

Filter the solids, slurry-wash the filter cake with MTBE (3 V) for 0.5 h, and filter again. Dry the isolated solid and collect the product L3-L2-L1. Test the purity and submit for MS analysis.

Table 7.6.1 Experiment 6: Synthesis of SC26 (C20diacid Des-tBu) impurity marker

Reaction and solid product purity for each step				
No.	**Impurity name**	**Purity of product solid**	**Product solid Chromatogram**	**MS Chromatogram**
Step 1	SC26(C20diacid Des-tBu) impurity Raw material purity：92.30% Feed amount: 4 g	purity：73.50% Step1 Product: 7.54 g	CDo140019-03-3 040-069-01	CDo140019-03-3 040-069-01.PNG
Step 2	Raw material purity：73.50% Feed amount: 6 g	purity：65.68% Step2 Product: 7.76 g	CDo140019-03-3 040-075-01	CDo140019-03-3 040-075-01.PNG

Step 3	Raw material purity：65.68% Feed amount: 6 g	Purity：45.87% Step3 Product: 4.53 g	CDo140019-03-3 040-081-01	CDo140019-03-3 040-081-01.PNG

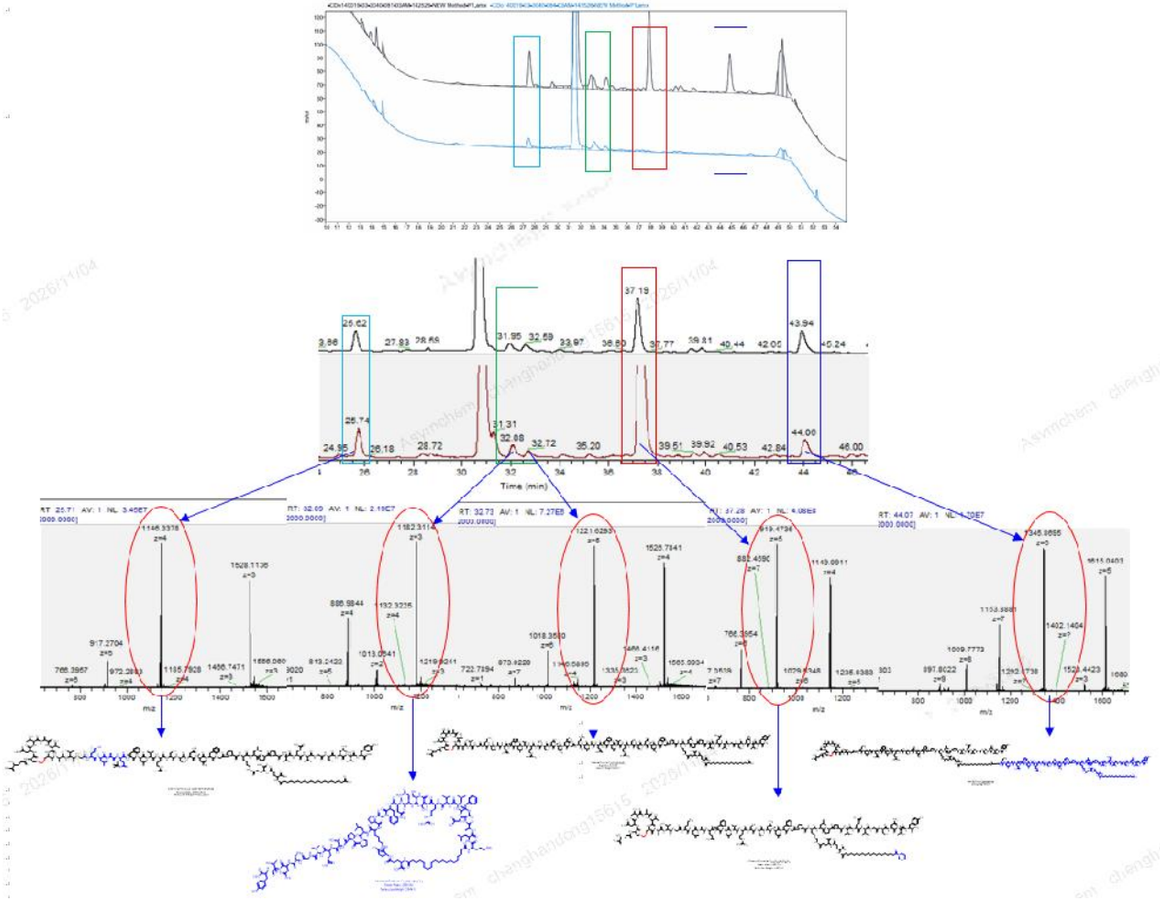
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The SC26 (Des-tBu) impurity underwent a 3-step reaction (step1, step2, step3) to afford des-tBu-API, with the main peak corresponding to crude API (ASYM-142526). Des-tBu-API was spiked into crude API (ASYM 142526) at a level of 50%. The purity data for the impurity, the crude API (ASYM-142526), and the crude API spiked with des-tBu-API impurity (ASYM-142526) are presented in Table 8.3.2.

Then the SC26 (Des-tBu)-API impurity-spiked API sample (ASYM-142526) was adjusted to pH 8~9 with NaOH solution, stirred at 10 ± 5 °C for 2 h to transfer the depsipeptide, then re-adjusted the pH value to 2.8-3.2 with citric acid solution.

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Five major spiked impurities were included: +tBu, L2+L1 cyclization, L3+L2+L2+L1, Piperidine-capped impurity, and L3+L2+L1+L2+L1, with corresponding RRT values of 0.886, 1.048, 1.077, 1.202 and 1.422, respectively.

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7.6.2 Expt.6: LC/MS data comparison of step 3 crude API:

RT	10.491	13.084	13.668	14.150	16.014	21.261	26.343	28.477	29.036	29.726	30.683	31.141	31.324	31.616	32.026	32.689	35.143	35.729	38.143	38.666	39.166	40.691	42.270	45.956	46.423	46.551	46.738	46.908
RRT	0.353	0.440	0.460	0.476	0.539	0.715	0.886	0.958	0.977	1.000	1.033	1.048	1.054	1.064	1.077	1.100	1.182	1.202	1.283	1.301	1.318	1.369	1.422	1.546	1.562	1.566	1.572	1.578
m/z	1285.634	789.423	-	-	-	1187.981	1528.449	-	1475.746	1509.761	1471.415	1444.727	1444.727	1504.758	1221.628	1503.754	1045.561	919.473	-	-	-	1191.994	1345.195	-	-	-	1082.090	1180.130
charge	2	2	-	-	-	3	3	-	3	3	3	3	3	3	5	3	2	5	-	-	-	3	6	-	-	-	2	2
Monoisotopic Mass mass	2568.252	1578.845	-	-	-	3559.919	4579.343	-	4422.210	4523.254	4409.239	4329.153	4329.153	4509.248	6103.969	4506.237	2088.104	4590.340	-	-	-	3571.956	8065.170	-	-	-	2161.162	2357.243
Delta mass	-1955.000	-2944.409	-	-	-	-963.340	56.090	-	-101.040	0	-114.020	-194.100	-194.100	-14.010	1580.715	-17.020	-2435.150	67.086	-	-	-	-951.300	3541.915	-	-	-	-2362.090	-2166.010
Structure	Glu1-Phe23]+Pip	unknown	unknown	unknown	unknown	Orn11-Tyr37	**+ tBu**	unknown	Des Thr9	142526	Des Asn21	**L2 + L1 cyclization**	DesPro^{28}Pro^{29}	-CH_2	**L3+L2+L2 +L1**	Asn22 cyclization	105+[Pro25 -Tyr37]	**Piperidine - capped impurity**	unknown	unknown	unknown	Formaldehyde+[Orn11-Tyr37]	**L3+L2+L1 +L2+L1**	unknown	unknown	unknown	178+[Pro25 -Tyr37]	Acetyl+[MeAsn22-Tyr37}
142526 Crude API CPo142526-02-03-01	0.72	0.01	0.01	-	0.74	0.69	0.50	0.16	0.72	72.29	1.37	0.24	0.70	0.53	1.08	0.12	0.77	0.38	0.27	0.36	0.32	0.99	-	-	0.11	-	0.10	-
Exp.6 CDo140019-03-3040-081-04	-	0.61	1.04	0.70	-	-	3.70	0.18	0.28	48.86	-	1.55	1.28	0.21	1 **L3+L2+L2 +L1**.60	0.50	0.29	9.13	0.59	0.86	0.66	0.20	5.13	1.23	5.27	2.97	3.22	1.03

7.6.3 The purification conditions and collection strategy of RP1

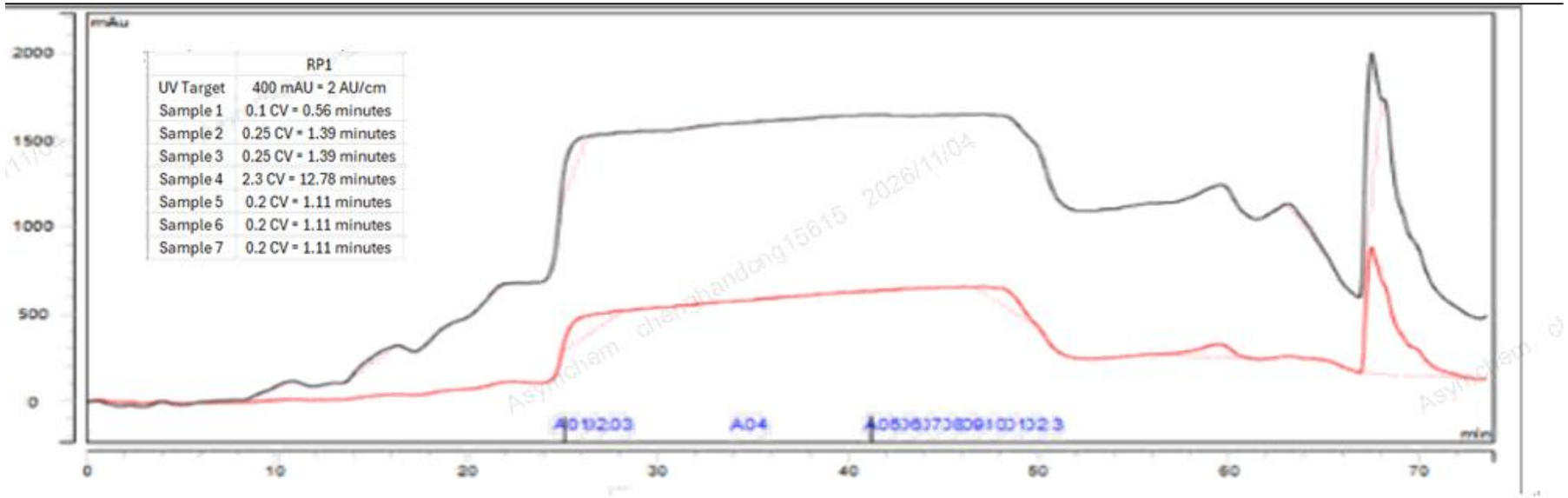
Column	10*250mm bed column=19.63 mL
Sample	CDo140019-03-3040-081-05 (Purity: 60.45%)

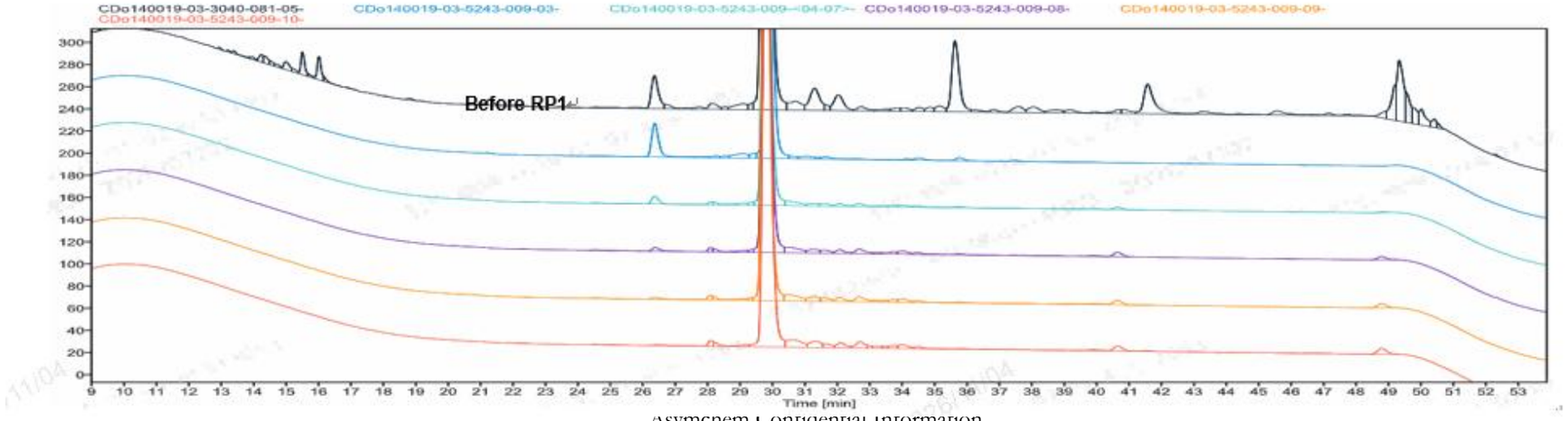
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Resin	Daisogel 120-10-ODS-RPS						
Mobile phase A	90% 50 mM Citric Acid, **pH 3.00**: 10% ACN			Mobile phase B	Acetonitrile		
Detection wavelength	230nm			Monitor wavelength	280nm		
Flowrate	3.53mL/min(270cm/h)						
Loading amount	**45.0mg/mL, 883 mg used.**						
Post-Load Wash	30~35.0(RP1)B% for 11.1min(2CV) at 3.53mL/min						
Elution	T(time)	0	62.5	62.5	73.6		
	B (%)	27.8	44.4	66.7	66.7		
UV	Start collection when UV reaches 350mAu						

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7.6.4 Expt.6: Purity and impurity profile of RP1 fractions

Batch	Fraction Number	Impurity profile (RRT, %area) Only the components with the main - peak purity greater than 80% were counted												
RRt		0.886	0.977	0.985	1.000	1.033	1.048	1.054	1.064	1.077	1.100	1.318	1.422	Spectrum Attachments
m/z		1528.449	1475.746	-	930.2891	1471.415	1444.727	1444.727	1504.758	1221.628	1503.754	-	1345.2059	-
charge		3	3	-	5	3	3	3	3	5	3	-	6	
Monoisotopic Mass mass		4579.3431	4422.2095	-	1509.761	4409.239	4329.153	4329.153	4509.248	6103.969	4506.2373	-	8071.235	
Delta mass		56.09	-101.04	-	3	-114.020	-194.100	-194.100	-14.010	1580.715	-17.02	-	3547.981	
Structure		+tBu	Des Thr9	unknown	142526	Des Asn21	**L2 + L1 cyclization**	DesPro^{28}Pro^{29}	-CH_2	**L3+L2+L2+L1**	Asn22 cyclization	unknown	L3+L2+L1+L2+L1	
Exp.6 CDo140019-03-3040-081-04		3.70	0.28	0.19	48.86	-	1.55	1.28	0.21	1.60	0.50	0.66	5.13	CDo140019-03-3040-081-04.pdf
Before RP1 CDo140019-03-3040-081-05		2.24	0.71	0.26	60.46	1.10	2.15		0.23	1.49	0.39	0.33	3.09	CDo140019-03-3040-081-05.pdf
CDo140019-5243-009	CDo140019-5243-009-03	6.89	1.74	0.77	83.74	0.57	-	0.20	0.64	0.16	-	-	-	CDo140019-03-5243-009-03.pdf
	CDo140019-5243-009-04-07	1.45	0.44	0.35	89.77	1.48	-	0.62	0.62	0.59	0.73	-	-	CDo140019-03-5243-009-04-07.pdf
	CDo140019-5243-009-08	0.71	0.36	0.27	88.45	1.80	-	0.86	0.64	0.73	1.02	0.60	0.64	CDo140019-03-5243-009-08.pdf
	CDo140019-5243-009-09	0.43	0.32	0.21	87.79	1.99	-	1.04	0.66	0.80	1.11	0.41	0.66	CDo140019-03-5243-009-09.pdf
	CDo140019-5243-009-10	-	0.33	-	86.52	2.31	-	1.41	0.63	0.91	1.09	0.46	0.72	CDo140019-03-5243-009-10.pdf
Mock pool (Combine fraction: CDo140019-5243-009-03 ~ 10)		2.15	0.48	0.40	90.26	0.66	-	0.65	0.62	0.47	0.53	0.27	0.33	CDo140019-03-5243-009-03-10, 16-25.pdf

Conclusion: RP1 exhibits poor removal capacity for the impurity with RRT 0.886, while it performs well in eliminating the impurity at RRT 1.054, 1.077, 1.202, 1.397.

7.6.5 Expt.6: The purification conditions and collection strategy of RP2

Column	10*250mm bed column=19.63 mL
Sample	CDo140019-03-5243-009-03-10-16-25 (Purity: 90.26)

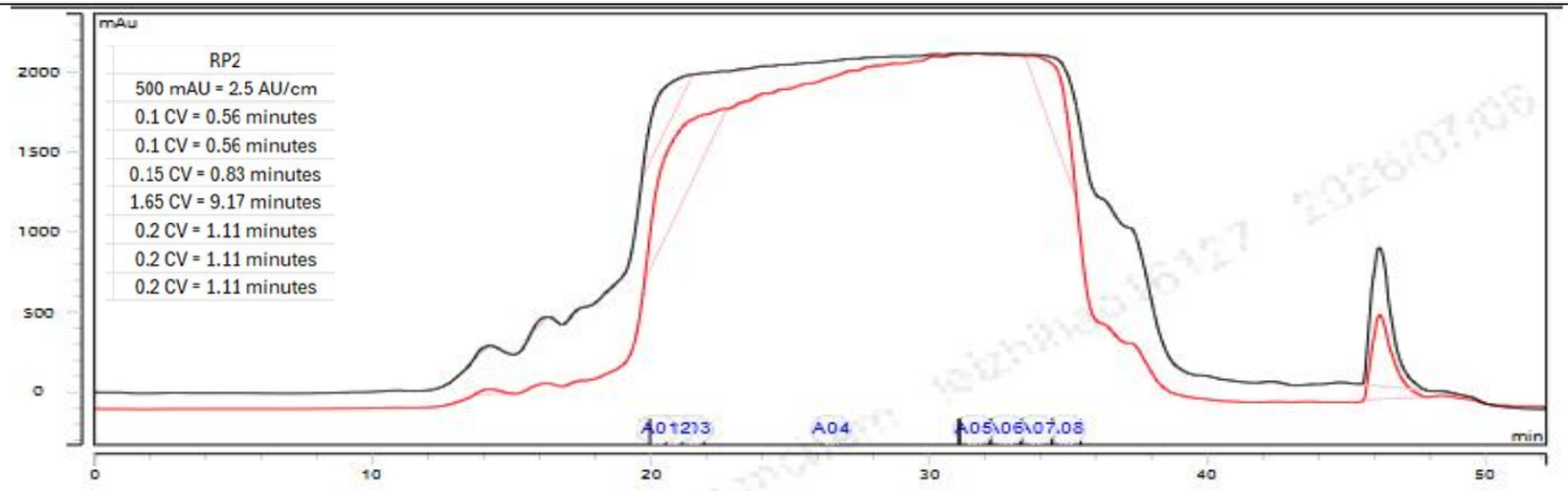
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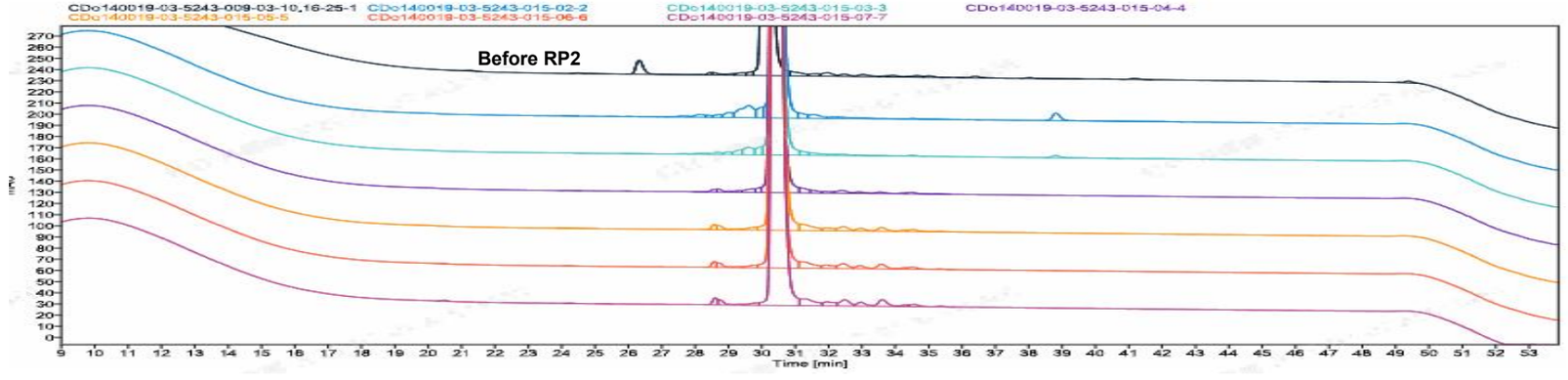
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Resin	Daisogel 120-10-ODS-RPS						
Mobile phase A	90% 50 mM Citric Acid, pH 6.50: 10% ACN			Mobile phase B	Acetonitrile		
Detection wavelength	230nm			Monitor wavelength	280nm		
Flowrate	3.53mL/min(270cm/h)						
Loadingamount	45.0 mg/mL						
Post-Load Wash	20.0(RP2)B% for 11.1min(2CV) at 3.53mL/min						
Elution	T(time)	0	41.1	41.1	73.6		
	B(%)	20	35.6	66.7	66.7		
Collection	Start collection when the UV280 reaches 450mAu						

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7.6.6 Expt.6: Purity and impurity profile of RP2 fractions

Batch	Fraction Number	Impurity profile (RRT, %area) Only the components with the main - peak purity greater than 90% were counted													
RRt		0.886	0.956	0.965	0.976	0.985	1.000	1.026	1.033	1.052	1.061	1.077	1.098	1.276	Spectrum Attachments
Before RP2 CDo140019-5243-009-03 ~ 10		2.15	0.10	0.23	0.48	0.40	90.26	0.66	0.60	0.65	0.62	0.47	0.53	-	CDo140019-03-5243-009-03-10，16-25.pdf
CDo140019-03-5243-015	CDo140019-03-5243-015-02	-	0.71	-	3.04	1.10	89.97	0.66	0.77	0.25	0.30	0.20	0.18	1.06	CDo140019-03-5243-015-02.pdf
	CDo140019-03-5243-015-03	-	0.30	0.54	1.12	0.77	94.79	0.45	0.45	0.11	0.25	0.11	-	0.30	CDo140019-03-5243-015-03.pdf
	CDo140019-03-5243-015-04	-	-	0.18	0.45	0.34	96.23	0.56	0.31	0.23	0.41	0.22	0.22	/	CDo140019-03-5243-015-04.pdf
	CDo140019-03-5243-015-05	-	-	0.11	0.32	-	95.70	1.07		0.38	0.55	0.33	0.45	-	CDo140019-03-5243-015-05.pdf
	CDo140019-03-5243-015-06	-	-	-	0.30	-	95.37	1.12		0.45	0.61	0.37	0.55	-	CDo140019-03-5243-015-06.pdf
	CDo140019-03-5243-015-07	-	-	-	0.27	-	94.40	1.26		0.58	0.78	0.45	0.78	-	CDo140019-03-5243-015-07.pdf
Mock pool (Combine fraction: CDo140019-03-5243-015-03 ~ 07)		ND	-	0.22	0.42	0.30	95.76	0.97		0.29	0.46	0.25	0.35	-	CDo140019-03-5243-015-03-07.pdf

Conclusion: After two-step purification of RP1 and RP2, the removal rate of the impurity at RRT 0.886 is 100%, while the removal rate of the impurity at RRT 1.077 is 93.8%

7.7. Expt.7: Spiking experiment study of RRT0.722 Ac-endcapping-8mer

- **Step3: Global cleavage Ac-endcapping-8mer impurity markers**

Charge toluene (6 V) and anisole (0.25 V) into the reaction flask and maintain the temperature at 20 °C. Add DTT (0.25 eq), purified water (0.25 V) and TIPS (0.193 eq). While holding at 20 °C, charge Ac-endcapping-8mer impurity (1.00 eq). Cool the batch to 10 °C, then add TFA (9.24 V) dropwise. Upon completion of addition, warm the mixture to

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20 °C and hold the reaction for 3 h.

Sample the batch to confirm full conversion of Ac-endcapping-8mer impurity. Cool the reaction mixture to −15 °C, then add pre-chilled MTBE (48 V) dropwise. After addition is complete, stir the slurry for 1 h.

Filter the solids, slurry-wash the filter cake with MTBE (3 V) for 0.5 h, and filter again. Dry the isolated solid and collect the product. Test the purity and submit for MS analysis.

Table 7.7.1 Experiment 7: Synthesis of Ac-endcapping-8mer impurity marker

Step 3 reaction and solid product purity				
No.	Impurity name	Purity of product solid	Product solid Chromatogram	MS Chromatogram
Step 3	Ac-endcapping-8mer impurity Raw material purity: 91.98% Feed amount: 2g	Purity : 75.69% Product: 0.92 g	CDo140019-03-3 040-051-02	CDo140019-03-3 040-051-02.PNG

The Ac-endcapping-8mer impurity underwent step3 reaction to afford Ac-endcapping-8mer impurity marker. Ac-endcapping-8mer impurity marker was spiked into crude API (ASYM 142526) at a level of 2.49% area.

Table 7.7.2 Summary of sipking experiment

Spiking study.					
Impurity name	Purity	UPLC	Suggested Spiking Amount	Real Spiking Amount	UPLC after spiking

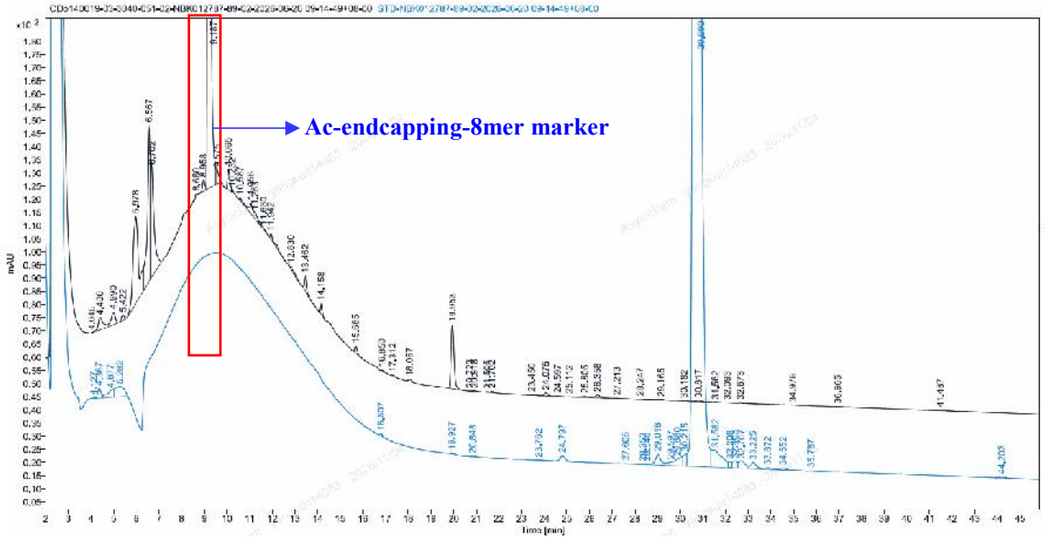
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Ac-endcapping-8mer marker	75.69%	CDo140019-03-3040-051-02	2.2%	2.5%	CDo140019-03-3040-051-07.pdf

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Then the Ac-endcapping-8mer impurity-spiked API sample (ASYM-142526) was adjusted to pH 8~9 with NaOH solution, stirred at 10 ± 5 °C for 2 h to transfer the depsipeptide, then re-adjusted the pH value to 2.8-3.2 with citric acid solution.

7.7.1 Expt.7: UPLC Overlay Diagram of Experimental Group and Control Group



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7.7.3 Expt.7: LC/MS data comparison of step 3 crude API:

RT	5.767	6.332	6.45	9.72	10.491	16.014	19.232	21.261	26.343	28.477	29.036	29.726	30.683	31.141	31.324	31.616	32.026	32.689	35.143	35.729	40.691	46.738	48.798	49.179	49.814	50.013
RRT	0.194	0.213	0.217	0.327	0.353	0.539	0.647	0.715	0.886	0.958	0.977	1.00	1.033	1.048	1.054	1.064	1.077	1.100	1.182	1.190	1.369	1.572	1.642	1.654	1.676	1.682
m/z	-	-	-	-	1285.634	-	-	1187.981	1528.449	-	1475.746	1509.761	1471.415	1444.727	1444.727	1504.758	1221.628	1503.754	1045.561	1131.823	1191.994	1082.09-	1592.791	-	-	-
charge	-	-	-	-	2	-	-	3	3	-	3	3	3	3	3	3	5	3	2	4	3	2	3	-	-	-
Monoisotopic Mass mass	-	-	-	-	2568.252	-	-	3559.918	4579.343	-	4422.210	4523.254	4409.239	4329.153	4329.153	4509.248	6103.969	4506.237	2088.104	4527.290	3571.956	2161.162	4775.374	-	-	-
Delta mass	-	-	-	-	-1955.000	-	-	-963.340	56.090	-	-101.040	0	-114.020	-194.100	-194.100	-14.010	1580.715	-17.020	-2435.150	4.036	-951.300	-2362.090	252.080	-	-	-
Structure	unknown	unknown	unknown	**Ac-endcapping-8mer marker**	Glu1-Phe23]+Pip	unknown	unknown	Orn11-Tyr37	+ tBu	unknown	Des Thr9	142526	Des Asn21	L2 + L1 cyclization	$DesPro^{28}Pro_{29}$	$-CH_2$	L3+L2+L2+L1	Asn22 cyclization	105+[Pro25-Tyr37]	unknown	Formaldehyde+[Orn11-Tyr37]	178+[Pro25-Tyr37]	+ Pbf	unknown	unknown	unknown
142526 Crude API CPo142526-02-03-01	-	-	-	-	0.72	0.74	-	0.69	0.5	0.16	0.72	72.29	1.37	0.24	0.70	0.53	1.08	0.12	0.77	0.38	0.99	0.1	0.55	4.44	0.51	1.35
Exp.7 CDo140019-03-3040-051-01	5.73	5.3	4.37	75.68	-	-	2.07	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

7.7.4 Expt.7: The purification conditions and collection strategy of RP1

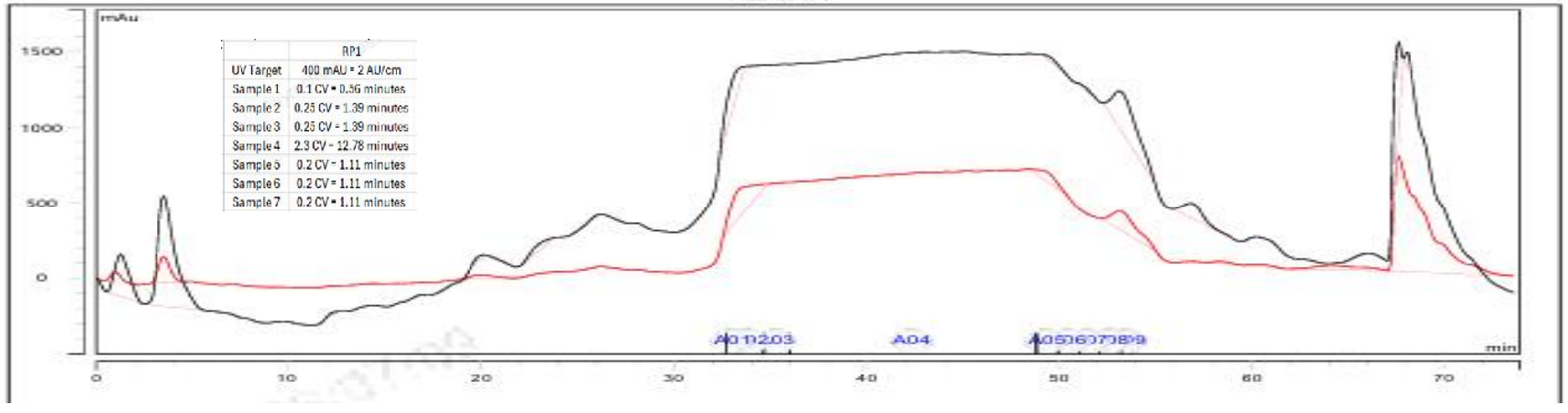
Column	10*250mm bed column=19.63 mL
Sample	CDo140019-03-3040-051-07 (Purity: 69.40%)
Resin	Daisogel 120-10-ODS-RPS

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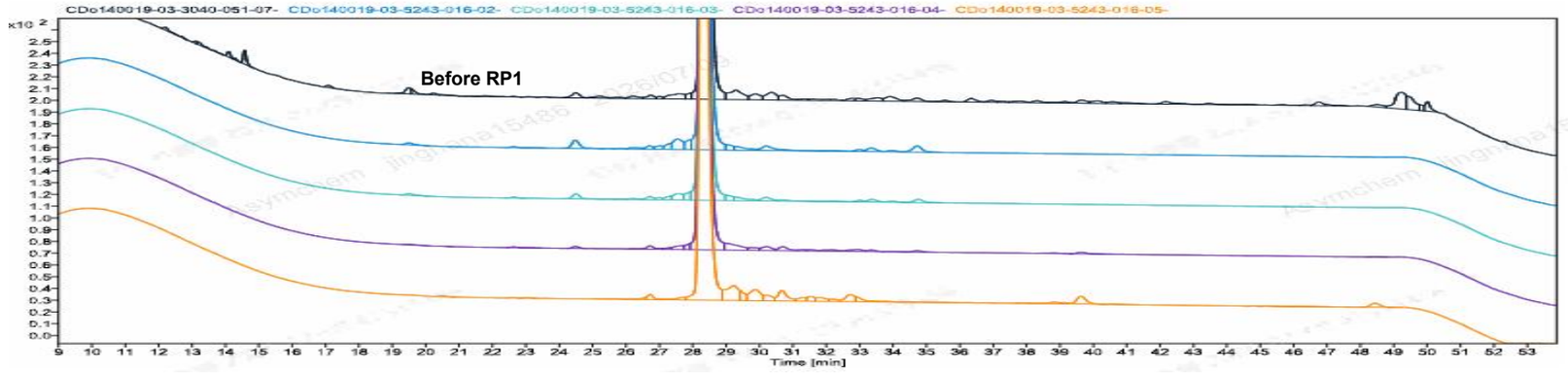
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Mobile phase A	90% 50 mM Citric Acid, **pH 3.00**: 10% ACN			Mobile phase B		Acetonitrile	
Detection wavelength	230nm			Monitor wavelength		280nm	
Flowrate	3.53mL/min(270cm/h)						
Loading amount	**45.0mg/mL, 883 mg used.**						
Post-Load Wash	30~35.0(RP1)B% for 11.1min(2CV) at 3.53mL/min						
Elution	T(time)	0	62.5	62.5	73.6		
	B(%)	27.8	44.4	66.7	66.7		
UV	Start collection when UV reaches 350mAu						

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7.7.5 Expt.7: Purity and impurity profile of RP1 fractions

Batch	Fraction Number	Impurity profile (RRT, %area) Only the components with the main - peak purity greater than 80% were counted																				
RRt		0.327	0.886	0.943	0.952	0.962	0.973	0.984	1.000	1.033	1.041	1.054	1.077	1.083	1.106	1.113	1.129	1.136	1.162	1.177	1.196	1.225	Spectrum Attachments
m/z		-	1528.449	-	-	919.0769	1475.746	-	1509.761	1471.415	1444.727	1444.727	1221.628	1503.754	1045.561	1131.8225	-	-	-	-	857.6394	-	-
charge		-	3	-	-	5	3	-	3	3	3	3	5	3	2	4	-	-	-	-	5	-	
Monoisotopic Mass mass		-	4579.343	-	-	4595.385	4422.210	-	4523.254	4409.239	4329.153	4329.153	6103.969	4506.237	2088.104	4527.29	-	-	-	-	4288.197	-	
Delta mass		-	56.090	-	-	72.130	-101.040	-	0	-114.020	-194.100	-194.100	1580.715	-17.020	-2435.150	4.036	-	-	-	-	- 235.057	-	
Structure		Ac-endcapping-8mer marker	+ tBu	unknown	unknown	unknown	Des Thr9	unknown	142526	Des Asn21	L2 + L1 cyclization	DesPro^{28}Pro^{29}	L3+L2+L2+L1	unknown	unknown	unknown	unknown	unknown	unknown	unknown	Piperidine - caped impurity	unknown	
Before RP1 CDo140019-03-3040-051-01		2.50	0.50	0.37	0.18	-	1.08	0.38	67.05	1.87	-	0.81	1.10	0.60	0.16	0.19	0.12	0.15	0.27	0.53	0.75	0.54	CDo140019-03-5243-015-03-07.pdf
CDo140019-03-5243-016	CDo140019-03-5243-016-02	ND	1.13	0.40	0.45	0.62	1.82	0.97	88.48	1.47	-	-	1.25	-	-	-	-	0.18	0.30	0.59	0.24	1.10	CDo140019-03-5243-016-02.pdf
	CDo140019-03-5243-016-03	ND	0.64	0.36	0.25	0.37	1.04	0.72	92.17	0.61	0.73	-	0.61	0.31	-	-	-	0.15	0.23	0.41	0.18	0.52	CDo140019-03-5243-016-03.pdf
	CDo140019-03-5243-016-04	ND	0.24	0.42	-	0.17	0.58	0.44	92.30	1.62	-	0.43	0.65	0.63	0.15	0.17	-	0.35	0.47	0.26	-	0.23	CDo140019-03-5243-016-04.pdf

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CDo140019-03-5243-016-05	ND	-	0.54	-	-	0.47	-	83.71	3.15	0.75	2.04	0.78	1.59	0.48	0.54	0.61	0.31	0.62	-	0.14	-	CDo140019-03-5243-016-05.pdf
Mock pool (Combine fraction: CDo140019-03-5243-016-02~05	ND	0.30	0.45	0.12	0.22	0.70	0.49	91.38	1.74	-	0.52	0.68	0.64	0.16	0.16	0.16	0.19	0.28	0.29	0.12	0.31	CDo140019-03-5243-016-02-05,11-14.p

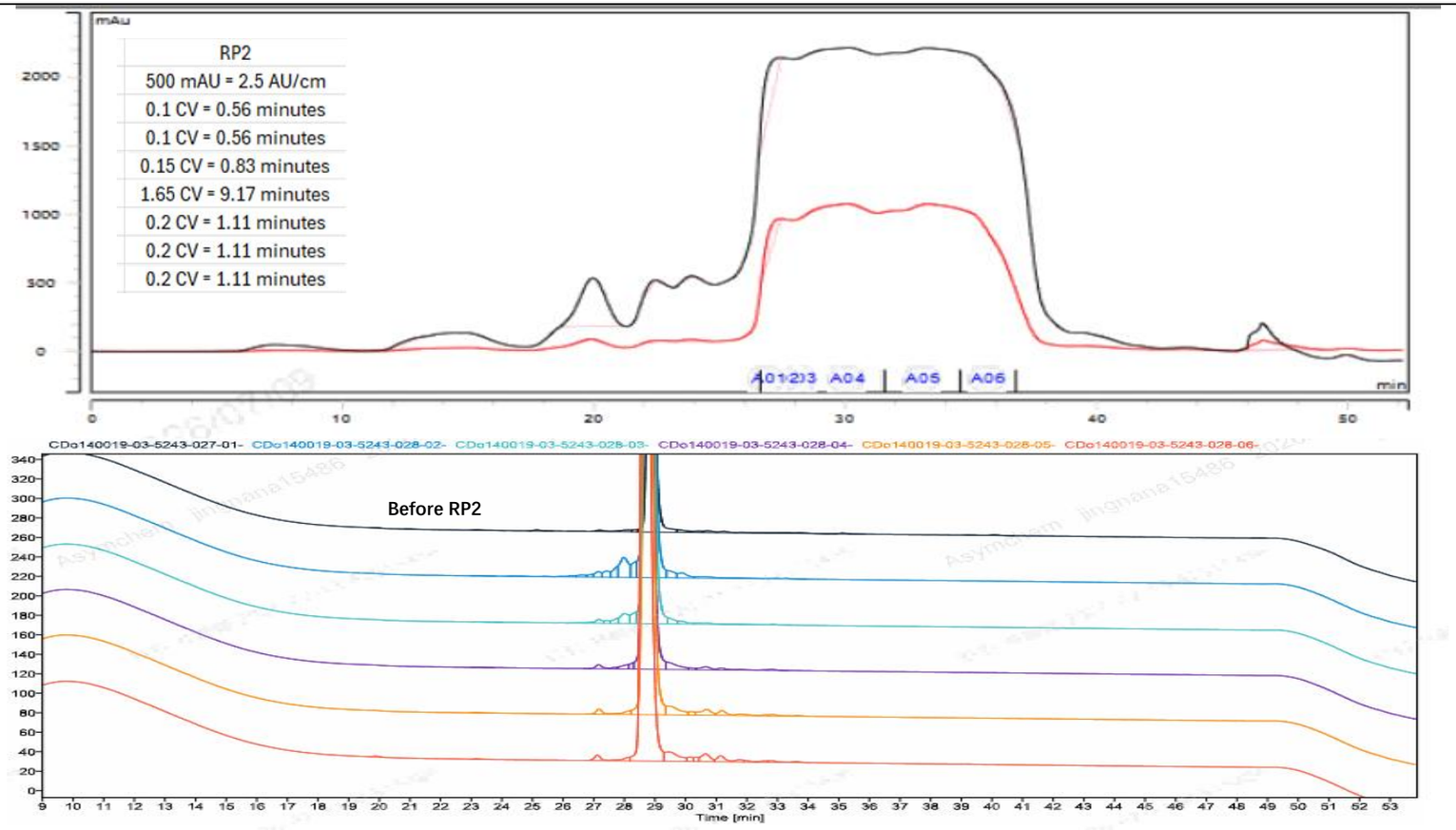
Conclusion: RP1 can completely remove the impurity at RRT 0.327 (Impurity Marker #7).

7.7.6 Expt.7: The purification conditions and collection strategy of RP2

Column	4.6*250mm bed column=4.15 mL						
Sample	CDo140021-03-6118-02-09						
Resin	Daisogel 120-10-ODS-RPS						
Mobile phase A	90% 50 mM Citric Acid, **pH 6.50**: 10% ACN			Mobile phase B		Acetonitrile	
Detection wavelength	230nm			Monitor wavelength		280nm	
Flowrate	0.75mL/min(270cm/h)						
Loadingamount	**45.0 mg/mL**						
Post-Load Wash	20.0(RP2)B% for 11.1min(2CV) at 3.53mL/min						
Elution	T(time)	0	41.1	41.1	52.2		
	B(%)	20	35.6	66.7	66.7		
Collection	Start collection when the UV280 reaches 450mAu						

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7.7.7 Expt.7: Purity and impurity profile of RP2 fractions

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Batch	Fraction Number	Impurity profile (RRT, %area) Only the components with the main - peak purity greater than 80% were counted																						
RRt		0.327	0.886	0.914	0.924	0.936	0.943	0.952	0.962	0.973	0.984	1.000	1.026	1.033	1.041	1.054	1.064	1.077	1.083	1.106	1.113	1.136	1.144	1.177	Spectrum Attachments
Before RP2 CDo140019-03-5243-016-02~05		ND	0.30	-	-	-	0.45	0.12	0.22	0.7	0.49	91.38	-	1.74	-	0.52	-	0.68	0.64	0.16	0.16	0.19	0.23	0.29	CDo140019-03-5243-016-02-05,11-14.pdf
CDo140019-03-5243-028	CDo140019-03-5243-028-02	ND	ND	0.15	0.29	0.24	0.57	0.70	0.89	2.85	1.29	90.17	0.93	0.82	-	-	0.73	-	-	-	-	-	0.10	-	CDo140019-03-5243-028-02.pdf
	CDo140019-03-5243-028-03	ND	ND	-	-	-	0.40	0.24	0.42	1.36	0.89	94.80	0.65	-	0.43	-	0.49	-	-	-	-	-	0.11	-	CDo140019-03-5243-028-03.pdf
	CDo140019-03-5243-028-04	ND	ND	-	-	-	0.41	-	0.13	0.45	0.36	96.24	1.22	-	-	0.15	0.47	-	0.24	0.12	-	-	0.12	-	CDo140019-03-5243-028-04.pdf
	CDo140019-03-5243-028-05	ND	ND	-	-	-	0.46	-	-	0.35	-	95.13	1.62	-	-	0.29	0.90	-	0.55	0.26	-	-	0.15	-	CDo140019-03-5243-028-05.pdf
	CDo140019-03-5243-028-06	ND	ND	-	-	-	0.49	-	-	0.31	-	94.15	1.83	-	-	0.34	0.96	-	0.68	0.24	0.14	0.11	0.19	0.10	CDo140019-03-5243-028-06.pdf
Mock pool (Combine fraction: 02~06)		ND	ND	/	/	/	0.40	0.12	0.20	0.74	0.47	95.21	1.01	/	0.40	0.26	0.45	/	0.31	0.15	/	/	0.14	/	CDo140019-03-3040-028-02-06.pdf

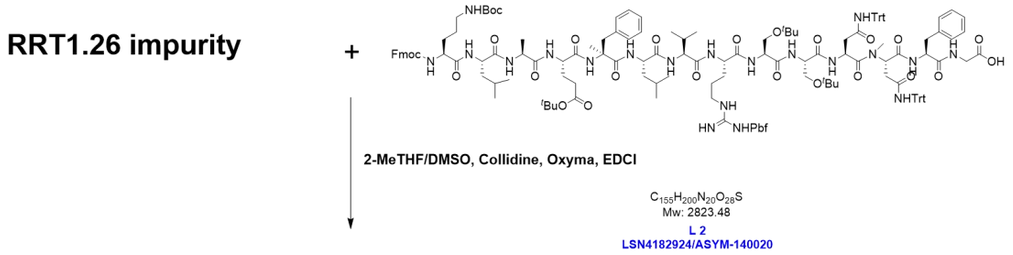
Conclusion:The spiked impurity can be completely removed after RP1 and RP2.

7.8 Expt.8: Spiking experiment study of RRT1.26 impurity

- **Downstream assessment with pure RRT1.26 impurity**

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Step1a coupling of RRT1.26 impurity + L2



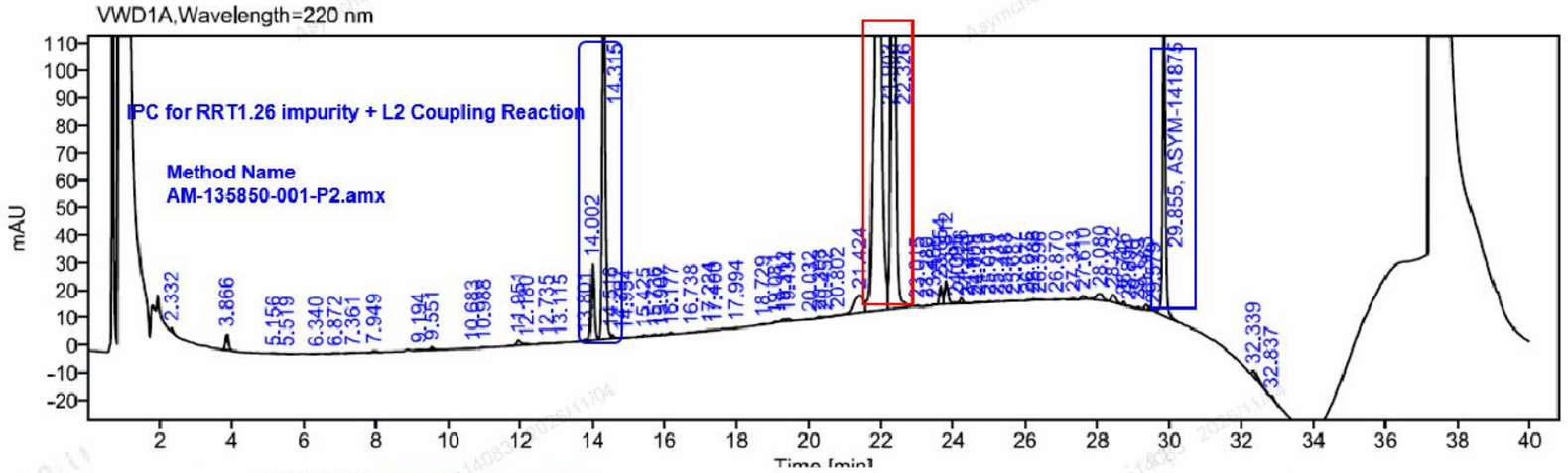
RRT1.26 impurity + L2

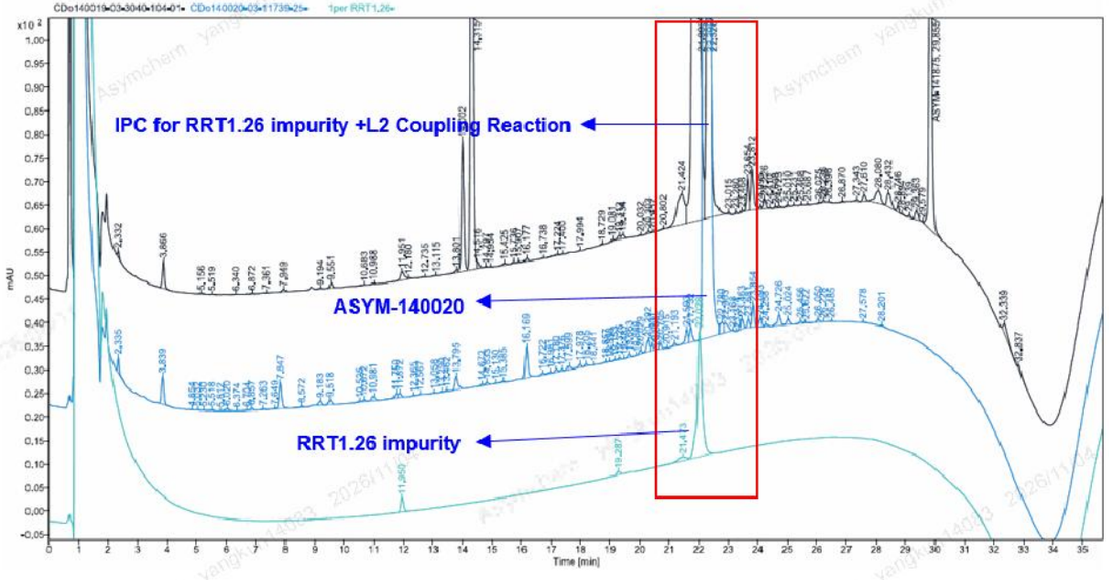
Table 7.8.1 Summary of spiking experiment

Coupling of RRT1.26 impurity + L2				
Impurity name	**Purity**	**Main impurity(ASYM-140019)**	**UPLC**	**Coupling of RRT1.26 impurity + L2: IPC**
RRT1.26 impurity	**80.30%**	**4.78%**	CDo140019-03-2699-038-01.pdf	CDo140019-03-3040-104-01.pdf

Step1a coupling of RRT1.26 impurity + L2 ： MS and IPC Analysis

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Table 7.8.2 Comparison of experimental group with production batch IPC data

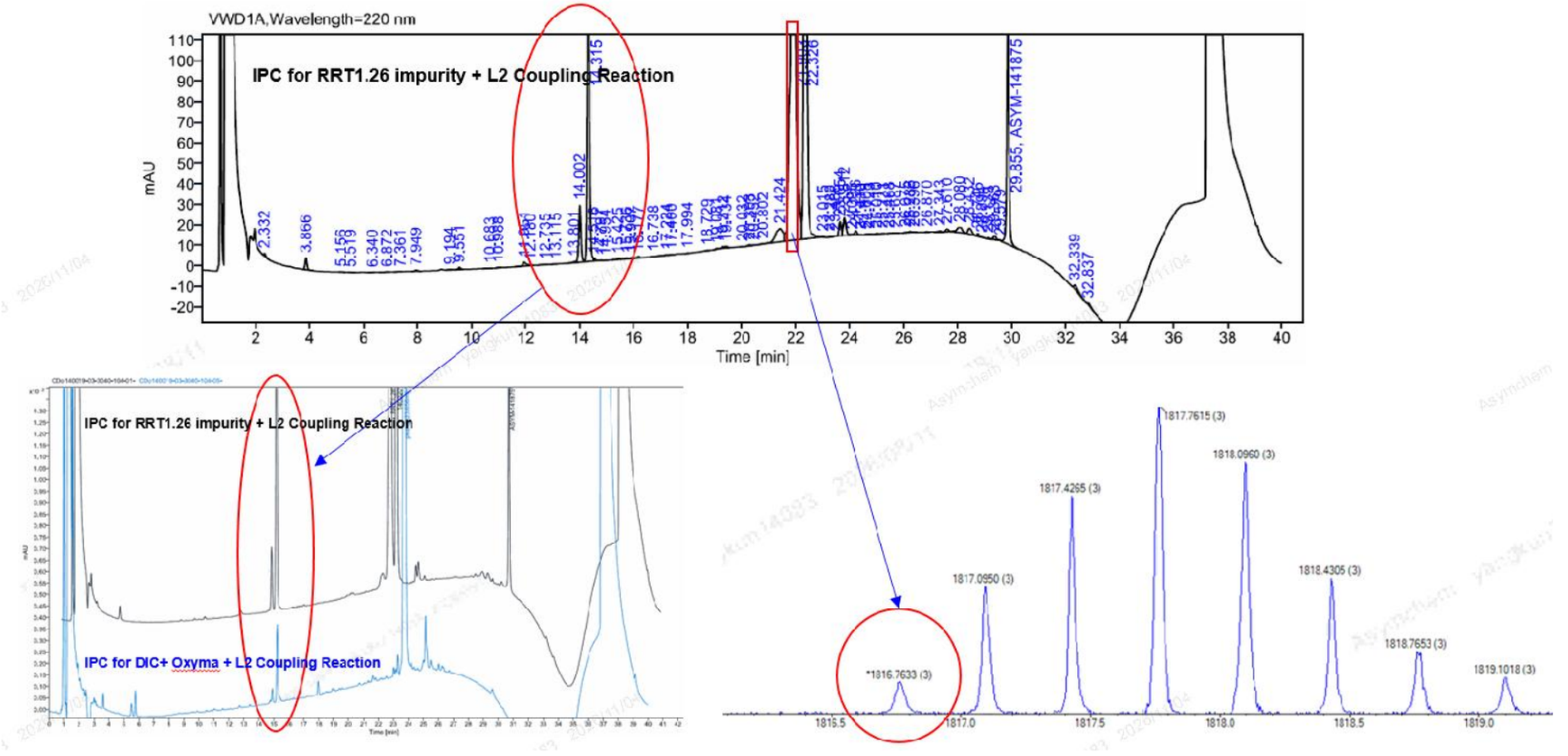
No.	1	2	3	4	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
RRT	0.138	0.478	0.487	0.512	0.741	0.753	0.80	0.813	0.902	0.947	0.953	0.967	0.972	0.980	0.987	0.994	141875	1.014	1.025	1.035	1.047
CPo142526-02-01-01-09-03（API-7）IPC	0.20	0.08	0.05	0.30	-	0.65	0.49	0.36	0.27	0.29	0.41	1.65	0.44	0.39	0.98	0.75	88.06	0.93	0.57	0.48	0.30
Experimental IPC	-	11.43	0.07	0.03	44.32 RRT 1.26 impurity	28.56 ASYM-140020	0.68	0.14	-	0.41	0.34	0.08	0.03	-	0.16	0.01	7.96	-	-	-	-

Conclusion: It can be demonstrated by IPC analysis that X does not participate in the downstream coupling reaction (Step 1 and Step 2). 7.96% of ASYM-141875 is generated by the reaction between L1 (contained in 2M+100.06 impurity) and L2. The peak at RRT 0.478 may correspond to an intermediate compound formed between ASYM-140020 and the coupling reagent.

Verification experiment: Only ASYM-140020, EDCI and Oxyma were charged into the reaction system to verify whether the peak at RRT 0.478 is the intermediate compound formed between ASYM-140020 and the coupling reagent.

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Conclusion: Comparison by UPLC shows that the impurity peak with an RRT of 0.478 was observed. It is therefore concluded that this impurity is an intermediate species formed from ASYM-140020 and the coupling reagents.

- **Step3: Global cleavage 2M+100.06 impurity marker**

Charge toluene (6 V) and anisole (0.25 V) into the reaction flask and maintain the temperature at 20 °C. Add DTT (0.25 eq), purified water (0.25 V) and TIPS (0.193 eq). While holding at 20 °C, charge 2M+100.06 impurity (1.00 eq). Cool the batch to 10 °C, then add TFA (9.24 V) dropwise. Upon completion of addition, warm the mixture to 20 °C and hold the reaction for 3 h.

Sample the batch to confirm full conversion of 2M+100.06 impurity Cool the reaction mixture to −15 °C, then add pre-chilled MTBE (48 V) dropwise. After addition is complete, stir the slurry for 1 h.

Filter the solids, slurry-wash the filter cake with MTBE (3 V) for 0.5 h, and filter again. Dry the isolated solid and collect the product. Test the purity and submit for MS analysis.

Table 7.8.3 Experiment 8: Synthesis of 2M+100.06 impurity marker

Step 3 reaction and solid product purity				
No.	Impurity name	Purity of product solid	Product solid Chromatogram	MS Chromatogram
Step 3	2M+100.06 impurity Raw material purity: 86.56% Feed amount: 300 mg	Purity : 81.38% Product: 180 mg	CDo140019-03-3040-104-13.pdf	CDo140019-03-3040-104-13.PNG

The **2M+100.06 impurity** underwent step3 reaction to afford 2M+100.06 impurity marker . 2M+100.06 impurity marker was spiked into crude API (ASYM 142526) at a level of 1.60 % Area.

Table 7.8.4 Summary of spiking experiment

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Spiking study.					
Impurity name	**Purity**	**UPLC**	**Suggested Spiking Amount**	**Real Spiking Amount**	**UPLC after spiking**
2M+100.06 impurity marker	**81.38%**	CDo140019-03-3040-104-13.pdf	**1.5%**	**1.60%**	CDo140019-03-3040-104-16.pdf

Then the Ac-endcapping-8mer impurity-spiked API sample (ASYM-142526) was adjusted to pH 8~9 with NaOH solution, stirred at 10 ± 5 °C for 2 h to transfer the depsipeptide, then re-adjusted the pH value to 2.8-3.2 with citric acid solution.

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7.8.1. Expt.8: UPLC Overlay Diagram of Experimental Group and Control Group

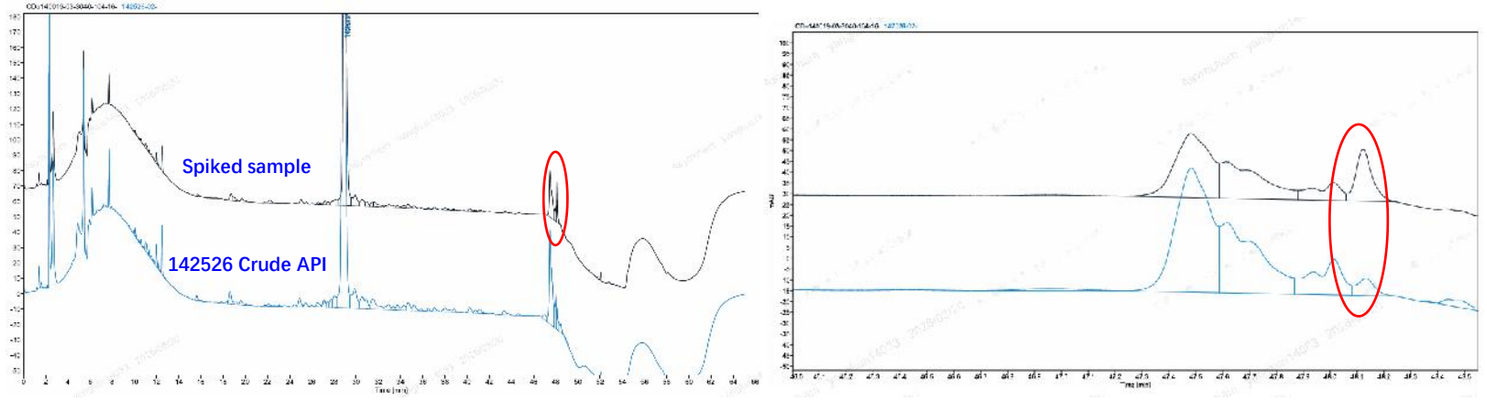


Table 7.8.5 Expt.8: LC/MS data comparison of step 3 crude API:

Items	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	27	28	29	30	31
RT	12.001	12.514	18.627	19.627	24.881	26.521	27.107	27.401	27.701	28.014	28.801	29.867	30.541	30.887	31.501	32.274	32.927	33.747	34.287	34.661	35.214	46.954	47.481	47.621	48.014	48.121
RRT	0.417	0.435	0.647	0.681	0.864	0.921	0.941	0.951	0.962	0.973	1.000	1.026	1.033	1.054	1.064	1.072	1.077	1.172	1.191	1.203	1.223	1.630	1.649	1.653	1.659	1.663
m/z	1285.634	789.423	1187.981		1528.449	-	-	919.077	-	1475.745	1509.761	1475.746	1471.415	1444.727	1504.758	-	1221.628	1456.086	-	-	1547.46	-	1592.791	-	-	-
charge	2	2	3		3	-	-	5	-	3	3	3	3	3	3	-	5	3	-	-	3	-	3	-	-	-
Monoisotopic Mass mass	2568.252	1578.845	3559.919	-	4579.343	-	-	4595.385	-	4422.209	4523.254	4422.210	4409.239	4329.153	4509.248	-	6103.969	4363.229	-	-	4636.356	-	4775.374	-	-	-

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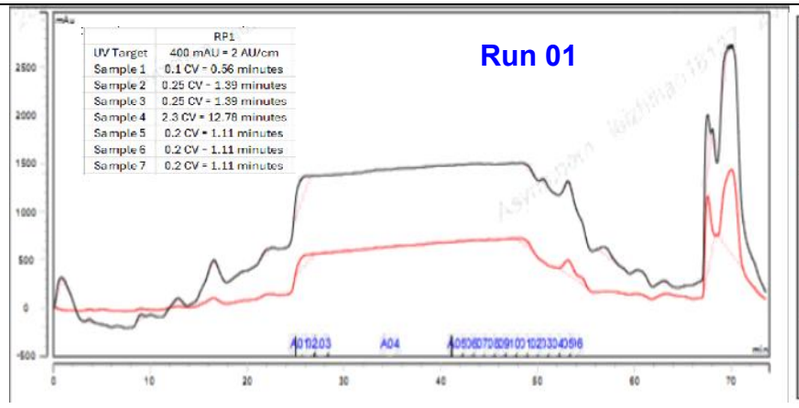
Delta mass	-1955.00	-2944.409	-963.340	-	56.090	-	-	72.130	-	-101.040	0	-101.040	-114.020	-194.100	-14.010	-	1580.715	-160.030	-	-	113.100	-	252.080	-	-	-
Structure	Glu1-Phe23]+Pip	unknown	Orn^{11}-Tyr^{37}	unknown	+ tBu	unknown	unknown	unknown	unknown	Des Thr9	142526	Des Thr30	Des Asn21	DesPro^{28}Pro^{29}	-CH_2	unknown	L3+L2+L2+L1	unknown	unknown	unknown	+Leu27	unknown	+ Pbf	unknown	unknown	unknown
142526 Crude API CPo142526-02-03-01	0.40	0.80	0.74	0.41	0.55	0.32	0.34	0.25	0.51	0.89	75.26	2.48	0.96	0.70	1.09	0.37	0.55	0.37	0.33	0.64	0.46	0.09	2.88	1.84	0.89	0.14
Spiked sample CDo140019-03-3040-104-16	0.33	0.78	0.65	0.48	0.64	0.30	0.32	0.33	1.31		73.52	2.45	1.04	0.68	1.05	0.37	0.78	0.36	0.30	0.59	0.37	3.09		2.25	0.89	1.48

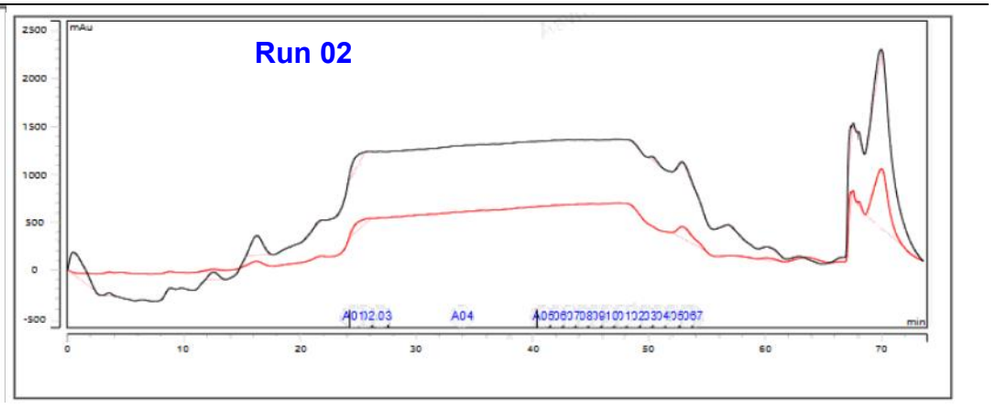
7.8.6. Expt.8: The purification conditions and collection strategy of RP1

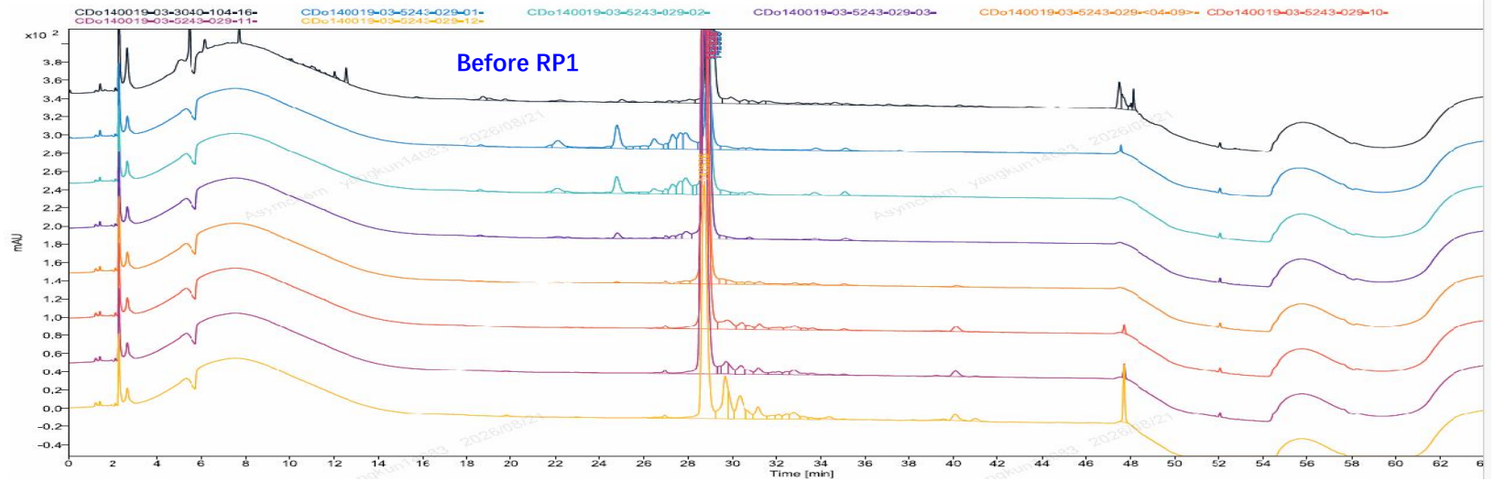
Column	10*250mm bed column=19.63 mL						
Sample	CDo140019-03-3040-104-16 (Purity: 73.88%)						
Resin	Daisogel 120-10-ODS-RPS						
Mobile phase A	90% 50 mM Citric Acid, **pH 3.00**: 10% ACN				Mobile phase B	Acetonitrile	
Detection wavelength	230nm				Monitor wavelength	280nm	
Flowrate	3.53mL/min(270cm/h)						
Loading amount	**45.0mg/mL**						
Post-Load Wash	22.2~27.8(RP1)B% for 11.1min(2CV) at 3.53mL/min						
Elution	T(time)	0	62.5	62.5	73.6		
	B(%)	27.8	44.4	66.7	66.7		
UV	Start collection when UV reaches 350mAu						

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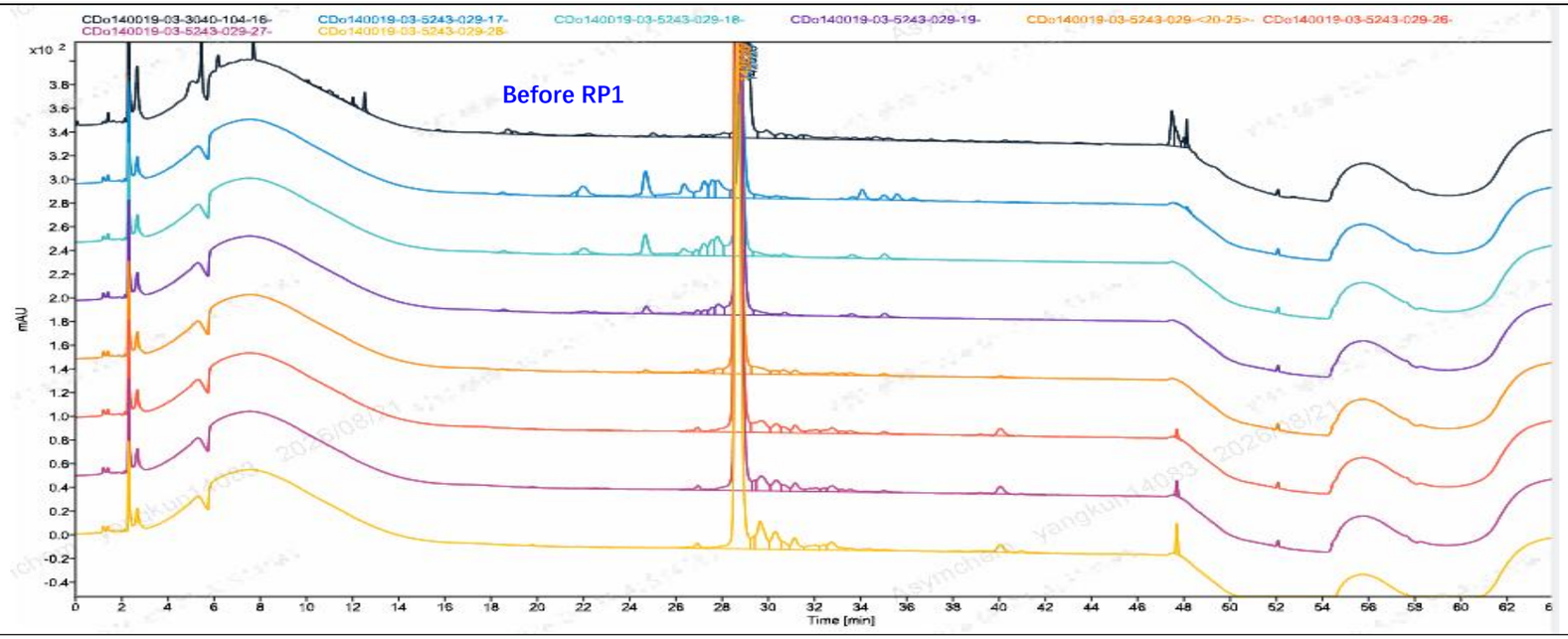
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7.8.7. Expt.8: Purity and impurity profile of RP1 fractions

Batch	Fraction Number	Impurity profile (RRT, %area)																						Spectrum Attachments
RRT		0.681	0.935	0.941	0.951	0.962	0.973	1.000	1.026	1.033	1.054	1.064	1.072	1.077	1.121	1.143	1.172	1.191	1.203	1.223	1.398	1.430	1.663	-
m/z		-	-	-	919.0769	-	1475.745	1509.761	1475.746	1471.415	1444.727	1504.758	-	1221.628	1095.095	1503.754	-	-	-	1547.46	-	-	-	
charge		-	-	-	5	-	3	3	3	3	3	3	-	5	2	3	-	-	-	3	-	-	-	
Monoisotopic Mass mass		-	-	-	4595.385	-	4422.210	4523.254	4422.210	4409.239	4329.153	4509.248	-	6103.969	2187.173	4506.237	-	-	-	4636.356	-	-	-	
Delta mass		-	-	-	72.130	-	-101.040	0	-101.040	-114.020	-194.100	-14.010	-	1580.715	-2336.080	-17.020	-	-	-	113.100	-	-	-	
Structure		unknown	unknown	unknown	unknown	unknown	Des Thr9	142526	Des Thr30	Des Asn21	DesPro28Pro29	-CH2	unknown	L3+L2+L2+L1	Phe23-Tyr37	Asn22 cyclization	unknown	unknown	unknown	+Leu27	unknown	unknown	unknown	
Spiked sample		0.48	-	0.32	0.33	1.31		73.52	2.45	1.04	0.68	1.05	0.37	0.78	0.37	0.78	000.36	0.30	0.59	0.37	0.36	0.26	1.48	CDo140019-03-3040-104-16.pdf
Loading: 45 mg/mL (CDo140019-03-5243-029)	CDo140019-03-5243-029-01	0.07	1.52	5.50	5.43	9.42		42.87	2.49		-	0.88	0.93	0.66	0.71	1.48	1.18	-	-	-	-	-	-	CDo140019-03-5243-029-01.pdf
	CDo140019-03-5243-029-02	0.03	0.95	2.15	2.57	4.35		72.04	1.31		-	0.60	0.50	0.58	0.16	-	1.07	-	-	1.11	-	-	-	CDo140019-03-5243-029-02.pdf
	CDo140019-03-5243-029-03	-	0.46	-	0.57	0.78	2.01	89.30	0.84	0.77	-	0.27	0.63	0.30	-	-	0.59	-	-	0.68	-	-	-	CDo140019-03-5243-029-03.pdf
	CDo140019-03-5243-029-04-09	-	-	0.41	0.17	0.31	0.81	90.90	1.02	1.05	-	0.65	0.72	0.74	0.42	0.56	0.50	-	-	0.34	0.27	-	0.03	CDo140019-03-5243-029-04-09.pdf
	CDo140019-03-5243-029-10	-	-	0.65	-	-	-	87.09	3.58		-	1.54	0.84	1.20	0.78	1.29	0.56	-	-	0.28	1.05	0.06	0.46	CDo140019-03-5243-029-10.pdf
	CDo140019-03-5243-029-11	-	-	0.71	-	-	-	84.89	0.83	2.12	1.30	2.03	0.86	1.39	0.90	1.29	0.37	-	0.21	0.20	1.11	0.15	0.75	CDo140019-03-5243-029-11.pdf
	CDo140019-03-5243-029-12	0.30	0.38	0.55	0.11	-	-	56.81	10.17		3.77	7.20	2.04	3.51	1.00	2.17	0.46	0.93	-	0.11	1.79	0.75	4.26	CDo140019-03-5243-029-12.pdf

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7.8.8 Expt.8: Objective:RP1 Run 02 data

Batch	FractionNumber	Impurity profile (RRT, %area)																									Spectrum Attachments
RRT		0.647	0.681	0.749	0.765	0.864	0.900	0.921	0.935	0.941	0.951	0.962	0.973	1.000	1.024	1.037	1.060	1.072	1.094	1.121	1.129	1.143	1.172	1.191	1.203	1.223	1.398	1.663	-
Spiked sample		0.65	0.48	-	-	0.64	-	0.30	-	0.32	0.33	1.31		73.52	-	2.45	1.04	0.68	1.05	0.37	-	0.78	0.36	0.30	0.59	0.37	0.36	1.48	
Loading: 45 mg/mL (CDo140019-03-5243-029)	CDo140019-03-5243-029-17	0.55	-	1.04	6.08	8.12	-	8.33	-	6.32	4.74	7.99		44.40	1.87		1.08	0.64	0.23	-	-	0.19	0.80	2.86	1.47	1.69	-	0.48	CDo140019-03-5243-029-17.pdf

带格式表格[Handong Cheng15615]

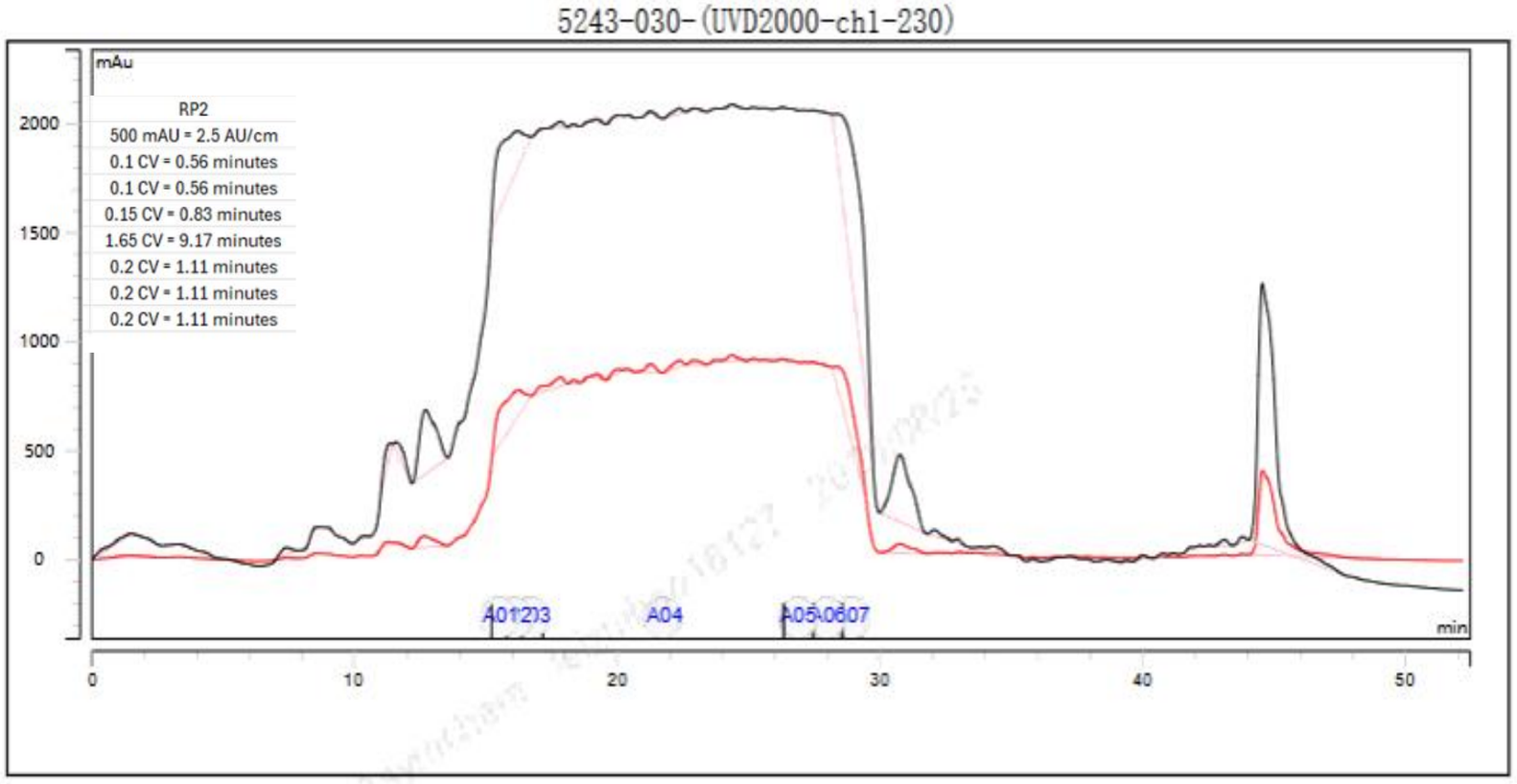
	CDo140019-03-5243-029-18	0.36	-	0.30	1.73	4.00	0.92	1.87	0.92	2.20	2.67	4.21		75.95	1.42		0.26	0.48	-	-	-	0.17	0.88	-	-	1.03	-	-	CDo140019-03-5243-029-18.pdf
	CDo140019-03-5243-029-19	0.24	-	-	0.58	1.23	0.33	0.52	0.54	-	0.65	1.03	2.10	88.86	1.36		0.24	0.52	-	-	-	-	0.53	-	-	0.66	-	-	CDo140019-03-5243-029-19.pdf
	CDo140019-03-5243-029-20-25	0.07	-	-	0.05	0.43	-	0.24	-	0.49	-	0.63	0.87	90.52	2.28		0.63	0.73	0.74	0.52	-	0.44	0.50	-	-	0.38	0.23	-	CDo140019-03-5243-029-20-25.pdf
	CDo140019-03-5243-029-26	-	-	-	-	-	-	-	-	0.64	-	-	-	87.72	3.40		1.43	0.81	1.15	0.75	-	1.21	0.55	-	-	0.30	1.02	0.34	CDo140019-03-5243-029-26.pdf
	CDo140019-03-5243-029-27	-	-	-	-	-	-	-	-	0.67	-	-	-	86.34	0.78	2.97	1.74	0.84	1.24	0.81	0.42	0.94	0.71	-	0.17	0.27	1.07	0.56	CDo140019-03-5243-029-27.pdf
	CDo140019-03-5243-029-28	-	0.12	-	-	-	-	-	-	0.87	-	-	-	77.76	1.09	5.63	3.39	1.09	2.01	1.24	0.69	1.80	0.40	-	0.39	0.15	1.26	1.33	CDo140019-03-5243-029-28.pdf
Mock pool (Combine fraction: 03-11，19-27)										0.44	0.25	0.35	1.06	91.23	1.96		0.59	0.59	0.61			0.69	0.39	-	-	0.33	0.29	0.08	CDo140019-03-5243-029-03-11，19-27.pdf

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删除[Handong Cheng15615]: Combine fractions 03-11 and 19-27 for RP2. Purity and MS of the combined sample are under testing.

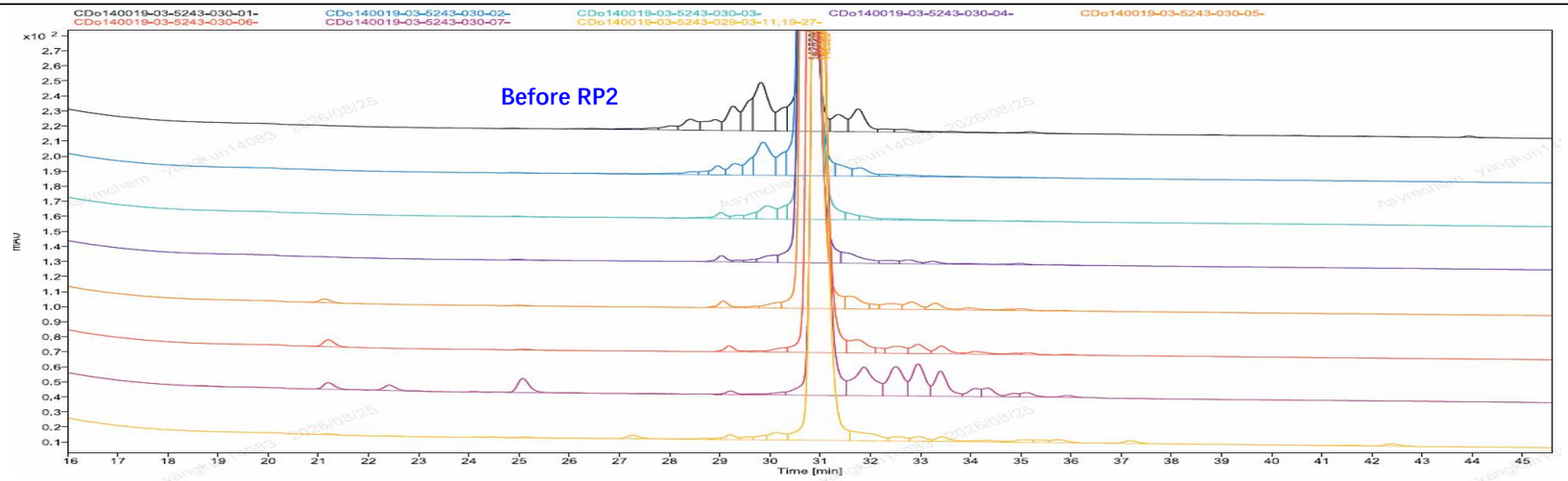
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7.8.9 Expt.8: Purity and impurity profile of RP2 fractions

Batch	Fraction Number	Impurity profile (RRT, %area)																Spectrum Attachments
RRT		0.921	0.935	0.941	0.951	0.962	0.973	0.984	1.000	1.024	1.037	1.050	1.060	1.072	1.094	1.121	1.129	1.143	-
Mock pool (Combine fraction: 03-11，19-27)		-	-	0.44	0.25	0.35	1.06	-	91.23	1.96		-	0.59	0.59	0.61	-	-	0.69	CDo140019-03-5243-029-03-11，19-27.pdf
Loading: 45 mg/mL (CDo140019-03-5243-030)	CDo140019-03-5243-030-01	1.99	-	2.25	3.82	3.55	8.86	2.78	65.28	2.98	4.18	0.51	0.63	0.21	0.25	-	-	0.28	CDo140019-03-5243-030-01.pdf
	CDo140019-03-5243-030-02	0.31	0.22	0.76	1.08	1.06	3.86	1.51	87.84	1.12	1.02	0.26	0.22	0.13	-	0.14	-	-	CDo140019-03-5243-030-02.pdf

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	CDo140019-03-5243-030-03	-	-	0.46	0.30	0.41	1.55	0.84	94.70	0.54	0.39	0.19	0.15	0.11	-	-	-	-	CDo140019-03-5243-030-03.pdf
	CDo140019-03-5243-030-04	-	-	0.40	0.10	0.16	0.67	-	96.06	1.33		0.32	0.35	0.24	0.10	-	-	-	CDo140019-03-5243-030-04.pdf
	CDo140019-03-5243-030-05	-	-	0.39	0.05	0.11	0.48	-	94.70	1.27	0.26	0.63	0.67	0.56	0.24	-	-	0.16	CDo140019-03-5243-030-05.pdf
	CDo140019-03-5243-030-06	-	-	0.37	0.05	0.08	0.41	-	93.50	1.10	0.27	0.82	0.85	0.69	0.33		0.10	0.16	CDo140019-03-5243-030-06.pdf
	CDo140019-03-5243-030-07	-	-	0.30	0.09	0.06	0.29	-	77.31	5.33		4.01	3.80	2.95	0.87	0.91	0.34	0.52	CDo140019-03-5243-030-07.pdf
Mock pool (Combine fraction: CDo140019-03-5243-030-03-06)		-	-	0.41	0.09	0.17	0.65	-	95.50	1.44		0.42	0.45	0.32	0.14	-	0.05	0.13	CDo140019-03-5243-030-03-06.pdf

Conclusion: The spiked impurity can be completely removed after RP1 and RP2.

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8. Revision Histor

Document No. and version	Document Changes	Effective Date
TR-1400190305.01	The first version	

9. Appendix I Abbreviation and Code Names

Abbreviation/Code name	Description
PAR	Proven Acceptable Range
IPC	In Process Control
NMT	No More Than
NLT	No Less Than
RRT	Relative Retention Time
RT	Retention Time
ND	No detected
N/A	Not applicable
Chengdu ZY	Critical Material Attributes
Hecheng	Tianjin Nankai Hecheng S&TCo.,Ltd;
Sunresin	Sunresin New Materials Co., Ltd, Xi'an;
Chengdu ZY	Chengdu ZY Biochemical Technoliogy Co.,Ltd;
Asymchem（Jilin）	Asymchem Laboratories（Jilin）Co.,Ltd;
Asymchem（Liaoning）	Asymchem Laboratories（Liaoning）Co.,Ltd;
Asymchem（TJ3）	Tianjin Asymchem Pharmaceuticals Co., Ltd.
Hecheng	Tianjin Nankai Hecheng S&TCo.,Ltd;

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10. Appendix II LSN4182913 /ASYM-140019 specification

TEST ITEMS		ANALYTICAL METHOD	ACCEPTANCE CRITERIA	
Characters	Appearance	AM-140019-002	White or practically white solid	
Identification	Identification by UHPLC	AM-140019-001	Retention time conforms to reference	
	Identification by MS	AM-013	Consistent with the exact mass	
Test	Purity and related substances by UHPLC	AM-140019-001	Purity	≥90.0%
			Total Impurities	≤10.0%
			ASYM-146983 (Des-tBu of Tyr)	≤2.0%
			Any unspecified impurity	≤1.0%
			Individual impurities ≥0.10%	Report by RRT
	Assay by UHPLC	AM-140019-001	≥80.0 and ≤110.0 (%w/w)	
	Water content by KF	AM-140019-003	≤3.0%	
	Trichloroacetic acid content by HPLC	AM-140019-004	≤2.0%	
	Residual 2,6-Lutidine by HS-GC	AM-140019-018	Report	
	Residual Piperidine by HS-GC	AM-140019-006	Report	
	Residual organic Solvents by HS-GC	AM-140019-005	Isopropanol	Report

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			t-Butanol	≤5000ppm
			Methyl tert-Butyl Ether	Report
			Isopropyl acetate	≤5000ppm
			n-Heptane	Report
			Toluene	Report
			N,N-Dimethylformamide	≤880ppm
			Acetonitrile	Report

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11. Appendix III Analytical method

Column	Waters Acquity UPLC Peptide CSH C18, 130 A, 2.1×150 mm, 1.7 µm, P/N: 186006938						
UV length	UV = 220 nm						
Mobile phase	MPA: 0.05% TFA in water (v/v)						
	MPB: 0.05% TFA in ACN (v/v)						
Gradient	Time (min)	0.0	1.0	40.0	43.0	43.1	48.5
	B (%)	40	40	99	99	40	40
Flow Rate	0.45 mL/min						

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Sampler Temperature	5 °C
Column Temperature	65 °C
Injection Volume	2.0 µL
Run Time	48.5 min
Needle Wash & Diluent	60 DMSO : 40 ACN (v/v)

13. Appendix IV Lot genealogy summary for peptidyl-on-resin and cleaved by TCA

Table 13.1 Lot genealogy summary for peptidyl-on-resin and cleaved by TCA

SPPS PAR Experiment ID	SPPS Lot # (Asymchem NB no)	Cleavage Lot #	L1 Final UV Analytical Lot #	L1 Final MS Analytical Lot #	Step 1 exp Lot	Step 1 analytical lot (LOD)	Step 2 exp Lot	Step 2 analytical lot
Run1	NBK011577-1	NBK013104-8	CDo140019-03-3104-008-15	CDo140019-03-3104-008-15	CDo140019-03-1577-001-01	CDo140019-03-1577-001-01	CDo140019-03-3104-008-15	CDo140019-03-3104-008-15
Run2	NBK011577-2	NBK013104-9	CDo140019-03-3104-009-15	CDo140019-03-3104-009-15	CDo140019-03-1577-002-01	CDo140019-03-1577-002-01	CDo140019-03-3104-009-15	CDo140019-03-3104-009-15
Run2-drying	NBK011577-2& NBK011577-17	NBK011577-17	CDo140019-03-1577-017-02	CDo140019-03-1577-017-02	CDo140019-03-1577-002-01	CDo140019-03-1577-002-01	CDo140019-03-1577-017-02	CDo140019-03-1577-017-02
Run3	NBK011577-3	NBK013104-10	CDo140019-03-3104-010-15	CDo140019-03-3104-010-15	CDo140019-03-1577-003-01	CDo140019-03-1577-003-01	CDo140019-03-3104-010-15	CDo140019-03-3104-010-15
Run4	NBK011577-4	NBK013104-11	CDo140019-03-3104-011-15	CDo140019-03-3104-011-15	CDo140019-03-1577-004-01	CDo140019-03-1577-004-01	CDo140019-03-3104-011-15	CDo140019-03-3104-011-15
Run5	NBK011577-5	NBK013104-12	CDo140019-03-3104-012-15	CDo140019-03-3104-012-15	CDo140019-03-1577-005-01	CDo140019-03-1577-005-01	CDo140019-03-3104-012-15	CDo140019-03-3104-012-15
Run6	NBK011577-6	NBK013104-1	CDo140019-03-3104-001-15	CDo140019-03-3104-001-15	CDo140019-03-1577-006-01	CDo140019-03-1577-006-01	CDo140019-03-3104-001-15	CDo140019-03-3104-001-15

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SPPS PAR Experiment ID	SPPS Lot # (Asymchem NB no)	Cleavage Lot #	L1 Final UV Analytical Lot #	L1 Final MS Analytical Lot #	Step 1 exp Lot	Step 1 analytical lot (LOD)	Step 2 exp Lot	Step 2 analytical lot
Run7	NBK011577-7	NBK013104-2	CDo140019-03-3104-002-15	CDo140019-03-3104-002-15	CDo140019-03-1577-007-01	CDo140019-03-1577-007-01	CDo140019-03-3104-002-15	CDo140019-03-3104-002-15
Run8	NBK011577-8	NBK013104-3	CDo140019-03-3104-003-15	CDo140019-03-3104-003-15	CDo140019-03-1577-008-01	CDo140019-03-1577-008-01	CDo140019-03-3104-003-15	CDo140019-03-3104-003-15
Run9	NBK011577-9	NBK013104-4	CDo140019-03-3104-004-15	CDo140019-03-3104-004-15	CDo140019-03-1577-009-01	CDo140019-03-1577-009-01	CDo140019-03-3104-004-15	CDo140019-03-3104-004-15
Run10	NBK011577-10	NBK013104-5	CDo140019-03-3104-005-15	CDo140019-03-3104-005-15	CDo140019-03-1577-010-01	CDo140019-03-1577-010-01	CDo140019-03-3104-005-15	CDo140019-03-3104-005-15
Run11	NBK011577-11	NBK013104-6	CDo140019-03-3104-006-15	CDo140019-03-3104-006-15	CDo140019-03-1577-010-01	CDo140019-03-1577-010-01	CDo140019-03-3104-006-15	CDo140019-03-3104-006-15
Run12	NBK011577-12	NBK013104-7	CDo140019-03-3104-007-15	CDo140019-03-3104-007-15	CDo140019-03-1577-011-01	CDo140019-03-1577-011-01	CDo140019-03-3104-007-15	CDo140019-03-3104-007-15
Run13	NBK011577-13	NBK013104-16	CDo140019-03-3104-016-15	CDo140019-03-3104-016-15	CDo140019-03-1577-012-01	CDo140019-03-1577-012-01	CDo140019-03-3104-016-15	CDo140019-03-3104-016-15
Run14	NBK011577-14	NBK013104-17	CDo140019-03-3104-017-15	CDo140019-03-3104-017-15	CDo140019-03-1577-013-01	CDo140019-03-1577-013-01	CDo140019-03-3104-017-15	CDo140019-03-3104-017-15
Run15	NBK011577-15	NBK013104-18	CDo140019-03-3104-018-15	CDo140019-03-3104-018-15	CDo140019-03-1577-014-01	CDo140019-03-1577-014-01	CDo140019-03-3104-018-15	CDo140019-03-3104-018-15
Run16	NBK011577-16	NBK013104-19	CDo140019-03-3104-019-15	CDo140019-03-3104-019-15	CDo140019-03-1577-015-01	CDo140019-03-1577-015-01	CDo140019-03-3104-019-15	CDo140019-03-3104-019-15

Note [5]: the final UV analysis chromatogram had been updated in a single experimental analysis in the above section 4 Table 4.0.1.

Table 13.2 Lot genealogy summary for Impurity Spiking of L1 preparation

Fate and Purge Experiment ID.	Step 1 (SPPS) (Asymchem NB no)	Step 2 Cleavage Lot #(Asymchem NB no)	Final UV Analytical Lot #	Final MS Analytical Lot #	Step 1 exp Lot	Step 1 analytical lot(LOD)	Step 2 analytical lot
RRT 0.910; Tyr37 (Des-tBu) of L1 preparation	RRT 0.910; Tyr37 (Des-tBu) of L1 preparation	NBK012423-024	NBK013040-012	CDo140019-03-3040-012-01  CDo140019-03-3040-012-01	CDo140019-03-3040-012-01  CDo140019-03-3040-012-01	CDo140019-03-2423-024-01	CDo140019-03-3040-012-01

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RRT 0.960; Ser34 (Des-tBu) of L1 preparation	RRT 0.960; Ser34 (Des-tBu) of L1 preparation	NBK012423-005	NBK013040-002	CDo140019-03-3040-002-12 CDo140019-03-3040-002-12	CDo140019-03-3040-002-12 CDo140019-03-3040-002-12	CDo140019-03-2423-005-01	CDo140019-03-3040-002-12
RRT 0.947; Thr36 (Des-tBu) of L1 preparation	RRT 0.947; Thr36 (Des-tBu) of L1 preparation	NBK012423-006	NBK013040-013	CDo140019-03-3040-013-01 CDo140019-03-3040-013-01	CDo140019-03-3040-013-02 CDo140019-03-3040-013-02	CDo140019-03-2423-006-01	CDo140019-03-3040-013-01
RRT 0.954; Thr30 (Des-tBu) of L1 preparation	RRT 0.954; Thr30 (Des-tBu) of L1 preparation	NBK012423-001	NBK013040-001	CDo140019-03-3040-001-01 CDo140019-03-3040-001-01	CDo140019-03-3040-001-01 CDo140019-03-3040-001-01	CDo140019-03-2423-001-01	CDo140019-03-3040-001-01
RRT 0.940; Glu31 (Des-tBu) of L1 preparation	RRT 0.940; Glu31 (Des-tBu) of L1 preparation	NBK012423-025	NBK013040-011	CDo140019-03-3040-011-01 CDo140019-03-3040-011-01	CDo140019-03-3040-011-02 CDo140019-03-3040-011-02	CDo140019-03-2423-025-01	CDo140019-03-3040-011-01
RRT 0.860; SC26 (Des-tBu) of L1 preparation	RRT 0.860; SC26 (Des-tBu) of L1 preparation	NBK012423-040	NBK013040-028	CDo140019-03-3040-028-01 CDo140019-03-3040-028-01	CDo140019-03-3040-028-01 CDo140019-03-3040-028-01	CDo140019-03-2423-040-01	CDo140019-03-3040-028-01
RRT 0.722; Ac-endcapping-8mer of L1 preparation	RRT 0.722; Ac-endcapping-8mer of L1 preparation	NBK012423-019	NBK013040-010	CDo140019-03-3040-010-01 CDo140019-03-3040-010-01	CDo140019-03-3040-010-01 CDo140019-03-3040-010-01	CDo140019-03-2423-019-01	CDo140019-03-3040-010-01
RTT 1.26; 2M+100 of L1 preparation	RTT 1.26; 2M+100 of L1 preparation	N/A	NBK013040-104	CDo140019-03-3040-104-13 CDo140019-03-3040-104-13 (1)	CDo140019-03-3040-104-13 CDo140019-03-3040-104-13	CDo140019-03-5243	CDo140019-03-3040-104-13

Note [6]: The final UV analysis chromatogram had been updated in a single experimental analysis in the above section 8

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Table 13.3 Lot genealogy summary for impurity spiking and purification

Table 13.4 Lot genealogy summary for stability data collection of Fragment 1's peptidyl-on-resin

Experiment ID.	SPPS Lot # (Asymchem NB no)	Cleavage Lot #	L2 Final UV Analytical Lot #	Step 1 exp Lot	Step 2 exp Lot	Step 2 analytical lot	Chromatogram
0 (month by R&D)	CDo140019-03-1577-002-01	NBK013104-9	CDo140019-03-3104-009-15	CDo140019-03-1577-002-01	CDo140019-03-3104-009-15	CDo140019-03-3104-009-15	PAR2
6 (month by R&D)	CDo140019-03-1577-002-01	NBK011870-148	CDo140019-03-1870-148-04	CDo140019-03-1577-002-01	CDo140019-03-1870-148-04	CDo140019-03-1870-148-04	CDo140019-03-1870-148-04.pdf

14. Appendix V Experimental Material information

Table 14.1 Summary of all Material Lots for Step 1 PAR Study

Materials	Runs	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Sieber Amide Resin	lot No.	L125060454	L125060454	L125060454	L125060454	L125060454	L125060454	L225120950	L225120836	L125060454	L125060454	L125060454	L125060454	L125060454	L125060454	L125060454	L125060454
	Supplier	Hecheng	Hecheng	Hecheng	Hecheng	Hecheng	Hecheng	Hecheng	Sunresin	Hecheng	Hecheng	Hecheng	Hecheng	Hecheng	Hecheng	Hecheng	Hecheng
Fmoc-Tyr(tBu)-	lot No.	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-002	FJL124070130-	FJL124070130-	FJL124070130-	FJL124070130-	FJL124070130-

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Materials	Runs	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
OH													002	002	002	002	002
	supplier	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY
Fmoc-L-Thr(tBu)-OH	lot No.	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L225111436*	L225111436	L225111436	CPr138389-05-04-01	CPr138389-05-04-01
	supplier	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)
Fmoc-L-Asn(Trt)-OH	lot No.	L125070301	L125070301	L125070301	L125070301	L125070301	L125070301	L125070301	L125070301	L125070301	L125070301	CPS138519-03-02P-02	L125070301	L225120049	L225120049	CPr138519-03-02p-02	CPr138519-03-02p-02
	supplier	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Asymchem (Liaoning)	Chengdu ZY	Chengdu ZY	Chengdu ZY	Asymchem (Liaoning)	Asymchem (Liaoning)
Fmoc-L-Ser(tBu)-OH	lot No.	L125040586	L125040586	L125040586	L125040586	L125040586	L125040586	L125040586	L125040586	L125040586	L125040586	L225120045	L125040586	L125040586	L125040586	L125040586	L125040586
	supplier	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Chengdu ZY	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)
Fmoc-Gly-OH	lot No.	L225120048	L225120048	L225120048	L225120048	L225120048	L225120048	L225120048	L225120048	L225120048	L225120048	L225111434	L225120048	L225111702	L225111702	CPr138200-04-01-01	CPr138200-04-01-01
	Supplier	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Asymchem	Asymchem	Asymchem	Asymchem

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Materials	Runs	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
														(Liaoning)	(Liaoning)	(Liaoning)	(Liaoning)
Fmoc-L-Val-OH	lot No.	CPS138202-02-01-01	CPS138202-02-01-01	CPS138202-02-01-01	CPS138202-02-01-01	CPS138202-02-01-01	CPS138202-02-01-01	CPS138202-02-01-01	CPS138202-02-01-01	CPS138202-02-01-01	CPS138202-02-01-01	L225120043	CPS138202-02-01-01	L126010746	L126010746	CPr138202-04-01-01	CPr138202-04-01-01
	Supplier	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Chengdu ZY	Asymchem (Liaoning)	Chengdu ZY	Chengdu ZY	Asymchem (Liaoning)	Asymchem (Liaoning)
Fmoc-L-Glu(OtBu)-OH.H_2O	lot No.	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053	L125070053
	Supplier	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)
Fmoc-L-Thr(tBu)-OH	lot No.	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L125120052	L225111436	L225111436	CPr138389-05-04-01	L124090070	L125120052
	Supplier	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Asymchem (Liaoning)	Asymchem (Liaoning)	Asymchem (Liaoning)	Sichuan Tongsheng Amino Acid Co., Ltd.
Fmoc-L-Leu-L-Pro-L-Pro-OH	lot No.	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534	L125090534
	Supplier	Asymchem (Jilin)	Asymchem (Jilin)	Asymchem (Jilin)	Asymchem (Jilin)	Asymchem (Jilin)	Asymchem (Jilin)	Asymchem (Jilin)	Asymchem (Jilin)	Asymchem (Jilin)	Asymchem (Jilin)	Asymchem (Jilin)	Asymchem (Jilin)	Asymchem (Jilin)	Asymchem (Jilin)	Asymchem (Jilin)	Asymchem (Jilin)

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Materials	Runs	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
ASYM-140400	lot No.	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02	CPo140400-01-03-02
	Supplier	Asymchem (TJ3)	Asymchem (TJ3)	Asymchem (TJ3)	Asymchem (TJ3)	Asymchem (TJ3)	Asymchem (TJ3)	Asymchem (TJ3)	Asymchem (TJ3)	Asymchem (TJ3)	Asymchem (TJ3)	Asymchem (TJ3)	Asymchem (TJ3)	Asymchem (TJ3)	Asymchem (TJ3)	Asymchem (TJ3)	Asymchem (TJ3)
Fmoc-Pro-OH	lot No.	L125070093	L225120917	L225120917	L125070093	L225111090	L225120917	L225120917	L225120917	L225120917	L225120917	L225121595	L225121595	L225121595	L225121595	CPr140730-01-01-03	CPr140730-01-01-03
	Supplier	Zenji Pharmaceuticals(SUZHOU) Ltd	ChengDu Chron Chemicals Co.,Ltd	ChengDu Chron Chemicals Co.,Ltd	Zenji Pharmaceuticals(SUZHOU) Ltd	Zenji Pharmaceuticals(SUZHOU) Ltd	ChengDu Chron Chemicals Co.,Ltd	ChengDu Chron Chemicals Co.,Ltd	ChengDu Chron Chemicals Co.,Ltd	ChengDu Chron Chemicals Co.,Ltd	ChengDu Chron Chemicals Co.,Ltd	Chengdu ZY	Chengdu ZY	Chengdu ZY	Chengdu ZY	Asymchem (Liaoning)	Asymchem (Liaoning)
Ac_2O	lot No.	L125080638	L125080638	L125080638	L125080638	L125080638	L125080638	L125080638	L125080638	L125080638	L125080638	L125080638	L125080417	L125080417	L125080417	L125080417	L125080417
	Supplier	TIANJIN YIFANG SCIENCE & TECHNOLOGY, CO., LTD.	TIANJIN YIFANG SCIENCE & TECHNOLOGY, CO., LTD.	TIANJIN YIFANG SCIENCE & TECHNOLOGY, CO., LTD.	TIANJIN YIFANG SCIENCE & TECHNOLOGY, CO., LTD.	TIANJIN YIFANG SCIENCE & TECHNOLOGY, CO., LTD.	TIANJIN YIFANG SCIENCE & TECHNOLOGY, CO., LTD.	TIANJIN YIFANG SCIENCE & TECHNOLOGY, CO., LTD.	TIANJIN YIFANG SCIENCE & TECHNOLOGY, CO., LTD.	TIANJIN YIFANG SCIENCE & TECHNOLOGY, CO., LTD.	TIANJIN YIFANG SCIENCE & TECHNOLOGY, CO., LTD.	TIANJIN YIFANG SCIENCE & TECHNOLOGY, CO., LTD.	Tianjin Rmwell Chemical Industry Limited Company	Tianjin Rmwell Chemical Industry Limited Company	Tianjin Rmwell Chemical Industry Limited Company	Tianjin Rmwell Chemical Industry Limited Company	Tianjin Rmwell Chemical Industry Limited Company
Pyridine	lot No.	L125080127	L125080127	L125080127	L125080127	L125080127	L125080127	L125080127	L125080127	L125080127	L125080127	L125110406	L125080127	L125080127	L125080127	L125080127	L125080127
	Supplier	Tianjin Rmwell Chemical Industry	Tianjin Rmwell Chemical Industry	Tianjin Rmwell Chemical Industry	Tianjin Rmwell Chemical Industry	Tianjin Rmwell Chemical Industry	Tianjin Rmwell Chemical Industry	Tianjin Rmwell Chemical Industry	Tianjin Rmwell Chemical Industry	Tianjin Rmwell Chemical Industry	Tianjin Rmwell Chemical Industry	Shandong Yuanrun Chemical Co., Ltd.	Tianjin Rmwell Chemical Industry	Tianjin Rmwell Chemical Industry	Tianjin Rmwell Chemical Industry	Tianjin Rmwell Chemical Industry	Tianjin Rmwell Chemical Industry

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Materials	Runs	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
		Limited Company	Limited Company	Limited Company	Limited Company	Limited Company	Limited Company	Limited Company	Limited Company	Limited Company	Limited Company		Limited Company	Limited Company	Limited Company	Limited Company	Limited Company
Piperidine	lot No.	L125110087-m-TF30	L125110087-m-TF30	L125110087-m-TF30	L125110087-m-TF30	L125110087-m-TF30	L125110087-m-TF30	L125110087-m-TF30	L125110087-m-TF30	L125110087-m-TF30	L125110087-m-TF30	L125100624	L125110087-m-TF30	L22602023	L22602023	L22602023	L22602023
	Supplier	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Hebei Nuorukang Pharmaceutical Technology Co., Ltd.	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Bohuachemical reagents Co.,Ld	Tianjin Bohuachemical reagents Co.,Ld	Tianjin Bohuachemical reagents Co.,Ld	Tianjin Bohuachemical reagents Co.,Ld
Oxyma	lot No.	L225111981	L225111981	L225111981	L225111981	L225111981	L225111981	L225111981	L225111981	L225111981	L225111984	L225111984	L225120161	L225111942	L225120161	L225120161	L225120161
	Supplier	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Huangganglu Pharmaceutical Co, Ltd.	Huangganglu Pharmaceutical Co, Ltd.	Liren Media Technology	GL Biochem (Shanghai) Lt	Liren Media Technology	Liren Media Technology	Liren Media Technology
DIC	lot No.	L225111983	L225111983	L225111983	L225111983	L225111983	L225111983	L225111983	L225111983	L225111983	L225111983	L125120162	L225120001	FJL124070059-012	FJL124070059-012	L225120001	L225120001
	Supplier	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Liren Medical Technology	Zheiang BulkChemical Co.,Ld	Suzhou Highfine Biotech Co., Ltd	Suzhou Highfine Biotech Co., Ltd	Zheiang Bulk Chemical Co.,Ld	Zheiang Bulk Chemical Co.,Ld
DMF	lot No.	L125120281-m-TF01	L125120281-m-	L125120281-m-	L125120281-m-TF01	L125120281-m-TF01	L125120281-m-	L125120281-m-	L125120281-m-	L125120281-m-	L125120281-m-	L125120281-m-	L125120281-m-	L125120281-m-	L125120281-m-	L226012197	L226012197

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Materials	Runs	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
			TF01	TF01			TF01	TF01	TF01	TF01	TF01	TF01	TF01	TF01	TF01		
	Supplier	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.
IPAC	lot No.	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L125120520	L125090783-m-TF03	L125090783-m-TF03	L125090783-m-TF03	L226011711	L226011711
	Supplier	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Tairong Chemical Trading Co., Ltd	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.
IPA	lot No.	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L125080512	L226010079	L226010079
	Supplier	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.	Tianjin Yubo Fine Chemical Co., Ltd.
MTBE	lot No.	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L125110298-m-TF05	L225111835	L225111835
	Supplier	Tianjin Tairong	Tianjin Tairong	Tianjin Tairong	Tianjin Tairong	Tianjin Tairong	Tianjin Tairong	Tianjin Tairong	Tianjin Tairong	Tianjin Tairong	Tianjin Tairong	Tianjin Tairong	Tianjin Tairong	Tianjin Tairong	Tianjin Tairong	Tianjin Yubo	Tianjin Yubo

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Materials	Runs	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
		Chemical Trading Co., Ltd	Chemical Trading Co., Ltd	Chemical Trading Co., Ltd	Chemical Trading Co., Ltd	Chemical Trading Co., Ltd	Chemical Trading Co., Ltd	Chemical Trading Co., Ltd	Chemical Trading Co., Ltd	Chemical Trading Co., Ltd	Chemical Trading Co., Ltd	Chemical Trading Co., Ltd	Chemical Trading Co., Ltd	Chemical Trading Co., Ltd	Chemical Trading Co., Ltd	Fine Chemical Co., Ltd.	Fine Chemical Co., Ltd.

Table 14.2 Summary of all Material Lots, Suppliers and CoA for Step 1 PAR Study

Raw material	Asymchem/ Supplier Lot	Supplier	COA
Sieber amide resin	L125060454	Tianjin Nankai Hecheng Science & Technology Co., Ltd	L125060454.pdf
	L225120836	Sunresin New Materials Co., Ltd	10-240205.pdf
	L225120950		L225120950.pdf
Fmoc-Tyr(tBu)-OH	L124070130	Chengdu ZY Biochemical Technoliogy Co.,Ltd	L124070130
Oxyma	L225111981	Suzhou Highfine Biotech Co., Ltd	L225111981.pdf
	L225111984	Huanggang Luban Pharmaceutical Co., Ltd.	L225111984.pdf
	L225120161	Liren Medical Technology	L225120161.pdf
DIC	L124070059	Suzhou Highfine Biotech Co., Ltd.	L124070059.pdf

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	L225111983		
	L225120001	Zhejiang Bulk Chemical Co.Ltd	DIC20251103..pdf
	L225120162	Liren Medical Technology	N,N'-二异丙基碳二亚胺（693-13-0
Fmoc-L-Thr(tBu)-OH	L225120052	Chengdu ZY Biochemical Technoliogy Co.,Ltd	Fmoc-L-Thr(tBu)-OH ZYF07025041
	L225111436	Liaoning Asymchem Laboratories Co.,Ltd	YL003563.pdf
	CPr138389-05-04P-01		CPr138389-05-04-01-a COA-01-生3
Fmoc-L-Asn(Trt)-OH	L125070301	Chengdu ZY Biochemical Technoliogy Co.,Ltd	L125070301.pdf
	L225120049		L225120049.pdf
	CPs138519-03-02p-02		L225111704.pdf
Fmoc-L-Ser(tBu)-OH	L225120045	Chengdu ZY Biochemical Technoliogy Co.,Ltd	L225120045.pdf
	L125040586	Liaoning Asymchem Laboratories Co.,Ltd	L125040586.pdf
Fmoc-L-Gly-OH	L225120048	Chengdu ZY Biochemical Technoliogy Co.,Ltd	L225120048.pdf
	L225111702	Liaoning Asymchem Laboratories Co.,Ltd	L225111702.pdf
	L225111434		

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	CPr138200-04-01-01		
Fmoc-L-Val-OH	CPs138202-02-01-01	Liaoning Asymchem Laboratories Co.,Ltd	L225111703.pdf
	CPr138202-04-01-01		CPr138202-04-01-01
	L126010746	Chengdu ZY Biochemical Technoliogy Co.,Ltd	L126010746.pdf
	L225120043		L225120043.pdf
Fmoc-L-Glu(OtBu)-OH.H2O	L125070053	Liaoning Asymchem Laboratories Co.,Ltd	L125070053.pdf
Fmoc-Leu-Pro-Pro-OH	L125090534	Jilin Asymchem Laboratories Co.,Ltd	L125090534.pdf
Ac_2O	L125080638	TIANJIN YIFANG SCIENCE & TECHNOLOGY, CO., LTD.	L125080638.pdf
	L125080417	Tianjin Rmwell Chemical Industry Limited Company	L125080417.pdf
pyridine	L125080127	Tianjin Rmwell Chemical Industry Limited Company	L125080127.pdf
ASYM-140400/LY2000	CPr140400-01-03-02	Asymchem Laboratories (Tianjin) Co.,Ltd (TJ3)	COA of LY2000 FMOC-LYS (CPr14
Fmoc-Pro-OH	CPr140730-01-01-03	Liaoning Asymchem Laboratories Co.,Ltd	CPr140730-01-01-03 COA-01-生效
	L225120917	ChengDu Chron Chemicals Co,.Ltd	L225120917.pdf
	L225121595	Chengdu ZY Biochemical Technoliogy Co.,Ltd	Package

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	L225120917	Zenji Pharmaceuticals(SU ZHOU) Ltd	L225120917
Piperidine	L125110087	Tianjin Tairong Chemical Trading Co., Ltd	L125110087.pdf
	L226020623	Tianjin Bohua chemical reagents Co., Ltd	L226020623.pdf
DMF	L125120281	Tianjin Tairong Chemical Trading Co., Ltd	L125120281.pdf
	L226012197	Tianjin Yubo Fine Chemical Co., Ltd	L226012197.png
IPAC	L125090783	Tianjin Yubo Fine Chemical Co., Ltd	L125090783.pdf
	L226011711		L226011711.png
IPA	L125080512	Tianjin Yubo Fine Chemical Co., Ltd	L125080512.pdf
	L226010079		L226010079.png
	L125120520	Tianjin Tairong Chemical Trading Co., Ltd	L125120520.pdf
MTBE	L125110298	Tianjin Tairong Chemical Trading Co., Ltd	L125110298.pdf
	L225111835	Tianjin Yubo Fine Chemical Co., Ltd	L225111835.png

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15. Appendix VI Yield Data Summary for Peptidyl-Resin and TCA-Cleaved Isolated Product

PAR study Number	Step 1					Step 2 cleaved by TCA					
	Scale (Resin amount*Sub)	Net weight of peptide-on-resin (M1)	Theoretical yield (M0)	LOD	Gain-in Yield Uncorrected with LOD	Input peptidyl-on-resin	Output L1/ ASYM-140019 (M2)	Assay	Two steps overall Theoretical yield (M3)	Product total weight in sample	Two steps overall yield
Run1	60*0.75 (Resin: Nankai)	176.33 g	169.63 g	7.58%	93.9%	30g	16.04 g	89.7248%	20.48 g	1.3170	76.69%
Run2	250*0.75 (Resin: Nankai)	726.6 g	706.79 g	7.36%	92.9%	30g	15.42 g	88.9648%	20.71 g	1.5096	73.52%
Run3	60*0.75 (Resin: Nankai)	181.9 g	169.63 g	7.14%	99.3%	30g	15.23 g	88.3014%	19.86 g	1.4136	74.85%
Run4	60*0.75 (Resin: Nankai)	174.08 g	169.63 g	7.41%	92.3%	30g	16.69 g	87.7379%	20.75 g	1.4867	77.74%
Run5	60*0.75 (Resin: Nankai)	177.19 g	169.63 g	6.67%	96.1%	30g	15.75 g	89.2358%	20.38 g	1.7488	77.53%
Run6	60*0.75 (Resin: Nankai)	175.44 g	169.63 g	6.93%	94.2%	30g	16.80 g	88.5117%	20.59 g	1.3044	78.56%
Run7	60*0.74 (Resin: Sunresin)	169.78 g	168.17 g	6.93%	90.6%	30g	16.87 g	89.3277%	20.99 g	1.4198	78.56%
Run8	60*0.88 (Resin: Sunresin)	193.22 g	188.63 g	7.01%	93.0%	30g	17.12 g	90.2204%	21.93 g	1.4211	76.90%
Run9	60*0.75 (Resin: Nankai)	177.9 g	169.63 g	7.14%	96.0%	30g	16.28 g	90.1435%	20.30 g	1.3882	79.12%
Run10	60*0.75 (Resin: Nankai)	178.92 g	169.63 g	5.99%	98.7%	30g	15.61 g	90.4294%	20.19 g	1.5769	77.74%
Run11	60*0.75 (Resin: Nankai)	182.15 g	169.63 g	7.09%	99.6%	30g	15.41 g	90.9275%	19.83 g	1.4945	78.20%
Run12	250*0.75 (Resin: Nankai)	748.68 g	706.79 g	6.73%	98.1%	30g	16.10 g	90.4415%	20.10 g	1.5708	80.25%

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PAR study Number	Step 1					Step 2 cleaved by TCA					
	Scale (Resin amount*Sub)	Net weight of peptide-on-resin (M1)	Theoretical yield (M0)	LOD	Gain-in Yield Uncorrected with LOD	Input peptidyl-on-resin	Output L1/ ASYM-140019 (M2)	Assay	Two steps overall Theoretical yield (M3)	Product total weight in sample	Two steps overall yield
Run13	60*0.75 (Resin: Nankai)	174.27 g	169.63 g	5.04%	96.2%	30g	16.33 g	91.9627%	20.73 g	1.5288	79.83%
Run14	60*0.75 (Resin: Nankai)	179.15 g	169.63 g	6.62%	97.9%	30g	15.91 g	87.2942%	20.16 g	1.5010	76.33%
Run15	60*0.75 (Resin: Nankai)	179.08 g	169.63 g	7.28%	96.7%	30g	16.29 g	90.1338%	20.17 g	1.4815	80.14%
Run16	60*0.75 (Resin: Nankai)	179.73 g	169.63 g	6.98%	97.8%	30g	16.67 g	91.7385%	20.10 g	1.6063	84.09%

Note [7]: 1. Step 1 Gain-in Yield Uncorrected with LOD = ((M1 (net weight of peptide-on-resin)(1-LOD)-Resin amount)/ (M0 (theoretical yield) -Resin amount) *100%*

*Step 1 Theoretical calculation formula (M0) = Resin amount + Resin amount*Sub × 2436.23/1000 kg/mol*

*2. Two steps overall yield (Step 1 & 2) = (M2 * Assay+ Product total weight in sample) / M2*100%*

*The theoretical yield of Step 1&2 M3 is calculated: M3= Resin amount*Sub * FW of L1 / 1000 = Input peptidyl-on-resin/ Net weight of peptide-on-resin (M1)* Resin amount*Sub*2675.51/1000 kg/mol*

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